

The Binary Totient Series

*Results, proof dependencies and remaining arithmetic conditions
for Erdős Problem #249*

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ABSTRACT

The dyadic sections of Euler’s totient through level $e \geq 1$ have an explicit rational basis of size $2^e + 1$. This record gives the detailed statements behind that result and examines what coefficient structure can say about $S = \sum_{n \geq 1} \varphi(n)2^{-n}$. Truncating a scaled tail difference produces an integer whose residue certifies nonintegrality when its distances from both endpoints of the residue interval exceed an explicit error bound. We develop the resulting finite denominator exclusions, equivalent residue criteria and sufficient exponential-sum estimates. Rational comparison sequences identify the limitations of particular size, parity, rank and finite-prefix arguments. The short paper supplies the all-base basis theorem and its integral relations. Here each conditional result is accompanied by the arithmetic hypothesis still needed to apply it; these hypotheses are not established in the form required to prove irrationality of S .

Reading guide

The short paper gives a direct proof of the basis theorem and its integral normal form. This longer record is organised for reference. The opening section introduces the finite residue test and its quantified form, Definition 1.2. Section 2 gives rational comparison sequences, and Section 3 states four possible arithmetic conditions for irrationality. The detailed formulas then begin with the series identities and denominator bound in Section 5.3. The basis and relation modules are revisited in Section 10.8.

Two distinctions recur. “Cofinal” means that a condition holds at some index beyond each threshold; it does not mean that it holds at every sufficiently large index. A counterexample excludes only the hypotheses it satisfies. Agreement with a finite totient prefix, for example, need not preserve multiplicativity.

The annotations identify the kind of support: [Lean] refers to a formal statement and its pinned source or supplied checking record; [Cert] to a finite computation; [Math] to an ordinary argument; [Cited] to a literature result; and [Open] to an unproved assertion. The Lean kernel was not rerun for this revision. In particular, a checked implication does not establish its hypothesis.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Contents

1	Totient sections and tail differences	3
2	Rational comparison sequences	6
3	Conditions that would imply irrationality	13
4	Guide to the detailed statements	18
5	Series identities and finite exclusions	18
6	Detailed statements	25
7	Parameter ranges and quantifiers	81
8	Relations between the representations	91
9	Index of unproved conditions	94
10	Counterexamples to proposed deductions	108
11	Why the stated hypotheses cannot be omitted	121
12	Additional mathematics required by the sufficient conditions	122
13	Remaining arithmetic questions	138

1 Totient sections and tail differences

Let φ be Euler's totient function, with $\varphi(0) = 0$, and use $\mathbb{N} = \{0, 1, 2, \dots\}$. The dyadic sections $n \mapsto \varphi(2^j n + r)$, for $0 \leq j \leq e$ and $0 \leq r < 2^j$, have rational rank $2^e + 1$ when $e \geq 1$. A basis consists of $n \mapsto \varphi(n)$, $n \mapsto \varphi(2n)$, and the sections with odd r at levels $1 \leq j \leq e$. For example, $\varphi(4n) = 2\varphi(2n)$ and $\varphi(4n + 2) = \varphi(2n + 1)$ recover the omitted sections at level two. The short paper proves the corresponding rank $k^e + 1$ in every integer base $k \geq 2$, together with the integral coordinates and relations. For composite k , the retained condition is $k \nmid r$, not $\gcd(k, r) = 1$.

The question about the series concerns more than these coefficient relations. Put

$$S := \sum_{n \geq 1} \frac{\varphi(n)}{2^n} = \frac{1}{2} + \frac{1}{4} + \frac{2}{8} + \frac{2}{16} + \frac{4}{32} + \dots = 1.3676308019850223 \dots$$

The series converges absolutely because $\varphi(n) \leq n$. Erdős Problem #249 asks whether S is irrational, which remains open. The exact rank theorem controls linear relations among coefficient sections. Irrationality requires information about their weighted infinite sum. The countermodels below explain several ways that a passage from coefficient structure to the sum can fail. Each excludes its stated assumptions; they do not establish a single obstruction theorem covering all possible methods.

Remark 1.1 (Earlier coefficient results). Erdős and Graham record the irrationality question for S [8, p. 61]. An integer sequence is k -regular when the $\mathbb{Z}$ -module generated by its sections $n \mapsto c(k^j n + r)$, $j \geq 0$ and $0 \leq r < k^j$, is finitely generated. Coons proved that φ is not k -regular for any $k \geq 2$ [2, Theorem 3.2]. Martin's Theorem 1 implies that the functions $n \mapsto \varphi(a_i n + b_i)$ are linearly independent for pairwise nonproportional affine forms with positive integer slopes [4, Theorem 1]. The exact ranks in §10.8, together with the all-base normal form in the short paper, identify the retained sections and every relation among them. The independence itself is not claimed as new. The CRT–Dirichlet separation method is already present in Yazdani's Theorem 2 and proof, which explicitly credit Shallit [28, pp. 652–653]; our evaluation matrix uses one such separation for each column. Yazdani also proves non-automaticity of $\varphi(n) \bmod m$ for every $m \geq 3$ in every base [28, Cor. 4]. Allouche, Shallit and Yassawi give a survey account [29, Thm. 3, Ex. 4 and Rem. 7].

The following identity relates the series to finite sums. It also explains which finite inequalities would suffice for irrationality. Write $2^N S = \Phi_N + R_N$ with $\Phi_N := \sum_{n=1}^N \varphi(n) 2^{N-n} \in \mathbb{N}$ and $R_N := \sum_{j \geq 0} \varphi(N+1+j)/2^{j+1} \in [0, \infty)$ (LEAN SOURCE, LEAN SOURCE). If $S = a/(2^c v)$ with $c \geq 0$ and $v \geq 1$ odd, Euler's theorem gives $h = \varphi(v) > 0$ and $N_0 = c$ such that $R_{N+h} - R_N \in \mathbb{Z}$ for every $N \geq N_0$ (LEAN SOURCE). Indeed,

$$R_{N+h} - R_N \equiv 2^N (2^h - 1) S \pmod{\mathbb{Z}},$$

and the right side is integral for those choices.

Thus it suffices, for every positive shift h , to find arbitrarily large N for which $R_{N+h} - R_N$ is not an integer. A finite test is obtained by truncating this difference. Throughout, $a \bmod m$ denotes the least nonnegative residue, and an unqualified index in a sum is a nonnegative integer. For $h, N, L \in \mathbb{N}$, write

$$D(h, N, L) := \sum_{j < L} (\varphi(N+h+1+j) - \varphi(N+1+j)) 2^{L-1-j} \in \mathbb{Z},$$

the depth- L truncation of $2^L(R_{N+h} - R_N)$, and

$$C(h, N, L) :\Leftrightarrow (N+h+L+2) < D(h, N, L) \bmod 2^L < 2^L - (N+h+L+2),$$

a decidable condition ([LEAN SOURCE](#), [LEAN SOURCE](#)).

The error bound $N+h+L+2$ comes from $\varphi(m) \leq m$, so a certificate is sound: it forces $R_{N+h} - R_N \notin \mathbb{Z}$, and in fact certificates are *complete* for non-integrality ([LEAN SOURCE](#)). The resulting quantified condition is the following.

Definition 1.2 (The quantified residue condition).

$$\text{Sep} :\Leftrightarrow \forall h \geq 1 \forall N_0 \exists N \geq N_0 \exists L, C(h, N, L).$$

$\text{Sep} \Leftrightarrow S \notin \mathbb{Q}$ ([LEAN SOURCE](#)). The forward implication is also [LEAN SOURCE](#). The equivalence identifies the missing quantified input; it supplies no certificate by itself. [cofinal] [Lean]

The order of the quantifiers matters. A certificate at one (h, N, L) excludes those hypothetical rational values whose denominator divides $2^N(2^h - 1)$; it does not exclude all denominators. The condition above asks for certificates beyond every threshold for each positive h , and is therefore exactly as strong as the original irrationality assertion. Its depth must grow: the two strict residue inequalities imply $2(N+h+L+2) < 2^L$. Keeping L fixed cannot provide certificates as N grows. Neither this necessary size condition nor a finite list of successful tests supplies the required residues. [LEAN SOURCE](#)

1.1 Unconditional results

The following results require no irrationality hypothesis. The finite denominator bound concerns rational values of S ; the rank and positivity results concern its coefficients and tails. The latter do not by themselves exclude rationality.

Theorem 1.3 (Denominator exclusion from a fixed Farey window). *If $S \in \mathbb{Q}$ then its reduced denominator exceeds $Q_0 := 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}$. Equivalently, S differs from every rational number whose reduced denominator is at most Q_0 . The bound is sharp for this window: $q = Q_0 + 1$ is the exact first failing denominator. It is the denominator of the mediant of two explicit unimodular Farey neighbours.* [bounded] [Lean] [LEAN SOURCE](#) [LEAN SOURCE](#)

This is obtained from a classical Stern–Brocot gap lemma ([LEAN SOURCE](#)) applied at window $K = 240$, the largest of the recorded bounds $4838, 2^{22}$, approximately 2.49×10^{17} , and Q_0 . It is proved without assuming Sep or using the tail-residue tests. To deduce irrationality by this argument, the analogous denominator bounds would have to be unbounded as the window grows; this has not been proved. [Open]

Proposition 1.4 (Finite certificate computations). *C has been verified at: the 28 diagonal instances of the least-common-multiple diagonal through $t = 64$* ([LEAN SOURCE](#), endpoint [LEAN](#)

[SOURCE](#)); all shifts $h \in [1, 16]$ simultaneously at $(N, L) = (14, 9)$, by decide ([LEAN SOURCE](#)); and eight further period examples at $N = 300$. These are historical subsets of the examples. The supplied source also proves a complete diagonal band for $1 \leq t \leq 82$, with no omitted scales ([LEAN SOURCE](#)). This is bounded coverage, not a cofinal diagonal family. Note the quantifier order: the simultaneous small-shift example uses one pair N, L for every $1 \leq h \leq 16$. In symbols, it proves $\exists N \exists L \forall h (1 \leq h \leq 16 \Rightarrow C(h, N, L))$. In contrast, Sep requires arbitrarily large N for each positive h ; neither the shift range nor the basepoint threshold is bounded. [fixed] [Lean]

Lean: [certifiedKill_diagonal_all_imported_through_t64](#), [certifiedKill_diagonal_t64](#), [certifiedKill_all_upto_sixteen](#), [certifiedKill_67_300](#), and 8 further declarations in the [coverage section](#).

Proposition 1.5 (Positivity does not exclude an integer). Put $H_a = \text{lcm}(1, \dots, 2^a)$. For every $a \geq 8$ and $J \geq 0$ with $J + (a + 6) < 2 \cdot 2^a$,

$$R_{2H_a+J} - R_{H_a+J} > 0.$$

This needs no irrationality hypothesis ([LEAN SOURCE](#), [LEAN SOURCE](#)). If this difference is an integer and $K \geq 0$ satisfies

$$J + K + (a + 6) < 2 \cdot 2^a, \quad 2H_a + J + K + 2 < 2^K,$$

then

$$D(H_a, H_a + J, K) \bmod 2^K = 2^K - (R_{2H_a+J+K} - R_{H_a+J+K})$$

lies strictly between $2^K - (2H_a + J + K + 2)$ and 2^K ([LEAN SOURCE](#)). The later tail difference is a positive integer. Its negative, not the carry itself, is the representative near zero. Thus positivity locates the residue near the upper endpoint; it does not establish the central-residue inequalities. Both restrictions on K are part of this conclusion. [cofinal] [Lean]

Lean: [actualLcmTailDiff_shift_pos](#), [actualLcm_integral_forces_topEdgeResidue](#), [carry-Orbit_eq_tail_diff](#).

For the next two propositions, a rational value $S = a/v$, with integers a and $v \geq 1$, would give the integer carry

$$u_N = vR_N = 2^N a - v\Phi_N, \quad u_{N+1} = 2u_N - v\varphi(N+1), \quad 0 \leq u_N \leq v(N+2).$$

The formal sources call this a tempered carry. Here v is a positive multiplier, not necessarily odd. Its dyadic sections are the sequences $n \mapsto u_{2^j n + r}$.

Proposition 1.6 (Rationality forces unbounded carry rank). If S is rational then, for every e , the carry sections $n \mapsto u_{2^j n + r}$ with $1 \leq j \leq e$ and $0 \leq r < 2^j$ span a rational vector space of dimension at least $2^e - 1$ ([LEAN SOURCE](#)). The lower bound holds at every depth. It comes from the linear independence of the $2^e + 1$ retained dyadic totient sections for $e \geq 1$, proved using the Chinese remainder theorem and Dirichlet's theorem ([LEAN SOURCE](#)), so the full family spans an infinite-dimensional space ([LEAN SOURCE](#)), the case $k = 2$ of Coons's non-regularity theorem (§10.8). [uniform] [Lean]

Proposition 1.7 (Periodicity modulo an integer does not bound rational rank). *If S is rational, the same carry u has unbounded rational section rank and one eventual period modulo v valid for all its dyadic sections. More precisely, there are $h \geq 1$ and N_0 such that, for every $j, r \geq 0$ and $n \geq N_0$,*

$$u_{2^j(n+h)+r} \equiv u_{2^j n+r} \pmod{v}.$$

The rank and periodicity assertions hold together (LEAN SOURCE). This conditional theorem alone is not a counterexample to a general periodicity-to-rank implication: its antecedent is not established. The separate 5/4 control supplies a concrete counterexample to the generic rationality-driven rank ceiling; see the detailed comparison in Section 10.8. [uniform] [Lean]

Proposition 1.8 (Equivalent certificate conditions). *Put $H_t = \text{lcm}(1, \dots, t)$. Several variations of Definition 1.2 are equivalent to $S \notin \mathbb{Q}$. One may allow a positive multiple of each prescribed shift, still requiring certificates beyond every basepoint threshold (Theorem 6.48; LEAN SOURCE). Alternatively, one may use the original quantified condition (LEAN SOURCE) or restrict to $h = N = H_t$ at arbitrarily large t (Theorem 6.49; LEAN SOURCE). Replacing the symmetric residue test by*

$$N + L + 2 \leq D(h, N, L) \pmod{2^L} \leq 2^L - (N + h + L + 2)$$

also gives equivalent quantified conditions, including its restriction to $h = N = H_t$ at arbitrarily large t (LEAN SOURCE). Finally, the counted phase-separation condition in Proposition 2.7(b) is equivalent to irrationality (LEAN SOURCE). These equivalences change the form of the arithmetic question, not its logical strength. [cofinal] [Lean]

Lean: `periodMultipleKillSupply_iff_irrational`, `irrational_totient_series_iff_certificate_supply`, `irrational_totient_series_iff_lcm_diagonal_certificate_supply`, `irrational_totientSeries_iff_cofinalDirectedLcmCertificateSupply`, and 1 further declaration in the [coverage section](#).

Proposition 1.9 (A rational sequence preserving size bounds, parity and aperiodicity). *There is $c : \mathbb{N} \rightarrow \mathbb{N}$ with $c(n) \leq 6$ and $c(n) \leq n$ for all n , $c(n) \equiv \varphi(n) \pmod{2}$ for every n , and c not eventually periodic; indeed for every N, G, K there are K explicit carry pulses beyond N , pairwise separated by more than G ; and yet $\sum_n c(n)/2^n = 3/2 \in \mathbb{Q}$ (LEAN SOURCE; sum at :359, aperiodicity at :487, parity match at :194; #print axioms clean). [uniform] [Lean]*

Lean: `exists_totientParity_arbitrarilyManySeparatedCarry_rational_countermodel`, `tsum_parityCoboundaryWeight_eq_three_halves`.

These propositions separate several logical obligations. Positivity leaves an upper-endpoint condition; a large rational carry rank is not a contradiction; and exact equivalences still require a quantified certificate condition. The stated size bounds, parity and aperiodicity also hold for a rational control. Each conclusion concerns its stated hypotheses, not all possible arithmetic information about φ .

2 Rational comparison sequences

The countermodels distinguish information that determines a finite calculation from information that forces nonintegrality at unbounded indices. A fixed prefix, parity at

every index, and a specified finite collection of linear relations are different hypotheses. The useful conclusion in each case is the precise information a rational control sequence can preserve.

2.1 What finite information does not determine

The comparison sequences below preserve different parts of the totient data. Theorem 2.4 preserves an arbitrary fixed prefix. Proposition 1.9 preserves parity at every index, together with size bounds and aperiodicity. Propositions 2.13 and 2.14 address particular carry descriptions and linear constructions, not every finite-state or finite-dimensional argument.

Two other limitations do not require a rational comparison sequence. A congruence construction can increase the index so much that its tail error exceeds the residue it produces (Proposition 2.11). An exact reformulation or a finite list of examples still needs the quantified arithmetic assertion (Proposition 2.7 and Remark 2.9).

The finite certificates in Proposition 1.4 and the $K = 240$ Farey calculation in Theorem 1.3 have bounded ranges. Theorem 2.4 excludes an inference from one such fixed range uniformly over the whole coefficient class. It does not exclude a proof using the arithmetic of φ at arbitrarily large horizons. Likewise, the cofinal positivity theorem does not by itself separate the residue from the top edge: that is the additional information identified by Proposition 1.5.

2.2 A rational sequence with any prescribed totient prefix

A finite prefix leaves the remaining coefficients free. We construct a rational continuation satisfying the same linear size bound, then show why it cannot satisfy the quantified residue condition. First we record the two tail facts needed for a general coefficient sequence. For $c : \mathbb{N} \rightarrow \mathbb{N}$ with $c(n) \leq n$, write $T_c := \sum_{n \geq 1} c(n)/2^n$, $R_N^c := \sum_{j \geq 0} c(N+1+j)/2^{j+1}$, $D_c(h, N, L) := \sum_{j < L} (c(N+h+1+j) - c(N+1+j))2^{L-1-j}$, and let $C_c(h, N, L)$ and Sep_c be C and Sep with D_c in place of D . For $c = \varphi$ these are the objects of Definition 1.2.

Lemma 2.1 (A residue certificate excludes an integral tail difference). *Let $c : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $c(n) \leq n$ for all n . For $h, N, L \in \mathbb{N}$, $C_c(h, N, L) \Rightarrow R_{N+h}^c - R_N^c \notin \mathbb{Z}$. [uniform] [Lean]*
Lean: `certificate_sound`, `scaled_difference`, `truncation_error_bound`.

Proof. Splitting both tails after L terms gives

$$2^L(R_{N+h}^c - R_N^c) - D_c(h, N, L) = R_{N+h+L}^c - R_{N+L}^c.$$

Both tails on the right are nonnegative, and $R_M^c \leq \sum_{j \geq 1} (M+j)2^{-j} = M+2$. Their difference therefore has absolute value at most $N+h+L+2$. The corresponding totient bound is formalised at [LEAN SOURCE](#) and used there to prove [LEAN SOURCE](#). If $R_{N+h}^c - R_N^c = k \in \mathbb{Z}$ then $2^L k \equiv 0 \pmod{2^L}$ and D_c lies within $N+h+L+2$ of $2^L k$, so $D_c \bmod 2^L$ lies within $N+h+L+2$ of 0 or of 2^L , contradicting C_c . $\square$

Lemma 2.2 (Generic tail-period law). *Let $c : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $c(n) \leq n$ for all n . If $T_c = p/(2^e m)$ with $p \in \mathbb{Z}$, $e \geq 0$ and m a positive odd integer, and if $h \geq 1$ satisfies $m \mid 2^h - 1$, then $R_{N+h}^c - R_N^c \in \mathbb{Z}$ for every $N \geq e$. [cofinal] [Lean]*

Lean: [generic_tail_period](#), [scaled_tail_split](#).

Proof. Since $2^N T_c - R_N^c = \sum_{n=1}^N c(n) 2^{N-n}$ is an integer,

$$R_{N+h}^c - R_N^c \equiv 2^N (2^h - 1) T_c \pmod{\mathbb{Z}}.$$

The right side is integral because $2^e \mid 2^N$ and $m \mid 2^h - 1$. $\square$

Remark 2.3. Lemmas 2.1 and 2.2 are also checked for every coefficient sequence satisfying $c(n) \leq n$. The formal proofs use the true-tail recurrence and the bound $0 \leq R_N^c \leq N + 2$; the residue certificate keeps the full finite discrepancy. The tail-period proof uses $m \mid 2^h - 1$ directly, so oddness of m is only needed when choosing a period from the denominator. The following rational continuation and its finite-prefix consequence are also checked. [LEAN SOURCE](#) [LEAN SOURCE](#)

Theorem 2.4 (A rational sequence agreeing with any finite totient prefix). *Let $B \geq 1$ and $P > B$ be integers. Define $\gamma : \mathbb{N} \rightarrow \mathbb{N}$ by*

$$\gamma(n) := \varphi(n) \quad (n \leq B), \quad \gamma(n) := \begin{cases} n-1, & P \mid n \\ n, & P \nmid n \end{cases} \quad (n > B).$$

Then:

- (i) $\gamma(n) \leq n$ for all n , so γ lies in the same coefficient class as φ ;
- (ii) $\gamma(n) = \varphi(n)$ for every $n \leq B$, and consequently $D_\gamma(h, N, L) = D_\varphi(h, N, L)$ for every (h, N, L) with $N + h + L \leq B$;
- (iii) $T_\gamma = 2 - \sum_{n \leq B} (n - \varphi(n)) / 2^n - 1 / (2^P - 1) \in \mathbb{Q}$, and the odd part of its reduced denominator is exactly $2^P - 1$;
- (iv) Sep_γ is false; indeed it fails already at $h = P$.

Hence for every B there is a coefficient sequence in the same class, agreeing with φ on all of $[1, B]$, whose quantified condition is false. [uniform] [Lean]

Lean: [gamma_le](#), [gamma_prefix](#), [discrepancy_prefix](#), [exact_series](#), and 4 further declarations in the [coverage section](#).

Proof. (i) is immediate; (ii) holds because $D_\varphi(h, N, L)$ reads φ only at indices $\leq N + h + L$. For (iii), since $P > B$ every multiple of P exceeds B , so

$$T_\gamma = \sum_{n \geq 1} \frac{n}{2^n} - \sum_{n \leq B} \frac{n - \varphi(n)}{2^n} - \sum_{k \geq 1} 2^{-kP} = 2 - \sum_{n \leq B} \frac{n - \varphi(n)}{2^n} - \frac{1}{2^P - 1}.$$

The first two terms have denominator dividing 2^B . On putting the expression over $2^B(2^P - 1)$, its numerator is congruent to -2^B modulo $2^P - 1$. Since $\gcd(2^B, 2^P - 1) = 1$, no odd factor cancels. Thus the reduced denominator has odd part exactly $2^P - 1$ and its power-of-two part divides 2^B . For (iv), apply Lemma 2.2 with $h = P$: $R_{N+P}^\gamma - R_N^\gamma \in \mathbb{Z}$ for every $N \geq B$, so by Lemma 2.1 no $C_\gamma(P, N, L)$ holds for any $N \geq B$ and any L . Taking $N_0 = B$ refutes the inner existential of Sep_γ at $h = P$. The formalisation checks the series value, the exact reduced denominator, and the failure of the separation condition for this same γ . [LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#) $\square$

Corollary 2.5 (The limit of a finite-prefix argument). *No proof rule uniform over all $c : \mathbb{N} \rightarrow \mathbb{N}$ with $c(n) \leq n$ can establish Sep from a single fixed prefix $\{c(n) : n \leq B\}$: Theorem 2.4 supplies a rational countermodel with that same prefix. This does not invalidate an argument that uses the fixed arithmetic sequence φ together with compatible information at arbitrarily large horizons; the theorem gives a different γ_B for each B , not one sequence agreeing with φ at every B .*

Lean: `no_uniform_prefix_rule, gamma_not_separation`.

[LEAN SOURCE](#)

Remark 2.6 (Four distinct finite restrictions). Theorem 2.4 concerns the information in a finite prefix. It does not identify φ with one of the comparison sequences. It also does not exclude a proof that combines finite computation with an arithmetic theorem applying at all larger indices. Four different restrictions must be kept separate: a bound on the indices inspected, a bound on the depth L , a bound on the carry states retained, and a bound on matrix rank. The depth inequality $2(N + h + L + 2) < 2^L$, for example, concerns the second restriction, not all four.

Proposition 1.9 gives another rational comparison sequence, this time preserving parity and specified aperiodicity properties. The two examples test different assumptions and must not be combined into a single stronger example.

2.3 Limits of specific reductions and estimates

The following results concern specified assumptions, not all possible proof methods. Statements labelled as an audit describe the finite set of arguments examined in that audit. They are not mathematical impossibility theorems.

Proposition 2.7 (Three particular equivalences). (a) Certificate completeness. *The complete residue tests considered here are equivalent to the corresponding nonintegrality assertions. Rewriting the quantified condition using one of these equivalences does not weaken it* ([LEAN SOURCE](#), forward at :132, converse at :143).

(b) A selectable two-point sample. *The condition is as follows. For every $h \geq 1$ and X_0 , choose $X, L \in \mathbb{N}$ with $X \geq \max(X_0, 1)$ and $16(2X + h + L + 2) \leq 2^L$, a nonempty set $T \subseteq [X, 2X) \cap \mathbb{N}$, a set $P \subseteq T \times T$ of ordered pairs, and a real number $\delta \geq 0$. With $E_N = \exp(2\pi i D(h, N, L)/2^L)$, require*

$$|E_i - E_j| \geq \delta \quad ((i, j) \in P), \quad \frac{2|T|^2}{5} \leq |P|\delta^2.$$

This condition is equivalent to irrationality. For the converse, irrationality and the doubling map allow $T = \{N, N + 1\}$ and $P = \{(N, N + 1), (N + 1, N)\}$ at arbitrarily large N , with $\delta = 9/10$. The numerical requirement is then $8/5 \leq 2(9/10)^2$. The freedom to select T matters: this is not an estimate on a sample prescribed in advance, such as all prime positions.

Formal sources: [LEAN SOURCE](#)

(c) Four integral tail differences. For $H \geq 0$ and positive integers p, q , the conjunction

$$R_{2kH} - R_{kH} \in \mathbb{Z} \quad (k \in \{1, p, q, pq\})$$

is equivalent to $R_{2H} - R_H \in \mathbb{Z}$ alone. The affine transport identity for $H \mapsto kH$ preserves integrality, so the three additional conditions add no restriction. Neither p nor q need be prime ([LEAN SOURCE](#)).

These three equivalences explain why these particular reformulations retain the original arithmetic question. They do not rule out useful weaker intermediate lemmas. The full-block and fixed-margin conditions in §3 ask for additional quantitative information that the three arguments above do not supply. [cofinal] [Lean]

Lean: [three_particular_equivalences](#), [two_point_sample_numerical_requirement](#).

Remark 2.8. Second-difference certificates need not occur at a smaller depth than first-difference certificates. At $(h, N) = (1, 8)$, the first-difference test holds at depth 8, whereas no second-difference certificate exists at a depth at most 8 (certificate kernel source:18762, [Lean]). This one example refutes a uniform claim that second differences always save depth; it does not give a uniform ordering in the opposite direction.

Separately, unbounded growth of the denominator exclusions produced by the Farey windows would imply irrationality. The record does not establish a converse for that specific sequence of windows, so the proposed growth condition is used only as a sufficient condition. [Math]

Remark 2.9 (A finite audit of proposed extensions). This is a report of a finite audit. The audit listed 56 candidate deductions for four stated goals and examined 18 proposed extensions. Its recorded verdict was 16 not accepted as routine extensions and 2 accepted only as restatements for a wider target family. One of the 16, labelled NM-12 in the audit, was not verified because the endpoint proofs were not read. Thus it is not correct to say that all 18 proof bodies were checked or that all 16 negative verdicts were proved. The recorded categories were hypothesis strength (6), parameter range (5), representation (4), and multiple issues (3). Neither accepted extension supplied the arithmetic assertion needed for irrationality. No conclusion about arguments outside this finite audit follows. [n/a] [Cert]

Observation 2.10 (What the finite audit observed). Several unsuccessful arguments in the audit sought one certificate at a specially chosen large index, whereas the exponential-sum criterion seeks cancellation across an entire block. This is a difference in the statements being attempted, not evidence that one approach is the only possible one. The older list ending at $t = 64$ is not the current finite record: the supplied sources certify every diagonal scale $1 \leq t \leq 82$. Neither list proves the required assertion for arbitrarily large scales.

Proposition 2.11 (Failure of a specified two-adic congruence construction). *A certificate requires the residue to lie farther than $N + h + L + 2$ from either endpoint modulo 2^L . In the specific construction below, a pulse places the residue at 2^{K-1} modulo 2^K , but the defining congruence is $p \equiv 1 + 2^{K-1} \pmod{2^K}$. Hence $p \geq 1 + 2^{K-1}$. At $N = p - K$, $h = H$, and $L = K$, the error bound is $p + H + 2 > 2^{K-1}$, so the residue does not satisfy the certificate inequalities. This*

calculation defeats this construction at every depth. It does not prove a corresponding statement for every use of the Chinese Remainder Theorem or every prescribed totient pattern. [cofinal] [Lean] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: [twoAdic_pulse_construction_never_certifies](#), [twoAdic_pulse_defining_congruence](#), [twoAdic_pulse_error_bound](#).

Remark 2.12 (Two further prescribed patterns). Two related results concern particular prescribed patterns. Under the stated room condition, the complete terminal divisibility pattern is impossible: its last totient difference would be positive, below a modulus, and divisible by that modulus. For the pattern with the last condition removed, the penultimate difference is forced to be 2^{m-1} , with $2^m < 2(2H + J + K + 2)$.

A different sufficient condition would prescribe the accumulated sum rather than an individual difference: $D(h, N, L) \equiv 2^{L-1} \pmod{2^L}$ and $2^{L-1} > N + h + L + 2$. These two inequalities would give a certificate. They have not been supplied at the required quantified scope. The failure of the earlier patterns does not prove that this replacement is necessary. [Math]

Formal sources: [LEAN SOURCE](#)

Proposition 2.13 (Information lost by specific carry descriptions). (i) Fix $m \geq 2$ and write $R = \lfloor (m + 1)/2 \rfloor$. For $r = 0, \dots, R$, let c_r vanish except at $c_r(m) = R - r$ and $c_r(m + 1) = 2r$. Each sequence satisfies $0 \leq c_r(n) \leq n$, has the same binary sum $R2^{-m}$, and has the same coefficients and scaled tails before position m . Its scaled tail at position m is r . The common history therefore does not determine that tail. A finite set of labels that determines the tail for each member of this family must contain at least $R + 1 = \lfloor (m + 1)/2 \rfloor + 1$ elements, one for each value of r ([LEAN SOURCE](#)).

(ii) Reduction modulo 2^L forgets the initial value after the same L recurrence steps: two affine binary orbits with different seeds satisfy $\text{orbit}_u(L) - \text{orbit}_v(L) = 2^L(u_0 - v_0)$, so the endpoint residue mod 2^L is independent of the initial carry ([LEAN SOURCE](#)).

(iii) Fix a precision $u \geq 1$, a finite list of nonnegative valuations v_j and odd integers a_j , and an initial integer e_0 . There are integers z_j such that

$$c_j = 2^{v_j}(a_j + 2^u z_j), \quad e_{j+1} = 2e_j + c_j, \quad |e_{j+1}| \leq 2^{v_j+u-1}.$$

At each step, choose the centred representative of $2e_j + 2^{v_j}a_j$ modulo 2^{v_j+u} ([LEAN SOURCE](#)). The unrestricted integers z_j are essential: the assertion does not prescribe the full coefficients c_j .

These statements exclude decoders using only the specified common history, distinctions using only the endpoint residue, and contradictions using only the stated local symbols. They do not prove that the actual totient expansion cannot be studied by automata. A proposed application must verify that it uses no additional arithmetic information that the comparison families fail to preserve. [uniform] [Lean]

Lean: [balancedPulse_common_history](#), [balancedPulse_tail_at](#), [balancedPulse_series](#), [balancedPulse_label_lower_bound](#), and 3 further declarations in the [coverage](#) section.

Proposition 2.14 (Four limits of particular linear constructions). *The following four constructions have different limitations. They must not be read as a claim that every finite family of totient sections is independent: the full family has the explicit relations described in the short paper.*

- (i) Dyadic sections and an integer identity. *The retained family of $2^e + 1$ dyadic sections is linearly independent over $\mathbb{Q}$ for $e \geq 1$ (Proposition 1.6). Separately, positive integers Q, v and integers A, b cannot satisfy*

$$A \neq 0, \quad QvA = b, \quad |b| < Qv,$$

because $|A| \geq 1$ gives $|b| \geq Qv$ (LEAN SOURCE). This elementary incompatibility concerns the displayed requirements; it does not exclude every argument using an adjugate matrix.

- (ii) Möbius incidence. $U_N(i, j) = \mu((i+1)/(j+1))$ when $(j+1) \mid (i+1)$ and 0 otherwise is lower triangular with unit diagonal, so $\det U_N = 1$ for every N . Hence multiplication by U_N is injective; the corresponding coefficient transformation cannot create a nonzero vector in its kernel (LEAN SOURCE, injectivity of the companion map at :77).
- (iii) Adjugate reconstruction. Let $w_i \in \mathbb{Q}$ and $x_i \in \mathbb{N}$ be finite families with $\sum_i w_i \varphi(x_i) = 1$. Reconstructing each value from two tails and bounding them separately gives the error bound $\sum_i |w_i|(3x_i + 4)$. It is at least 3, since

$$1 \leq \sum_i |w_i| \varphi(x_i) \leq \sum_i |w_i| x_i.$$

Consequently this particular bound cannot be less than 1, whatever the size of the evaluation matrix (LEAN SOURCE, closure at :307).

- (iv) Finite combinations of shifts. *The synthetic sequence in Observation 6.115 has the prescribed differences $a_{(q-1)H-1} = \varphi(H)$ for $2 \leq q < t$, where $H = \text{lcm}(1, \dots, t)$. It is of the form $a_i = 2c_i - c_{i+1}$, and every finite integer combination of its shifts has the same form with a correspondingly shifted state. Thus these linear operations alone do not remove the compatible carry recurrence. Their uniform bounds depend on the absolute coefficient sum, as made explicit in that observation. The construction does not assert that a_i equals the actual totient difference at other indices.*

A further limitation concerns quotients of the finite sums $t(Y, r) = \sum_{d=1}^Y \mu(d)/(2^d - 1)^r$. For $e \geq 1$ and $Y \geq 4$,

$$\frac{t(Y, e+2)^2}{t(Y, 2e+2)} - (S - \frac{1}{2}) > \frac{1}{480}.$$

The proof bounds each infinite sum $\sum_{d \geq 1} \mu(d)/(2^d - 1)^r$, $r \geq 3$, between 1429/1512 and 1, and its truncation error after $Y \geq 4$ terms by 1/3584. Thus the lower bound holds for every stated pair e, Y , not just a finite list of computed examples (LEAN SOURCE). These results concern the listed matrices, estimates, and comparison sequence. The fourth item has the explicit construction and source theorem given in Observation 6.115; it has not been replayed here. A

different finite-dimensional argument, or one using further arithmetic assumptions, is not excluded. Proposition 1.7 is conditional on rationality of S and is not, by itself, a counterexample. The separate rational $5/4$ comparison supplies the counterexample to the generic rank bound. [uniform] [Lean]

Lean: `b6_retained_dyadic_sections_independent`, `b6_compressed_adjoint_identity_impossible`, `b6_mobius_incidence_unimodular_and_injective`, `b6_adjugate_tail_cost_floor`, and 6 further declarations in the [coverage section](#).

Proposition 2.15 (The coefficient properties of the rational example). *The rational sequence in Proposition 1.9 satisfies uniform boundedness, $c(n) \leq n$, agreement with $\varphi(n)$ modulo 2 at every index, and the stated separated-carry form of aperiodicity. Consequently, those properties alone cannot imply irrationality of a dyadic series. The example does not exclude arguments using further totient identities, including its multiplicative relations.* [uniform] [Lean]

Lean: `exists_rational_parityComparison`, `parityComparisonProperties_do_not_imply_irrational`.

2.4 What the counterexamples establish

The counterexamples isolate several distinctions. Finite prefix data are not a statement about all larger indices. A congruence construction can produce a large residue while producing an even larger error bound. Periodicity modulo an integer does not control rational rank. An exact reformulation leaves the original arithmetic assertion to be proved.

These are limitations of specified arguments, not an exhaustive classification of possible methods. In particular, an averaging estimate is one possible source of the missing information, not the only possible source. A new arithmetic identity or a propagation theorem could also be useful. The next section describes the precise assumptions of several approaches considered here.

3 Conditions that would imply irrationality

Four conditions considered below would imply irrationality. The first three ask for quantitative estimates; the fourth is an exact reformulation of irrationality. We also record a generic rank assertion that is false, to prevent its being mistaken for an open sufficient hypothesis. None of this is an exhaustive list of approaches. For each condition the question is whether its particular arithmetic hypothesis can be proved.

3.1 Cancellation on a full block

Definition 3.1 (The full-block norm condition). Write $e(x) := \exp(2\pi ix)$ and $E(h, N, L) := e((D(h, N, L) \bmod 2^L)/2^L)$. The block norm condition asserts that

$$\forall h \geq 1 \forall X_0 \exists X, L : \max(X_0, 1) \leq X, \quad 16(2X + h + L + 2) \leq 2^L, \quad \left\| \sum_{N \in [X, 2X)} E(h, N, L) \right\| \leq \frac{21}{25} X.$$

It implies $S \notin \mathbb{Q}$. [cofinal] [Lean] [LEAN SOURCE](#) [LEAN SOURCE](#)

A bound on the real part also suffices: it is enough that $\sum_{N \in [X, 2X)} \operatorname{Re} E(h, N, L) \leq \frac{9}{10} X$ ([LEAN SOURCE](#), the elementary real-part estimate; the implication from 21/25 to 9/10 is [LEAN SOURCE](#)). Unpacked, the required object is a constant-saving cancellation estimate for the dyadically weighted totient-difference exponential sum

$$\sum_{X \leq N < 2X} e\left(\frac{\sum_{j < L} (\varphi(N+h+1+j) - \varphi(N+1+j)) 2^{L-1-j}}{2^L}\right), \quad L \text{ subject to the displayed room inequality.}$$

No theorem in the supplied record establishes this estimate with the required quantifiers. The displayed room inequality imposes a lower bound on L ; it does not require $L = \log_2 X + O(1)$. A depth of that order would be a possible regime for the estimate, not an additional proved conclusion. A finite instance would not suffice. The norm hypothesis requires cancellation: if all phases are 1, the norm is X and the bound fails; a balanced set of opposite phases has sum zero. If instead every phase is -1 , the real part is $-X$ and satisfies the real-part bound, while the norm is X and fails the norm bound. Thus a real-part estimate is strictly weaker for general phase families; these examples do not distinguish the two quantified conditions for the actual totient sequence. The same depth L must work across the whole block. [Open]

Comparison with the preceding counterexamples. The earlier counterexamples do not establish or refute this block estimate. The following distinctions explain why those particular arguments do not settle it.

- The parameters grow. The room condition forces the truncation depth to increase with X , so this is not a test of a fixed prefix.
- The sample is prescribed: all $N \in [X, 2X)$ are summed at one depth. Choosing two favourable indices does not establish this block estimate.
- The sum uses the accumulated totient differences. It does not prescribe individual values through a congruence construction.
- The estimate concerns the actual phases. Neither a bound on carry rank nor agreement only in parity gives it.
- The proved implication uses the real-part threshold 9/10 and the room constant 16. The stronger norm bound 21/25 implies that real-part bound. These constants do not assert any distribution theorem for the totient phases.

Formal sources for these implications: [LEAN SOURCE](#) [LEAN SOURCE](#)

Remark 3.2 (The finite-prefix example does not choose a method). The finite-prefix example does not select an analytic method. It shows only that a proof valid for every sequence $0 \leq c(n) \leq n$ cannot deduce the required assertion from one fixed prefix. An argument using additional identities of φ , a theorem propagating a finite base case, a Farey estimate, or a suitable exponential sum could provide information not contained in that prefix. The earlier finite audit cannot rank all such possibilities.

Remark 3.3 (Finite certificate observations). In the historical table of 28 diagonal certificates through $t = 64$, the reported depths exceed the necessary depth bound by a

small additive constant. A separate enumeration at the indices where the least common multiple increases records a closest margin from the central interval's boundary of approximately 0.000221 of the modulus, at $t = 100$. Both are finite observations. Neither establishes nor refutes equidistribution or the estimate at arbitrarily large scales. [Cert]

3.2 A four-term bound for the real part

The next condition separates one totient value from each phase, then subtracts a mean on each group with the same cofactor. The definitions below make all four sums explicit.

Fix $h, s \geq 1$, $L \geq s + h$, $X \geq 1$ and $0 < \eta < 1$, and put $t = L - s + 1$. For $X \leq N < 2X$, let p_N be the largest prime factor of $N + t$ and put $m_N = (N + t)/p_N$. Assign the index N to this factorisation when

$$m_N \leq \left\lfloor \frac{\sqrt{X}}{2} \right\rfloor, \quad p_N > 2\lfloor \sqrt{X} \rfloor.$$

Write A for the assigned indices and $\mathcal{G} = \{N \in A : \varphi(m_N) \geq \eta m_N\}$. The two inequalities ensure $p_N > m_N \geq 1$, so $\varphi(N + t) = \varphi(m_N)(p_N - 1)$.

The offset t lies in the overlap of the two truncated tails. Their net coefficient at $\varphi(N + t)$ is $(2^h - 1)2^{-t}$. On the assigned indices, set

$$z_N = e\left(\frac{(2^h - 1)\varphi(m_N)(p_N - 1)}{2^t}\right), \quad w_N = \frac{E(h, N, L)}{z_N}.$$

Both factors have absolute value 1. Let $\bar{z}_m$ be the average of z_N over the assigned indices with cofactor m , taking the average to be zero when the group is empty. The four contributions are then defined by the identity

$$\begin{aligned} \sum_{N=X}^{2X-1} E(h, N, L) &= \underbrace{\sum_{N \in \mathcal{G}} w_N(z_N - \bar{z}_{m_N})}_{\text{mean-subtracted contribution}} + \underbrace{\sum_{N \in \mathcal{G}} w_N \bar{z}_{m_N}}_{\text{mean contribution}} \\ &+ \underbrace{\sum_{N \in A \setminus \mathcal{G}} E(h, N, L)}_{\text{excluded-cofactor contribution}} + \underbrace{\sum_{\substack{X \leq N < 2X \\ N \notin A}} E(h, N, L)}_{\text{unassigned contribution}}. \end{aligned}$$

This follows by partitioning the indices and using $E(h, N, L) = w_N z_N$. Subtracting the group mean is exact algebra; it does not make the two factors independent. The chosen prime need not be confined to the selected argument: with $X = 16$, $h = s = 1$, $L = 20$ and $N = 18$, the argument is $N + t = 38 = 2 \cdot 19$, which satisfies both assignment inequalities, but the same prime also divides $N + 1 = 19$. Thus the factorisation at $N + t$ alone gives no separation from the other totient values in the phase.

Definition 3.4 (Four simultaneous inequalities). For every $h > 0$ there are $s > 0$ and $\eta \in (0, 1)$ such that for every X_0 there are X, L with $\max(X_0, 1) \leq X$, $h \leq L - s$, $16(2X + h + L + 2) \leq 2^L$, and all four of

$$\begin{aligned}
 \operatorname{Re} \sum_{N \in \mathcal{G}} w_N (z_N - \bar{z}_{m_N}) &\leq \frac{14}{25} X, \\
 \left\| \sum_{N \in \mathcal{G}} w_N \bar{z}_{m_N} \right\| &\leq \frac{1}{100} X, \\
 \left\| \sum_{N \in A \setminus \mathcal{G}} E(h, N, L) \right\| &\leq \frac{1}{100} X, \\
 \left\| \sum_{\substack{X \leq N < 2X \\ N \notin A}} E(h, N, L) \right\| &\leq \frac{8}{25} X.
 \end{aligned}$$

It implies $S \notin \mathbb{Q}$. [cofinal] [Lean] [LEAN SOURCE](#) [LEAN SOURCE](#)

The four bounds must hold for the same X, L, s, η , not along four unrelated sequences of blocks. Their constants add to $14/25 + 1/100 + 1/100 + 8/25 = 9/10$, so they imply the real-part criterion. They do not establish the stronger $21/25$ norm criterion. The exact partition proves an identity, not any one of these estimates; in particular, grouping by a prime factor does not justify statistical independence between the mean-subtracted phase and its weight.

Like the block criterion, this condition concerns actual totient values at arbitrarily large indices and at a common truncation depth. The four bounds are a sufficient way to prove the needed real-part estimate, not an equivalent restatement of the norm estimate.

The decomposition is an exact identity. The supplied formal record contains its unconditional proof and an axiom report ([LEAN SOURCE](#), budget implication [LEAN SOURCE](#)). The mean being subtracted is the mean of z_N , not of the residual weight w_N . On each cofactor group, $\sum_N (z_N - \bar{z}_m) = 0$, but $\sum_N w_N (z_N - \bar{z}_m)$ need not vanish because w_N varies with N . Thus centring the phase does not estimate the weighted correlation. Only its real part is required: $14X/25$ suffices in place of the earlier proposed norm bound $X/2$. For $X \geq 4$ and $s = L - h$, the assigned indices *with cofactor* $m_N = 1$ are exactly those for which $N + h + 1$ is prime. This describes one cofactor group, not all assigned indices. The source proves this membership equality ([LEAN SOURCE](#)), using $\varphi(mp) = \varphi(m)(p - 1)$ and $p > m$. This special choice has $s = L - h$ varying with L ; it illustrates the arithmetic of the groups but does not establish the condition with s fixed before X_0 .

The four inequalities are not proved with their required common parameters. Propositions 12.7 and 12.8 below do establish the unassigned and excluded-cofactor bounds at minimal admissible depth, with one sufficiently small fixed η . The mean and mean-subtracted estimates remain to be established at that same depth and for that same η . Only the real part of the mean-subtracted contribution is required.

A nonzero determinant alone does not prove the needed correlation estimate. The comparison in the linked source preserves such a determinant while fixing an entire row of weighted phases to 1. A successful application must distinguish that comparison, for example by additional arithmetic information, rather than only by column rescaling. The exact four-term identity supplies no prime-distribution estimate. [Open] [LEAN SOURCE](#)

3.3 A fixed separation at prime positions

Definition 3.5 (A fixed gap at prime positions). For every $h \geq 1$ and every N_0 there is a prime p with $\max(N_0+h+1, h+5) \leq p$ and $\operatorname{Re} e(R_{p-1} - R_{p-h-1}) < \frac{9}{10}$, where $e(x) = \exp(2\pi i x)$. Equivalently: cofinally many primes p at which the totient tail difference across the shift h stays a *fixed* distance $> \arccos(9/10)/(2\pi) \approx 0.0718$ from every integer. It implies $S \notin \mathbb{Q}$. [cofinal] [Lean] [LEAN SOURCE](#) [LEAN SOURCE](#)

This asks for more than nonintegrality. A nonintegral real number can be arbitrarily close to an integer, whereas the displayed condition requires a fixed positive distance, along prime positions and for each shift h . For example, phases at a half-integer satisfy the numerical gap and phases converging to an integer eventually fail it. These are illustrations of the hypothesis, not assertions about the totient tails. No estimate for these tail differences at prime positions with the stated quantifiers is proved here. Certificate completeness alone does not supply it. [Open]

3.4 Certificates with depth equal to the shift

Definition 3.6 (Depth equal to a shift multiple). For every $d \geq 1$ and every N there is $t \geq 1$ with $C(td, N, td)$; unpacked,

$$N + 2td + 2 < D(td, N, td) \bmod 2^{td} < 2^{td} - (N + 2td + 2).$$

It is equivalent to $S \notin \mathbb{Q}$. [cofinal] [Lean] [LEAN SOURCE](#) [LEAN SOURCE](#)

For the actual totient series, the condition in Definition 3.6 is equivalent to irrationality, by the full-depth amplification argument of the short note (Theorem 3.1 there; [LEAN SOURCE](#)). For a fixed pair (d, N) , a nonintegral tail difference supplies a certificate in every sufficiently late adjacent pair of multipliers. The missing arithmetic input is a nonintegral tail difference for every relevant pair, not a converse implication for the condition with depth equal to the shift. This remains an exact reformulation of the open problem, not a proof of it. An older one-direction lemma ([LEAN SOURCE](#)) is still in the cited source; it is not the current classification.

The finite examples establish the hypothesis for individual pairs (d, N) , not for all pairs. Thus the universally quantified condition is not a mild regularity assumption: it has the full strength of the irrationality question. [Open]

Formal sources: [LEAN SOURCE](#)

3.5 A false generic rank bound

Definition 3.7 (A counterfactual rank criterion). The proposed assertion is a constant C bounding the rank for every $c : \mathbb{N} \rightarrow \mathbb{N}$ with $c(n) \leq n$ and $T_c \in \mathbb{Q}$, every integer $v > 0$, and every integer sequence u satisfying

$$u(n+1) = 2u(n) - vc(n+1), \quad u(n)/2^n \rightarrow 0.$$

Specifically, the proposed bound is

$$\dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}} \{n \mapsto u(2^j n + r) : 1 \leq j \leq e, 0 \leq r < 2^j\} \leq C \quad (e \geq 0).$$

The alternative replaces C by a function $g(e)$ growing strictly slower than $2^e - 1$, with the same quantifiers. Either assertion would contradict the lower bound of Proposition 1.6 under rationality of S . Both assertions are false, as the following comparison shows. [uniform]

The generic assertion in Definition 3.7 is false, not a live open proposition. The 5/4 control in Section 10.8 has $0 \leq c(n) \leq n$ and a tempered integral carry of rank at least $2^e - 1$ at every depth (the unbounded-rank theorem for the rational comparison sequence, line 478). It refutes both proposed generic ceilings. A bound restricted by additional totient-specific hypotheses would be a different assertion.

Periodicity modulo v alone does not provide the additional rank bound. [LEAN SOURCE](#)

4 Guide to the detailed statements

The remaining sections are reference material. They restate the identities and implications in enough detail to compare their hypotheses. A repeated statement has the same meaning as before; its repetition does not supply a missing assumption.

Quantifiers and hypotheses. The direct certificate computations cover individual parameters or a finite range. Other results hold uniformly or at arbitrarily large scales, but their conclusions are different: positivity, for example, is not nonintegrality. The finite-prefix example, rank bounds, and residue estimates must each be used only under their stated hypotheses.

Dependencies. Sections 2 and 3 explain the principal counterexamples and sufficient conditions. The following catalogue supplies the associated formulas and source references. The later tables compare logical implications and the limits of particular estimates; their source labels are not extra mathematical hypotheses.

5 Series identities and finite exclusions

The identities in this section express S through finite prefixes, scaled tails and Möbius sums. The finite calculation in Section 5.3 then excludes a specified range of rational denominators. Equivalent series representations do not by themselves extend that finite exclusion to all denominators.

5.1 The series

Definition 5.1 (The Erdős–249 constant). Let φ denote Euler’s totient function. Define

$$S := \sum_{n \geq 1} \frac{\varphi(n)}{2^n} = \sum_{n: \mathbb{N}} \frac{\varphi(n)}{2^n},$$

the two forms coinciding because $\varphi(0) = 0$; the second, $\mathbb{N}$ -indexed form is the one carried in the Lean source. The series converges absolutely since $\varphi(n) \leq n$. The irrationality of this value is the question under consideration.

Remark 5.2 (Status). No proof or disproof of $S \notin \mathbb{Q}$ exists anywhere in the record, formal or informal. What exists is: one large unconditional finite denominator exclusion (§5.3), several exact reformulations of S that relocate the same open question onto different coordinates without touching its truth value (§5.4–§5.6), and a certificate apparatus whose cofinal quantified condition is the precise open target (Definition 1.2; only its finite, checked instances are catalogued here, §5.7).

5.2 The tail-shift identity

Proposition 5.3 (Digit-shift identity). *For every $N : \mathbb{N}$,*

$$2^N \cdot S = \Phi_N + R_N,$$

where $\Phi_N := \sum_{n \leq N} \varphi(n) \cdot 2^{N-n} \in \mathbb{N}$ (the integer prefix) and $R_N := \sum_{j \geq 0} \varphi(N+1+j)/2^{j+1}$ (the scaled tail, R in Lean). [uniform] [Lean] [LEAN SOURCE](#)

Consequence for rationality. Subtracting the prefix identities at $N+h$ and N gives

$$R_{N+h} - R_N \equiv 2^N(2^h - 1)S \pmod{\mathbb{Z}}.$$

If $S = a/(2^c v)$ with $v \geq 1$ odd, Euler’s theorem supplies $h = \varphi(v) > 0$ and $N_0 = c$, for which the difference is integral for all $N \geq N_0$ ([LEAN SOURCE](#), [Lean]). Conversely, an integral difference for even one $h > 0, N \geq 0$ forces S to be rational, because $2^N(2^h - 1)$ is a nonzero integer. Thus the unresolved question is not this converse, but whether every such tail difference is nonintegral. Definition 1.2 expresses that question through finite residues. $\square$

5.3 The finite denominator bound

The finite calculation gives a lower bound on its denominator should it happen to be rational. It is obtained by a Farey mediant argument that does not assume the quantified residue condition or use the tail-period test.

Lemma 5.4 (Farey gap, fully general). *For integers a, b, c, d, r, s with $b > 0, d > 0, bc - ad = 1$ (i.e. a/b and c/d are unimodular Farey neighbours), and $as < rb, rd < cs$ (i.e. r/s lies strictly between them): $b + d \leq s$. [n/a] [Lean] [LEAN SOURCE](#)*

Indeed, the two strict inequalities make $cs - dr$ and $br - as$ positive integers, and

$$s = b(cs - dr) + d(br - as) \geq b + d,$$

using $bc - ad = 1$. This also proves $s > 0$, which the formulation did not need to assume separately. The lemma is the classical Stern–Brocot denominator bound. The mediant $(a+c)/(b+d)$ lies between the two neighbours and is reduced: any common divisor of $a+c$ and $b+d$ divides $b(a+c) - a(b+d) = bc - ad = 1$. It attains denominator $b+d$, so the general interval bound is sharp. A particular tail interval still requires its own endpoint and first-failure checks.

Proposition 5.5 (The Farey certificate at $K = 240$). *For the window $(N, K) = (1, 240)$, put*

$$V = \left(\sum_{r=1}^{240} \varphi(1+r) 2^{240-r} \right) \bmod 2^{240}.$$

For every $q : \mathbb{N}$ with

$$0 < q \leq Q_0 := 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \quad (\approx 7.96 \times 10^{34}),$$

$$(q \cdot V) \bmod 2^{240} + 243q < 2^{240}.$$

This bound is sharp: $q = Q_0 + 1 = 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,054$ is the exact first failing denominator, obtained as the denominator of the mediant of two explicit unimodular Farey neighbours. [bounded] [Lean] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: `totient_carry_residue_window_1_240_eq`, `gap_check_window_1_240_le_79639646646701375323355774875831053`, `gap_check_window_1_240_first_failure`.

Theorem 5.6 (Denominator exclusion). *For every $p \in \mathbb{Q}$ whose reduced denominator is at most Q_0 ,*

$$S \neq p.$$

Equivalently: if S is rational, its reduced denominator exceeds $Q_0 \approx 7.96 \times 10^{34}$. [bounded] [Lean] [LEAN SOURCE](#)

Remark 5.7 (What this does and does not say). Theorem 5.6 is a complete, unconditional finite fact: no rational of denominator at most Q_0 equals S . A single such bound does not exclude arbitrarily large denominators. Proving unbounded exclusions by the same mechanism would, however, establish irrationality. The docstring behind Proposition 5.5 states the question this leaves: whether $\sup_K (b+d)(K)$ (the growing analogue of Q_0 as the window K grows) is unbounded as $K \rightarrow \infty$; which would close #249 through this theorem's implication, entirely independently of the certificate-supply obligation of Proposition 5.3. [Open]; not claimed or proved anywhere in the record.

Remark 5.8 (Earlier finite denominator bounds). The recorded Farey computations gave bounds 4838, 2^{22} , approximately 2.49×10^{17} at $K = 120$, and Q_0 at $K = 240$. Each applies the finite mechanism of Lemma 5.4 and Proposition 5.5. These values do not prove that the bounds are unbounded as K increases.

5.4 Equivalent expressions and their denominator bounds

The exact finite Farey record behind Theorem 5.6 is not tied to the $\varphi(n)/2^n$ presentation of S ; it transfers verbatim to two other exact reformulations of the same constant, at exactly half the bound. Amiram Eldar posted both reformulations to OEIS A256936 on 15 March 2026: an equivalent coprimality interpretation of Proposition 5.9 in revision 28, and the Möbius-square formula of Proposition 5.14 in revision 31 [6]. Steve Fan posted the Möbius-square formula on the problem's forum thread on 16 May 2026 [7]. Both identities follow here from Möbius inversion and geometric sums, with the proofs recorded in the linked declarations.

Proposition 5.9 (Fair-coin coprimality form). *Let X, Y be independent random variables with $\Pr(X = n) = \Pr(Y = n) = 2^{-n}$ for $n \geq 1$ (independent fair-coin waiting times). Then*

$$S = \frac{1}{2} + \Pr(\gcd(X, Y) = 1) = \frac{1}{2} + \sum_{\substack{a, b \geq 1 \\ \gcd(a, b) = 1}} 2^{-(a+b)}.$$

Equivalently, on the visible lattice: summing $2^{-(a+b)}$ over the half-open coprime pairs $(a \geq 1, b \geq 0, \gcd(a, b) = 1)$ recovers $\sum_n \varphi(n)/2^n$ exactly, with no boundary correction, because the visible-point count on the half-open antidiagonal at height n equals $\varphi(n)$ for every n , including $n = 0, 1$. [n/a] [Lean] [LEAN SOURCE](#) [LEAN SOURCE](#)

Proposition 5.10 (The gcd-layer normalisation). *For independent fair-coin waiting times as above, $\sum_{g \geq 1} \Pr(\gcd(X, Y) = g) = 1$ exactly; and for every $d > 0$, $\Pr(d \mid X \wedge d \mid Y) = 1/(2^d - 1)^2$. [uniform] [Lean] [LEAN SOURCE](#) [LEAN SOURCE](#)*

Lean: `tsum_gcd_layer_pos_coprime_half_eq_one`, `tsum_pos_pair_both_dvd_half_eq_inv_mersenne_sq`.

Theorem 5.11 (Denominator exclusion for the coprimality-probability form). *Let*

$$Q_1 := \left\lfloor \frac{Q_0}{2} \right\rfloor = 39\,819\,823\,323\,350\,687\,661\,677\,887\,437\,915\,526.$$

For every $a \in \mathbb{Z}$, $d \in \mathbb{N}$ with $0 < d \leq Q_1$: the visible coprime-pair probability $\Pr(\gcd(X, Y) = 1)$ is not equal to a/d . [bounded] [Lean] [LEAN SOURCE](#)

Theorem 5.12 (Denominator exclusion for the Möbius-square form). *With Q_1 as in Theorem 5.11: for every $a \in \mathbb{Z}$, $d \in \mathbb{N}$ with $0 < d \leq Q_1$, the signed series $T := \sum_{m \geq 1} \mu(m)/(2^m - 1)^2 = S - \frac{1}{2}$ (see §5.5) is not equal to a/d . [bounded] [Lean] [LEAN SOURCE](#)*

Remark 5.13 (Why the bound halves, and why this is not new information). $Q_1 = \lfloor Q_0/2 \rfloor$ arithmetically: Q_0 is odd ($Q_0 = 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053$), so $Q_0/2 = 39\,819\,823\,323\,350\,687\,661\,677\,887\,437\,915\,526.5$ and Q_1 is its floor. The halving is the cost of transporting the known bound through the affine shift $S = \frac{1}{2} + T$ (resp. $S = \frac{1}{2} + \Pr(\gcd(X, Y) = 1)$) on a denominator-exclusion statement: excluding all denominators $\leq Q_1$ for T follows from excluding all denominators $\leq Q_0$ for S (a denominator- d value of T with $d \leq Q_1$ yields a denominator dividing $2d \leq Q_0$ for S). Theorems 5.11 and 5.12 are therefore the *same* finite Farey record as Theorem 5.6, transported through Proposition 5.9 and Proposition 5.14 respectively; not independent evidence. The converse finite implication is not asserted: subtracting $1/2$ can double a denominator, so the Q_1 exclusion for T alone need not recover the full Q_0 exclusion for S .

5.5 The Möbius-weighted identity

Proposition 5.14 (Möbius-square reduction).

$$S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n} = \frac{1}{2} + \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2},$$

where μ is the Möbius function, so $\mu(d) \in \{-1, 0, 1\}$ for every d . Consequently Erdős #249 $\iff T := \sum_{d \geq 1} \mu(d)/(2^d - 1)^2 \notin \mathbb{Q}$, since adding a rational number does not change irrationality. [n/a] [Lean] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: [mobius_square_reduction](#), [irrational_totient_iff_mobius_square](#), [moebius_three_values](#).

Remark 5.15 (Why the bounded coefficients matter; the Erdős-1948 regime). The identity of Proposition 5.14 rewrites S (equivalently T) as a Lambert-type series with squared denominators and bounded Möbius weights. To see the constant $1/2$ directly, expand

$$\sum_{d \geq 1} \frac{\mu(d)}{2^d - 1} = \sum_{n \geq 1} \frac{\sum_{d|n} \mu(d)}{2^n} = \frac{1}{2}.$$

Combining this identity with $\varphi = \mu * \text{Id}$ and $2^d = (2^d - 1) + 1$ proves the formula for T ; absolute convergence justifies the rearrangements. Thus the corresponding first-power sum $L(\mu) := \sum_d \mu(d)/(2^d - 1)$ is rational, whereas $L(1) = \sum_{d \geq 1} 1/(2^d - 1) = E$ is the Erdős–Borwein constant. Erdős proved its irrationality in 1948 [9] and the supplied formal source also proves it irrational ([LEAN SOURCE](#), [Lean]).

In contrast, the coefficients of the original power series satisfy $0 \leq \varphi(n) \leq n$ and are unbounded. Bounded Möbius weights do not make the new denominators powers of a fixed base, so they do not turn this representation into a bounded-digit expansion. The function φ has average order $6n/\pi^2$, equivalently $\sum_{k \leq x} \varphi(k) \sim 3x^2/\pi^2$, but there is no pointwise estimate $\varphi(n) = \Theta(n)$. The near-integer irrationality criterion (formalised generically as [LEAN SOURCE](#) and its base-power specialisation [LEAN SOURCE](#), both [Lean]) applies to either representation if its nonzero approximation errors can be established. Boundedness of the coefficients does not itself establish those errors. Erdős’s 1948 argument is itself a proof for the specific Lambert series: it chooses arguments by a system of congruences so that the divisor counts along a block are divisible by increasing powers of the base, and a tail bound then forces a long run of zero digits [9, pp. 63–66]. Boundedness of a signed weight supplies none of that divisibility input.

The rational identity $L(\mu) = 1/2$ is already a warning: bounded signed coefficients alone do not imply irrationality. The later denominator calculations retain the Möbius terms whose indices do not divide the chosen LCM; those terms cannot be discarded when transferring an estimate.

5.6 A general squared-Lambert identity

Proposition 5.16 (The squared-Lambert identity). *For $w : \mathbb{N} \rightarrow \mathbb{R}$ with $|w(d)| \leq d$ for all $d > 0$, and $0 \leq r < 1$:*

$$\sum_{d \geq 1} w(d) \left(\frac{r^d}{1 - r^d} \right)^2 = \sum_{n \geq 1} \left(\sum_{e|n} w(e) \left(\frac{n}{e} - 1 \right) \right) r^n.$$

[uniform] [Lean] [LEAN SOURCE](#)

The identity follows by expanding $r^d/(1 - r^d)$ as a geometric series and interchanging absolutely convergent sums. It is the squared analogue of the classical divisor-sum Lambert identity $\sum_n a_n q^n / (1 - q^n) = \sum_m (\sum_{d|m} a_d) q^m$ [14, (1), p. 2].

At $r = 1/2$, the choices $w = 1$ and $w = \varphi$ give the following two identities. They help distinguish the role of the weight from that of the squared denominator.

Proposition 5.17 (The divisor-sum identity).

$$\sum_{d \geq 1} \frac{1}{(2^d - 1)^2} = \sum_{n \geq 1} \frac{\sigma(n) - \tau(n)}{2^n} = \zeta_q(2) - \zeta_q(1) \text{ at } q = \frac{1}{2},$$

where σ is the sum-of-divisors function and τ the number-of-divisors function. The identity is machine-checked ([Lean]). With $\zeta_q(s) = \sum_{n \geq 1} n^{s-1} q^n / (1 - q^n)$, Postelmans and Van Assche prove that $1, \zeta_q(1), \zeta_q(2)$ are linearly independent over $\mathbb{Q}$ for $q = 1/p$ with an integer $p \geq 2$ [10, Theorem 1.3, p. 3]; at $q = \frac{1}{2}$ this gives the irrationality of the value $\zeta_q(2) - \zeta_q(1)$ ([Cited], not formalised in this corpus). Two irrational numbers can have a rational difference, so the separate irrationality of the two values would not suffice. [n/a] [LEAN SOURCE](#)

Lean: `divisor_sum_identity`, `qZeta`, `qZeta_half_two_sub_one`, `irrational_qZeta_half_difference_of_linearIndependent`, and 1 further declaration. Conditional on the linear independence theorem of Postelmans and Van Assche; see the [coverage section](#).

Proposition 5.18 (The gcd-moment identity).

$$\sum_{d \geq 1} \frac{\varphi(d)}{(2^d - 1)^2} = \sum_{n \geq 1} (P(n) - n) \cdot 2^{-n} = \mathbb{E}[\gcd(X, Y)],$$

where $P(n) := \sum_{e|n} \varphi(e) \cdot (n/e) = (\varphi * \text{Id})(n)$ is Pillai's gcd-sum function $P(n) = \sum_{k \leq n} \gcd(k, n)$ [13, §1, (1)–(2)] and X, Y are the independent fair-coin waiting times of Proposition 5.9. The identity itself is machine-checked ([Lean]); irrationality of $\mathbb{E}[\gcd(X, Y)]$ is not proved here. This is a different series from S . [n/a] [LEAN SOURCE](#)

Lean: `gcd_moment_identity_three_members`, `tsum_pos_pair_gcd_half_eq_totient_div_mersenne_sq`, `sum_gcd_lcc_eq_pillaiP`, `pillaiP_eq_totient_mul_id`, and 1 further declaration in the [coverage section](#).

Remark 5.19 (Related series). Writing $L(f) := \sum_d f(d) / (2^d - 1)$ and $L_2(f) := \sum_d f(d) / (2^d - 1)^2$, the preceding identities give $L(\mu) = \frac{1}{2}$ (rational), $L(1) = E$ (Erdős–Borwein, irrational, [Lean]), $L(\varphi) = 2$ (rational); $L_2(\mu) = S - \frac{1}{2}$ ([Open], this is #249), $L_2(1) = \zeta_q(2) - \zeta_q(1)$ ([Cited] irrational, Proposition 5.17), $L_2(\varphi) = \mathbb{E}[\gcd(X, Y)]$ ([Open], Proposition 5.18). These examples show why the denominator shape alone does not determine irrationality: changing the weight can produce either rational or irrational values.

5.7 Diagonal certificates through scale 82

This table records finite instances of the residue test introduced in Definition 1.2. The aggregate result covers every integer scale $t \leq 82$. It does not assert an instance at scale 83 or an unbounded family. The individual source links identify the computations used.

Definition of the finite residue test For $h, N, L : \mathbb{N}$, recall the truncated difference $D(h, N, L) := \sum_{j < L} (\varphi(N + h + 1 + j) - \varphi(N + 1 + j)) \cdot 2^{L-1-j} \in \mathbb{Z}$ ([LEAN SOURCE](#), [Lean]), and

$$C(h, N, L) := (N + h + L + 2) < D(h, N, L) \bmod 2^L < 2^L - (N + h + L + 2),$$

([LEAN SOURCE](#), [Lean], decidable). By completeness ([LEAN SOURCE](#), [Lean]),

$$(\exists L, C(h, N, L)) \iff R_{N+h} - R_N \notin \mathbb{Z}$$

for every h, N . Requiring such certificates beyond every basepoint threshold for each fixed positive h is precisely Sep and is equivalent to $S \notin \mathbb{Q}$. The table records finite instances, indexed either by (h, N, L) or by t ; it does not establish that quantified condition.

Anchor	Exact statement of what was checked	Scale	Site
Fixed-window example, depth 16	$C(h, 12, 16)$ holds for every $h \in \{1, \dots, 8\}$ (by decide, 256 evaluations with totient arguments below 37); consequently $S \neq r$ for every $r \in \mathbb{Q}$ with $1 \leq h \leq 8$ and $r.\text{den} \mid 2^{12}(2^h - 1)$.	[fixed]	LEAN SOURCE LEAN SOURCE
Fixed-window example, depth 9, wider net	$C(h, 14, 9)$ holds for every $h \in \{1, \dots, 16\}$: $S \neq r$ for every $r \in \mathbb{Q}$ with $1 \leq h \leq 16$ and $r.\text{den} \mid 2^{14}(2^h - 1)$.	[fixed]	LEAN SOURCE
Diagonal-pincer certificates, base 12 scales	$C(H_t, H_t, L_t)$, $H_t := \text{lcm}(1, \dots, t)$, holds (by explicit decide/norm_num on checked <code>Nat.totient</code> values, via factored prime-power blocks with Lucas-primality certificates) for $t \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17\}$ with certificate depths $L_t \in \{6, 5, 7, 7, 9, 14, 15, 14, 21, 22, 23, 26\}$ respectively (in the same order).	[fixed]	LEAN SOURCE LEAN SOURCE LEAN SOURCE
Diagonal-pincer certificates, extended scales through $t = 64$	The same predicate $C(H_t, H_t, L_t)$ is additionally checked, one sibling module per scale, at $t \in \{19, 23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49, 53, 59, 61, 64\}$ (16 further explicit values). Together with the base 12 scales above this is 28 explicit values of t in total , the complete list being $\{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49, 53, 59, 61, 64\}$. This does <i>not</i> establish $C(H_t, H_t, \cdot)$ for infinitely many t ; these 28 examples are a historical strict subset of the contiguous band in the next row.	[fixed]	Per-scale Lean modules; endpoint LEAN SOURCE

Anchor	Exact statement of what was checked	Scale	Site
Contiguous lcm-diagonal band through $t \leq 82$	For every natural $t \leq 82$ there exists a depth L with $\mathcal{C}(H_t, H_t, L)$. This closes every scale in the finite interval, including plateau transfers, with no holes. The next lcm jump is at the prime 83; no certificate at $t = 83$ is claimed, and any such certificate must have depth at least 125.	[bounded]	LEAN SOURCE LEAN SOURCE
Farey denominator floor	Every $q \in \mathbb{N}$ with $0 < q \leq Q_0 = 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053$ satisfies the window- $(N, K) = (1, 240)$ gap certificate; $q = Q_0 + 1$ is the exact first failure.	[bounded]	LEAN SOURCE
Coprimality-probability / Möbius-square Farey floor	Every $d \in \mathbb{N}$ with $0 < d \leq Q_1 = 39\,819\,823\,323\,350\,687\,661\,677\,887\,437\,915\,526$ (exactly $\lfloor Q_0/2 \rfloor$) is excluded as a denominator of $\Pr(\gcd(X, Y) = 1)$ and, separately, of $T = \sum_d \mu(d)/(2^d - 1)^2$.	[bounded]	LEAN SOURCE LEAN SOURCE

Remark 5.20 (Scope of the table). No row above is cofinal in its indexing parameter (h , t , or the denominator bound), and no row is claimed to extrapolate. The historical diagonal-pincer depths are irregular and nonmonotone $(6, 5, 7, 7, 9, 14, 15, 14, 21, 22, 23, 26, \dots)$; no closed-form growth rate for L_t is proved or conjectured in the record. This table records the historical 28-scale bank through $t = 64$; the later aggregate theorem for every $t \leq 82$ is stated above and is not itemised row by row here. Neither bounded record supplies the cofinal obligation stated in Definition 1.2.

6 Detailed statements

The following statements collect the dependencies used in the earlier arguments. We begin with unconditional identities, rank bounds and finite examples. The next subsection supplies the definitions and elementary implications, followed by the quantified conditions that would be needed to prove irrationality. A source link supports the indicated statement, not an unproved hypothesis appearing in it.

6.1 Unconditional structure and finite examples

The distinction between a finite example and a family at arbitrarily large scales is important here. The rank and denominator statements below hold with their displayed quantifiers; the residue computations establish only the explicitly stated finite ranges.

Theorem 6.1 (Rationality gives an eventual tail period). *If $S = a/(2^c v)$ with $a \in \mathbb{Z}$, $c \in \mathbb{N}$ and v a positive odd integer, then*

$$R_{N+h} - R_N \in \mathbb{Z} \quad (N \geq c), \quad h = \varphi(v) > 0.$$

Euler's theorem gives $v \mid 2^h - 1$, and the prefix-tail identity then makes $2^N(2^h - 1)S$ integral. Thus rationality supplies a positive shift for which all sufficiently late tail differences are integral. [Lean] [uniform] [LEAN SOURCE](#)

Lean: [specified_euler_tail_period](#).

Theorem 6.2 (Rationality forces unbounded carry rank). *If $S \in \mathbb{Q}$, there are an integer $v > 0$ and an integer sequence u such that*

$$u(N+1) = 2u(N) - v\varphi(N+1), \quad u(N)/2^N \rightarrow 0,$$

and, for every $e \geq 0$, the sequences $n \mapsto u(2^j n + r)$ with $1 \leq j \leq e$ and $0 \leq r < 2^j$ span a space of $\mathbb{Q}$ -dimension at least $2^e - 1$.

The recurrence is the one obtained from the scaled tails after clearing a hypothetical denominator of S . The formal term **tempered** means exactly $u(N)/2^N \rightarrow 0$ here, not a separate assumption of subexponential growth. This rank lower bound is a necessary consequence of rationality, not a contradiction. An upper bound using additional properties of the actual totient coefficients would be needed; the later rational comparison rules out such a bound for arbitrary rational coefficient series.

[Lean] [uniform] [LEAN SOURCE](#)

Theorem 6.3 (Independence of the retained dyadic sections). *For every integer $e \geq 0$, the family*

$$\varphi(n), \quad \varphi(2n), \quad \varphi(2^j n + r) \quad (1 \leq j \leq e, 0 < r < 2^j, r \text{ odd})$$

is linearly independent over $\mathbb{Q}$ and has $2^e + 1$ members. At $e = 0$ this family still contains $\varphi(2n)$; it is not the full level-zero truncation, which consists only of $\varphi(n)$. For $e \geq 1$ it is the basis of the full truncated dyadic family.

The proof separates one section at a time: the Chinese remainder theorem forces suitable prime divisors in the other affine arguments, while Dirichlet's theorem makes the selected argument prime. This proves independence, not just the displayed cardinality. In particular, the span of all dyadic sections is infinite-dimensional; that conclusion alone says nothing about rationality of their weighted sum.

[Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: [displayed_canonical_and_full_dyadic](#).

Proposition 6.4 (Denominator divisibility forces tail integrality). *If $S = a/q$ in lowest terms with $q > 0$ and $q \mid 2^N(2^h - 1)$, then $R_{N+h} - R_N \in \mathbb{Z}$. Indeed, the prefix identity expresses that difference as $2^N(2^h - 1)S$ minus an integer. This is the contradiction used by a nonintegrality certificate to exclude the rational value a/q .*

[Lean] [uniform] [LEAN SOURCE](#)

The denominator results use the following polynomial, defined for $r \geq 1$:

$$P_r(X) = \sum_{d \mid r} \mu(d) \frac{r}{d} \sum_{j=0}^{r/d-1} X^{dj}.$$

For example, $P_p(X) = p(1 + X + \cdots + X^{p-1}) - 1$ when p is prime. Thus $P_p(2) = p(2^p - 1) - 1$ is coprime to $2^p - 1$. For general $r \geq 1$ and $H \geq 1$, write

$$b_r = \frac{P_r(2)}{2^r - 1}, \quad \beta_H = \frac{b_{\text{rad}(H)}}{\text{rad}(H)}, \quad r_t = \text{rad}(H_t).$$

Here $\text{rad}(H)$ is the product of the distinct prime factors of H , and $\text{den}(x)$ denotes the positive reduced denominator of $x \in \mathbb{Q}$. These rational numbers are finite rational contributions to the Möbius decomposition; no convergence to S is asserted for them. Also put $\mathcal{D}_t = \{p \text{ prime} : t/2 < p \leq t\}$. The hypotheses below specify which factors of $2^{r_t} - 1$ remain in the reduced denominator after cancellation.

Proposition 6.5 (A cyclotomic factor remains after cancellation). *If r is squarefree, then*

$$\gcd(|P_r(2)|, |\Phi_r(2)|) = 1, \quad |\Phi_r(2)| \mid \text{den}(b_r).$$

Here Φ_r is the r -th cyclotomic polynomial. Proposition 6.40 gives $P_r \equiv \mu(r) \pmod{\Phi_r}$, and $\mu(r) = \pm 1$ because r is squarefree. This proves that the indicated cyclotomic factor cannot cancel from b_r .

[Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: `mobiusNumeratorPolynomial_eval_two`, `mobiusNumerator_gcd_cyclotomicValue`, `cyclotomicValue_dvd_baseMobiusShadow_den`.

Proposition 6.6 (Primes in the upper half remain after cancellation). *For $t \geq 5$ the whole product*

$$\prod_{p \in \mathcal{D}_t} (2^p - 1) \mid \text{den}(H_t \beta_{H_t})$$

remains after cancellation. More generally, the source gives the same product-divisibility conclusion for a multiplier coprime to that product; it also supplies a sufficient condition on the multiplier's prime factors. For the multiplier H_t/r_t , those prime factors are at most t , and the LCM result above follows.

[Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: `lcmHeight_upperHalf_product_dvd_den`, `upperHalfChannel_product_dvd_den_of_coprime_scale`, `upperHalfChannel_product_dvd_den_of_scale_primeFactors_le`.

The restriction $t \geq 5$ belongs to the sufficient prime-factor condition, not just to the LCM example. At $t = 3$, one has $r_t = 6$, $b_6 = 298/63$ and $\mathcal{D}_3 = \{2, 3\}$. The multiplier 9 has only prime factors at most 3, but $\text{den}(9b_6) = 7$ is not divisible by $(2^2 - 1)(2^3 - 1) = 21$. Thus the prime-factor condition alone does not imply survival of the product at every scale. This is a denominator statement about a finite rational contribution, not a nonintegrality assertion about a real tail.

Proposition 6.7 (A lower bound for the reduced denominator). *For every integer $t \geq 5$,*

$$2^{\lfloor t/2 \rfloor} \leq \prod_{p \in \mathcal{D}_t} (2^p - 1) \leq \text{den}(H_t \beta_{H_t}).$$

Bertrand's postulate supplies a prime in $\mathcal{D}_t$; its factor is at least $2^{\lfloor t/2 \rfloor}$. Proposition 6.6 then gives the second inequality. The complementary term in the real tail decomposition can still cancel this rational contribution, so denominator growth alone does not prove irrationality of S .

[Lean] [uniform] [LEAN SOURCE](#)

Lean: [upper_half_product_denominator_bounds](#).

Proposition 6.8 (The exact reduced denominator). *At every integer scale $t \geq 0$,*

$$\text{den}(H_t \beta_{H_t}) = \frac{2^{r_t} - 1}{\gcd\left(2^{r_t} - 1, \frac{H_t}{r_t} \prod_{\substack{p \mid r_t \\ p \text{ odd prime}}} (p^2 - 1)\right)}.$$

The product is 1 when there is no odd prime divisor. This gives the exact cancellation, including the multiplier H_t/r_t , rather than merely a lower bound for the denominator. For example, $H_5 = 60$ and $r_5 = 30$, and the denominator is $(2^{30} - 1)/3$.

[Lean] [uniform] [LEAN SOURCE](#)

Lean: [lcmHeight_scaledMobiusShadow_den_exact](#), [lcmHeight_five_scaledMobiusShadow_den_exact](#).

Proposition 6.9 (Nonvanishing of a signed dyadic sum). *Let I be finite, let $u_i \in \mathbb{Z}$ and $e_i \in \mathbb{N}$, and suppose that $m \in I$ is the unique index with maximal exponent e_m . If u_m is odd, then*

$$2^{e_m} \sum_{i \in I} \frac{u_i}{2^{e_i}} = \sum_{i \in I} u_i 2^{e_m - e_i} \equiv 1 \pmod{2}.$$

Indeed, every summand other than u_m is even. The sum is therefore nonzero. In the determinant application, the rectangular Cauchy–Binet formula produces such finite signed sums from truncated moments. The result tests a specified configuration; it does not show that a suitable maximal-exponent term exists at arbitrarily large scales.

[Lean] [bounded] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: [signed_dyadic_clearing](#), [signed_dyadic_sum_ne_zero](#), [scaled_dyadic_sum_odd](#).

Proposition 6.10 (Small certificates). *For each integer $1 \leq h \leq 8$, the finite test $C(h, 12, 16)$ holds. Each discrepancy uses two 16-term windows. Across all eight shifts, only the 24 distinct totient values at $13 \leq n \leq 36$ are needed, since the windows overlap. The source verifies the eight integer residue inequalities by exact computation.*

Consequently, if a/b is a reduced fraction with $b > 0$ and $b \mid 2^{12}(2^h - 1)$ for at least one $1 \leq h \leq 8$, then $S \neq a/b$. This follows from Propositions 6.18 and 6.4.

[Cert] [fixed] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: [small_certificate_windows](#), [small_certificates_and_exclusions](#).

Proposition 6.11 (A common certificate for sixteen shifts). *For each integer $1 \leq h \leq 16$, one has $C(h, 14, 9)$. The common basepoint is 14 and the depth is 9; these parameters have different roles. The two windows for each shift use only the 25 distinct totient values at $15 \leq n \leq 39$ across the whole family. Thus $S \neq a/b$ for every reduced fraction with $b > 0$ such that $b \mid 2^{14}(2^h - 1)$ for at least one $1 \leq h \leq 16$.*

[Cert] [fixed] [LEAN SOURCE](#)

Lean: [sixteen_certificate_windows](#), [sixteen_certificates_and_exclusions](#).

Proposition 6.12 (Historical diagonal examples and the complete band through 82). *Let $H_t = \text{lcm}(1, \dots, t)$ and let*

$$P(t) \quad :\Longleftrightarrow \quad \exists L, C(H_t, H_t, L).$$

The checked certificate table proves $P(t)$ at 28 explicit indices, ending at $t = 64$. For the initial indices

$$t = 1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, \dots,$$

the corresponding depths begin 6, 5, 7, 7, 9, 14, 15, 14, 21, 22, 23, 26, Each entry is a finite kernel computation on explicit totient values. The factorisations use checked prime-power blocks and Lucas primality certificates. Thus the table proves a finite list of instances of $P(t)$; it does not prove that $P(t)$ holds for infinitely many t .

[Cert] [fixed] [LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#)

The historical 28 examples are now a strict subset of the aggregate theorem $\forall t \leq 82, P(t)$. That theorem closes the finite interval without holes but supplies neither $P(83)$ nor a cofinal family.

[Cert] [bounded] [LEAN SOURCE](#)

Lean: [historical_table_size_and_initial_depths](#), [historical_table_and_complete_band](#).

6.2 Definitions and elementary identities

The finite test rests on two identities: the prefix-tail decomposition and the truncation of a tail difference. We state those first, then give the soundness and completeness arguments, the LCM specialisation and the corresponding rational-approximation criteria. The subsequent Möbius identities describe the same series with different coefficient weights.

6.2.1 Initial implications

Definition 6.13 (The scaled tail). For $N \in \mathbb{N}$,

$$R_N = \sum_{j \geq 0} \frac{\varphi(N + 1 + j)}{2^{j+1}}.$$

The series converges because $\varphi(n) \leq n$. It is the tail of S multiplied by 2^N , not the fractional part of $2^N S$; indeed, R_N need not lie in $[0, 1)$.

[Lean] [LEAN SOURCE](#)

Definition 6.14 (The truncated tail difference). For $h, N, L \in \mathbb{N}$, define the integer

$$D(h, N, L) = \sum_{j=0}^{L-1} (\varphi(N + h + 1 + j) - \varphi(N + 1 + j)) 2^{L-1-j}.$$

The sum is empty when $L = 0$. Dividing it by 2^L gives the first L terms of $R_{N+h} - R_N$. Computing the finitely many totients therefore determines $D(h, N, L)$ exactly; the residue inequality below is the decidable condition.

[Lean] [n/a] [LEAN SOURCE](#)

Definition 6.15 (The central-residue inequality). For $h, N, L \in \mathbb{N}$, the finite condition $C(h, N, L)$ is

$$N + h + L + 2 < D(h, N, L) \bmod 2^L < 2^L - (N + h + L + 2).$$

It puts the residue outside both endpoint intervals of width $N + h + L + 2$. After division by 2^L , the excluded width tends to zero as L grows with h, N fixed. No analytic hypothesis is part of this finite test; the separate condition Sep quantifies over such tests.
[Lean] [LEAN SOURCE](#)

Proposition 6.16 (The prefix-tail identity). For every $N \in \mathbb{N}$,

$$2^N S = \Phi_N + R_N, \quad \Phi_N = \sum_{n \leq N} \varphi(n) 2^{N-n} \in \mathbb{Z}.$$

Thus R_N and $2^N S$ have the same fractional part. Subtracting the identities at $N + h$ and N gives the tail-difference congruence used in the irrationality criterion.

[Lean] [uniform] [LEAN SOURCE](#)

Lean: `prefix_fractional_part, two_pow_mul_totient_series_eq, tail_diff_mem_int_iff_scaled_series_mem_int`.

Lemma 6.17 (The necessary depth inequality). For $h, N, L \in \mathbb{N}$, the condition $C(h, N, L)$ implies

$$2(N + h + L + 2) < 2^L.$$

Indeed, its lower residue bound must be smaller than its upper bound. Thus $L > 1 + \log_2(N + h + L + 2)$ whenever a certificate holds. In particular, a fixed depth cannot accommodate unbounded $N + h$. This is a necessary depth bound, not an upper bound for finding a certificate.

[Lean] [uniform] [LEAN SOURCE](#)

Lean: `certificate_logarithmic_depth, fixed_depth_bounds_indices, certifiedKill_depth_floor`.

Proposition 6.18 (A certificate implies nonintegrality). For all $h, N, L \in \mathbb{N}$,

$$C(h, N, L) \implies R_{N+h} - R_N \notin \mathbb{Z}.$$

The scaled truncation error satisfies $|2^L(R_{N+h} - R_N) - D(h, N, L)| \leq N + h + L + 2$. If the tail difference were integral, $D(h, N, L)$ would therefore lie within that distance of a multiple of 2^L , contrary to the two strict residue inequalities.

[Lean] [uniform] [LEAN SOURCE](#)

Lean: `totient_scaled_truncation_error, tail_diff_notMem_int_of_certifiedKill`.

Theorem 6.19 (Nonintegrality gives a certificate at some depth). For every $h, N \in \mathbb{N}$,

$$(\exists L \in \mathbb{N}, C(h, N, L)) \iff R_{N+h} - R_N \notin \mathbb{Z}.$$

For the converse to Proposition 6.18, fix a positive distance from the nonintegral tail difference to the nearest integer. The normalised error $(N + h + L + 2)/2^L$ tends to zero, so a sufficiently large truncation preserves that distance. This is a pointwise equivalence; irrationality requires the stated quantifiers over the shift and basepoint.

[Lean] [uniform] [LEAN SOURCE](#)

Definition 6.20 (The least common multiple of the first integers). Put $H(0) = 1$ and $H(t+1) = \text{lcm}(H(t), t+1)$. Then $H(t) = \text{lcm}(1, \dots, t)$, $t \leq H(t)$, and every positive integer $h \leq t$ divides $H(t)$. This is why a single LCM parameter can absorb any fixed hypothetical period once t is large enough.

[Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#)

Proposition 6.21 (The Farey gap lemma). Let $a, b, c, d, r, s \in \mathbb{Z}$, with $b, d > 0$ and $bc - ad = 1$. If

$$as < rb \quad \text{and} \quad rd < cs,$$

then $b + d \leq s$. The two cross-products $rb - as$ and $cs - rd$ are positive integers, and

$$s = d(rb - as) + b(cs - rd) \geq d + b.$$

In particular, a rational r/s strictly between the Farey neighbours a/b and c/d , with positive denominator, has denominator at least $b + d$.

[Lean] [uniform] [LEAN SOURCE](#)

Proposition 6.22 (An irrationality criterion from rational approximations). Let $x \in \mathbb{R}$ and let $u_k = p_k/q_k \in \mathbb{Q}$ be in lowest terms, with $q_k > 0$. If $u_k \neq x$ for every sufficiently large k and $q_k|x - u_k| \rightarrow 0$, then x is irrational. Indeed, if $x = a/b$ with $a \in \mathbb{Z}$ and $b \geq 1$, a nonzero integer numerator gives $q_k|x - u_k| = |aq_k - bp_k|/b \geq 1/b$, a contradiction. This elementary criterion requires both nonvanishing and the scaled error estimate. Convergence $u_k \rightarrow x$ alone is insufficient, and the approximants need not be continued-fraction convergents.

[Lean] [n/a] [LEAN SOURCE](#)

Proposition 6.23 (An irrationality criterion from near integers). Let $\xi \in \mathbb{R}$. Suppose that for every integer $q \geq 1$ there are integers m, z with

$$0 < |m\xi - z| < 1/q.$$

Then ξ is irrational: if $\xi = a/b$ with $b \geq 1$, every nonzero such difference is at least $1/b$. Restricting the multiplier to powers $m = b_0^n$ of a fixed integer base $b_0 \geq 2$ gives a sufficient special case, not a hypothesis satisfied by every irrational number. The strict lower bound excludes exact integer hits; the upper bound must be available for arbitrarily large q .

For a binary example, form a number by concatenating the blocks 10 at square indices $k \geq 1$ and 01 at the other indices. Its binary expansion is not eventually periodic, since the block sequence has increasingly long gaps between the square indices, so the number is irrational. There are no three consecutive equal digits. Every fractional part after a binary shift therefore lies in $[1/8, 7/8]$, and the multiples $2^n\xi$ do not approach the integers. This verifies that the restriction to base powers is genuinely stronger. This example is an ordinary mathematical argument, not an additional claim about the linked formalisation.

[Lean] [n/a] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: `irrational_of_int_mul_near_int`, `irrational_of_pow_mul_near_int`, `irrational_of_near_integer_multiples`, `irrational_of_near_integer_base_powers`, and 16 further declarations in the [coverage section](#).

Proposition 6.24 (A nonzero evaluation minor gives independence). *Let I be a finite index set and let $f_j : \mathbb{N} \rightarrow \mathbb{Q}$ for $j \in I$. If there are evaluation points $n_i \in \mathbb{N}$ such that*

$$\det(f_j(n_i))_{i,j \in I} \neq 0,$$

then the family $(f_j)_{j \in I}$ is linearly independent over $\mathbb{Q}$. A relation among the functions, evaluated at the n_i , gives a vector in the kernel of this nonsingular matrix, so all coefficients vanish. Finiteness is required to form the displayed determinant. In the totient application the arithmetic work is to construct these evaluation points; the implication itself is ordinary linear algebra.

[Lean] [n/a] [LEAN SOURCE](#) [LEAN SOURCE](#)

Proposition 6.25 (The gap between distinct rational numbers). *If $a/b < c/d$ are reduced rational numbers with $b, d > 0$, then*

$$\frac{c}{d} - \frac{a}{b} = \frac{bc - ad}{bd} \geq \frac{1}{bd}.$$

The numerator $bc - ad$ is a positive integer. Thus an upper bound on a positive rational error gives a lower bound on the product of the reduced denominators. A large denominator in an unreduced displayed expression need not be the reduced denominator appearing in this conclusion.

[Lean] [n/a] [LEAN SOURCE](#)

Lean: [rational_difference_exact](#), [rational_cross_numerator_positive](#), [rational_error_denominator_bound](#), [positive_rational_difference_lower_bound](#).

Proposition 6.26 (The Möbius identity for S). *The absolutely convergent series satisfy*

$$S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n} = \sum_{d \geq 1} \frac{\mu(d)2^d}{(2^d - 1)^2} = \frac{1}{2} + \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2}.$$

For the first rearrangement use $\varphi(n) = \sum_{d|n} \mu(d)n/d$. For the second use $2^d = (2^d - 1) + 1$ and the classical identity $\sum_{d \geq 1} \mu(d)/(2^d - 1) = 1/2$. These are identities of values, not transfers of irrationality from a Lambert series with a different weight.

[Lean] [n/a] [LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: [tsum_moebius_lambert_sq](#), [totient_series_eq_half_add_moebius_mersenne_square](#).

Corollary 6.27 (An equivalent irrationality question). *Subtracting the rational number $1/2$ gives*

$$S \notin \mathbb{Q} \iff \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2} \notin \mathbb{Q}.$$

This is the same irrationality question in a different series representation. The notation $L_2(\mu)$ for the right-hand series is introduced in Definition 6.29.

[Lean] [n/a] [LEAN SOURCE](#)

Lean: [irrational_totient_iff_moebius_square](#).

Proposition 6.28 (The squared-Lambert identity). *Let $w : \mathbb{N} \rightarrow \mathbb{R}$ satisfy $|w(d)| \leq d$ for $d \geq 1$, and let $0 \leq r < 1$. Then*

$$\sum_{d \geq 1} w(d) \left(\frac{r^d}{1 - r^d} \right)^2 = \sum_{n \geq 1} \left(\sum_{d|n} w(d) \left(\frac{n}{d} - 1 \right) \right) r^n.$$

Expand $(x/(1-x))^2 = \sum_{j \geq 1} (j-1)x^j$ and group terms by $n = dj$. Absolute convergence follows from the bound on w and geometric decay. The cases $w = \mu$, $w = 1$ and $w = \varphi$ give the next three series; the same algebra applies to each, but their arithmetic values differ.

[Lean] [uniform] [LEAN SOURCE](#)

Definition 6.29 (The squared-Lambert series with weight μ). Put

$$L_2(\mu) = \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2} = S - \frac{1}{2}.$$

The equality follows from Proposition 6.26 and Proposition 6.28 with $w = \mu$. Subtracting $1/2$ preserves rationality, so the irrationality of this series is exactly the question about S , not a separate easier claim.

[Cited] [n/a] [LEAN SOURCE](#)

Proposition 6.30 (Weight one and divisor sums). $\sum'_{d:\mathbb{N}^+} 1/(2^d - 1)^2 = \sum'_{n:\mathbb{N}^+} (\sigma(n) - \tau(n)) \cdot (1/2)^n = \zeta_q(2) - \zeta_q(1)$ at $q = 1/2$. The displayed identity is formalised. Irrationality of its value follows from the cited linear independence result of Postelmans and Van Assche, as explained in Proposition 5.17; that literature result is not formalised here. Replacing the weight 1 by μ changes the value to that of Definition 6.29. Observation 6.32 compares the two choices of weight.

[Lean] [n/a] [LEAN SOURCE](#)

Lean: `divisor_sum_identity`, `qZeta`, `qZeta_half_two_sub_one`, `irrational_qZeta_half_difference_of_linearIndependent`, and 1 further declaration. Conditional on the linear independence theorem of Postelmans and Van Assche; see the [coverage section](#).

Proposition 6.31 (Totient weight and gcd moments). $\sum'_{d:\mathbb{N}^+} \varphi(d)/(2^d - 1)^2 = \sum'_{n:\mathbb{N}^+} (P(n) - n) \cdot (1/2)^n$, where $P = \varphi * \text{Id}$ (Pillai's gcd-sum function). It also equals $\mathbb{E}[\gcd(X, Y)]$ when X, Y are independent and $\mathbb{P}(X = n) = \mathbb{P}(Y = n) = 2^{-n}$ for $n \geq 1$; see Proposition 5.18. This is a different weighted series. Its rationality is not settled by the identities proved here.

[Lean] [n/a] [LEAN SOURCE](#)

Lean: `gcd_moment_identity_three_members`, `tsum_pos_pair_gcd_half_eq_totient_div_mersenne_sq`, `pillaiP_eq_totient_mul_id`, `sum_gcd_Icc_eq_pillaiP`, and 1 further declaration in the [coverage section](#).

Observation 6.32 (Comparison of weighted series). For $L(f) = \sum_{d \geq 1} f(d)/(2^d - 1)$, the three weights give

$$L(\mu) = \frac{1}{2}, \quad L(1) = E, \quad L(\varphi) = 2,$$

where E is the irrational Erdős–Borwein constant. For the squared denominators, they give

$$L_2(\mu) = S - \frac{1}{2}, \quad L_2(1) = \zeta_{1/2}(2) - \zeta_{1/2}(1), \quad L_2(\varphi) = \mathbb{E}[\gcd(X, Y)].$$

The first equality is Definition 6.29; the second and third are Propositions 6.30 and 6.31. The middle value is irrational by the cited linear independence theorem. The other two rationality questions are not settled in this record. These examples show that the choice of weight matters: an identity with the same denominators does not transfer an irrationality theorem from one weight to another. They do not classify all weighted Lambert series.

[Lean] [n/a]

Proposition 6.33 (The weight of pairs divisible by a fixed integer). *Let X, Y be independent random variables with $\mathbb{P}(X = n) = \mathbb{P}(Y = n) = 2^{-n}$ for $n \geq 1$. For every $d \geq 1$,*

$$\mathbb{P}(d \mid X, d \mid Y) = \frac{1}{(2^d - 1)^2}.$$

This follows by multiplying the two geometric sums $\sum_{k \geq 1} 2^{-dk} = 1/(2^d - 1)$. It explains the squared denominator in the preceding Möbius identity.

[Lean] [uniform] [LEAN SOURCE](#)

Lean: [pair_divisibility_mass](#), [tsum_geometric_multiples](#), [tsum_pos_pair_both_dvd_half_eq_inv_mersenne_sq](#).

Proposition 6.34 (The sum over coprime directions). *The sum over positive coprime pairs satisfies*

$$\sum_{\substack{a, b \geq 1 \\ \gcd(a, b) = 1}} \frac{1}{2^{a+b} - 1} = 1.$$

To see the normalisation, write every pair of positive integers uniquely as (ka, kb) with $\gcd(a, b) = 1$, and sum $2^{-k(a+b)}$ over $k \geq 1$. The total is $(\sum_{n \geq 1} 2^{-n})^2 = 1$. This base-two normalisation is used in Proposition 6.35.

[Lean] [n/a] [LEAN SOURCE](#)

Proposition 6.35 (A Stern–Brocot recursion with stopping). *For positive integers a, b , put*

$$M(a, b) = \frac{1}{(2^a - 1)(2^b - 1)}.$$

Then

$$M(a, b) = \frac{1}{2^{a+b} - 1} + M(a + b, b) + M(a, a + b).$$

Multiplication by the common denominator proves the identity. At the root $(1, 1)$, the three terms on the right are each $1/3$, and $M(1, 1) = 1$.

The first term is a mass retained at the current node, not passed to either child. Thus the probabilistic interpretation is a branching process with stopping: after division by $M(a, b)$, the stopping and two transition probabilities are, respectively,

$$\frac{(2^a - 1)(2^b - 1)}{2^{a+b} - 1}, \quad \frac{2^a - 1}{2^{a+b} - 1}, \quad \frac{2^b - 1}{2^{a+b} - 1}.$$

They sum to one. The stopping probability is at least $1/3$, which gives the next proposition's geometric error bound. Also $M(a, b) = \mathbb{P}(a \mid X)\mathbb{P}(b \mid Y)$ for the independent geometric variables used above. The identity does not describe a conservative mass split between the two children alone.

[Lean] [uniform] [LEAN SOURCE](#)

Lean: [divisibility_mass](#), [cylinderMass_eq_divisibility_mass_mul](#), [cylinder_mediant_split](#), [cylinder_root_values](#), and 3 further declarations in the [coverage section](#).

Proposition 6.36 (Convergence with an explicit error). *In the splitting identity of Proposition 6.35, the sum of the two child terms is at most $2/3$ of the parent term. If $M_d(a, b)$ is the sum of the contributions removed during the first d levels, then*

$$|M(a, b) - M_d(a, b)| \leq (2/3)^d M(a, b).$$

In particular, $M_d(a, b) \rightarrow M(a, b)$. This convergence statement is separate from identifying the limit with a sum over any independently defined infinite set of descendants; no additional identification is asserted here.

[Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: [cylinderMass_children_le](#), [sternBrocotDepthMass_error](#), [tendsto_sternBrocotDepthMass](#).

Proposition 6.37 (A numerator polynomial and its coefficients). *For squarefree $r \geq 1$, the polynomial defined above has coefficients*

$$[X^k]P_r(X) = \begin{cases} \frac{r}{\gcd(r, k)} \varphi(\gcd(r, k)), & 0 \leq k < r, \\ 0, & k \geq r. \end{cases}$$

For $k < r$, extracting the coefficient gives $\sum_{d \mid \gcd(r, k)} \mu(d) r/d$; the usual divisor formula for φ evaluates this sum. All coefficients in the range $0 \leq k < r$ are positive. Thus a signed sum of geometric polynomials has an explicit positive-coefficient expression. The squarefree hypothesis is retained as in the cited formal statements.

[Lean] [n/a] [LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#)

Proposition 6.38 (Evaluation at two). *For squarefree $r \geq 1$, evaluation of P_r at two gives the integer*

$$P_r(2) = \sum_{d \mid r} \mu(d) \frac{r}{d} \frac{2^r - 1}{2^d - 1}.$$

Each quotient is an integer because $d \mid r$. The formal definition sums over subsets of the prime divisors of r ; for squarefree r these are exactly its divisors. This identifies the polynomial calculation with the numerator used in b_r .

[Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: `mobiusNumeratorPolynomial_eval_two, numerator_eval_two_divisors`.

Proposition 6.39 (Decomposition by the radical). *For $H \geq 1$ and $r \geq 1$,*

$$H\beta_H = \frac{H}{\text{rad}(H)} b_{\text{rad}(H)}, \quad \text{den}(b_r) = \frac{2^r - 1}{\gcd(|P_r(2)|, 2^r - 1)}.$$

The first equality follows from the definition of β_H ; the second is ordinary reduction of an integer fraction. Multiplication by $H/\text{rad}(H)$ can cause further cancellation. No coprimality assumption is included in either identity.

[Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: `radical_decomposition`.

The constant in Proposition 6.40 is the value of P_r at a primitive m -th root of unity. Since the X^k coefficient of P_r depends only on $\gcd(k, r)$, the value is a discrete Fourier coefficient of a function of the greatest common divisor, which Schramm expresses through Ramanujan sums [12, Theorem and (2), p. 2]; Proposition 6.40 specialises this to the present numerator polynomial and proves the integral cyclotomic consequence.

Proposition 6.40 (A cyclotomic congruence). *Let r be squarefree and $m \mid r$. In $\mathbb{Z}[X]$,*

$$P_r(X) \equiv \mu(m)J_2(r/m) \pmod{\Phi_m(X)},$$

where $J_2(n) = n^2 \prod_{p \mid n} (1 - p^{-2})$ is the second Jordan totient. Evaluation at two gives

$$\Phi_m(2) \mid P_r(2) - \mu(m)J_2(r/m).$$

[Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: `cyclotomic_dvd_mobiusNumeratorPolynomial_sub, mobiusNumerator_mod_cyclotomicEval, jordanTotientTwo_eq_prod_primeFactors`.

Proof. To prove the congruence, let ζ be a primitive m -th root of unity. Since $m \mid r$, the geometric sum in the definition of P_r is r/d when $m \mid d$ and zero otherwise. Hence

$$P_r(\zeta) = \sum_{\substack{d \mid r \\ m \mid d}} \mu(d)(r/d)^2.$$

Write $r = mu$. Squarefreeness gives $\gcd(m, u) = 1$, so substituting $d = me$ reduces this to $\mu(m) \sum_{e \mid u} \mu(e)(u/e)^2 = \mu(m)J_2(u)$. The minimal polynomial Φ_m therefore divides the difference in $\mathbb{Q}[X]$; because it is monic, polynomial division gives the quotient in $\mathbb{Z}[X]$. Taking $m = r$ gives the coprimality in Proposition 6.5. $\square$

Squarefreeness cannot simply be dropped from this constant formula. For $r = 4$ and $m = 2$, $P_4(X) = 2 + 4X + 2X^2 + 4X^3$, so $P_4(-1) = -4$, whereas $\mu(2)J_2(2) = -3$. Here $P_4(2) = 50$ and $\Phi_4(2) = 5$; the latter factor cancels from $b_4 = 50/15 = 10/3$. Positive polynomial coefficients alone therefore do not prevent denominator cancellation.

Proposition 6.41 (Adjoining a prime: new divisors). *Let $r \geq 1$, let $p \nmid r$ be prime, and let $m \mid r$. Then*

$$P_{rp}(X) \equiv -P_r(X^p) \pmod{\Phi_{mp}(X)}.$$

[Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#)

Proof. Splitting the divisors of rp into d and pd , with $d \mid r$, gives the exact polynomial identity

$$P_{rp}(X) = p \left(\sum_{j=0}^{p-1} X^{jr} \right) P_r(X) - P_r(X^p).$$

At a primitive mp -th root ζ , the number ζ^r is a nontrivial p -th root, because $m \mid r$ and $p \nmid r$. The geometric factor vanishes, proving the stated congruence. No squarefree hypothesis on r is needed for this identity. If r is squarefree, ζ^p is a primitive m -th root, and Proposition 6.40 gives the constant remainder $-\mu(m)J_2(r/m)$, as in the second linked theorem. $\square$

Proposition 6.42 (Adjoining a prime: existing divisors). *Let r be squarefree, let $p \nmid r$ be prime, and let $m \mid r$. For the existing divisors m , the corresponding identity is*

$$P_{rp}(X) \equiv (p^2 - 1)P_r(X) \pmod{\Phi_m(X)}.$$

[Lean] [uniform] [LEAN SOURCE](#)

Proof. Apply Proposition 6.40 to both r and rp . Since $p \nmid r/m$, multiplicativity gives $J_2(rp/m) = (p^2 - 1)J_2(r/m)$, which proves the identity. The squarefree hypothesis is retained as in the linked theorem. This compares the remainders at the existing divisors m ; the preceding proposition treats the new divisors mp . $\square$

Proposition 6.43 (Lambert-series identities). *For an arithmetic function f for which the sums converge absolutely, write $L(f) = \sum_{n \geq 1} f(n)/(2^n - 1)$. Expanding each geometric series gives*

$$L(f) = \sum_{m \geq 1} \frac{(f * 1)(m)}{2^m}.$$

*In particular, if $\alpha = \varphi * \mu$, then $\alpha * 1 = \varphi$ and $L(\alpha) = S$. The table distinguishes this value from four other Lambert-series values. The identity changes the coefficients and the form of the denominator; it does not transfer an irrationality theorem for one weight to another.*

[Lean] [n/a] [LEAN SOURCE](#)

Lean: `lambertValue_eq_divisor_sum_series`, `alpha_divisor_sum_eq_totient`, `lambertValue_alpha_eq_totientSeries`.

Table 2: Five Lambert-series values and the support for their stated arithmetic status.

Weight f	Value $L(f)$	Status
μ	$L(\mu) = 1/2$	rational, trivial
φ	$L(\varphi) = 2$	rational
Id	$L(\text{Id}) = \sum_m \sigma(m)/2^m$	transcendental, since $1 - 24L(\text{Id})$ is Ramanujan's $P(1/2)$ (Nesterenko [11, Cor. 2, p. 1320], [Cited], not formalised)
1	$L(1) = E$, Erdős–Borwein constant	irrational, [Lean] (irrational_erdosBorwein_series, CertificateKernel.lean:8007); matches #257's full-support case
$\alpha = \varphi * \mu$	$L(\alpha) = S$	OPEN ; this is #249

Remark 6.44 (How the identities differ). The coefficients matter: a result for the constant weight 1 does not settle the series with weight $\alpha = \varphi * \mu$. A theorem expressed in Lambert-series form applies here only after its assumptions have been checked for that weight.

The supplied source review checked the equality between the two indexing conventions for S directly: [LEAN SOURCE](#). For the other rows it consulted the summary in the source rather than every proof body. The table retains the distinction between a cited theorem, an ordinary identity, and a named formal declaration.

Proposition 6.45 (An affine combination annihilates the specified divisor terms). *Let $H \geq 1$ and $d \mid H$, with $d > 0$. The d th term in the Möbius expansion of R_{mH} is $\mu(d)K_d(mH)$, where*

$$K_d(mH) = \frac{mH}{d(2^d - 1)} + \frac{2^d}{(2^d - 1)^2}.$$

This is affine in m . Consequently, for any finite family of real coefficients c_i and nonnegative integers m_i with $\sum_i c_i = \sum_i c_i m_i = 0$, one has $\sum_i c_i K_d(m_i H) = 0$.

For the four multipliers $(1, 3, 5, 15)$ and coefficients $(4, -3, -2, 1)$,

$$K_d(15H) - 3K_d(3H) - 2K_d(5H) + 4K_d(H) = 0.$$

The zeroth and first moments are zero, while the second is $15^2 - 3 \cdot 3^2 - 2 \cdot 5^2 + 4 = 152$. Equivalently, $XY - 3X - 2Y + 4$ takes the values $0, 0, 152$ at $(1, 1), (3, 5), (9, 25)$. This explains the coefficients: they cancel the affine contribution of each divisor of H , without cancelling a general quadratic function of the multiplier.

For a finite truncation of depth $L \geq 0$, write

$$U = \sum_{j=1}^L (\varphi(15H + j) - 3\varphi(3H + j) - 2\varphi(5H + j) + 4\varphi(H + j))2^{L-j}.$$

The difference between $2^L(R_{15H} - 3R_{3H} - 2R_{5H} + 4R_H)$ and U has absolute value at most $19H + 5L + 5$. Indeed, the error equals

$$(R_{15H+L} + 4R_{H+L}) - (3R_{3H+L} + 2R_{5H+L}),$$

and both parenthesised terms lie between 0 and $19H + 5L + 5$ by $0 \leq R_n \leq n + 1$ for $n \geq 1$. Thus

$$19H + 5L + 5 < U \bmod 2^L < 2^L - (19H + 5L + 5)$$

is sufficient for this four-tail combination to be nonintegral. The bound is derived from these tail enclosures; no optimality for the actual totient tails is asserted. Obtaining such certificates at the required unbounded family of LCM heights remains unproved, as in Theorems 6.60 and 6.52. [Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: `transportResidueKernel_eq_mobiusTermKernel`, `mobiusTermKernel_affine_in_multiplier`, `joint35ConeWindow_eq_totientTail_enclosure`, and 5 further declarations in the [coverage](#) section.

Proposition 6.46 (The error under composite dilation). For $A \subseteq \mathbb{N}$ and $n \geq 1$, let $d_A(n) = \#\{d \geq 1 : d \mid n, d \in A\}$. If $a \in A$ and $a, x \geq 1$, then

$$d_A(ax) = d_A(x) + \mathbf{1}_{a \nmid x} + \#\{d \mid ax : d \in A, d \nmid x, d \neq a\}.$$

Indeed, partition the divisors of ax into those already dividing x , the possible new divisor a , and all other new divisors. If every element of A is prime, the last set is empty, since a prime dividing ax but not x must equal the prime a . For composite a it can be nonempty: at $A = \mathbb{N}$, $a = 6$, $x = 1$, the other new divisors are 2 and 3.

This is an unweighted counting identity. The associated Lambert series satisfies

$$\sum_{\substack{a \in A \\ a \geq 1}} \frac{1}{2^a - 1} = \sum_{n \geq 1} \frac{d_A(n)}{2^n}.$$

For the totient series the required divisor-convolution weight is instead $\alpha = \varphi * \mu$, since $\alpha * 1 = \varphi$, as in Proposition 6.43. Weighting all divisors by φ would give $\sum_{d \mid n} \varphi(d) = n$, not $\varphi(n)$; already at $n = 2$ these values are 2 and 1. A weighted version must retain the weights of the new divisors, so it cannot be obtained by substituting $A = \mathbb{N}$ in this unweighted formula. Also $\alpha(p) = p - 2$ for primes p , so α is unbounded and is not a periodic weight.

[Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: `composite_dilation_divisor_count`, `composite_dilation_defect_eq_zero_of_prime_support`, `composite_dilation_divisor_count_prime_support`, `composite_dilation_defect_univ_six_one`, and 8 further declarations in the [coverage](#) section.

Iterated prime dilations of divisor counts. Multiplying an argument by a new prime adds exactly one new family of divisors. For the divisor count d_A above and $n \geq 1$, put $\Delta_p f(n) = f(pn) - f(n)$. If $p_1, \dots, p_r$ are distinct primes, $P = p_1 \cdots p_r$ and $\gcd(n, P) = 1$, then

$$\Delta_{p_1} \cdots \Delta_{p_r} d_A(n) = \#\{d \mid n : Pd \in A\}.$$

For one prime, the divisors of pn not dividing n are exactly pd with $d \mid n$, because $p \nmid n$. Iteration gives the displayed identity: distinctness ensures that each new prime is coprime to the arguments produced at the earlier steps. Equivalently, inclusion–exclusion cancels every divisor not containing all the new prime factors.

Let $\Omega(a)$ count the prime factors of $a \geq 1$ with multiplicity, and let $K \geq 0$ be an integer. If $\Omega(a) \leq K$ for every positive $a \in A$ and $r > K$, the set on the right is empty, since $\Omega(Pd) = r + \Omega(d) > K$. This proves the stated vanishing with all its hypotheses. [LEAN SOURCE](#) The source also proves vanishing of the right-hand divisor count for a list with repeated primes. That is a statement about the condition $Pd \in A$, not an identity with repeated dilation differences.

For example, with $A = \{2\}$ and $n = 1$,

$$\Delta_2^2 d_A(1) = d_A(4) - 2d_A(2) + d_A(1) = -1, \quad \#\{d \mid 1 : 4d \in A\} = 0.$$

Thus the bound $\Omega(a) \leq 1$ and two prime factors do not by themselves make the repeated difference vanish. The strict inequality $r > K$ is also material: $\Delta_2 d_{\{2\}}(1) = 1$ when $r = K = 1$. Finally, at $n = 2$ the one-prime identity without the coprimality hypothesis would give $\Delta_2 d_{\{2\}}(2) = 0$ on the left and $\#\{d \mid 2 : 2d \in \{2\}\} = 1$ on the right. This last example concerns the identity, not the necessity of coprimality for every possible vanishing theorem. These divisor counts belong to the unweighted support problem; they are not the totient coefficients.

6.2.2 Equivalent quantified certificate conditions

Each statement in this block relates a quantified certificate condition to irrationality. The implications are proved; the assertions that certificates occur beyond every threshold remain unproved. Completeness gives equivalences for the base condition and the LCM-diagonal condition. The one-directional implications remain useful proof components, but do not by themselves describe the full logical relation.

Theorem 6.47 (The quantified certificate condition is equivalent to irrationality). $(\forall h : \mathbb{N}, 0 < h \rightarrow \forall N_0 : \mathbb{N}, \exists N \geq N_0, \exists L, C(h, N, L)) \leftrightarrow S \notin \mathbb{Q}$. The quantified-condition side is exactly $C(h, N, L)$ quantified as $\forall h \geq 1 \forall N_0 \geq 0 \exists N \geq N_0 \exists L$. The universally quantified assertion remains unproved. The equivalence identifies the finite witnesses that suffice; it does not establish their existence beyond every threshold.

[Lean] (equivalence proved; supply [Open]) [cofinal] [LEAN SOURCE](#)

Theorem 6.48 (It suffices to use multiples of a period). $(\forall h_0 > 0, \forall N_0, \exists m > 0, \exists N \geq N_0, \exists L, C(m \cdot h_0, N, L)) \rightarrow S \notin \mathbb{Q}$. The shift may be any positive multiple of a prescribed period. This enlarges the choice of finite witness: Theorem 6.47 gives the condition with $m = 1$.

Conversely, rationality would make all such tail differences integral after a fixed starting index. The displayed implication therefore makes this quantified condition equivalent to irrationality; its truth remains unproved.

[Lean] (implication proved; hypothesis [Open]) [cofinal] [LEAN SOURCE](#)

Lean: [irrational_of_period_multiple_certificate_supply](#), [period_multiple_certificate_at_one](#), [rational_forces_period_multiple_integrality](#), [period_multiple_certificate_supply_iff](#).

Theorem 6.49 (The diagonal condition is equivalent to irrationality). $(\forall t_0 : \mathbb{N}, \exists t \geq t_0, \exists L, C(H(t), H(t), L)) \leftrightarrow S \notin \mathbb{Q}$. For a fixed hypothetical rational value, sufficiently large t makes both its dyadic denominator and its odd-part period admissible at $N = h = H(t)$. Conversely, irrationality and pointwise completeness supply a witness for every prescribed t . This is an equivalent condition with one scale parameter; the verified cases $t \leq 82$ do not establish it at arbitrarily large scales.

[Lean] (equivalence proved; supply [Open]) [cofinal] [LEAN SOURCE](#)

Lemma 6.50 (Nondivisors in a short LCM window). Let $t \geq 1$ and $1 \leq j < 2t$ be integers. If $j \nmid H(t)$, then $j = p^a > t$ for a prime p and an integer $a \geq 1$. Indeed, a prime-power divisor of j must exceed t , and $j < 2t$ leaves no room for a cofactor larger than 1. This classifies the exceptional offsets in this short window; it does not bound their contribution to a weighted sum of totient differences.

[Lean] [uniform] [LEAN SOURCE](#)

Proposition 6.51 (Totient factorisation on an LCM progression). Let $t \geq 1, j \geq 1$ and $q \geq 0$ be integers. Suppose $j \mid H(t)$ and every prime divisor of j also divides $H(t)/j$. Then

$$\varphi(qH(t) + j) = \varphi(j) \varphi(q(H(t)/j) + 1).$$

The two factors in $qH(t) + j = j(q(H(t)/j) + 1)$ are coprime: the second is 1 modulo each prime dividing j . Totient multiplicativity gives the identity. The extra prime-divisor hypothesis is not automatic from $j \mid H(t)$; for example, it fails at $t = j = 2$, where $\varphi(H(2) + 2) = 2$ but the proposed product would be 1.

[Lean] [uniform] [LEAN SOURCE](#)

Lean: [clean_lcm_ray_factorisation](#), [unclean_lcm_ray_counterexample](#).

Theorem 6.52 (Rationality forces tail integrality on an LCM grid). If $S \in \mathbb{Q}$, there is $t_1 \in \mathbb{N}$ such that for all $t \geq t_1, q \geq 1$ and $m \geq 0$,

$$R_{(q+m)H(t)} - R_{qH(t)} \in \mathbb{Z}.$$

Thus all tails at positive multiples of $H(t)$ have the same fractional part once $t \geq t_1$. A certificate at one fixed scale does not refute rationality: the threshold t_1 depends on the hypothetical rational value. At a shift $mH(t) > 0$ and basepoint $qH(t)$, such a certificate excludes rational values whose reduced denominator divides $2^{qH(t)}(2^{mH(t)} - 1)$. Certificates at arbitrarily large scales, as required in Theorem 6.53, exclude every hypothetical rational value.

[Lean] [uniform] [LEAN SOURCE](#)

Lean: `lcm_grid_flatness`, `lcm_grid_fractional_parts`, `certificate_denominator_exclusion`, `lcm_grid_supply_iff`.

Theorem 6.53 (A sufficient nonintegrality condition on the grid). *Suppose that for every $t_0 \in \mathbb{N}$ there are integers $t \geq t_0$, $q \geq 1$, $m \geq 0$ and $L \geq 0$ such that*

$$C(mH_t, qH_t, L).$$

Then $S \notin \mathbb{Q}$. Rationality would make the corresponding tail difference integral at every sufficiently large scale, contradicting Proposition 6.18. A successful certificate necessarily has $m > 0$, since the discrepancy vanishes at shift zero.

The diagonal choice is $q = m = 1$; consecutive grid points have $m = 1$, and for $p \geq 2$ the pair (H_t, pH_t) has $(q, m) = (1, p - 1)$. The diagonal equivalence in Theorem 6.49 also gives the converse from irrationality to the displayed quantified condition.

[Lean] (implication proved; hypothesis [Open]) [cofinal] [LEAN SOURCE](#)

Lean: `lcm_grid_supply_iff`, `lcm_grid_multiplier_positive`.

Corollary 6.54 (Equivalent conditions stated without certificates). *The series S is irrational if and only if*

$$R_{2H_t} - R_{H_t} \notin \mathbb{Z} \quad \text{for arbitrarily large integers } t.$$

Equivalently, for arbitrarily large t there are integers $q \geq 1$ and $m \geq 1$ with $R_{(q+m)H_t} - R_{qH_t} \notin \mathbb{Z}$. Apply the pointwise equivalence of Theorem 6.19 to Theorems 6.49 and 6.53, respectively. These are equivalent statements of the irrationality question; the required unbounded sets of scales are not established.

[Lean] (implication proved; hypothesis [Open]) [cofinal] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: `irrational_totient_series_iff_all_tail_diffs_nonintegral`, `periodLcm_diagonal_kill_iff_tail_diff_notMem_int`, `irrational_totient_series_of_lcm_diagonal_nonintegrality_supply`, `irrational_totient_series_of_lcm_cone_nonintegrality_supply`.

Proposition 6.55 (Soundness of a second-difference certificate). *Let $h, N, L \in \mathbb{N}$. If*

$$\begin{aligned} 2(N + 2h + L + 2) &< (D(h, N + h, L) - D(h, N, L)) \bmod 2^L \\ &< 2^L - 2(N + 2h + L + 2), \end{aligned}$$

then $R_{N+2h} - 2R_{N+h} + R_N \notin \mathbb{Z}$. The numerator is the second difference of the finite windows. Subtracting the two tail identities leaves an error of absolute value at most $2(N + 2h + L + 2)$, which explains the radius in the hypothesis. At $(h, N) = (1, 8)$, the first-difference test holds at depth 8, whereas the second-difference test fails at every depth $L \leq 8$ and holds at $L = 9$. A historical probe over $t \leq 20$ reported that the first-difference test was at least as shallow in 30 of 40 sampled cases. This finite comparison is not a uniform ordering of the depths.

[Lean] [uniform] [LEAN SOURCE](#) (the certified $(1, 8)$ comparison: [LEAN SOURCE](#)).

Lean: `abs_tail_diff_scaled_sub_window_le`, `second_difference_error_bound`, `second_difference_certificate_sound`, `second_difference_cell_one_eight`.

For $H, L \in \mathbb{N}$ and a finite set Q of multipliers, put

$$A_q = \sum_{j=1}^L \varphi(qH + j)2^{L-j}, \quad B_q = qH + L + 2 \quad (q \in Q).$$

The finite-grid test compares these integer numerators modulo 2^L .

Theorem 6.56 (A finite-grid condition gives a nonintegral pair). *Let $Q \subseteq \mathbb{N}_{>0}$ be finite and nonempty. Suppose that $B_q < 2^L$ for every $q \in Q$ and that*

$$\forall q_i \in Q \exists q_j \in Q, \quad B_{q_j} < (A_{q_i} - A_{q_j}) \bmod 2^L.$$

Then $R_{q_i H} - R_{q_j H} \notin \mathbb{Z}$ for some $q_i, q_j \in Q$.

[Lean] [uniform] [LEAN SOURCE](#)

Lean: [paperGridNumerator_eq](#), [finite_grid_nonintegral_pair](#).

Proof. The tail identity and its elementary bound give

$$2^L R_{qH} = A_q + R_{qH+L}, \quad 0 \leq R_{qH+L} \leq B_q.$$

Suppose all pairwise tail differences are integers. Choose $q_* \in Q$ minimising R_{qH+L} . For every $q \in Q$, the difference $R_{qH+L} - R_{q_*H+L}$ is then an integer in $[0, B_q]$, congruent to $A_{q_*} - A_q$ modulo 2^L . Since $B_q < 2^L$, it is the least nonnegative residue. This contradicts the displayed condition at q_* . $\square$

For example, take $H = 840$, $L = 12$ and $Q = \{1, 2, 3, 4\}$. The following exact values verify one failed arc containment in each row:

q_i	$A_{q_i} \bmod 4096$	B_{q_i}	q_j	$(A_{q_i} - A_{q_j}) \bmod 4096$
1	2920	854	3	3484
2	2024	1694	1	3200
3	3532	2534	4	3832
4	3796	3374	1	876

The last column exceeds B_{q_i} , not necessarily B_{q_j} . On the circle $\mathbb{R}/4096\mathbb{Z}$, every two arcs $[A_q, A_q + B_q]$ intersect, but the four arcs have no common point. Thus a test using all four points can exclude a common fractional part without any pair of these arcs being disjoint. The theorem does not require pairwise compatibility, or require $|Q| \geq 3$.

The size hypothesis is $2^L > \max_{q \in Q} B_q$. In contrast, the symmetric test C for a pair $q_i < q_j$ needs $2^L > 2B_{q_j}$ before its two-sided interval can contain a residue. These are inequalities for the modulus 2^L . They do not say that the new test uses half as many binary digits, or give a uniform comparison of the first depths at which the tests succeed.

Theorem 6.57 (A sufficient quantified finite-grid condition). *Suppose that for every t_0 there are $t \geq t_0$, a depth L , and a finite nonempty $Q \subseteq \mathbb{N}_{>0}$ such that, with $H = H_t$, all the hypotheses of Theorem 6.56 hold. Then $S \notin \mathbb{Q}$. Indeed, rationality would make all tail differences on every sufficiently large LCM grid integral, whereas the finite-grid theorem supplies a nonintegral pair on such a grid. The example above verifies one grid; it does not establish the quantified hypothesis.*

[Lean] (implication proved; hypothesis [Open]) [cofinal] [LEAN SOURCE](#)

Lean: [paperGridNumerator_eq](#), [finite_grid_supply_irrational](#).

Proposition 6.58 (A finite carry test implies nonintegrality). *For each integer z with $|z| \leq N + h + 1$, define*

$$c_z(0) = z, \quad c_z(i+1) = 2c_z(i) - (\varphi(N+h+i+1) - \varphi(N+i+1)).$$

Suppose that for every such z there is an $i \leq K$ with

$$|c_z(i)| \geq N + i + h + 2.$$

Then $R_{N+h} - R_N \notin \mathbb{Z}$. Indeed, an integral value of the tail difference would be one of the $2(N+h+1)+1$ initial candidates. Its recurrence would satisfy $c_z(i) = R_{N+h+i} - R_{N+i}$ and hence $|c_z(i)| < N + i + h + 2$ at every step, contrary to the finite test. The excluded strip is open: equality at its boundary already suffices. This proposition proves soundness for a specified h, N, K ; an irrationality argument still needs the quantified family of tests.

[Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: [finite_carry_test_sound](#), [finite_carry_candidate_count](#), [finite_carry_true_or_bit](#).

Theorem 6.59 (The block norm condition implies irrationality). *Suppose that for every integer $h \geq 1$ and every threshold $X_0 \in \mathbb{N}$ there are $X, L \in \mathbb{N}$ with*

$$X \geq \max(X_0, 1), \quad 16(2X + h + L + 2) \leq 2^L,$$

and

$$\left\| \sum_{X \leq N < 2X} E(h, N, L) \right\| \leq \frac{21}{25} X.$$

Then $S \notin \mathbb{Q}$. Indeed, the norm bound implies the real-part bound $\operatorname{Re} \sum E(h, N, L) \leq (9/10)X$. Under the displayed size condition, this latter bound already gives a finite residue certificate, by $\cos(\pi/8) > 9/10$ and averaging. The argument uses this specific numerical gap; an unspecified saving below X is not the stated hypothesis.

[Lean] (conditional theorem proved; hypothesis [Open]) [cofinal] [LEAN SOURCE](#) [LEAN SOURCE](#)

Lean: [irrational_of_first_harmonic_norm_gap](#), [first_harmonic_re_bound_of_norm_bound](#), [exists_certificate_of_first_harmonic_real_bound](#), [exists_certificate_of_first_harmonic_norm_bound](#), and 1 further declaration in the [coverage section](#).

Theorem 6.60 (A four-tail residue criterion). *For nonnegative integers H, p, L , define the four-tail combination, its finite numerator and its error bound by*

$$\begin{aligned} J(H, p) &= R_{2pH} - R_{pH} - pR_{2H} + pR_H, \\ W(H, p, L) &= D(pH, pH, L) - pD(H, H, L), \\ B(H, p, L) &= 3pH + (p+1)(L+2). \end{aligned}$$

Then

$$|2^L J(H, p) - W(H, p, L)| \leq B(H, p, L).$$

Consequently the finite condition

$$B(H, p, L) < W(H, p, L) \bmod 2^L < 2^L - B(H, p, L)$$

implies $J(H, p) \notin \mathbb{Z}$. If, for every $t_0 \in \mathbb{N}$, there are $t \geq t_0$, $p \geq 1$ and $L \geq 0$ satisfying this condition with $H = H(t)$, then $S \notin \mathbb{Q}$. Rationality would instead make both diagonal differences in $J(H(t), p)$ integral for every sufficiently large t .

The error bound groups the remainder as

$$(R_{2pH+L} + pR_{H+L}) - (R_{pH+L} + pR_{2H+L}).$$

Each parenthesis is nonnegative and at most $B(H, p, L)$, by $0 \leq R_n \leq n+2$. Their difference therefore has absolute value at most $B(H, p, L)$. This proves the bound without asserting optimality for the actual totient tails. Primality of p is not required by the criterion; a prime p is useful when comparing consecutive LCM heights. The checked instance is $(H, p, L) = (12, 5, 15)$, for which $W(H, p, L) \bmod 2^L = 18834$ and $B(H, p, L) = 282$.

[Lean] (conditional theorem + [Cert]witness; supply [Open]) [cofinal] [LEAN SOURCE](#) [LEAN SOURCE](#)
Lean: [four_tail_combination_eq](#), [four_tail_window_eq](#), [windowDiscrepancy_diagonal_eq](#), [four_tail_error_bound](#), and 5 further declarations in the [coverage section](#).

6.3 Consequences

The next conditions concern a single LCM-diagonal tail difference. For $t, j, a \in \mathbb{N}$, write

$$\delta_t(j) = \varphi(2H(t) + j) - \varphi(H(t) + j), \quad \Omega_a = R_{2H(2^a)} - R_{H(2^a)}.$$

For $h, N \in \mathbb{N}$ and $z \in \mathbb{Z}$, define the integer recurrence

$$c_{h,N,z}(0) = z, \quad c_{h,N,z}(k+1) = 2c_{h,N,z}(k) - (\varphi(N+h+k+1) - \varphi(N+k+1)).$$

If $z = R_{N+h} - R_N$, then $c_{h,N,z}(k) = R_{N+h+k} - R_{N+k}$. This separates an arbitrary integer initial value from the actual tail.

Two finite quantities will also recur. With $H = H_{2^a}$, put

$$\rho_{a,q} = \frac{D(H, H, 2q+1) + \delta_{2^a}(2q+2)}{2^{2q+1}}, \quad \varepsilon_{a,q} = \frac{2H + 2q + 3}{2^{2q+1}}.$$

The prescribed odd depth used below is $2q_a + 1$, where

$$q_a = \left\lfloor \frac{\lfloor \log_2 H \rfloor + 10}{2} \right\rfloor.$$

Thus $2q_a + 1$ is the least odd integer at least $\lfloor \log_2 H \rfloor + 10$. The error bound is $|\Omega_a - \rho_{a,q}| < \varepsilon_{a,q}$. For $a \geq 2$, let $u_{a,q}$ be the centred integer representative of

$$\frac{D(H, H, 2q+1) + \delta_{2^a}(2q+2)}{2} \pmod{4^q}, \quad -\frac{4^q}{2} < u_{a,q} \leq \frac{4^q}{2}.$$

The numerator is even, since the totient arguments exceed 2; at the midpoint this convention chooses the positive representative. The normalised block $\rho_{a,q}$ approximates Ω_a , whereas $u_{a,q}$ records a residue. They must not be confused with the real tail difference or with the integer recurrence.

Proposition 6.61 (An equivalent diagonal nonintegrality condition). *The following statements are equivalent:*

$$S \notin \mathbb{Q} \iff \forall a_0 \in \mathbb{N} \exists a \geq a_0, \Omega_a \notin \mathbb{Z}.$$

This is an exact reformulation of irrationality, not a weaker theorem about a finite set of scales. The later sufficient conditions seek a finite inequality that establishes the right-hand side.

[Lean] (equivalence) [LEAN SOURCE](#)

Lean: [irrational_iff_diagonal_orbit_nonintegrality](#).

Proposition 6.62 (A sufficient short-window condition). *Suppose that for every $a_0 \in \mathbb{N}$ there are $a \geq a_0$ and $L < 2 \cdot 2^a$ with $C(H_{2^a}, H_{2^a}, L)$. Then $S \notin \mathbb{Q}$. Writing out the certificate, the required inequality is*

$$2H_{2^a} + L + 2 < D(H_{2^a}, H_{2^a}, L) \bmod 2^L < 2^L - (2H_{2^a} + L + 2).$$

It places the residue away from both endpoints of the modulus interval. The examples at $a = 4, 6$ establish the bounded statement in Proposition 6.67, not the condition for every threshold. Pointwise completeness allows a sufficiently large depth but does not supply a depth below $2 \cdot 2^a$.

[cofinal] [Open]

[LEAN SOURCE](#)

Lean: [irrational_of_short_window_diagonal_supply](#), [diagonal_certificate_unfolded](#), [pointwise_completeness_supplies_some_depth](#).

Proposition 6.63 (A sufficient approximation condition). *Use the prescribed index q_a and the error ε_{a,q_a} defined above. Suppose*

$$\forall a_0 \in \mathbb{N} \exists a \geq \max(2, a_0) \forall z \in \mathbb{Z}, \quad |\Omega_a - z| \geq \frac{1}{32} + \varepsilon_{a,q_a}.$$

Then $S \notin \mathbb{Q}$. The estimate $|\Omega_a - \rho_{a,q_a}| < \varepsilon_{a,q_a}$ implies $|\rho_{a,q_a} - z| > 1/32$ for every integer z , and hence the finite residue separation used by the linked implication. This is a fixed separation condition along an unbounded family of LCM scales. It is not an irrationality-measure theorem for arbitrary rational approximants, and nonintegrality alone does not give the stated margin. The depth is prescribed, not an additional free witness.

[cofinal] [Open]

[LEAN SOURCE](#)

Lean: [irrational_of_diagonal_orbit_separation_supply](#), [oddGuarded_depth_eq_prescribed](#), [abs_orbit_sub_rawApprox_lt](#), [rawApprox_separation_of_orbit_separation](#).

Proposition 6.64 (A sufficient upper-endpoint separation). *Let $a, J, K, m \in \mathbb{N}$, $a \geq 8$, and $H = H(2^a)$. Assume*

$$J + K + (a + 6) < 2 \cdot 2^a.$$

Define the finite condition

$$\begin{aligned} \mathcal{G}_a(J, K, m) \quad &:\iff m \leq K, \quad 2H + J + K + 2 < 2^m, \\ &D(H, H + J, K) \bmod 2^m \leq 2^m - (2H + J + K + 2). \end{aligned}$$

Then $G_a(J, K, m)$ implies $R_{2H+J} - R_{H+J} \notin \mathbb{Z}$. In particular, irrationality follows if for every a_0 there are $a \geq \max(a_0, 8)$ and K, m satisfying $K + (a + 6) < 2 \cdot 2^a$ and $G_a(0, K, m)$.

The sign theorem forces an integral tail to give a residue in the upper endpoint interval. This one-sided test excludes that interval; it does not require a lower residue bound. Both the sign-range hypothesis and the strict modulus bound are essential premises of this implication. The required unbounded family is not established.

[Lean] (implication; quantified hypothesis [Open]) [LEAN SOURCE](#)

Lean: `upper_endpoint_condition_iff`, `upper_endpoint_gap_nonintegral`, `integral_tail_forces_upper_endpoint_residue`, `irrational_of_upper_endpoint_gap_supply`.

Proposition 6.65 (Relations among the sufficient conditions). *The sufficient conditions do not form a single linear chain. The proved implications are*

$$\begin{aligned} \text{guarded odd-prefix band} &\iff \text{guarded centred-magnitude bound,} \\ \text{guarded centred-magnitude bound} &\implies \text{flexible centred-magnitude bound,} \\ \text{flexible centred-magnitude bound} &\implies \text{adjacent-suffix band} \implies \text{upper-endpoint test.} \end{aligned}$$

There is also a separate sufficient argument: terminal dominance and flexible centred magnitude each imply the two-sided exclusion in Proposition 6.66, which implies diagonal nonintegrality. These statements refer to the quantified versions, with the bounds on a, q and the depth retained. The detailed conditions and sources appear in Proposition 9.14. No reverse implication between terminal dominance and flexible magnitude is asserted, and none of the required unbounded families is proved here.

[Lean] (implications; hypotheses [Open]) [LEAN SOURCE](#)

Lean: `te_chain_relations`.

Proposition 6.66 (An exact endpoint formula). *Let $a, q \in \mathbb{N}$, $a \geq 8$, and $H = H(2^a)$. Assume*

$$2q + 2 + (a + 6) < 2 \cdot 2^a, \quad 2(H + q + 2) \leq 4^q.$$

If $z \in \mathbb{Z}$ and $z = \Omega_a$, then

$$2u_{a,q} = \delta_{2^a}(2q + 2) - c_{H,H,z}(2q + 1).$$

Consequently either of the inequalities

$$\begin{aligned} 2u_{a,q} &\leq \delta_{2^a}(2q + 2) - (2H + 2q + 3), \\ \delta_{2^a}(2q + 2) &\leq 2u_{a,q} \end{aligned}$$

is sufficient for $\Omega_a \notin \mathbb{Z}$. Under integrality the recurrence equals a positive tail difference strictly smaller than $2H + 2q + 3$, and the identity rules out both displayed inequalities. The identity is exact; the resulting nonintegrality criterion is only sufficient. Its converse is not asserted.

[Lean] [LEAN SOURCE](#)

Lean: `endpoint_identity`, `integral_carry_strictly_between`, `endpoint_criterion_nonintegral`, `oddHalfCenteredLift_spec`, and 1 further declaration in the [coverage section](#).

Proposition 6.67 (A sufficient extension of the examples through exponent 6). *The supplied finite result is*

$$\forall a_0 \leq 6 \exists a, L \in \mathbb{N}, \quad a \geq a_0, \quad L < 2 \cdot 2^a, \quad \mathcal{C}(H_{2^a}, H_{2^a}, L).$$

The witnesses at $(a, L) = (4, 23)$ and $(6, 93)$ give this bounded family. A witness with $a \geq 7$ would extend the verified range, but any finite extension would still leave the assertion for arbitrarily large thresholds in Proposition 6.62 unproved.

[bounded] [Lean (finite prefix only)]

LEAN SOURCE

Lean: [short_window_diagonal_through_six](#), [short_window_diagonal_witnesses](#).

Proposition 6.68 (A sufficient extremal-order condition). *Let $H \geq 1$ and $j \geq 0$. If*

$$\varphi(2H + j) < \min\{\varphi(H + j), \varphi(3H + j)\},$$

then

$$\varphi(3H + j) - 2\varphi(2H + j) + \varphi(H + j) > 0.$$

If instead the middle value is strictly larger than both outer values, the displayed second difference is negative. In each case the claim follows by adding the two strict differences. No lower bound on their sizes is needed.

For $H = H_{2^a}$, this reduces nonvanishing of one second difference to an ordering of three totient values. For a prescribed j , the remaining question is whether either strict ordering occurs for arbitrarily large a . The elementary implication does not establish those occurrences. [uniform] [Lean]

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Lean: [extremal_order_curvature_pos](#), [extremal_order_curvature_neg](#), [extremal_order_curvature_ne_zero](#).

Proposition 6.69 (A directed certificate condition). *For fixed $h, N \in \mathbb{N}$, a depth $L \in \mathbb{N}$ satisfying*

$$N + L + 2 \leq D(h, N, L) \bmod 2^L \leq 2^L - (N + h + L + 2)$$

exists if and only if $R_{N+h} - R_N \notin \mathbb{Z}$. The endpoint inequalities are non-strict.

Lean: [directed_certificate_iff](#), [directed_certificate_example](#).

Proof. The exact truncation identity leaves an error $R_{N+h+L} - R_{N+L}$ strictly between $-(N + L + 2)$ and $N + h + L + 2$. An integral tail difference would therefore put the residue in $[0, N + L + 2)$ or $(2^L - (N + h + L + 2), 2^L)$, contrary to the displayed test. Conversely, a nonintegral difference has a symmetric certificate by Theorem 6.19, and every symmetric certificate satisfies this wider interval condition. $\square$

At $H_3 = 6$ and $L = 6$, the discrepancy is $D(6,6,6) = 270$, with residue 14 modulo 64. The directed interval is $[14, 44]$, so its lower endpoint succeeds. The symmetric interval is $(20, 44)$ and fails; its first successful depth is 7. Thus the wider test can improve the first successful depth, but no uniform depth saving is asserted. Replacing $\mathcal{C}$ by the displayed inequality in Theorem 6.49 gives the equivalent directed diagonal condition, whose occurrence at arbitrarily large scales remains unproved. [uniform] [Lean]

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Proposition 6.70 (A sufficient condition at the prescribed mod-four pulses). *Suppose that for every positive integer h and every $B \in \mathbb{N}$ there are a prime $p > B$ and $K \in \mathbb{N}$ such that*

$$\varphi(p + 4h) - \varphi(p) \equiv 2 \pmod{4}$$

and every integer z with $|z| \leq p + 4h + 1$ and $z \equiv 2 \pmod{4}$ has an index $0 \leq i \leq K$ satisfying

$$|c_{4h,p,z}(i)| \geq p + i + 4h + 2.$$

Then $S \notin \mathbb{Q}$. Under a hypothetical eventual period h , the proved congruence argument at a sufficiently large such prime forces $R_{p+4h} - R_p$ to be an integer in the tested residue class. The finite test excludes every candidate in that class, a contradiction.

The mod-four congruence is supplied by Proposition 6.91; exclusion at the same prime is not. Restricting to $2 \pmod{4}$ retains one residue class out of four, not necessarily exactly one quarter of the candidates in a finite interval. The quantifier over every bound B cannot be replaced by one fixed prime for each h .

[cofinal] [Lean]

[LEAN SOURCE](#)

Lean: [irrational_of_modFour_pulse_supply](#), [rational_forces_pulse_class_integrality](#).

Proposition 6.71 (The full quantified condition).

$$(\forall h \geq 1, \forall N_0, \exists N \geq N_0, \exists L, \mathcal{C}(h, N, L)) \iff S \notin \mathbb{Q}.$$

This is exactly the quantifier structure $\forall h \geq 1 \forall N_0 \exists N \geq N_0 \exists L \mathcal{C}(h, N, L)$. The quantified condition is not supplied here. The equivalence specifies exactly what a residue-based proof would need beyond the finite examples. [cofinal] [Lean] (equivalence; supply [Open])

[LEAN SOURCE](#)

Proposition 6.72 (One diagonal parameter suffices). *The following condition is equivalent to irrationality:*

$$(\forall t_0 \in \mathbb{N} \exists t \geq t_0 \exists L \geq 0, \mathcal{C}(H(t), H(t), L)) \iff S \notin \mathbb{Q}.$$

For a hypothetical rational value, let h_0 and N_0 be its eventual tail period and starting index. Taking $t \geq \max(h_0, N_0)$ ensures $h_0 \mid H(t)$ and $H(t) \geq N_0$. The corresponding diagonal difference is then integral, contradicting a certificate at that scale. Conversely, irrationality and pointwise completeness provide a witness at every prescribed t . The reduction uses one scale parameter; it neither establishes the certificate condition nor makes one fixed scale sufficient for all rational values.

[cofinal] [Lean] (equivalence; supply [Open])

[LEAN SOURCE](#)

Proposition 6.73 (A sufficient condition on the LCM grid).

$$(\forall t_0, \exists t \geq t_0, \exists q m L, 0 < q \wedge C(m \cdot H t)(q \cdot H t) L) \implies S \notin \mathbb{Q}.$$

One residue certificate anywhere on the two-multiplier LCM cone, at arbitrarily large t , suffices; the diagonal choice $q = m = 1$ recovers Proposition 6.72. [cofinal] [Lean (implication; premise unproved)]

[LEAN SOURCE](#)

Proposition 6.74 (A finite-grid certificate condition). Let $H, L \in \mathbb{N}$ and let $Q \subseteq \mathbb{N}_{>0}$ be finite and nonempty. Define $A_q = \sum_{j=1}^L \varphi(qH + j)2^{L-j}$ and $B_q = qH + L + 2$. If $B_q < 2^L$ for every $q \in Q$ and

$$\forall q_i \in Q \exists q_j \in Q, \quad B_{q_j} < (A_{q_i} - A_{q_j}) \bmod 2^L,$$

then $R_{q_j H} - R_{q_i H} \notin \mathbb{Z}$ for some $q_i, q_j \in Q$. Such a test at $H = H_t$ for arbitrarily large t implies $S \notin \mathbb{Q}$. Neither $|Q| \geq 3$ nor compatibility of every pair is a hypothesis. The proof, by choosing a minimum tail in the finite set, is given in Theorem 6.56. [cofinal] [Lean (implication; premise unproved)]

[LEAN SOURCE](#)

Lean: [paperGridNumerator_eq](#), [finite_grid_nonintegral_pair](#), [finite_grid_supply_irrational](#).

Proposition 6.75 (Unbounded Farey bounds imply irrationality). The Farey-gap denominator bound at window K is currently $\sim 7.96 \times 10^{34}$ at $K = 240$ (Prop. 6.99). If

$$\sup_K (b + d)(K) = \infty$$

(that is, the proved exclusion bounds are arbitrarily large), then #249 follows via the denominator-exclusion implication, without assuming the certificate supply in Proposition 6.71. No unboundedness claim is proved or attempted in the record. [cofinal] [Open]

[LEAN SOURCE](#)

Lean: [gapCertificate_window_1_240](#), [irrational_totientSeries_of_fareyGapExclusionUnbounded](#), [gapFareyBound_window_1_240](#).

Proposition 6.76 (The lower bound and a false proposed upper bound). By Prop. 6.98, the canonical dyadic totient-kernel family is unconditionally $(2^e + 1)$ -dimensional at every level $e \geq 1$. Rationality of S forces an associated tempered carry orbit with $\mathbb{Q}$ -rank $\geq 2^e - 1$ at every level (Prop. 6.90). An upper bound independent of e for these same carry ranks would contradict the lower bound. More generally, a bound $g(e)$ with $g(e) < 2^e - 1$ at some level $e \geq 1$ would suffice, provided it applies to the actual carry under the hypothetical rationality of S . A bound that merely grows with e need not contradict anything. The generic assertion for arbitrary rational coefficient series is false: the 5/4 comparison sequence in Section 10.8 has an integer carry satisfying the growth condition and the same rank lower bound. The incompatible integer identity and the finite-shift counterexample (Observation 6.115) remain separate counterexample results. None is a generic rationality-driven rank ceiling. The evidence label below applies to the proved rank floor and counterexample results, not to the counterfactual upper bound. [uniform] [Lean]

[LEAN SOURCE](#)

Lean: `rank_floor_and_false_proposed_carryRank_ceiling`, `fiveQuarter_comparison_rational_with_carryRank_floor`, `no_generic_rationality_carryRank_ceiling`.

Proposition 6.77 (Soundness of the finite carry test). *For fixed $h, N, K \in \mathbb{N}$, use the integer recurrences $c_{h,N,z}$ defined above. Test each of the $2(N + h + 1) + 1$ candidates $z \in \mathbb{Z}$ with $|z| \leq N + h + 1$. If every candidate has some $i \leq K$ for which*

$$|c_{h,N,z}(i)| \geq N + i + h + 2,$$

then $R_{N+h} - R_N \notin \mathbb{Z}$. An integral tail difference would give one of the initial candidates and would remain in the open strip $|c_{h,N,z}(i)| < N + i + h + 2$ at every step. Equality at a boundary already excludes a candidate. This is the soundness implication for a specified finite test; no bound for a successful search depth is asserted.

[n/a] [Lean]

LEAN SOURCE

Lean: `finite_carry_test_sound`, `finite_carry_candidate_count`, `finite_carry_true_orbit`.

6.4 Further exact identities

The identities in this subsection hold independently of the proposed residue tests. Each gives either an exact expression or a bound that can be used when studying those tests.

Proposition 6.78 (The exact totient difference on an LCM progression). *Put $H = H(t)$. For every $j \geq 0$, the arithmetic expression in the linked source equals $\delta_t(j) = \varphi(2H+j) - \varphi(H+j)$. For a divisor $j \mid H$, this assertion contains a useful product formula, not merely a change of notation. Set*

$$a = H/j, \quad g_1 = \gcd(j, a + 1), \quad g_2 = \gcd(j, 2a + 1).$$

Then

$$\delta_t(j) = \varphi(j) \left(\frac{g_2 \varphi(2a + 1)}{\varphi(g_2)} - \frac{g_1 \varphi(a + 1)}{\varphi(g_1)} \right).$$

Here $j \geq 1$ because $H > 0$, so both denominators are nonzero. The identity follows from $\varphi(jx) = \varphi(j)\varphi(x)\gcd(j, x)/\varphi(\gcd(j, x))$, which retains all primes shared by j and x . If every prime dividing j also divides a , then $g_1 = g_2 = 1$, and the formula reduces to $\varphi(j)(\varphi(2a + 1) - \varphi(a + 1))$. For example, at $H = j = 2$ the two gcds are $g_1 = 2$ and $g_2 = 1$; the exact difference is $\varphi(6) - \varphi(4) = 0$, whereas discarding the gcd factors would give 1. For $j \nmid H$, the source retains the literal totient difference. Thus no coprimality assumption is imposed on the full expression.

[uniform] [Lean]

LEAN SOURCE

Lean: `lcmRayArithmeticLetter_eq_totient_difference`, `totient_mul_eq_totient_mul_gcd_div_totient_gcd`, `lcmRay_divisor_denominators_pos`, `lcmRay_divisor_product_formula`, and 3 further declarations in the [coverage section](#).

Proposition 6.79 (A lower bound for the totient of a rough integer). *Let $a \geq 8$, set $t = 2^a$, and let $n > 0$ be an integer all of whose prime factors exceed t . If $n < 2^{2t}$, then n has fewer than $t/4$ distinct prime factors and*

$$\varphi(n) > \frac{3}{4}n.$$

If n has r distinct prime factors, then $n \geq t^r = 2^{ar}$, so $ar < 2t$ and $r < t/4$ because $a \geq 8$. For the second, use $\varphi(n)/n = \prod_{p|n} (1 - 1/p) \geq 1 - \sum_{p|n} 1/p > 3/4$. These are bounds for rough integers in the stated size range; they are not pointwise bounds for arbitrary totient values.

[uniform] [Lean]

LEAN SOURCE

Lean: `rough_integer_prime_count_and_totient_bound`.

Proposition 6.80 (Positivity of the short-window differences). *For $a \geq 8$ and $1 \leq j < 2 \cdot 2^a$,*

$$\delta_{2^a}(j) = \varphi(2H_{2^a} + j) - \varphi(H_{2^a} + j) > 0.$$

The proof treats divisor offsets and the exceptional prime-power offsets separately, using multiplicativity and the rough-integer estimate of Proposition 6.79. No rationality hypothesis is used. Positivity of these finitely many coefficients contributes to the later sign theorem; a bound for the remaining infinite tail is still needed to obtain that theorem.

[uniform] [Lean]

LEAN SOURCE

Lean: `shortWindow_totient_difference_pos`.

Proposition 6.81 (The accumulated sum is the discrepancy). *For all $t, L \in \mathbb{N}$, the weighted diagonal sum is exactly*

$$\sum_{j=1}^L \delta_t(j) 2^{L-j} = D(H(t), H(t), L).$$

Thus its two-sided residue inequality is precisely $C(H(t), H(t), L)$. The equality follows term by term from the definition of δ_t . A separate restriction on L , when present, remains part of the hypothesis; this identity does not supply such a restricted-depth witness.

[n/a] [Lean]

LEAN SOURCE

Lean: `weighted_shortWindow_sum_eq_windowDiscrepancy`, `weighted_shortWindow_band_iff_certifiedKill`.

Proposition 6.82 (A global-to-local identity).

$$\Omega_a = 2^H(2^H - 1) \left(\sum_n' \varphi(n)/2^n \right) - (\Phi_{2^H} - \Phi_H), \quad H = H(2^a).$$

Thus a separation estimate for Ω_a is an estimate for this integer translate of a multiple of S . The identity does not require other rational-approximation arguments to use this particular translate. [n/a] [Lean]

LEAN SOURCE

Lean: [actualLcmTailOrbit_global_to_local](#).

Proposition 6.83 (A rational approximation with an error bound). *For all $a, q \in \mathbb{N}$, with $H = H_{2^a}$ and the finite $\rho_{a,q}$ defined above,*

$$|\Omega_a - \rho_{a,q}| < \varepsilon_{a,q} = \frac{2H + 2q + 3}{2^{2q+1}}.$$

The triangle inequality transfers a separation estimate between the real tail and its rational approximation at the cost of this error. The bound tends to zero with q for fixed a ; no assertion of optimality is made. [uniform] [Lean]

LEAN SOURCE

Lean: [abs_actualLcmTailOrbit_sub_rawApprox_lt_explicit](#), [tendsto_actualLcmRawError-Radius_atTop_nhds_zero](#).

Proposition 6.84 (Unconditional positivity). *For $a \geq 8$, $J + (a + 6) < 2 \cdot 2^a$:*

$$0 < R_{2H+J} - R_{H+J}, \quad H = H(2^a),$$

with no irrationality hypothesis; the true, infinite, real translated tail difference is strictly positive throughout almost the entire short window, proved unconditionally from Prop. 6.80 plus a directed one-sided tail bound. Any argument about the sign of the actual orbit (not merely its residue mod 2^L) must agree with this; specialised at $J = 0$ this gives $0 < \Omega_a$. [uniform] [Lean]

LEAN SOURCE

Lean: [actualLcm_tailDiff_shift_pos_paper](#), [actualLcmTailOrbit_pos](#).

Proposition 6.85 (The residue forced by integrality). *Let $a, J, K \in \mathbb{N}$, $a \geq 8$, $H = H(2^a)$, and assume*

$$J + K + (a + 6) < 2 \cdot 2^a, \quad 2H + J + K + 2 < 2^K.$$

If $R_{2H+J} - R_{H+J} \in \mathbb{Z}$, then

$$\begin{aligned} D(H, H + J, K) \bmod 2^K &= 2^K - (R_{2H+J+K} - R_{H+J+K}), \\ 0 < R_{2H+J+K} - R_{H+J+K} &< 2H + J + K + 2. \end{aligned}$$

The later tail difference is a positive integer. Its negative is the representative near zero, while the least nonnegative residue lies near 2^K . Positivity alone therefore does not prove nonintegrality; a separate estimate must exclude this upper interval.

[Lean] [uniform] LEAN SOURCE

Lean: [actualLcm_integral_forces_topEdgeResidue_paper](#).

Proposition 6.86 (The penultimate term of a partial divisibility pattern). *Let $a, J, K, m \in \mathbb{N}$, $a \geq 8$, $H = H_{2^a}$ and $B = 2H + J + K + 2$. Suppose that*

$$\begin{aligned} 0 < m &\leq K, & J + K + a + 6 &< 2 \cdot 2^a, \\ B &< 2^m, & \delta_{2^a}(J + K) &\leq 2^m - B, \\ 2^{r+1} &\mid \delta_{2^a}(J + K - m + r + 1) & (0 \leq r, r + 1 &< m). \end{aligned}$$

Then

$$\delta_{2^a}(J + K - 1) = 2^{m-1}, \quad 2^m < 2B.$$

Indeed, $B \geq 4$ and $B < 2^m$ force $m \geq 3$. The penultimate divisibility, positivity, and the bound $\delta_{2^a}(J + K - 1) < B < 2^m$ leave 2^{m-1} as its only possible value. Thus the modulus lies in the strict interval $B < 2^m < 2B$; there is at most one possible power of two. The value fixed at 2^{m-1} is the penultimate difference. The terminal difference $\delta_{2^a}(J + K)$ instead satisfies the separate upper bound in the hypotheses. These restrictions are necessary for the specified pattern; the proposition does not construct such patterns at arbitrarily large a . [bounded] [Lean]

LEAN SOURCE

Lean: `penultimate_shortWindow_difference_eq_half`, `dyadicScale_unique_in_open_interval`.

Proposition 6.87 (An equivalent test using two residue bits). *For every $h, N \in \mathbb{N}$, existence of a certificate at some depth is equivalent to the following explicit condition:*

$$\exists s, b \in \mathbb{N}, \quad C(h, N + s, b + 1) \quad \text{or} \quad \begin{cases} N + s + h + b + 4 < 2^b, \\ 2^b \leq D(h, N + s, b + 2) \bmod 2^{b+2} < 3 \cdot 2^b. \end{cases}$$

The last interval says that the two leading bits are 01 or 10. The proof uses the first successful certificate of depth L and may take $b = \lfloor \log_2(N + h + L + 2) \rfloor + 1$. Since b depends on that unknown L , the equivalence is not an a priori bound on the search depth. The proof uses the finite tail recurrence and its size bound; it supplies no totient-specific assertion that a witness exists. [uniform] [Lean]

LEAN SOURCE

Lean: `exists_certifiedKill_iff_twoBitResidueTest`, `dyadicMixedGuard_iff_twoBitBand`.

Proposition 6.88 (The factor in a second difference). *Let $a \geq 4$ and $j \geq 1$ be integers with $j^2 \leq 2^a$, and put $H_a = H(2^a) = \text{lcm}(1, \dots, 2^a)$. Then*

$$2\varphi(j) \mid \varphi(3H_a + j) - 2\varphi(2H_a + j) + \varphi(H_a + j).$$

To see the factor, set $A = H_a/j$. The square bound ensures that $j \mid H_a$ and that every prime divisor p of j also divides A : indeed, $pj \leq j^2 \leq 2^a$ implies $pj \mid H_a$. Consequently $\gcd(j, qA + 1) = 1$ for $q = 1, 2, 3$, and the displayed difference equals

$$\varphi(j)(\varphi(3A + 1) - 2\varphi(2A + 1) + \varphi(A + 1)).$$

Also $2j \leq 2^a$, so $2j \mid H_a$ and $A \geq 2$ is even. The three arguments $qA + 1$ are therefore odd and greater than two, and all three totients are even. This proves the extra factor of two. The divisibility is a lower bound, not an exact valuation at each LCM height. The different family in Observation 6.108 does not satisfy these LCM and square-window hypotheses, so it does not establish sharpness within this restricted family. [bounded] [Lean]

LEAN SOURCE

Lean: `two_mul_totient_dvd_totient_second_difference`, `fixedRank_cleanWindow_structure`.

Proposition 6.89 (Carry displacement and tail integrality). *For a positive integer v and a tempered integral totient carry u ,*

$$v \mid u(N+k) - u(N) \iff R_{N+k} - R_N \in \mathbb{Z}.$$

Indeed, temperedness identifies $u(N) = vR_N$, so the displacement is $v(R_{N+k} - R_N)$. This is a divisibility test for that displacement, not a dimension bound for the carry sections. The same proof works for another coefficient sequence once its tempered carry has been identified with its scaled tail; those hypotheses must be checked in each application. [uniform] [Lean]

LEAN SOURCE

Lean: `carryShift_dvd_iff_tailDiff_integral, temperedCarry_eq_scaledTail_and_shift`.

Proposition 6.90 (A consequence of rationality and its limitation). *If $S \in \mathbb{Q}$, there are a positive integer v and a tempered integer carry orbit u such that its retained sections through every level e have rational rank at least $2^e - 1$. At the same time, those sections are uniformly eventually periodic modulo v . Periodicity after reduction modulo v is a statement in a finite quotient; it is not a rank upper bound over $\mathbb{Q}$. Proposition 6.76 gives the lower bound, while the rational comparison sequence explains why a general rank upper bound cannot be deduced from these recurrence assumptions alone. [uniform] [Lean]*

LEAN SOURCE

Lean: `rationality_forces_mod_period_and_unbounded_rank`.

Proposition 6.91 (A totient difference congruent to two modulo four). *For every positive integer h and every $B \in \mathbb{N}$, there is a prime $p > B$ such that*

$$\varphi(p+4h) - \varphi(p) \equiv 2 \pmod{4}.$$

To construct p , set $H = 4h$, choose a prime $r > H$ with $r \equiv 1 \pmod{4}$, and use Dirichlet's theorem in the progression $p \equiv 3r - H \pmod{4r}$. This progression is reduced: its residue is odd and is $-H \not\equiv 0 \pmod{r}$. Then $p \equiv 3 \pmod{4}$, so $\varphi(p) = p - 1 \equiv 2 \pmod{4}$, whereas $r \mid p+H$ gives $4 \mid \varphi(p+H)$. Subtracting gives the stated congruence. This supplies the arithmetic congruence at arbitrarily large primes; the additional finite exclusion in Proposition 6.70 remains a separate hypothesis.

[cofinal] [Lean]

LEAN SOURCE

Lean: `exists_prime_totient_shift_four_mul_congr_two_mod_four`.

Proposition 6.92 (An arbitrarily long zero prefix followed by a two-adic pulse). *Let $K \geq 2$ and $H > K$ be integers. For every $B \in \mathbb{N}$ there is a prime $p > \max(B, H+K)$ such that*

$$\begin{aligned} p &\equiv 1 + 2^{K-1} \pmod{2^K}, & 2^K \mid \varphi(p+H), \\ 2^K \mid \varphi(p-j), & & 2^K \mid \varphi(p-j+H) \quad (1 \leq j < K). \end{aligned}$$

The construction uses $2K - 1$ distinct auxiliary primes congruent to 1 $\pmod{2^K}$: one for $p+H$, and two for each of the $K-1$ preceding positions. Choose them larger than $\max(H+K, 2^K)$. For the first prime impose $p \equiv -H$; for the pair at position j impose $p \equiv j$ and $p \equiv j - H$,

respectively. These nonzero residues, together with the displayed odd residue modulo 2^K , combine by the Chinese remainder theorem into a reduced progression. Dirichlet's theorem then supplies arbitrarily large prime values of p . Each auxiliary prime contributes the required factor 2^K to the corresponding totient.

Thus the first $K - 1$ totient differences in the window starting at $p - K$ vanish modulo 2^K , while the last is 2^{K-1} modulo 2^K . Consequently

$$D(H, p - K, K) \equiv 2^{K-1} \pmod{2^K}.$$

If $R_{N+H} - R_N \in \mathbb{Z}$ for every $N \geq N_0$, choose such a prime with $p \geq N_0 + K$. Iterating the carry recurrence from $p - K$ then gives

$$R_{p+H} - R_p \in \mathbb{Z}, \quad R_{p+H} - R_p \equiv 2^{K-1} \pmod{2^K}.$$

The initial integer is multiplied by 2^K and disappears only after reduction modulo 2^K . This does not bound its higher quotient. The condition $H > K$ is part of the construction; it does not provide arbitrarily large K for one fixed shift H .

[cofinal] [Lean]

LEAN SOURCE

Lean: `exists_prime_twoAdic_pulse_block`, `pulse_delta_of_divisor_data`, `exists_prime_integral_tailDiff_half_pulse`.

Proposition 6.93 (A Möbius-inversion formula for the tail). For $N \in \mathbb{N}$ and $d \geq 1$, let $r_d(N) = d - (N \bmod d)$ and $q_d(N) = \lfloor N/d \rfloor + 1$. Thus $r_d(N)$ is the distance to the next strictly larger multiple of d , so $1 \leq r_d(N) \leq d$. Möbius inversion gives

$$R_N = \sum_{d \geq 1} \mu(d) 2^{d-r_d(N)} \left(\frac{q_d(N)}{2^d - 1} + \frac{1}{(2^d - 1)^2} \right).$$

Indeed, substitute $\varphi(n) = \sum_{d|n} \mu(d)n/d$ and group by d . Its multiples after N have quotients $q_d(N) + \ell$ and forward distances $r_d(N) + d\ell$, for $\ell \geq 0$; summing the two geometric series gives the formula. Absolute convergence justifies the regrouping, since the sum of the absolute divisor contributions at n is at most $\sum_{d|n} n/d \leq n^2$, and $\sum_{n > N} n^2 2^{N-n} < \infty$. This identity is available for later transformations; it does not assert that every Möbius or Lambert-series argument must use it.

[uniform] [Lean]

LEAN SOURCE

Lean: `forwardMultiple_spec`, `forwardMultipleShift_least`, `forwardMultiple_enumeration`, `totientTail_eq_tsum_mobius_inversion`.

Proposition 6.94 (Explicit finite examples). The finite certificates $\mathcal{C}(H_{16}, H_{16}, 23)$ and $\mathcal{C}(H_{64}, H_{64}, 93)$ imply $\Omega_4, \Omega_6 \notin \mathbb{Z}$. Both satisfy the short-window restriction, since $23 < 32$ and $93 < 128$. They use the pre-existing certificates `certifiedKill_diagonal_t16/t64`. These are two explicit instances of the desired condition, not the start of a proved induction or a claim that no other finite instances can be checked.

[fixed] [Cert]

LEAN SOURCE

Lean: `shortWindow_certificates_kill_omega_four_and_six`, `shortWindow_inequalities`, `actualLcmTailOrbit_eq_tail_difference`.

Proposition 6.95 (The necessary depth). *Every certificate $C(h, N, L)$ satisfies*

$$2(N + h + L + 2) < 2^L.$$

The two endpoint margins must fit inside an interval of length 2^L . In particular, successful depths grow at least logarithmically with $N + h$. This is a necessary lower bound, not an upper bound for the first successful depth. The later two-bit reformulation applies at its stated modulus and does not reverse this implication.

[n/a] [Lean]

LEAN SOURCE

Lean: `certificate_logarithmic_depth`, `fixed_depth_bounds_indices`, `certifiedKill_depth_floor`.

Proposition 6.96 (Rationality gives an eventual tail period). *Suppose $S = a/(2^c v)$ in lowest terms, where $a \in \mathbb{Z}$, $c \geq 0$ and $v \geq 1$ is odd. Take $h = \varphi(v)$, including $h = 1$ when $v = 1$. Euler's theorem gives $v \mid 2^h - 1$, so*

$$2^N(2^h - 1)S \in \mathbb{Z} \quad (N \geq c).$$

The identity relating S to its scaled tails therefore gives $R_{N+h} - R_N \in \mathbb{Z}$ for every $N \geq c$. These are explicit witnesses for the eventual period asserted in Proposition 6.71. To use the finite residue test against a hypothetical rational value, one must obtain a nonintegral difference for its period at an index beyond its preperiod. Soundness proves that the quantified certificate condition rules out rationality. Conversely, the prefix-tail identity makes every positive-shift tail difference nonintegral when S is irrational, and pointwise completeness (Theorem 6.19) then supplies a depth. Proposition 6.81 only identifies the weighted diagonal sum with D ; it is not the completeness argument and supplies no depth bound. [uniform] [Lean]

LEAN SOURCE

Lean: `rational_tail_period_explicit_witnesses`, `totient_one_eq_one`, `eventual_tail_period_of_not_irrational`, `irrational_of_certificate_supply`, and 2 further declarations in the [coverage section](#).

Proposition 6.97 (Tail integrality on an LCM grid). *If $S \in \mathbb{Q}$, there is $t_1 \in \mathbb{N}$ such that for all $t \geq t_1$, $q \geq 1$ and $m \geq 0$,*

$$R_{(q+m)H(t)} - R_{qH(t)} \in \mathbb{Z}.$$

The threshold may depend on the hypothetical rational value. A certificate at one fixed scale therefore gives a finite denominator exclusion, not irrationality. Proposition 6.73 requires certificates at arbitrarily large scales, so that one lies beyond the threshold supplied by any such rational value.

[n/a] [Lean]

LEAN SOURCE

Lean: `lcm_grid_flatness`, `certificate_denominator_exclusion`, `lcm_grid_supply_iff`.

Proposition 6.98 (Independence of the retained dyadic family). *For every $e \geq 0$, the auxiliary canonical dyadic family has $2^e + 1$ indexed sections, and these sections are linearly independent over $\mathbb{Q}$. The CRT–Dirichlet argument is explained in Section 10.8; the declaration below counts its index set. For $e \geq 1$ the family is a basis for all sections through level e . At $e = 0$ the auxiliary family still contains both $\varphi(n)$ and $\varphi(2n)$, so it is not the actual level-zero truncation, whose dimension is one. No rationality hypothesis on S is used.*

[n/a] [Lean]

LEAN SOURCE

Lean: `card_canonical_dyadic_index`, `linearIndependent_canonical_dyadic_family`, `canonical_family_basis_through_level`, `canonical_level_zero_channels`, and 1 further declaration in the [coverage section](#).

Proposition 6.99 (The exact range of the stated Farey-gap inequality). *Let V be the totient residue at $(N, K) = (1, 240)$ defined in Section 5.3. For every integer q satisfying*

$$1 \leq q \leq 79639646646701375323355774875831053,$$

one has

$$(qV) \bmod 2^{240} + 243q < 2^{240}.$$

The first failure occurs at

$$q = 79639646646701375323355774875831054,$$

obtained from the stated Farey mediant. Thus the range is exact for this particular inequality. It is not an optimality claim among all denominator-exclusion tests. The preceding range gives the exclusion in Proposition 6.100, approximately 7.96×10^{34} . [fixed] [Cert]

LEAN SOURCE

Lean: `farey_window_1_240_exact_range`, `farey_window_1_240_range_magnitude`.

Proposition 6.100 (The resulting denominator exclusion). *For every reduced fraction a/q , with $a \in \mathbb{Z}$ and $q \geq 1$,*

$$q \leq 79639646646701375323355774875831053 \implies S \neq a/q.$$

This is an unconditional finite denominator exclusion. It does not assume certificates at arbitrarily large scales, as required by Proposition 6.71. Conversely, excluding this finite range does not rule out rational values with larger reduced denominators. [fixed] [Lean]

LEAN SOURCE

Lean: `totient_series_ne_reduced_fraction_of_small_denominator`, `totient_series_ne_int_div_of_small_denominator`.

Proposition 6.101 (Two general irrationality criteria). *Let $x \in \mathbb{R}$. One sufficient condition is a sequence of reduced fractions $u_j = a_j/q_j$, $q_j \geq 1$, with $u_j \neq x$ for all sufficiently large j and*

$$q_j |x - u_j| \longrightarrow 0.$$

Another is that, for every integer $Q \geq 1$, there exist $m, z \in \mathbb{Z}$ such that

$$0 < |mx - z| < 1/Q.$$

Indeed, if $x = a/b$ is reduced with $b > 0$, every nonzero error in the first condition is at least $1/(bq_j)$, and every nonzero error in the second is at least $1/b$. The strict lower bounds exclude exact rational hits. Requiring $m = b_0^n$ for a fixed integer base $b_0 \geq 2$ is a stronger sufficient condition, not a reformulation valid for every irrational number. The binary example in Proposition 6.23 explains this distinction. The linked declaration proves the first criterion. [n/a] [Lean]

LEAN SOURCE

Lean: `irrational_of_den_mul_abs_sub_tendsto_zero`, `one_div_den_mul_den_le_abs_sub`, `irrational_of_int_mul_near_int`, `irrational_of_pow_mul_near_int`, and 8 further declarations in the [coverage section](#).

Proposition 6.102 (Lambert identities involving S). For $L(f) = \sum_{n \geq 1} f(n)/(2^n - 1)$, the identities are $L(\mu) = 1/2$, $L(\varphi) = 2$, $L(1) = E$ and $L(\varphi * \mu) = S$. The last identity rewrites the same unknown value; it does not deduce its irrationality from that of E . The value $L(\text{Id}) = \sum_{m \geq 1} \sigma(m)/2^m$ is transcendental by Nesterenko [11, Cor. 2, p. 1320] (cited, not formalised). These examples show that Lambert-series form alone does not determine arithmetic status. The accompanying source reference checks the equality between the two indexing conventions for S , not every status claim in this paragraph; the individual comparisons are in Proposition 6.43 and its table. [n/a] [Lean, Cited]

LEAN SOURCE

Lean: `lambert_ladder_four_values`, `primWeight_is_totient_conv_moebius`, `totient_series_index_bridge'`, `lambert_id_rung_eq_sigma_series`, and 2 further declarations. Conditional on Nesterenko's algebraic independence theorem; see the [coverage section](#).

Proposition 6.103 (A general rational gap bound). If $a/b < c/d$ are reduced fractions with $b, d > 0$, then

$$\frac{1}{bd} \leq \frac{c}{d} - \frac{a}{b},$$

since $bc - ad$ is a positive integer. In particular, if $S = a/b$ and $p_N/q_N < S$ is a reduced rational approximation with positive error at most ε_N , then

$$b \geq \frac{1}{q_N \varepsilon_N}.$$

These lower bounds tend to infinity precisely when $q_N \varepsilon_N \rightarrow 0$; a subsequence with this property is enough to contradict a fixed b . A tail bound by itself does not imply this. For example, the ordinary dyadic prefix has denominator dividing 2^N and tail at most $(N+2)2^{-N}$. Using only these two bounds yields $b \geq 1/(N+2)$, which is vacuous. The rational-spacing lemma can convert sharper error and denominator information into an exclusion; it does not automatically improve the Farey bound in Proposition 6.99. [n/a] [Lean]

LEAN SOURCE

Lean: `rational_gap_lower_bound`, `den_lower_bound_of_positive_error`, `denominator_bound_tendsto_atTop_iff`, `irrational_of_den_mul_error_product_tendsto_zero`, and 4 further declarations in the [coverage section](#).

Proposition 6.104 (Diagonal certificates through 82). *The current aggregate establishes*

$$\forall t \in \mathbb{N}, \quad t \leq 82 \implies P t,$$

where $P t$ is the diagonal certificate predicate in Proposition 6.72. The earlier aggregate covered 28 explicit cases through 64, including 1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17; the later theorem fills the gaps and extends the range through 82. Each instance reduces to exact finite arithmetic with the displayed totient values, using the prime-power factorisations and Lucas primality certificates in the source. This proves neither $P 83$ nor infinitely many instances, and therefore does not establish the quantified conditions in Propositions 6.71, 6.72 and 6.73. [bounded] [Cert]

LEAN SOURCE LEAN SOURCE

Lean: `historical_table_size_and_initial_depths`, `historical_table_and_complete_band`.

Proposition 6.105 (A coprime-pair expression for S). *For $n \geq 0$, the number of integer pairs $a \geq 1, b \geq 0$ with $a + b = n$ and $\gcd(a, b) = 1$ is $\varphi(n)$, with $\varphi(0) = 0$. Indeed, $b = n - a$ and $\gcd(a, n - a) = \gcd(a, n)$. The boundary case $n = 1$ contributes $(a, b) = (1, 0)$; it disappears when both coordinates are required to be positive. Consequently,*

$$\sum_{\substack{a, b \geq 1 \\ \gcd(a, b) = 1}} 2^{-(a+b)} = S - \frac{1}{2}.$$

For independent random variables X, Y with $\Pr(X = n) = \Pr(Y = n) = 2^{-n}$, $n \geq 1$, the left side is $\Pr(\gcd(X, Y) = 1)$. Thus $S - 1/2$ has a coprimality-probability interpretation. This identity does not supply an estimate for the binary digits or tail residues used elsewhere in the paper. [n/a] [Lean]

LEAN SOURCE

Lean: `card_visible_antidiagonal`, `totient_zero_eq_zero`, `visible_antidiagonal_one`, `positive_antidiagonal_one`, and 2 further declarations in the [coverage section](#).

6.5 Counterexamples and limitations

Each example below is a counterexample to a stated deduction, not to irrationality of S . The assumptions that fail to suffice are given with the example, so that its conclusion is not extended to other arguments.

Observation 6.106 (The sign of the actual endpoint representative). Assume $a \geq 8$, $H = H_{2^a}$ and $J + K + a + 6 < 2 \cdot 2^a$, as in Proposition 6.84. If $d = R_{2H+J} - R_{H+J}$ is an integer, define

$$c_0 = d, \quad c_{j+1} = 2c_j - (\varphi(2H + J + j + 1) - \varphi(H + J + j + 1)).$$

The tail recurrence gives $c_K = R_{2H+J+K} - R_{H+J+K} > 0$. Its negative is an integer representative satisfying

$$|-c_K| \leq 2H + J + K + 2, \quad -c_K \equiv D(H, H + J, K) \pmod{2^K}.$$

Thus the representative corresponding to the actual tail is negative. The stated size bound allows it to remain in the endpoint interval, so its sign alone does not contradict integrality. Other integers in that interval with the same residue need not represent the actual tail. Proposition 6.85 gives the further residue separation that would exclude this case. [uniform] [Lean]

[LEAN SOURCE](#)

Observation 6.107 (The complete divisibility pattern is impossible). Let $a \geq 8$, $H = H_{2^a}$ and $J, K, m \in \mathbb{N}$. Assume

$$m \leq K, \quad J + K + a + 6 < 2 \cdot 2^a, \quad 2H + J + K + 2 < 2^m.$$

The simultaneous divisibilities

$$2^{r+1} \mid \delta_{2^a}(J + K - m + r + 1) \quad (0 \leq r < m)$$

cannot hold. The terminal difference would be positive by Proposition 6.80, smaller than 2^m , and divisible by 2^m , a contradiction. The obstruction applies to this complete pattern under the stated size bounds. Omitting its last divisibility condition leads to the necessary restrictions in Proposition 6.86; the residue inequality in Proposition 6.64 is another sufficient condition. Neither statement here supplies such witnesses at arbitrarily large scales. [uniform] [Lean]

[LEAN SOURCE](#)

Observation 6.108 (An exact valuation in a dyadically scaled family). For every integer $n \geq 0$, take $H = 2^{n+2}$ and $j = 3 \cdot 2^{n+1}$. Then

$$\varphi(3H + j) - 2\varphi(2H + j) + \varphi(H + j) = -2^{n+1}.$$

Indeed, the three arguments are $9 \cdot 2^{n+1}$, $7 \cdot 2^{n+1}$ and $5 \cdot 2^{n+1}$. Their totients are respectively $6 \cdot 2^n$, $6 \cdot 2^n$ and $4 \cdot 2^n$, which give the identity. Thus the difference has 2-adic valuation exactly $n + 1$. This example limits a valuation bound for arbitrary even heights and scaled odd offsets. Here $j > H$, whereas the offset in Proposition 6.88 is at most $2^{a/2}$ and the height is $H(2^a)$. The example therefore does not exclude additional divisibility on that LCM family. [uniform] [Lean]

[LEAN SOURCE](#)

Observation 6.109 (A rational sequence with matching parity). Start with the coefficient sequence $0, 1, 1, 2, 4, 4, 4, \dots$, indexed from 0. For each $m = 2^{k+3}$, $k \geq 0$, replace the pair at positions $m, m + 1$ from $(4, 4)$ to $(6, 0)$, leaving all other coefficients unchanged. These pairs are disjoint. Each replacement preserves the binary sum because

$$2 \cdot 2^{-m} - 4 \cdot 2^{-(m+1)} = 0.$$

The resulting sequence has the same sum $3/2$ and the same parity at every index as φ . It is not eventually periodic, since the gaps between its occurrences of 6 tend to infinity. This construction uses the particular initial parity pattern and the binary weights; it does not establish an analogous result for arbitrary prescribed parities or other summand weights. [n/a] [Lean]

[LEAN SOURCE](#)

Observation 6.110 (What the parity example disproves). The sequence c in [Observation 6.109](#) satisfies

$$0 \leq c(n) \leq \min(n, 6), \quad c(n) \equiv \varphi(n) \pmod{2}, \quad \sum_{n \geq 0} c(n)2^{-n} = \frac{3}{2},$$

and is not eventually periodic. More precisely, for every $N, G, K \in \mathbb{N}$, it has K of the displayed $(6, 0)$ pairs starting beyond N , with successive starting positions separated by more than G . Choose sufficiently late powers of two in the construction to obtain these pairs. Hence the growth bound, the exact parity pattern, nonperiodicity and these arbitrarily separated occurrences are compatible with a rational sum. They cannot by themselves prove irrationality. The construction does not preserve the complete totient sequence or its multiplicativity, and does not specify which additional arithmetic hypothesis would suffice. [LEAN SOURCE](#)

Observation 6.111 (A Mersenne factor does not force tail-difference divisibility). For the completely multiplicative sequence $c(n) = n$, the scaled tail is

$$T_N = \sum_{j \geq 1} (N + j)2^{-j} = N + 2, \quad T_{N+K} - T_N = K.$$

If $q > K > 0$ and $q \mid 2^K - 1$, then every such tail difference is integral but is not divisible by q . For example, take $K = 3$ and $q = 7$; the multiplicative order of 2 modulo 7 is 3. Thus even this prime divisor of the homogeneous recurrence multiplier does not force divisibility of the inhomogeneous tail difference. An application to the totient sequence needs a further relation between the prime and the coefficient residues or the tail's size. This example supplies no general obstruction for [Problem #257](#) merely because that problem also involves Mersenne denominators. Compare the specific totient construction in [Proposition 6.91](#). [uniform] [Lean]

[LEAN SOURCE](#)

Observation 6.112 (A lower bound for the stated reconstruction cost). Let I be finite, $w_i \in \mathbb{Q}$ and $x_i \in \mathbb{N}$, and assume $\sum_{i \in I} w_i \varphi(x_i) = 1$. Then

$$\sum_{i \in I} |w_i| (2(x_i + 1) + (x_i + 2)) \geq 3.$$

Indeed, $1 \leq \sum_i |w_i| \varphi(x_i) \leq \sum_i |w_i| x_i$, and the displayed cost is $3 \sum_i |w_i| x_i + 4 \sum_i |w_i|$. For $x \geq 1$, the reconstruction $\varphi(x) = 2R_{x-1} - R_x$ and the separate estimates $R_M \leq M + 2$ give exactly this cost. Terms at $x = 0$ contribute nothing to the normalisation, since $\varphi(0) = 0$. This particular termwise triangle bound cannot be smaller than one. The conclusion is about that bound, not the true combined tail error; cancellation or a different estimate is not excluded. The same elementary calculation applies to any nonnegative sequence $c(n) \leq n$ under the corresponding normalisation $\sum_i w_i c(x_i) = 1$. [uniform] [Lean]

[LEAN SOURCE](#)

Observation 6.113 (A second-difference test need not use less depth). For $h, N, L \in \mathbb{N}$, put

$$D_2 = D(h, N + h, L) - D(h, N, L), \quad B_2 = 2(N + 2h + L + 2).$$

The second-difference certificate is the finite inequality

$$B_2 < D_2 \bmod 2^L < 2^L - B_2.$$

It implies $R_{N+2h} - 2R_{N+h} + R_N \notin \mathbb{Z}$. The two first-difference truncation errors have bounds $N + 2h + L + 2$ and $N + h + L + 2$ after multiplication by 2^L ; their sum is at most the stated B_2 . This explains the radius used by the linked test, without claiming it is optimal. At $(h, N) = (1, 8)$, the first-difference test holds at depth 8, whereas the second-difference test fails at every depth at most 8 and first holds at 9. The earlier 40-cell probe for $t \leq 20$ reported the first test at least as shallow in 30 cells; that aggregate is a reported computation, not part of the single-cell theorem linked below. These comparisons do not give a uniform ordering of the depths. [n/a] [Cert]

[LEAN SOURCE](#)

Observation 6.114 (The prime-power restriction in the unit-gap refinement). The ordinary interval test excludes a rational denominator when its possible numerator interval contains no integer. Reduced fractions permit a refinement: every numerator is coprime to its denominator, so an interval containing only nonunits can also be excluded. For a candidate $S = a/(2^c u)$ in lowest terms, with $u > 0$ odd and $N \geq c$, the integer $uR_N = 2^{N-c}a - u\Phi_N$ is coprime to u . For such an N and any $K \geq 0$, define

$$\begin{aligned} V &= \left(\sum_{r=1}^K \varphi(N+r) 2^{K-r} \right) \bmod 2^K, \\ uV &= J2^K + W, \quad 0 \leq W < 2^K, \\ L &= \left\lfloor \frac{W + u(N+K+2)}{2^K} \right\rfloor. \end{aligned}$$

After subtracting an integer multiple of u , the finite prefix of uR_N is $J + W/2^K$ and its remaining tail lies in $(0, u(N+K+2)/2^K]$. Its possible integer values are therefore $J+1, \dots, J+L$. The refined condition is $\gcd(J+t, u) > 1$ for all $1 \leq t \leq L$, contradicting the required coprimality of uR_N . For a prime p and $e \geq 1$, nonunits modulo p^e are exactly the multiples of p . Since two consecutive integers cannot both be multiples of p , the condition for $u = p^e$ is equivalent to

$$L = 0 \quad \text{or} \quad L = 1 \text{ and } p \mid J + 1.$$

Thus a *successful* refined certificate at a prime-power denominator has at most one candidate integer. This is not a bound for an arbitrary window: $(u, N, K) = (3, 3, 3)$ gives $V = 2, J = 0, W = 6$ and $L = 3$, and the test fails because 1, 2, 3 are not all nonunits modulo 3. The prime-power restriction also matters. With the actual totient window $(u, N, K) = (15, 13, 8)$, one gets

$$V = 148, \quad J = 8, \quad W = 172, \quad L = 2.$$

Both candidates 9, 10 are nonunits modulo 15. This interval passes the refined test but not the empty-interval test. The prime-power bound therefore neither applies to all

composite denominators nor bounds the possible improvement in the final denominator cutoff. [bounded] [Math; prime-power criterion in Lean]

[LEAN SOURCE](#)

Observation 6.115 (A synthetic sequence preserving the stated LCM differences). Here is the recurrence behind the comparison. Fix $t \geq 3$, put $H = \text{lcm}(1, \dots, t)$ and $A = \varphi(H)$, and define

$$c_i = \begin{cases} -A, & i = (q-1)H \text{ for some } 2 \leq q < t, \\ 0, & \text{otherwise,} \end{cases} \quad a_i = 2c_i - c_{i+1}.$$

The index i here corresponds to the argument $H + i + 1$ of a totient difference. For $2 \leq q < t$ this gives

$$a_{(q-1)H-1} = \varphi((q+1)H) - \varphi(qH) = A.$$

The equality follows because all prime divisors of q and $q+1$ divide H : hence $\varphi(qH) = qA$ and $\varphi((q+1)H) = (q+1)A$. Every a_i is divisible by A , and $|c_i| \leq A$, $|a_i| \leq 2A$. For finitely supported integer weights $(\lambda_j)_{j \geq 0}$, put $d_i = \sum_j \lambda_j c_{i+j}$ and $b_i = \sum_j \lambda_j a_{i+j}$. Linearity and telescoping give

$$b_i = 2d_i - d_{i+1}, \quad \sum_{r=0}^{L-1} 2^{L-1-r} b_{i+r} = 2^L d_i - d_{i+L}.$$

The triangle inequality gives the precise uniform bounds

$$|d_i| \leq A \sum_j |\lambda_j|, \quad |b_i| \leq 2A \sum_j |\lambda_j|.$$

These are uniform in i , not in the weights. The factor $\sum_j |\lambda_j|$ cannot be omitted: taking only $\lambda_0 = 2$ doubles the nonzero state and coefficient maxima. Divisibility by A and by $\varphi(d)$ for $d \mid H$ is preserved, since $\varphi(d) \mid \varphi(H)$.

Finite shift polynomials therefore preserve the recurrence, but not the original numerical bounds independently of their coefficients. This comparison does not identify a_i with actual totient differences away from the prescribed indices, or bound the terms discarded by a proposed projection. The construction and the weight-dependent bounds are in [LEAN SOURCE](#). The original source introduction and the decomposition of the actual totient into divisor contributions remain at [LEAN SOURCE](#) and [LEAN SOURCE](#). [uniform] [Math; formal source cited]

Observation 6.116 (Compatible carries for fixed-precision local symbols). For example, prescribing an odd coefficient modulo 8 permits any coefficient of the form $a + 8z$ with a odd and $z \in \mathbb{Z}$. Whatever the incoming integer carry e , one can choose z so that $2e + a + 8z$ lies between -4 and 3 . More generally, fix $u \geq 1$ and a finite list of integers $v_j \geq 0$ and odd integers a_j . Starting from any integer e_0 , there are integers z_j and coefficients

$$c_j = 2^{v_j}(a_j + 2^u z_j), \quad e_{j+1} = 2e_j + c_j, \quad |e_{j+1}| \leq 2^{v_j+u-1}.$$

At each step, reduce $2e_j + 2^{v_j}a_j$ to its centred representative modulo 2^{v_j+u} , choosing the interval

$$[-2^{v_j+u-1}, 2^{v_j+u-1}).$$

Solving for z_j gives the stated next carry. Iterate this choice. The bounds apply to the resulting states, not to the arbitrary initial state e_0 . The unrestricted higher quotients z_j are essential: the construction prescribes a valuation and a finite unit residue, not the full coefficients or their signs. It therefore does not realise a prescribed totient-difference sequence. A contradiction using only these local symbols cannot exclude all compatible integer carries; further restrictions on the coefficients need a separate argument. [n/a] [Lean]

[LEAN SOURCE](#)

Observation 6.117 (Correction-free factorisation does not ensure nonvanishing). Consider the two finite differences

$$\varphi(65) - \varphi(65) = 0, \quad \varphi(78) - \varphi(70) = 24 - 24 = 0.$$

They arise from the square-congruence choice $52 \equiv 52 \pmod{13^2}$ and $52 \equiv 2 \pmod{5^2}$. The four arguments factor as $13 \cdot 5$, $5 \cdot 13$, $13 \cdot 6$ and $5 \cdot 14$; in each product the prime and cofactor are coprime. Thus all four totients have the correction-free form $\varphi(pm) = (p-1)\varphi(m)$, yet both differences vanish. Nonvanishing is also possible: the choices $27 \equiv 0 \pmod{3^2}$ and $27 \equiv 2 \pmod{5^2}$ give

$$\varphi(30) - \varphi(30) = 0, \quad \varphi(33) - \varphi(35) = 20 - 24 = -4,$$

again with coprime factors at every term. The square-CRT construction removes the non-coprime multiplicativity correction on the prescribed finite range. That property alone does not imply nonvanishing. These are explicit witnesses of the two possibilities, not a claim that the first is a smallest counterexample. [n/a] [Lean (docstring-sourced)]

[LEAN SOURCE](#)

Observation 6.118 (No fixed integer clears every primitive denominator). Let $A = \varphi * \mu$ be the Dirichlet convolution used in Proposition 6.102. At prime powers,

$$A(p) = p - 2, \quad A(p^r) = p^{r-2}(p-1)^2 \quad (r \geq 2).$$

These follow by subtracting consecutive prime-power values of φ . For every odd prime p , the reduced denominator of $A(p)/p$ is p . For every prime p , the reduced denominator of $A(p^2)/p^2$ is p^2 , including the denominator 4 at $p = 2$. Consequently any positive integer D for which $DA(n)/n \in \mathbb{Z}$ for all $1 \leq n \leq N$ must be divisible by every odd prime $p \leq N$, every prime square $p^2 \leq N$, and by 4 once $N \geq 4$. No fixed D can do this for all n , since there are infinitely many odd primes. In periodic-point counting, the quantities $A(n)/n$ would count orbits of length n ; their integrality and nonnegativity characterise realisable periodic-point sequences [26, Lemma 3.1, p. 5]. Here only the denominator obstruction is used. It excludes a fixed integer clearing all these particular coordinates, not every possible Euler-product representation or an argument with a multiplier growing with N . It is not an irrationality proof for S . [n/a] [Lean]

[LEAN SOURCE](#)

Observation 6.119 (A finite exclusion is not a membership test). A successful residue certificate proves the exclusion stated in its soundness theorem. Failure of that sufficient condition at finitely many depths does not prove integrality of the tail difference. In particular, a finite unsuccessful search cannot be substituted for the eventual-period assertion needed to prove rationality. The same logical distinction occurs for the linked greedy test in a separate Mersenne achievement-set problem: its finite inequality implies nonmembership. The linked object is a definition, with a soundness theorem in the same file, not a theorem asserting a universal impossibility of finite rationality proofs for Problem #249. Membership or rationality established by some additional exact identity is not excluded by this observation. Likewise, the successful finite ranges in Propositions 6.94 and 6.104 remain finite ranges until an unbounded supply is proved. [n/a] [Math; related Lean definition]

[LEAN SOURCE](#)

6.6 Related structural results

The following results concern the Möbius-weighted series, integer carry recurrences, rational approximations, and factorizations used above. Each statement retains its pinned formal-source reference. The statements are grouped by their mathematical objects, rather than by the order in which files were added to the repository.

6.6.1 Log-concavity of the Möbius-weighted series

The series $\Theta_r = \sum_{n \geq 0} \mu(n+1)/(2^{n+1} - 1)^r$ includes $S - 1/2$ at $r = 2$. The next result proves strict log-concavity for every integer $r \geq 1$. This is an order property of these values, not an irrationality criterion.

Definition 6.120. For each integer $r \geq 1$, put

$$\Theta_r = \sum_{n \geq 0} \frac{\mu(n+1)}{(2^{n+1} - 1)^r} = 1 - 3^{-r} + \sum_{n \geq 0} \frac{\mu(n+3)}{(2^{n+3} - 1)^r}.$$

The first two summands isolate the dominant contribution. The remaining absolute sum is at most $1/(2^r(2^r - 1))$, by comparison with $\sum_{n \geq 2} 2^{-rn}$. In particular, $\Theta_r > 0$ for $r \geq 1$: $1 - 3^{-r} \geq 2/3$ and this tail bound is at most $1/2$. [LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#) [Lean] [uniform] .

Theorem 6.121 (The values at exponents one and two). *The first two values are $\Theta_1 = 1/2$ and $\Theta_2 = S - 1/2$.* [LEAN SOURCE](#) [LEAN SOURCE](#) [Lean] [fixed] .

Lean: `mobiusMersenneTheta_one_and_two`.

The second equality is the Möbius-square identity credited in §5.4; whether Θ_2 is rational is #249 itself.

Theorem 6.122 (The first two summands give a positive Hankel gap). $(1 - 3^{-(r+1)})^2 - (1 - 3^{-r})(1 - 3^{-(r+2)}) = 4/3^{r+2}$, so the rational sequence $1 - 3^{-r}$ is strictly log-concave for every integer $r \geq 1$. [LEAN SOURCE](#) [Lean] [uniform] .

Lean: `twoAtom_hankel_gap`, `twoAtom_strict_logConcave`.

Theorem 6.123 (Strict log-concavity for all integer $r \geq 1$). *For every integer $r \geq 1$,*

$$\Theta_r \Theta_{r+2} < \Theta_{r+1}^2.$$

Equivalently, the determinant of $\begin{pmatrix} \Theta_r & \Theta_{r+1} \\ \Theta_{r+1} & \Theta_{r+2} \end{pmatrix}$ is negative. The two-summand comparison gives an elementary proof, detailed below; the linked declarations prove the same strict inequality.

[LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Lean: `theta_strict_logConcave`, `theta_hankel_det_neg`, `theta_hankel_two_neg`.

A quantitative comparison. For this proof put $a_r = 1 - 3^{-r}$ and $\varepsilon_r = 1/(2^r(2^r - 1))$. The preceding bound gives $|\Theta_r - a_r| \leq \varepsilon_r$. Since $0 < a_r < 1$, expansion of the two products gives

$$\Theta_{r+1}^2 - \Theta_r \Theta_{r+2} \geq \frac{4}{3^{r+2}} - B_r, \quad B_r = 2\varepsilon_{r+1} + \varepsilon_{r+1}^2 + \varepsilon_r + \varepsilon_{r+2} + \varepsilon_r \varepsilon_{r+2}.$$

Here $\varepsilon_{r+1} < \varepsilon_r/4$, so $B_{r+1} < B_r/4$. The rational comparison

$$B_5 = \frac{50106593}{32001951744} < \frac{4}{3^7}$$

therefore proves the inequality for all $r \geq 5$: the error decreases faster than the positive main term. For $r = 1, 2, 3, 4$, truncate each Θ_j at $n = 8$. Its omitted absolute sum is at most

$$\frac{1}{(2^9 - 1)^j (1 - 2^{-j})}.$$

Using $\Theta_1 = 1/2$ and these rational intervals, the four gaps $\Theta_{r+1}^2 - \Theta_r \Theta_{r+2}$ are respectively greater than $1/4$, $1/16$, $1/64$ and $1/200$. These finite rational comparisons complete the ordinary proof. $\square$

Observation 6.124. The inequality excludes a representation $\Theta_r = \int_0^\infty x^r d\nu(x)$ by a positive measure with these finite moments: Cauchy–Schwarz would give $\Theta_{r+1}^2 \leq \Theta_r \Theta_{r+2}$. It does not by itself imply irrationality of Θ_2 . Indeed, the rational sequence $1 - 3^{-r}$ satisfies the same strict log-concavity inequalities. Thus the determinant sign supplies an obstruction to a positive-moment representation, not a rational-approximation estimate.

The accompanying formal results include a rectangular determinant expansion derived from the Leibniz formula ([LEAN SOURCE](#), [\[Lean\]](#), [\[uniform\]](#),), and the following elementary parity fact: in a finite sum of dyadic fractions, if the largest denominator exponent occurs once and its numerator is odd, clearing that denominator gives an odd numerator ([LEAN SOURCE](#), [\[Lean\]](#), [\[uniform\]](#),).

6.6.2 Integer recurrences and rational binary series

For a binary series with integer coefficients, rationality can be expressed by an integer recurrence whose growth is smaller than 2^N . The next equivalence is elementary algebra and convergence; no novelty is claimed for it.

Theorem 6.125 (Integer recurrence criterion for a binary series). *Let $c : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $c(n) \leq n$. Then $\sum_{n \geq 1} c(n)2^{-n}$ is rational if and only if there are an integer $v > 0$ and an integer sequence u such that*

$$u(N+1) = 2u(N) - v c(N+1), \quad \frac{u(N)}{2^N} \rightarrow 0.$$

For each fixed v , such a sequence is unique and equals

$$u(N) = vR_N^c, \quad R_N^c = \sum_{j \geq 1} \frac{c(N+j)}{2^j}.$$

[LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Proof. Iterating the recurrence gives the finite identity

$$\frac{u(N)}{2^N} = u(0) - v \sum_{n=1}^N \frac{c(n)}{2^n}.$$

The boundary condition therefore forces the series to equal $u(0)/v$; subtracting the prefix gives $u(N) = vR_N^c$. Conversely, when the series is rational, choose an integer $v > 0$ making its product with v an integer, use this product as $u(0)$, and iterate the recurrence. The displayed identity identifies the resulting integer sequence with vR_N^c . Since $0 \leq R_N^c \leq N+2$, its ratio to 2^N tends to zero. $\square$

Without the boundary condition, any integer $u(0)$ gives an integer solution of the recurrence, whether the series is rational or irrational. Two solutions with the same v differ by 2^N times their initial difference. Thus the limit, not the recurrence alone, determines the initial value. Changing v rescales the canonical solution; uniqueness is asserted only for a fixed multiplier.

Proposition 6.126 (Uniqueness under the growth condition). *A real sequence d with $d(N+1) = 2d(N)$ and $d(N) = o(2^N)$ is identically zero.* [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Lean: [doubling_tempered_sequence_eq_zero](#).

Observation 6.127 (The specified finite-state decoder is impossible). Fix $m \geq 2$, put $R = \lfloor (m+1)/2 \rfloor$, and for $0 \leq r \leq R$ define

$$c_{m,r}(n) = \begin{cases} R-r, & n = m, \\ 2r, & n = m+1, \\ 0, & \text{otherwise.} \end{cases}$$

The two nonzero terms have total contribution $(R-r)2^{-m} + 2r2^{-(m+1)} = R2^{-m}$, independent of r . All coefficients and scaled tails before m are therefore the same, whereas the scaled tail at m is r . No map from a state common to this whole family can recover that tail. More generally, exact recovery of all its possible values requires at least $R+1$ distinct labels. [LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) . The sequences in this family have identical histories before the isolated change but different subsequent tails. Therefore that finite history alone cannot determine the exact tail in this class of sequences. This is not a conclusion about the totient sequence.

There is also an infinite family with a common rational value. Let $a : \mathbb{N} \rightarrow \{0, 1\}$ satisfy $a(0) = 0$, and put $c(2k) = k - a(k)$ and $c(2k + 1) = 2a(k)$. The adjacent terms contribute $k4^{-k}$, so

$$\sum_{n \geq 1} \frac{c(n)}{2^n} = \sum_{k \geq 1} \frac{k}{4^k} = \frac{4}{9},$$

regardless of the binary sequence a . The cited result also gives an integer recurrence satisfying the growth condition ([LEAN SOURCE](#), [Lean], [uniform],). The assumptions $a(0) = 0$ and $a(k) \in \{0, 1\}$ are part of that formal statement, not an arbitrary bounded-input hypothesis. For comparison, the identity coefficients $c(n) = n$ have the explicit multiplier-one recurrence $u(N) = N + 2$ ([LEAN SOURCE](#), [Lean], [fixed],).

6.6.3 Rational approximation with a squared-Mersenne remainder

The first-order Lambert term in the diagonal tail difference can be summed exactly. The approximation below leaves only a squared-Mersenne tail to estimate.

Proposition 6.128 (An exact rational approximation formula). *For integers $H, D \geq 0$, with the integer prefixes Φ_N defined above,*

$$\begin{aligned} R_{2H} - R_H - \left[\Phi_H - \Phi_{2H} + 2^H(2^H - 1) \left(\frac{1}{2} + \sum_{d=1}^D \frac{\mu(d)}{(2^d - 1)^2} \right) \right] \\ = 2^H(2^H - 1) \sum_{d > D} \frac{\mu(d)}{(2^d - 1)^2}. \end{aligned}$$

This follows by substituting the squared-denominator identity for S into $R_{2H} - R_H = 2^H(2^H - 1)S + \Phi_H - \Phi_{2H}$. There is no required ordering between D and H . [LEAN SOURCE](#) [Lean] [uniform] .

Lean: [tailDifference_sub_rationalApproximation](#), [tailDifference_eq_coefficient_mul_series](#), [totientSeries_pnat_form](#), [totientSeries_eq_half_add_moebius_sq](#).

Proposition 6.129 (A geometric tail bound). *For every integer $D \geq 0$,*

$$\left| \sum_{d > D} \frac{\mu(d)}{(2^d - 1)^2} \right| \leq \frac{4}{3(2^{D+1} - 1)^2}.$$

Indeed, $|\mu(d)| \leq 1$ and $2^{D+1+j} - 1 \geq 2^j(2^{D+1} - 1)$ for $j \geq 0$. The sum of the resulting geometric majorant is $\sum_{j \geq 0} 4^{-j} = 4/3$. [LEAN SOURCE](#) [Lean] [uniform] .

Lean: [abs_moebiusSquareTail_le_paper](#), [abs_moebius_cast_le_one](#), [mersenne_geometric_shift](#), [tsum_quarter_geometric](#).

Observation 6.130 (One-sided and symmetric error bounds). Suppose $n > D$, $\mu(d) = 0$ for $D < d < n$, and $|\mu(n)| = 1$. Removing that first nonzero summand gives the sharper enclosure

$$\left| \mu(n) \sum_{d > D} \frac{\mu(d)}{(2^d - 1)^2} - \frac{1}{(2^n - 1)^2} \right| \leq \frac{4}{3(2^{n+1} - 1)^2}.$$

The radius is smaller than the positive centre, so the tail has sign $\mu(n)$. The interval's width, divided by the width of the preceding symmetric enclosure, is

$$\left(\frac{2^{D+1} - 1}{2^{n+1} - 1} \right)^2 < \frac{1}{4}.$$

The cited theorem gives this signed enclosure ([LEAN SOURCE](#), [Lean], [uniform],). Multiplication by $2^H(2^H - 1)$ and translation by the rational approximation above enclose $R_{2H} - R_H$. If the resulting interval contains no integer, the tail difference is nonintegral. For the symmetric bound, the linked statement requires the distance from its rational centre to every integer to exceed its error radius ([LEAN SOURCE](#)). The enclosure does not establish that strict separation. In particular, shrinking an interval around a rational approximation does not, without a denominator comparison, show that the interval misses every integer.

6.6.4 Divisor terms and their complement

At $H = 6$ and $s = 2$, the difference $\varphi(2H + s) - \varphi(H + s) = \varphi(14) - \varphi(8)$ is 2, whereas the terms with divisor index $d \mid H$ sum to 3. The other indices contribute -1 . The following decomposition keeps both contributions. It holds for every positive integer H , not only a least common multiple; an estimate is needed before its second sum can be discarded.

Proposition 6.131 (The sum over divisor indices). *For integers $H > 0$ and $s \geq 0$, the part of the Möbius expansion indexed by divisors of H is*

$$\sum_{\substack{d \mid H \\ d \mid s}} \mu(d) \frac{H}{d} = H \frac{\varphi(\gcd(H, s))}{\gcd(H, s)}.$$

This is the standard identity $\sum_{d \mid g} \mu(d)/d = \varphi(g)/g$ at $g = \gcd(H, s) > 0$. The formal statement writes the same value after extracting $H/\text{rad}(H)$, where $\text{rad}(H) = \prod_{p \mid H} p$. [LEAN SOURCE](#) [Lean] [uniform] .

Lean: `sum_divisorIndices_mobius`, `sum_divisors_moebius_div_eq_totient_div`, `sum_divisorIndices_radical_form`, `squarefreeKernel_eq_prod_primeFactors`.

Proposition 6.132 (The divisor sum and its complement). *For integers $H > 0$ and $s \geq 0$,*

$$\begin{aligned} \varphi(2H + s) - \varphi(H + s) &= H \frac{\varphi(\gcd(H, s))}{\gcd(H, s)} \\ &+ \sum_{\substack{1 \leq d \leq 2H+s \\ d \nmid H}} \mu(d) \left(\frac{2H + s}{d} \mathbf{1}_{d \mid 2H+s} - \frac{H + s}{d} \mathbf{1}_{d \mid H+s} \right). \end{aligned}$$

Here $\mathbf{1}$ is the indicator of the stated divisibility condition. The identity follows by expanding both totients as $\varphi(n) = \sum_{d \mid n} \mu(d)n/d$ and separating the indices $d \mid H$. Such an index divides either endpoint precisely when it divides s , and its difference is then $\mu(d)H/d$. [LEAN SOURCE](#) [Lean] [uniform] .

Lean: [totientDifference_eq_divisorPart_add_complement](#), [totient_eq_mobius_divisor_sum](#), [divisorIndex_endpoint_behaviour](#), [complementSummand_eq_phaseTerm](#).

Theorem 6.133 (Doubling the full totient difference). *For nonnegative integers H, r with H even,*

$$\varphi(4H + 2r) - \varphi(2H + 2r) = \begin{cases} 2(\varphi(2H + r) - \varphi(H + r)), & r \text{ even,} \\ \varphi(2H + r) - \varphi(H + r), & r \text{ odd.} \end{cases}$$

This applies to the full difference, not just its divisor contribution. It follows from $\varphi(2n) = 2\varphi(n)$ for even n and $\varphi(2n) = \varphi(n)$ for odd n . [LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Lean: [totientDifference_doubling_seam](#), [totient_two_mul_even](#), [totient_two_mul_odd](#).

Observation 6.134 (Squarefree support of the complementary terms). A nonzero summand in the complementary sum must have a squarefree index d dividing exactly one of $2H + s$ and $H + s$. Indeed, otherwise $\mu(d) = 0$ or both indicators vanish. Both endpoints cannot be divisible by d : their difference is H , and $d \nmid H$. [LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) . The restriction identifies which indices can contribute. It does not control the sign or magnitude of their sum.

Proposition 6.135 (A doubling identity for two portions of the sum). *Let $H > 0, s \geq 0$ and $d \geq 1$, with $d \nmid H$ and $d \mid H + s$. The d -summands in the complementary sums at offsets s and $2s$ are, respectively,*

$$-\mu(d) \frac{H + s}{d} \quad \text{and} \quad 2\mu(d) \frac{H + s}{d}.$$

For the second expression, $d \mid 2H + 2s$ and $d \nmid H + 2s$; otherwise d would divide their difference H . Thus the second contribution is -2 times the first, including the zero case $\mu(d) = 0$. The two terms have different binary weights in a window sum, so this identity alone is not a cancellation of their weighted contributions. [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Lean: [complementSummand_low_double_echo](#).

6.6.5 A numerator polynomial with explicit positive coefficients

Theorem 6.136 (Positive coefficients of the numerator polynomial). *For a squarefree integer $r \geq 1$, the numerator polynomial has the explicit expression*

$$\sum_{d \mid r} \mu(d) \frac{r}{d} \sum_{j=0}^{r/d-1} X^{dj} = \sum_{k=0}^{r-1} \frac{r}{\gcd(r, k)} \varphi(\gcd(r, k)) X^k.$$

Every coefficient for $0 \leq k < r$ is positive, and the higher coefficients are zero. Extracting X^k on the left gives $\sum_{d \mid \gcd(r, k)} \mu(d) r/d$, which proves the formula. The squarefree assumption identifies this divisor sum with the subset-of-primes definition in the formal source. [LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Lean: `paperNumerator_eq_gcdWordForm`, `paperNumerator_coeff_pos`, `paperNumerator_coeff_eq_zero`, `paperNumerator_coeff_eq_gcd_divisor_sum`, and 1 further declaration in the [coverage section](#).

Corollary 6.137. *For the same squarefree r , evaluation at $X = 2$ gives the integer*

$$\sum_{d|r} \mu(d) \frac{r}{d} \frac{2^r - 1}{2^d - 1}.$$

Each quotient is an integer because $d \mid r$. [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Lean: `paperNumerator_eval_two`, `mersenne_dvd_of_dvd`, `paperNumerator_eval_two_prime-SubsetForm`.

6.6.6 Exact dyadic comparisons and tail bounds

The next two formulas concern exact comparisons with a dyadic rational and an explicit tail bound. They are useful when a residue is computed using a finite prefix: the error must remain smaller than its distance from the boundary.

Definition 6.138. For $p \in \mathbb{Z}$, $n \geq 0$ and an integer $L > 0$, the difference between $p/(2L)$ and the dyadic point $2^{-(n+1)}$ is

$$\frac{p}{2L} - 2^{-(n+1)} = \frac{2^n p - L}{2^{n+1} L}.$$

Its sign is therefore the sign of the integer numerator $2^n p - L$. For fixed p, L , the next numerator satisfies $2^{n+1} p - L = 2(2^n p - L) + L$. [LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Proposition 6.139 (The geometric remainder bound). *The remainder after the first two geometric terms of $1/(2^n - 1)$ is*

$$\frac{1}{2^n - 1} - 2^{-n} - 4^{-n} = \frac{8^{-n}}{1 - 2^{-n}} \leq \frac{4}{3} 8^{-n} \quad (n \geq 2).$$

Indeed, $1 - 2^{-n} \geq 3/4$. Summing over $n > m$, for an integer $m \geq 1$, gives

$$\sum_{n>m} \left(\frac{1}{2^n - 1} - 2^{-n} - 4^{-n} \right) \leq \frac{4}{3} \sum_{n>m} 8^{-n} = \frac{4}{21} 8^{-m}.$$

[LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Lean: `mersenneRemainder_identity_and_bound`, `one_sub_half_pow_ge`, `mersenneRemainder-Tail_le`, `tsum_eighth_pow_tail`, and 1 further declaration in the [coverage section](#).

6.6.7 Squared distances between complex phases

For a finite set of unit complex phases, distance from 1 is related exactly to the real part of their sum. A second elementary identity bounds the total squared distance between pairs. These facts turn suitable separation estimates into the real-part inequality; neither supplies that separation for the totient phases.

Proposition 6.140 (Squared distance from the phase one).

$$\sum_{N \in T} \|E(h, N, L) - 1\|^2 = 2|T| - 2 \sum_{N \in T} \operatorname{Re} E(h, N, L).$$

[LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Lean: `sum_sq_dist_from_phase_one`.

Lemma 6.141 (Squared-distance bound for separated pairs). *For a finite family $z : T \rightarrow \mathbb{C}$, a real number $\delta \geq 0$, and any set of pairs $P \subseteq T \times T$ each separated by $\geq \delta$, $|P| \cdot \delta^2 \leq \sum_{i,j \in T} \|z_i - z_j\|^2$.* [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Lean: `card_mul_sq_le_pairwise_energy`.

6.6.8 Approximating the nondivisor terms

The terms with $d \mid H$ form a finite rational contribution to $R_{2H} - R_H$. To use it, we must also approximate the terms with $d \nmid H$. The approximation and its error are explicit; for a cutoff $D \geq 2H$, only separation of the resulting rational centre from the integers remains to be checked.

For $d \geq 1$ and $N \geq 0$, let $a_d(N) = d - (N \bmod d)$ be the least positive shift from N to a multiple of d . Summing the multiples of d in the totient tail gives the rational term

$$\kappa_d(N) = \mu(d) 2^{d-a_d(N)} \left(\frac{N + a_d(N)}{d(2^d - 1)} + \frac{1}{(2^d - 1)^2} \right), \quad R_N = \sum_{d \geq 1} \kappa_d(N).$$

Indeed, substitute $\varphi(n) = \sum_{d \mid n} \mu(d) n/d$ into R_N . For a fixed d , the admissible shifts are $a_d(N) + kd$, $k \geq 0$; their contribution is $\mu(d) \sum_{k \geq 0} ((N + a_d(N))/d + k) 2^{-a_d(N) - kd}$, which is the displayed expression. The interchange of sums is justified by

$$\sum_{j \geq 1} 2^{-j} \sum_{d \mid N+j} |\mu(d)| \frac{N+j}{d} \leq \sum_{j \geq 1} (N+j)^2 2^{-j} < \infty.$$

Thus subtraction and finite splitting of these series are valid. The terms are signed: for example, $\kappa_2(0) = -4/9$. This is the kernel identity already present in the cited source ([LEAN SOURCE](#)).

Proposition 6.142 (The finite divisor sum). *For an integer $H > 0$, the contribution of the divisor indices to $R_{2H} - R_H$ is*

$$A_H = H \sum_{d \mid H} \frac{\mu(d)}{d(2^d - 1)}.$$

For $d \mid H$, both $a_d(H)$ and $a_d(2H)$ equal d . Hence $\kappa_d(2H) - \kappa_d(H) = H\mu(d)/(d(2^d - 1))$. Summing over the divisors gives A_H , not the whole tail difference: the nondivisor terms must still be included. [LEAN SOURCE](#) [Lean] [uniform] .

Lean: [divisorChannels_sum_eq](#), [residueOffset_of_dvd](#), [residueKernel_increment_of_dvd](#), [scaleExplicitShadow_eq_divisorChannels](#).

For $H > 0$ and an integer cutoff $D \geq 0$, set

$$P_{H,D} = \sum_{\substack{1 \leq d \leq D \\ d \nmid H}} (\kappa_d(2H) - \kappa_d(H)),$$

$$\varepsilon_{H,D} = 2^H(2^H - 1) \left(\frac{2}{2^D} + \frac{4}{3 \cdot 4^D} \right).$$

Here A_H includes *all* divisor indices, even when $D < H$. The following implication therefore keeps the error estimate as an explicit hypothesis; it will be proved below when $D \geq 2H$.

Proposition 6.143 (Separation larger than the error implies exclusion). *Suppose*

$$|(R_{2H} - R_H) - (A_H + P_{H,D})| \leq \varepsilon_{H,D}, \quad |A_H + P_{H,D} - z| > \varepsilon_{H,D} \quad \text{for every } z \in \mathbb{Z}.$$

Then $R_{2H} - R_H \notin \mathbb{Z}$. [LEAN SOURCE](#) [Lean] [uniform] .

Lean: [tailDifference_not_integral_of_separation](#), [projectedForeignDefect_paper](#), [foreignComplementBound_paper](#).

Proof. If $R_{2H} - R_H = z \in \mathbb{Z}$, the two hypotheses give opposite inequalities for $|A_H + P_{H,D} - z|$. $\square$

Why the error hypothesis holds for $D \geq 2H$. Every divisor of H then lies below the cutoff. For $d > D$, both H and $2H$ are smaller than d , so the omitted increment simplifies to

$$\kappa_d(2H) - \kappa_d(H) = \mu(d)2^H(2^H - 1) \left(\frac{1}{2^d - 1} + \frac{1}{(2^d - 1)^2} \right).$$

Since $|\mu(d)| \leq 1$ and $2^d - 1 \geq 2^{d-1}$,

$$\begin{aligned} |(R_{2H} - R_H) - (A_H + P_{H,D})| &\leq 2^H(2^H - 1) \sum_{d>D} \left(\frac{2}{2^d} + \frac{4}{4^d} \right) \\ &= \varepsilon_{H,D}. \end{aligned}$$

Thus for $H > 0$ and $D \geq 2H$ the error bound needs no additional convergence hypothesis. The same estimate is proved in the pinned source ([LEAN SOURCE](#)); together with the displayed separation inequality, it gives the cited nonintegrality consequence ([LEAN SOURCE](#)).

For example, at $H = 1$ and $D = 5$ the rational centre and radius are

$$A_1 + P_{1,5} = \frac{304282}{423801}, \quad \varepsilon_{1,5} = \frac{49}{384}, \quad \frac{1}{2} < A_1 + P_{1,5} - \varepsilon_{1,5} < A_1 + P_{1,5} + \varepsilon_{1,5} < 1.$$

Thus this finite calculation excludes an integral value of $R_2 - R_1$. The cutoff restriction alone does not ensure success: at $H = 1, D = 4$, the centre $346/441$ is within the radius $25/96$ of 1. For fixed H the radius tends to zero as $D \rightarrow \infty$, but an irrationality argument still needs the required family of separated centres. Neither a single example nor convergence of the error supplies it.

When $D \geq 2H$, the first-order part of the omitted sum can also be included exactly: $\sum_{d>D} \mu(d)/(2^d - 1) = 1/2 - \sum_{d \leq D} \mu(d)/(2^d - 1)$. Adding $2^H(2^H - 1)$ times this rational correction to $A_H + P_{H,D}$ gives the squared-Mersenne approximation above ([LEAN SOURCE](#)). This explains the relation between the two approximations and the improved squared-denominator remainder; it does not provide their integer separation.

6.6.9 Coprime-pair sums

Counting coprime pairs gives another expression for the totient series. The adjacent Lambert identity is classical and uses different weights. Its sum is an elementary rational function of the parameter; it does not decide irrationality of S .

Theorem 6.144 (Coprime-pair counting and the totient). *For every $n \in \mathbb{N}$, $\#\{(a, b) \in \mathbb{N}^2 : a + b = n, a > 0, \gcd(a, b) = 1\} = \varphi(n)$. Here $b = 0$ is allowed: the boundary pair $(1, 0)$ accounts for the value $\varphi(1) = 1$. [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .*

Lean: `card_coprime_antidiagonal, boundary_pair_at_one`.

Proposition 6.145 (Two lattice sums). *For $0 \leq r < 1$, the two choices of boundary give*

$$\begin{aligned} \sum_{\substack{a \geq 1, b \geq 0 \\ \gcd(a, b) = 1}} r^{a+b} &= \sum_{n \geq 1} \varphi(n) r^n, \\ \sum_{\substack{a, b \geq 1 \\ \gcd(a, b) = 1}} r^{a+b} &= \sum_{n \geq 1} \varphi(n) r^n - r. \end{aligned}$$

The removed pair is $(1, 0)$. Partitioning all strictly positive pairs by their greatest common divisor gives

$$\sum_{g \geq 1} \sum_{\substack{a, b \geq 1 \\ \gcd(a, b) = 1}} r^{g(a+b)} = \left(\frac{r}{1-r} \right)^2.$$

This total is 1 exactly when $r = 1/2$. [LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Lean: `tsum_totient_pow_shift, coprimeLattice_halfOpen_sum, coprimeLattice_positive_sum, coprimeLattice_removed_pair`, and 2 further declarations in the [coverage section](#).

Theorem 6.146 (The classical coprime-pair Lambert identity). *For every $0 \leq r < 1$, $\sum_{(a, b) \text{ coprime}, a, b \geq 1} \frac{r^{a+b}}{1 - r^{a+b}} = \left(\frac{r}{1-r} \right)^2$, an elementary rational function of r , hence rational at every rational r including $r = 1/2$. This is the classical visible-point identity, and the Lean declaration is a formalisation of it rather than a new result: writing each pair (A, B)*

of positive integers uniquely as $g \cdot (a, b)$ with $\gcd(a, b) = 1$ converts the quadrant sum $\sum_{A, B \geq 1} r^{A+B} = (r/(1-r))^2$ into the displayed sum over visible points. With the plain weight r^{a+b} , the same strictly positive index set instead sums to $\sum_{n \geq 1} \varphi(n)r^n - r$. Thus at $r = 1/2$ the Lambert-weighted sum is 1, whereas the plain-weight sum is $S - 1/2$, not S . Adding the boundary pair $(1, 0)$ recovers S in the plain-weight sum. [LEAN SOURCE](#) [Lean] [uniform] .

Lean: `coprimeLattice_lambert_identity`, `coprimeLattice_lambert_rational`, `coprimeLattice_lambert_half_eq_one`, `coprimeLattice_positive_sum`, and 2 further declarations in the [coverage section](#).

Observation 6.147. The Lambert-weighted sum is rational at every rational $r \in [0, 1)$. Consequently, coprimality of the indices, the decomposition by greatest common divisor and geometric decay alone do not imply its irrationality. This comparison does not decide the arithmetic nature of the differently weighted sum $S - 1/2$, or of S . A proof using the plain-weight sum must exploit information not contained in those three properties.

6.6.10 The single-parameter diagonal condition

The period can be chosen to be a least common multiple. This reduces the required certificate condition to a single scale t . The reduction does not remove the demand for arbitrarily large scales, which is the remaining arithmetic assertion.

Observation 6.148 (The diagonal implication). The irrationality of S follows if, for every integer $t_0 \geq 0$, there are $t \geq t_0$ and $L \geq 0$ such that $\mathcal{C}(H_t, H_t, L)$, where $H_t = \text{lcm}(1, \dots, t)$ and $H_0 = 1$. The declaration below gives this implication. Theorem 6.49 also records the converse. The quantified condition is therefore equivalent to irrationality, not a weaker hypothesis. [LEAN SOURCE](#) [Lean] (implication proved; antecedent [Open]) [cofinal] .

Theorem 6.149 (Nondivisors in a short LCM window). *Let $t \geq 1$. If $1 \leq j < 2t$ and $j \nmid H_t$, then $j = p^a > t$ for a prime p and an integer $a \geq 1$. Every integer $1 \leq j \leq t$ divides H_t . To see the first claim, some prime-power divisor p^a of j exceeds t ; otherwise every prime-power divisor would divide H_t . Since $j < 2t < 2p^a$, its remaining cofactor is 1.*

If $j \mid H_t$ and every prime divisor of j also divides H_t/j , then, for every integer $q \geq 0$,

$$\varphi(qH_t + j) = \varphi(j)\varphi(q(H_t/j) + 1).$$

Indeed, the second factor's argument is coprime to j , so totient multiplicativity applies. The hypothesis is essential to this factorisation: at $t = 2$, $j = 2$, $q = 1$ the left side is $\varphi(4) = 2$ but the displayed product would be 1. [LEAN SOURCE](#) [LEAN SOURCE](#) [Lean] [uniform] .

Lean: `short_lcm_window_nondivisor`, `clean_lcm_ray_factorisation`, `unclean_lcm_ray_counterexample`, `dvd_periodLcm`.

Observation 6.150 (Finite diagonal examples). The cited file proves $\mathcal{C}(H_t, H_t, L)$ at the following depths:

t	1	2	3	4	5	6	7	8
H_t	1	2	6	12	60	60	420	840
L	6	5	7	7	9	9	14	15

The largest totient argument is $2H_6 + 9 = 129$ for $t \leq 6$, and $2H_8 + 15 = 1695$ for the full table. [LEAN SOURCE](#) [LEAN SOURCE](#) [Cert] [fixed] ; a finite verification, not a proof that such certificates occur at arbitrarily large scales.

6.6.11 Prime divisors of Mersenne factors and finite exclusions

For the polynomial $X - 2$, elementary order arguments give unbounded prime support in the Mersenne factors. This settles the particular abstract prime-support hypothesis used below. It does not give the tail-residue inequality needed for irrationality. The separate finite exclusions retain their stated periods and basepoints.

Theorem 6.151 (Unbounded prime support in Mersenne factors). *Every prime divisor p of $2^q - 1$, for prime q , satisfies $q \mid p - 1$ (the order of $2 \bmod p$ is exactly q , by Fermat/Lagrange in $(\mathbb{Z}/p)^\times$); consequently the prime divisors appearing in the layers $\{2^n - 1\}$ are unbounded, unconditionally, with no cyclotomic resultant hypothesis left open. [LEAN SOURCE](#) [LEAN SOURCE](#) [Lean] [uniform] .*

Lean: [mersenneLayer_prime_divisor_order](#), [mersenneLayer_unbounded_prime_support](#).

Theorem 6.152 (A prime satisfying the stated cyclotomic conditions). *For every period $h > 0$ and threshold N_0 , there exist a prime q and a prime factor p of $|\Phi_{hq}(2)|$ (the binary cyclotomic layer) with p coprime to hq , $hq \mid p - 1$, and $p - 1 \geq N_0$. The characteristic-prime exceptional case in the cyclotomic order decomposition is eliminated directly, by choosing $q > 2^h$ (rules out $p = q$) and $q > h$ (rules out $p \mid h$). [LEAN SOURCE](#) [LEAN SOURCE](#) [Lean] [uniform] .*

Lean: [exists_clean_cyclotomic_anchor_paper](#), [cyclotomic_layer_prime_order_decomposition_paper](#).

Observation 6.153. The first theorem uses the multiplicative order of 2 modulo a prime. For the second, choose a prime $q > \max(h, N_0, 2^h)$ and then a prime divisor of $|\Phi_{hq}(2)| > 1$. The cyclotomic order formula, together with the stated inequalities on q , excludes prime divisors of hq and gives $\text{ord}_p(2) = hq$. Existence of arbitrarily large primes suffices here; no prime-in-progression theorem is needed for this construction.

These results supply prime divisors with the required orders. They do not supply the tail-residue inequality at the resulting periods. The all-prime and eventual forms of the order hypothesis must also be distinguished, as explained in §6.6.12.

Observation 6.154 (Finite exclusions at period 30). The kernel decides two independent certificates at the composite cyclotomic anchor period $h = 30$ (natural basepoint $N = 300$, and the prime-anchored basepoint $N = 330$). The certificate at $N = 300$ gives the denominator exclusion $S \neq r$ for every $r \in \mathbb{Q}$ with $r.\text{den} \mid 2^{300}(2^{30} - 1)$. [LEAN SOURCE](#) [LEAN SOURCE](#) [Cert] [fixed] ; a finite exclusion, not a proof of the corresponding condition at arbitrarily large scales.

6.6.12 A sufficient hypothesis for unbounded prime support

The residue group matters in this criterion. The bounded exponent k in $p^k \equiv 1 \pmod{mq}$ is a finite-field extension degree, or equivalently a bound for the order of

p modulo mq . It is not a bound for the order of 2 modulo p in the binary examples. The displayed condition forces a prime divisor of $C(mq)$ to grow with q . A separate assumption that these values exceed 1 ensures that prime divisors exist.

Theorem 6.155 (A sufficient order hypothesis for unbounded prime support). *Let $C : \mathbb{N} \rightarrow \mathbb{N}$, and fix integers $m \geq 1$ and $d \geq 0$. Suppose that for every pair of primes q, p with $p \mid C(mq)$ there is an integer k such that*

$$1 \leq k \leq d, \quad mq \mid p^k - 1.$$

The divisibility forces $\gcd(p, mq) = 1$. For such p , existence of k in the displayed range is equivalent to $\text{ord}_{mq}(p) \leq d$. It also gives $mq < p^d$. Hence every fixed finite set of primes is disjoint from the prime divisors of $C(mq)$ for all sufficiently large prime q . If, in addition, for all sufficiently large prime q one has $C(mq) > 1$ and $\gcd(C(mq), mq) = 1$, then for every B, N_0 there are primes $q \geq N_0$ and $p > B$ with $p \mid C(mq)$.

Indeed, the divisibility gives $mq \leq p^k - 1 < p^k \leq p^d$. For a finite set of primes, take q larger than all their d -th powers; for the last assertion, take a prime divisor of the nontrivial value $C(mq)$ after this threshold. The stated coprimality is included in the formal source's layer hypothesis, although this last extraction argument uses only $C(mq) > 1$. [LEAN SOURCE](#) [LEAN SOURCE](#) [Lean] [uniform] . For $C(n) = 2^n - 1$, the first theorem above supplies the hypothesis with $m = d = 1$: here $\text{ord}_q(p) = 1$, whereas $\text{ord}_p(2) = q$. For $C(n) = |\Phi_n(2)|$ and a general fixed $m > 0$, the source uses the eventual version: the divisibility is required only for prime q above a fixed threshold. The same proof then applies after that threshold. It is not valid to replace this by an all-prime assertion: $\Phi_6(2) = 3$, but $6 \nmid 3 - 1$. For the binary cyclotomic family, choosing $q > \max(m, 2^m)$ removes these exceptional indices. Other families require their own proof of the divisibility and nontriviality assumptions.

Lean: `boundedDegreeOrderConsumer_unfolded`, `coprime_of_dvd_pow_sub_one`, `dvd_pow_sub_one_iff_orderOf_dvd`, `boundedOrder_witness_iff_orderOf_le`, and 8 further declarations in the [coverage section](#).

6.6.13 The lower bound for rank-one quotients

The following finite-sum quotients do not approach $\Theta_2 = S - 1/2$. Their lower bound is uniform in both the exponent and the truncation length. This matters for rational approximation: varying either parameter cannot make the associated linear form tend to zero.

Theorem 6.156 (A uniform positive gap for rank-one quotients). *For integers $r \geq 2$ and $Y \geq 1$, write*

$$\Theta_r = \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^r}, \quad t(Y, r) = \sum_{d=1}^Y \frac{\mu(d)}{(2^d - 1)^r}.$$

For every $e \geq 1$ and $Y \geq 4$,

$$\frac{t(Y, e+2)^2}{t(Y, 2e+2)} - \Theta_2 > \frac{1}{480}.$$

Here $\Theta_2 = S - 1/2$. For $r \geq 3$, the proof uses $1429/1512 \leq \Theta_r < 1$ and $|t(Y, r) - \Theta_r| \leq 1/3584$ when $Y \geq 4$. In particular, these inequalities make the denominator positive. [LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Lean: `rankOneSubrankQuotient_sub_theta_two_gt`, `mobiusMersenneTheta_ge_alpha`, `mobiusMersenneTheta_lt_one`, `abs_mobiusMersenneTheta_sub_prefix_le`, and 2 further declarations in the [coverage section](#).

Proposition 6.157 (The bound is preserved by positive normalised averaging). *For a nonempty finite family of admissible quotients with positive weights summing to 1, the weighted average still exceeds Θ_2 by more than $1/480$. If an admissible quotient is p/q in lowest terms, with $q > 0$, then $|q\Theta_2 - p| > q/480$.* [LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Observation 6.158. Writing an admissible quotient in lowest terms as p/q , with $q > 0$, gives $|q\Theta_2 - p| > q/480 \geq 1/480$. Such forms cannot tend to zero. The conclusion concerns exactly the displayed quotients and their positive normalised averages. It makes no assertion about other rational approximants or about signed averages.

6.6.14 Multiples of a period

Allowing positive multiples of a prescribed period gives another criterion equivalent to irrationality of S . Sufficiency follows by telescoping the tail-period identity, and necessity follows from completeness of the residue test. The equivalence is proved, but its quantified arithmetic condition is not established. The finite examples below give denominator exclusions whose overlap with the diagonal certificates is described explicitly.

Theorem 6.159 (Concatenation and specified period multiples). *For nonnegative integers a, b, N , define the integer block sum*

$$Q_{a,N} = \sum_{j=1}^a \varphi(N+j)2^{a-j}.$$

Splitting a block after its first a terms gives

$$Q_{a+b,N} = 2^b Q_{a,N} + Q_{b,N+a}.$$

In particular, $Q_{2h,N} = 2^h Q_{h,N} + Q_{h,N+h}$. The corresponding cyclotomic identity is $\Phi_4(2^h) = 2^{2h} + 1 = \Phi_2(2^{2h})$: the order-four factor at height h is the order-two factor at height $2h$.

The order-three factor is different: $\Phi_3(2^h) = 2^{2h} + 2^h + 1$ comes from tripling the period. It is not generally part of the doubling sequence. For example, $\Phi_3(2) = 7$ divides none of $2^{2^j} - 1$, because the order of 2 modulo 7 is 3, which does not divide 2^j . Thus the block identities relate specified period multiples; they do not put all the order-two, order-three and order-four factors into one doubling chain. [LEAN SOURCE](#) [LEAN SOURCE](#) [\[Lean\]](#) [\[uniform\]](#) .

Lean: `totientBlock_eq_paper_indexed_sum`, `totientBlock_concatenation`, `totientBlock_doubling`, `cyclotomic_four_eval`, and 5 further declarations in the [coverage section](#).

Observation 6.160 (An equivalent quantified condition). [LEAN SOURCE](#) [Lean] (both implications proved; the quantified condition itself is not established) [cofinal] . The paper's current aggregate diagonal table gives certificates at every H_t for $t \leq 82$; this module's equivalence contributes no new information about $t = 83$ or any condition at arbitrarily large scales. [LEAN SOURCE](#)

Observation 6.161 (Eight finite period certificates). The cited declarations give certificates at basepoint $N = 300$ for

$$h \in \{67, 81, 97, 101, 121, 125, 127, 128\}.$$

Each excludes every reduced fraction a/b , with $b > 0$ and $b \mid 2^{300}(2^h - 1)$. These exclusions overlap with those already obtained from the diagonal. In particular, 67 and 81 divide H_{82} , and $H_{82} > 300$, so

$$2^{300}(2^h - 1) \mid 2^{H_{82}}(2^{H_{82}} - 1) \quad (h = 67, 81).$$

Their entire denominator classes are already covered by Proposition 6.104. The eight cases therefore cannot all be described as new denominator classes relative to that proposition. The exact comparison of the odd factors is

$$\gcd(2^h - 1, 2^{H_{82}} - 1) = 2^{\gcd(h, H_{82})} - 1,$$

by the Euclidean algorithm on the exponents. None of the remaining six periods divides H_{82} , so its full odd factor is not covered by that diagonal denominator. This compares with the $t = 82$ certificate, not with every other exclusion in the record; the case $h = 127$ includes the Mersenne prime $2^{127} - 1$.

The case $h = 67$ also succeeds at depth $L = 67$. It illustrates the restriction $L = h$ in the full-depth condition, while retaining a small basepoint. That universally quantified condition is equivalent to irrationality; one finite instance is not. [LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#) [Cert] [fixed] ; eight finite exclusions, not a proof of the condition at arbitrarily large scales.

6.6.15 The strict prime-position criterion

A sufficient prime-index condition is that, for every integer $h \geq 1$ and every $N_0 \geq 0$, there is a prime $p \geq \max(N_0 + h + 1, h + 5)$ such that

$$\operatorname{Re} e(2^{p-h-1}(2^h - 1)S) < \frac{9}{10}.$$

Here $e(x) = \exp(2\pi ix)$. The positive difference between $9/10$ and this real part may depend on the chosen prime; the truncation depth is then chosen so that its error fits within that difference. This is a weaker phase requirement than the earlier bound $4/5$, not a larger saving from 1. The implication to irrationality is proved, but the displayed prime-index condition remains unproved. [LEAN SOURCE](#) [Lean] (implication proved; antecedent [Open]) [cofinal] .

6.6.16 Finite Euler-product identities

These are identities for the Dirichlet-convolution square $\mu * \mu$, not the pointwise square $\mu(n)^2$. For each prime p , its coefficients at $1, p, p^2$ are $1, -2, 1$, and the higher prime-power coefficients vanish. The local factors at exponents 1 and 2 are

$$1 - \frac{2}{p} + \frac{1}{p^2} = \left(1 - \frac{1}{p}\right)^2, \quad 1 - \frac{2}{p^2} + \frac{1}{p^4} = \left(1 - \frac{1}{p^2}\right)^2.$$

For a prime p and $e \geq 0$, the corresponding convolution with the divisor-sum function σ gives

$$\sigma(p^{e+2}) - 2\sigma(p^{e+1}) + \sigma(p^e) = p^{e+1}(p-1) = \varphi(p^{e+2}).$$

This follows by subtracting the finite geometric sums. These finite identities do not assert a transcendence result. [LEAN SOURCE](#) [LEAN SOURCE](#) [Lean] [uniform] .

7 Parameter ranges and quantifiers

The table records parameter ranges and quantifiers. “Fixed” refers to listed examples, “bounded” to a finite range, and “cofinal” to a statement of the form $\forall N_0 \exists N \geq N_0$. A bound of the form $\forall a \geq 8$ is not finite-range evidence: it gives a conclusion at all sufficiently large parameters. “Uniform” describes the quantification of a statement, not whether all hypotheses needed to apply it are proved. A bound on a secondary variable, such as $j < 2 \cdot 2^a$, does not restrict the scale a to a finite range. Such parameter-uniform results are grouped with the uniform statements, not with finite calculations.

Some identities have no distinguished scale parameter. They are listed with the general identities. An equivalence can hold for all parameters without establishing either of the equivalent arithmetic assertions.

The LCM formulation of Problem 249 requires arbitrarily large scales: [LEAN SOURCE](#)

$$\text{Irrational}\left(\sum_{n \geq 1} \varphi(n)/2^n\right) \iff \forall a_0 \exists a \geq a_0, \Omega_a \notin \mathbb{Z},$$

a cofinal statement in the exponent a indexing $H = H(2^a)$. A finite calculation alone does not establish it; a further theorem could propagate finite information to arbitrarily large scales.

The table groups statements by their parameter ranges, not by proof status. In particular, some assertions with cofinal quantifiers are already proved, such as the prime constructions and rational comparison sequences. Others are implications whose quantified hypotheses remain open. The horizontal rule separates the cofinal group for ease of reference; it does not mark a boundary between proved and unproved mathematics. The descriptions identify this distinction in that group. The prefixes summarise the quantifiers; complete hypotheses are given in the cited statements. A definition records an object or condition; it does not by itself prove the associated soundness or existence assertion.

In the periodic-weight rows, write $w : \mathbb{N} \rightarrow \mathbb{Z}$ and $c_w(n) = \sum_{d|n} w(d)$ for $n \geq 1$. Positivity and nonvanishing refer to these divisor coefficients, not to the weight w itself.

Here “nonzero arbitrarily far out” means that for every N there is $n > N$ with $c_w(n) \neq 0$; it does not mean nonvanishing on every arithmetic progression. The nonprincipal character χ_4 modulo four is 0 on even integers and $(-1)^{(n-1)/2}$ on odd integers.

Table 3: Results and unproved conditions, grouped by their quantifiers. The dividing rule separates the cofinal group, not proof status. Complete hypotheses are given in the cited statements.

Result	Source	Scale	Quantifier prefix	Representation
<i>Uniform statements, including unconditional identities and implications</i>				
scaled totient tail (definition)	LEAN SOURCE	uniform	$\forall N$	binary-weighted series
integer prefix and real tail	LEAN SOURCE	uniform	$\forall N$	binary-weighted series
$D(h, N, L)$ (definition)	LEAN SOURCE	uniform	$\forall h \forall N \forall L$	finite binary sums
$C(h, N, L)$ (definition)	LEAN SOURCE	uniform	$\forall h \forall N \forall L$	finite binary sums
necessary depth inequality	LEAN SOURCE	uniform	$\forall h \forall N \forall L$, given $C(h, N, L)$	finite binary sums
residue test soundness	LEAN SOURCE	uniform	$\forall h \forall N \forall L$, given C	finite binary sums
completeness of the residue test	LEAN SOURCE	uniform	$\forall h \forall N$	finite binary sums
integrality from denominator divisibility	LEAN SOURCE	uniform	$\forall N \forall h \forall r: \mathbb{Q}$, given $S = r, r.\text{den} \mid 2^N(2^h - 1)$	binary-weighted series
tail-period law	LEAN SOURCE	uniform	$\neg \text{Irrational } S \rightarrow \exists h > 0 \exists N_0 \forall N \geq N_0$	binary-weighted series
least common multiple $H(t)$	LEAN SOURCE	uniform	$\forall t$	LCM periods
a nondivisor below $2t$ is a prime power	LEAN SOURCE	uniform	$\forall t \forall j (0 < j < 2t)$, given $j \nmid H(t)$	LCM offsets
totient factorisation at divisor offsets	LEAN SOURCE	uniform	$j \mid H(t)$, with the prime-divisor hypotheses of Proposition 6.48	totient factorisation
integrality on the LCM grid under rationality	LEAN SOURCE	uniform	$\neg \text{Irrational } S \rightarrow \exists t_1 \forall t \geq t_1 \forall q > 0 \forall m$	LCM grids
second-difference residue test	LEAN SOURCE	uniform	$\forall h \forall N \forall L$, given second-difference residue condition	finite binary sums
a nonintegral pair on a finite grid	LEAN SOURCE	uniform	finite $Q \neq \emptyset$; all size and residue hypotheses of Theorem 6.56	finite LCM grids
finite exclusion of integer carries	LEAN SOURCE	uniform	$\forall h \forall N \forall K$, given finite carry exclusion	integer recurrences
Farey neighbour denominator law	LEAN SOURCE	uniform	$a, b, c, d, r, s \in \mathbb{Z}; b, d > 0, bc - ad = 1, as < br, dr < cs$	rational intervals
rational-approximation criterion	LEAN SOURCE	uniform	$u_j \in \mathbb{Q}, x \in \mathbb{R}$; eventually $u_j \neq x$, and $\text{den}(u_j) u_j - x \rightarrow 0$	general criterion

continued on next page

Result	Source	Scale	Quantifier prefix	Representation
near-integer irrationality criterion	LEAN SOURCE	uniform	$\forall \xi: \mathbb{R}, \forall q > 0 \exists m z 0 < m\xi - z < 1/q$	general criterion
independence from a nonsingular evaluation matrix	LEAN SOURCE	uniform	a finite family with a square evaluation matrix of nonzero determinant	general criterion
dyadic totient-kernel rank = $2^e + 1$	LEAN SOURCE	uniform	$\forall e \geq 1$	dyadic sections
rationality forces unbounded carry-kernel rank	LEAN SOURCE	uniform	$\neg \text{Irrational } S \rightarrow \exists v > 0 \exists u \forall e$	carry sections
no fixed D clears all $A(n)/n$ for $A = \varphi * \mu$	LEAN SOURCE	uniform	$\forall D > 0$	Möbius sums
Lambert-series value identities	LEAN SOURCE	uniform	value identities, $L(\mu) = \frac{1}{2}, L(\varphi) = 2, L(1) = \text{Erdős-Borwein}$	Möbius sums
prime-power unit-gap criterion	LEAN SOURCE	uniform	$\forall p \text{ prime}, e > 0, N, K \geq 0$	rational intervals
positive rational-difference lower bound	LEAN SOURCE	uniform	$\forall \text{ whole pfx: } \mathbb{Q}, \text{ pfx} < \text{whole}$	general criterion
$L_2(\mu) = S - \frac{1}{2}$	LEAN SOURCE	uniform	unconditional	Möbius sums
weighted squared-denominator identity	LEAN SOURCE	uniform	$\forall w: \mathbb{N} \rightarrow \mathbb{R} (w(d) \leq d) \forall r \in [0, 1)$	Möbius sums
$L_2(\mu) = S - \frac{1}{2}$	Proposition 5.14	uniform	unconditional	Möbius sums
$L_2(1) = \zeta_q(2) - \zeta_q(1)$ identity	LEAN SOURCE	uniform	unconditional	Möbius sums
$L_2(\varphi) = \text{gcd moment}$	LEAN SOURCE	uniform	unconditional	gcd moments
comparison of the weights	Propositions 5.14–5.18	uniform	unconditional	Möbius sums
gcd-divisibility factorizes	LEAN SOURCE	uniform	$\forall d: \mathbb{N}, d > 0$	geometric weights
sum over positive coprime pairs equals 1	LEAN SOURCE	uniform	unconditional (sum over all coprime pairs)	geometric weights
Stern–Brocot recursion with stopping	LEAN SOURCE	uniform	positive integers a, b	Stern–Brocot tree
positive numerator coefficients	LEAN SOURCE	uniform	$\forall r (\text{Squarefree } r) \forall k < r$	divisor and cyclotomic sums
evaluation at 2 gives the numerator	LEAN SOURCE	uniform	$\forall r (\text{Squarefree } r)$	divisor and cyclotomic sums
reduction to the radical of H	LEAN SOURCE	uniform	$\forall H > 0 \forall r > 0$	Möbius sums

continued on next page

Result	Source	Scale	Quantifier prefix	Representation
numerator congruence modulo Φ_m	LEAN SOURCE	uniform	$\forall r$ (Squarefree r) $\forall m$ ($m r$)	divisor and cyclotomic sums
gcd of the numerator and $\Phi_r(2)$	LEAN SOURCE	uniform	$\forall r$ (Squarefree r)	divisor and cyclotomic sums
Mersenne factors retained in the denominator	LEAN SOURCE	uniform	$t \geq 5$, squarefree r ; finite P of primes dividing r in $(t/2, t]$; prime factors of r and the multiplier are at most t	divisor and cyclotomic sums
denominator lower bound / exact value	LEAN SOURCE	uniform	$\forall t$ (exact-value); $t \geq 5$ (lower bound)	Möbius sums
numerator congruence after adjoining a prime	LEAN SOURCE	uniform	$r \geq 1$, p prime, $p \nmid r$, $m \mid r$	divisor and cyclotomic sums
squared-Mersenne series: partial sum plus tail	LEAN SOURCE	uniform	$\forall D: \mathbb{N}$	Möbius sums
cancellation of the affine divisor terms	LEAN SOURCE	uniform	$\forall d > 0$ ($d H$); general: finite affine annihilator	LCM diagonal
new-divisor counting identity	LEAN SOURCE	uniform	$\forall a \in A$ ($a > 0$) $\forall x > 0$	divisor counts
bound on zero blocks in divisor counts	LEAN SOURCE	uniform	$\forall \varepsilon > 0 \exists B$, given rationality of the base-2 support series	divisor counts
equivalent shift condition for the target $1/2$ (Problem 257)	LEAN SOURCE	uniform	iff, no extra parameter	Mersenne subsums
totient multiplication with common prime factors	LEAN SOURCE	uniform	$\forall x > 0$ (j fixed)	totient identities
totient relative-Euler-product split	LEAN SOURCE	uniform	$\forall j x > 0$	totient identities
formula for $\varphi(2H(t) + j) - \varphi(H(t) + j)$	LEAN SOURCE	uniform	$\forall t \forall j$	Möbius sums
the finite diagonal sum equals $D(H(t), H(t), L)$	LEAN SOURCE	uniform	$\forall t \forall L$	Möbius sums
#249 $\iff$ cofinal actual-orbit nonintegrality	LEAN SOURCE	uniform	iff	Möbius sums
the LCM tail difference is an integer translate of a multiple of S	LEAN SOURCE	uniform	$\forall a$	Möbius sums
explicit finite-block approximation	LEAN SOURCE	uniform	$\forall a \forall q$	Möbius sums
an equivalent residue-interval condition	LEAN SOURCE	uniform	$\forall h N$, iff	integer recurrences

continued on next page

Result	Source	Scale	Quantifier prefix	Representation
sign of the three-term totient difference under the stated middle-value bound	LEAN SOURCE	uniform	$\forall H j$, given MiddleRankTotientExtremal	totient differences
exact value on a specified dyadic family	LEAN SOURCE	uniform	$\forall n$ (closed-form)	totient differences
the rational parity comparison (definition)	LEAN SOURCE	uniform	$\forall n$ (definition)	binary-weighted series
R_N as a sum over positive divisor indices	LEAN SOURCE	uniform	$\forall N$	divisor and cyclotomic sums
R_N as a shifted Möbius sum	LEAN SOURCE	uniform	$\forall N$	divisor and cyclotomic sums
absolute convergence and regrouping of Lambert sums	LEAN SOURCE	uniform	$\forall w v: \mathbb{N} \rightarrow \mathbb{R} \ (w(d) \leq d, v(m) \leq m)$	divisor and cyclotomic sums
carry divisibility and integral tail differences	LEAN SOURCE	uniform	$\forall N k$, given tempered orbit	integer recurrences
rationality $\Rightarrow$ periodicity & unbounded rank	LEAN SOURCE	uniform	$\neg \text{Irrational } S \rightarrow \exists v u \forall e$	integer recurrences
the stated reconstruction cost is at least 3	LEAN SOURCE	uniform	$\forall$ finite $w: \iota \rightarrow \mathbb{Q} \ x: \iota \rightarrow \mathbb{N}$, given $\sum w_i \varphi(x_i) = 1$	weighted tail bounds
directed residue interval (definition)	LEAN SOURCE	uniform	$h, N, L \geq 0$; soundness and the existential-depth equivalence are proved in Proposition 6.69	integer recurrences
rationality and an integer carry with vanishing boundary term	LEAN SOURCE	uniform	$\forall c: \mathbb{N} \rightarrow \mathbb{N} \ (c(n) \leq n)$, iff	binary-weighted series
doubling-orbit rigidity	LEAN SOURCE	uniform	$\forall d: \mathbb{N} \rightarrow \mathbb{R}$, doubling + tempered $\rightarrow d \equiv 0$	integer recurrences
number of labels needed for exact tail recovery	LEAN SOURCE	uniform	$\forall m \forall \text{State}$, collapsing-state $\rightarrow$ no decoder	binary-weighted series
endpoint residues do not determine the initial carry	LEAN SOURCE	uniform	$\forall a: \mathbb{N} \rightarrow \mathbb{Z} \ \forall u_0 v_0 \forall L$	binary-weighted series
integer-coefficient block-certificate dichotomy	LEAN SOURCE	uniform	$\forall b \geq 2 \forall c \forall q$, given full-block certificate at q	binary-weighted series
divisibility under iterated forward differences	LEAN SOURCE	uniform	$d \mid f(n)$ for every n ; every finite list of shifts	integer sequences
totient series as a coprime-pair sum	LEAN SOURCE	uniform	$\forall n; \forall r \in [0, 1)$	geometric weights

continued on next page

Result	Source	Scale	Quantifier prefix	Representation
factorisation through a rational-valued linear functional	LEAN SOURCE	uniform	rational vector spaces V, W ; surjective linear $\text{ev} : V \rightarrow \mathbb{Q}$, linear $L : V \rightarrow W$, $\ker(\text{ev}) \subseteq \ker L$	divisibility and linear algebra
divisibility of a reduced denominator	LEAN SOURCE	uniform	$\forall D > 0 \forall m D \forall a$	divisibility and linear algebra
the complementary denominator divides the integer multiplier	LEAN SOURCE	uniform	$\forall x : \mathbb{Q} \forall H (H x.\text{den}) \forall c ((cx).\text{den} H)$	divisibility and linear algebra
largest-exponent nonvanishing criterion	LEAN SOURCE	uniform	$\exists m \in s$ (strictly maximal, odd coeff), $\forall i \neq m e(i) < e(m)$	divisibility and linear algebra
column rescaling and determinant nonvanishing	LEAN SOURCE	uniform	$\forall d \forall e : \text{Fin } d \rightarrow \mathbb{N} \forall z : \text{Fin } d \rightarrow \mathbb{C} (z \text{ nonzero})$	complex matrices
determinant 1 for the divisor-incidence matrix	LEAN SOURCE	uniform	$\forall N : \mathbb{N}$	Möbius sums
error bound from convergence	LEAN SOURCE	uniform	$2H \leq D$; convergence is supplied for $H > 0$ in Section 6.6.8	Möbius sums
carries realising fixed-precision unit residues	LEAN SOURCE	uniform	$\forall u > 0$ (fixed precision), $\forall$ finite carry word	divisibility and linear algebra
vanishing of iterated prime dilation differences	LEAN SOURCE	uniform	distinct primes $p_1, \dots, p_r, r > K$, $\gcd(n, \prod p_i) = 1$; $\Omega(a) \leq K$ on A	divisor counts
$\varphi(n) > \frac{3}{4}n$ for the stated rough integers	LEAN SOURCE	uniform	$\forall a \geq 8 \forall n > 0$ (n is t -rough, $t = 2^a, n < 2^{2 \cdot 2^a}$)	Möbius sums
positivity of short-window totient differences	LEAN SOURCE	uniform	$\forall a \geq 8 \forall j (0 < j < 2 \cdot 2^a)$	Möbius sums
the negative representative of the true endpoint	LEAN SOURCE	uniform	$\forall a \geq 8 \forall J K$ (room bound)	Möbius sums
integrality forces upper-endpoint residue	LEAN SOURCE	uniform	$\forall a \geq 8 \forall J K$ (room, $2H + J + K + 2 < 2^K$)	Möbius sums
terminal dyadic staircase impossible with room	LEAN SOURCE	uniform	$\forall a \geq 8 \forall J K m$ (room, modulus-wide)	Möbius sums
penultimate difference in a partial divisibility pattern	LEAN SOURCE	uniform	$\forall a \geq 8 \forall J K m$ (hypotheses of Proposition 6.86)	Möbius sums
upper-endpoint residue-gap $\Rightarrow$ nonintegral	LEAN SOURCE	uniform	$\forall a \geq 8 \forall J K m$ (room, one-sided)	Möbius sums

continued on next page

Result	Source	Scale	Quantifier prefix	Representation
relation between the centred residue and the actual carry	LEAN SOURCE	uniform	$\forall a \geq 8 \forall q \text{ (room, fit bound } 2(H+q+2) \leq 4^q)$	Möbius sums
divisibility of the three-term totient difference	LEAN SOURCE	uniform	$\forall a \geq 4 \forall j > 0 (j^2 \leq 2^a)$	totient differences
equality of two finite rational approximations	LEAN SOURCE	uniform	$H > 0, D \geq 2H$; no separation hypothesis for this equality	Möbius sums
a Mersenne divisor need not divide a tail difference	LEAN SOURCE	uniform	$\forall K, q (K > 0, K < q, q 2^K - 1)$	Möbius sums
finite numerator separation implies nonintegrality	LEAN SOURCE	uniform	$\forall H, D$ fixed, given uniform numerator-gap bound	Möbius sums
periodic-weight Lambert dichotomy	LEAN SOURCE	uniform	$\forall b \geq 2 \forall m > 0 \forall w (w(n+m) = w(n))$	Möbius sums
periodic weight with nonnegative divisor coefficients	LEAN SOURCE	uniform	$b \geq 2, m > 0, w(n+m) = w(n); c_w(n) \geq 0$ and $c_w(n) \neq 0$ arbitrarily far out	Möbius sums
zeros on one progression for the character modulo four	LEAN SOURCE	uniform	$c_{\chi_4}(n) = 0$ for every $n \equiv 3 \pmod{4}$; the weight is fixed	Möbius sums
positivity of the translated LCM tail difference	LEAN SOURCE	uniform [†]	$\forall a \geq 8 \forall J (J + (a+6) < 2 \cdot 2^a)$, no rationality hypothesis	Möbius sums
<i>Finite ranges and parameter-bounded checks</i>				
Farey window $K=240$ denominator bound	LEAN SOURCE	bounded	$\forall q: \mathbb{N}, 0 < q \leq 79,639,646,646,701, 375,323,355,774,875,831,053$	rational intervals
denominator lower bound $S \neq p/q, q \leq Q_0$	LEAN SOURCE	bounded	$\forall p: \mathbb{Q}, p.\text{den} \leq Q_0$	rational intervals
finite LCM certificates covering thresholds $a_0 \leq 6$	LEAN SOURCE	bounded	$\forall a_0 \leq 6 \exists a L$	Möbius sums
contiguous diagonal certificate band through $t \leq 82$	LEAN SOURCE	bounded	$\forall t \leq 82 \exists L$; no $t=83$ certificate claimed	LCM diagonal
<i>Listed examples</i>				
residue certificates for $1 \leq h \leq 8, N = 12, L = 16$	LEAN SOURCE	fixed	$\forall h \in [1, 8], C h 12 16$ (by decide)	finite binary sums

continued on next page

Result	Source	Scale	Quantifier prefix	Representation
denominator exclusions for $1 \leq h \leq 16$, binary exponent 14	LEAN SOURCE	fixed	$\forall h \in [1, 16]$, denominator exponent 14 (by decide)	finite binary sums
the second-difference test uses more depth at $(h, N) = (1, 8)$	LEAN SOURCE	fixed	first-difference certificate at $L = 8$; no second-difference certificate for $1 \leq L \leq 8$, and one at $L = 9$	finite binary sums
28 listed diagonal certificates with $t \leq 64$	LEAN SOURCE	fixed	$t \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, \dots\}$ (28 explicit values)	LCM diagonal
LCM certificates at $a = 4$ and $a = 6$	LEAN SOURCE	fixed	$a=4, a=6$ (concrete, formally checked)	Möbius sums
square-CRT coprime-factor examples: zero / nonzero differences	LEAN SOURCE	fixed	witness $n=52$ (vanishes); witness $n=27$ (nonzero)	totient factorisation
<i>Statements with cofinal quantifiers: proved constructions and conditional implications</i>				
<i>Cofinal statements; status specified in the description</i>				
certificate criterion (proved implication; unproved premise)	LEAN SOURCE	cofinal	$\forall h \geq 1 \forall N_0 \geq 0 \exists N \geq N_0 \exists L, C h N L$	finite binary sums
period-multiple criterion (equivalent; unproved condition)	LEAN SOURCE	cofinal	$\forall h_0 > 0 \forall N_0 \exists m > 0 \exists N \geq N_0 \exists L$	multiples of periods
LCM-diagonal criterion (unproved condition)	LEAN SOURCE	cofinal	$\forall t_0 \exists t \geq t_0 \exists L$	LCM diagonal
LCM-grid certificate criterion (unproved premise)	LEAN SOURCE	cofinal	$\forall t_0 \exists t \geq t_0 \exists q > 0 \exists m \exists L$	LCM grids
LCM nonintegrality criterion (unproved condition)	LEAN SOURCE	cofinal	$\forall t_0 \exists t \geq t_0, R(2H_t) - R(H_t) \notin \mathbb{Z}$	real tail differences
finite-grid residue criterion (unproved premise)	LEAN SOURCE	cofinal	for every t_0 , data $t \geq t_0, L, Q$ satisfying Theorem 6.57	finite LCM grids
first-harmonic norm criterion (unproved premise)	LEAN SOURCE	cofinal	$\forall h > 0 \forall X_0 \exists X \geq \max(X_0, 1) \exists L$, room $16(2X+h+L+2) \leq 2^L$	binary-weighted series
four-tail residue criterion (unproved premise)	LEAN SOURCE	cofinal	$\forall t_0 \exists t \geq t_0 \exists p, L > 0$	binary-weighted series
short LCM-window criterion (unproved premise)	LEAN SOURCE	cofinal	$\forall a_0 \exists a L, a_0 \leq a \wedge L < 2 \cdot 2^a$	Möbius sums

continued on next page

Result	Source	Scale	Quantifier prefix	Representation
LCM-tail separation criterion (unproved premise)	LEAN SOURCE	cofinal	cofinal $1/32$ +error separation at guarded odd ranks	Möbius sums
five endpoint criteria (unproved premises)	LEAN SOURCE	cofinal	$\forall a_0 \exists a \geq a_0 \exists \langle \text{supply predicate} \rangle$	Möbius sums
proved rational counterexample to bound, parity and aperiodicity	LEAN SOURCE	cofinal	$\exists c: \mathbb{N} \rightarrow \mathbb{N}, c \leq 6, c \leq n, c \equiv \varphi \pmod{2}$, arbitrarily-separated blocks	binary-weighted series
proved mod-4 prime construction (CRT and Dirichlet)	LEAN SOURCE	cofinal	$\forall h > 0 \forall B \exists \text{prime } p > B$	Möbius sums
sufficient mod-4 condition (unproved premise)	LEAN SOURCE	cofinal	the displayed mod-four residue condition at arbitrarily large scales	Möbius sums
proved two-adic prime construction	LEAN SOURCE	cofinal	$\forall K \geq 2 \forall H > K \forall B \exists \text{cofinally many primes } p > B$	divisibility and linear algebra
proved implication from eventual integrality	LEAN SOURCE	cofinal	$\forall K \geq 2$, given the stated two-adic prime construction and eventual integrality	divisibility and linear algebra

[†] In the positivity statement, only the window parameter J is bounded relative to $2 \cdot 2^a$. The exponent a ranges over all $a \geq 8$, and the constants 4, 8 and 32 are independent of a . The bound on J is a hypothesis of a uniform theorem; it is not a finite verification of the exponent range.

7.1 What extending a finite calculation would require

A finite computation can be extended by further computation or by a uniform argument. The following examples distinguish these possibilities from parameter-uniform theorems that are already proved. They do not assert that a computational bound is intrinsic to a method.

1. **Finite certificate tables.** Results such as [LEAN SOURCE](#) (28 explicit values of t through $t=64$, now a strict subset of the aggregate $\forall t \leq 82$ band), [LEAN SOURCE](#) ($h \in [1, 8]$ at fixed depth 16), and [LEAN SOURCE](#) ($h \in [1, 16]$ at exponent 14) are each a finite *list* of independently verified rows, not a single argument instantiated at a free parameter. The existing lists do not establish further rows; an extension needs either additional computations or an argument for a family of parameters.
2. **A bounded window inside a uniform theorem.** The results [LEAN SOURCE](#) and [LEAN SOURCE](#) hold for every $a \geq 8$, subject to their stated bounds on the other parameters. In the sign theorem, for example, $J + a + 6 < 2 \cdot 2^a$. Their proofs are therefore not merely enumerations of a fixed set of exponents. The missing conclusion is a suitable residue inequality at arbitrarily large scales, not extension of the range of a in the positivity theorem. No complexity lower bound follows from this audit.
3. **Parameter dependence in an approximation estimate.** For reduced approximants p_K/q_K to S , the conditions $0 < |S - p_K/q_K| \leq \varepsilon_K$ and $q_K \varepsilon_K \rightarrow 0$ imply irrationality. A growing denominator alone does not control this product. In particular, the exact reduced-denominator formula [LEAN SOURCE](#) concerns a specified finite Möbius sum; it does not estimate the complementary tail. Likewise, a bound proved with a constant depending on a fixed auxiliary parameter is not automatically uniform when that parameter grows. Both the dependence of the constant and the error estimate must be retained in any passage to an unbounded family.

The missing arithmetic inputs include the stated constant-saving exponential-sum estimate and an upper-endpoint residue inequality. For the positivity theorem, the unresolved step is exclusion of that residue interval, not positivity at more exponents.

8 Relations between the representations

The formulas below express the same series through binary tails, Möbius sums, rational approximations and exponential sums. Each identity permits a change of variables or representation. A limitation of one proposed argument does not automatically apply after that change: its hypotheses must be checked for the new objects. The last subsection illustrates this distinction with finite-state decoding and polynomial denominator identities.

Binary-weighted series and tails.

The coefficients in $S = \sum_{n \geq 1} \varphi(n)2^{-n}$ are unbounded; this expression is not the base-two digit expansion of S . The tail R_N , finite difference numerator $D(h, N, L)$ and test $C(h, N, L)$ retain this coefficient information. The identity

$$S = \frac{1}{2} + \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2}$$

[LEAN SOURCE](#) and its partial-sum version [LEAN SOURCE](#) express the same value by Möbius sums. An equality of values does not by itself transfer a claim about the original coefficient subsequences.

Lambert sums and finite numerator polynomials.

The Lambert identities for μ , 1 and φ give the series relations in Section 5. For a finite divisor sum, the polynomial evaluation [LEAN SOURCE](#) and the definition [LEAN SOURCE](#) put its numerator into integer polynomial arithmetic. This evaluates a finite contribution, not the complementary infinite tail.

Coprimality probability.

Let X, Y be independent with $\Pr(X = n) = \Pr(Y = n) = 2^{-n}$ for $n \geq 1$. Then $S - \frac{1}{2} = \Pr(\gcd(X, Y) = 1)$. The divisibility probabilities are $\Pr(d \mid X, d \mid Y) = (2^d - 1)^{-2}$ [LEAN SOURCE](#). Möbius inversion converts these to the coprimality probability. The decomposition by the greatest common divisor also gives the weighted coprime-pair identity [LEAN SOURCE](#). Counting coprime pairs by their sum gives the totient expression directly [LEAN SOURCE](#).

Cyclotomic factors of finite denominators.

The coefficient formula for P_r [LEAN SOURCE](#) and the cyclotomic divisibility statement [LEAN SOURCE](#) identify factors in the reduced denominators of the specified finite sums. The product-divisibility theorem [LEAN SOURCE](#) is used in the exact denominator formula [LEAN SOURCE](#). Growing finite denominators alone do not exclude cancellation with the remaining tail.

Numerator divisibility and denominator bounds.

Write $x = a/b$ in lowest terms, with $b > 0$. For $c \in \mathbb{Z}$ and a positive divisor H of b , if the reduced denominator of cx divides H , then $b/H \mid c$ [LEAN SOURCE](#). In particular, $|c| \geq b/H$ when $c \neq 0$. A related bound is

$$0 < \frac{a}{b} < \frac{2}{2^n - 1}, \quad 2^r \mid a, \quad n \geq 1 \quad \Rightarrow \quad 2^r(2^n - 1) < 2b$$

for integers $r \geq 0$ [LEAN SOURCE](#). Here positivity gives $a \geq 2^r$; the size inequality then gives the conclusion. These are arithmetic bounds on a reduced fraction, not irrationality criteria without an additional approximation estimate.

Farey intervals.

The denominator bound [LEAN SOURCE](#) applied to the explicit window $(N, K) = (1, 240)$ gives [LEAN SOURCE](#) and hence $S \neq p/q$ for integers p and $1 \leq q \leq Q_0$. The

finite input is a totient numerator modulo 2^{240} . Its exact first failing denominator is $Q_0 + 1$, the denominator of the relevant Farey mediant [LEAN SOURCE](#). This last assertion concerns this fixed window.

Differences at least-common-multiple shifts.

For $H_t = \text{lcm}(1, \dots, t)$ [LEAN SOURCE](#), a jump in H_t can occur only at a prime power [LEAN SOURCE](#). The identities [LEAN SOURCE](#) and [LEAN SOURCE](#) identify the coefficient difference as $\delta_t(j) = \varphi(2H_t + j) - \varphi(H_t + j)$ and give $D(H_t, H_t, L) = \sum_{j=1}^L \delta_t(j) 2^{L-j}$. Thus these arithmetic formulas concern particular instances of the same finite residue test; additional window restrictions remain in force.

Exponential sums of the window residues.

For integers $h, X \geq 1$, the relevant block statistic is

$$\sum_{N=X}^{2X-1} \cos\left(2\pi \frac{D(h, N, L)}{2^L}\right).$$

A bound by $9X/10$, together with $L \geq h$ and $16(2X + h + L + 2) \leq 2^L$, yields a certificate in the block [LEAN SOURCE](#). The linked irrationality implication uses the stronger complex norm bound $21X/25$, for every h on arbitrarily late blocks satisfying the same size inequality [LEAN SOURCE](#). The finite implication uses $\cos(\pi/8) > 9/10$; the required cancellation bound is still unproved. The norm bound on the complex exponential sum is sufficient for the displayed real-part bound, but is stronger.

8.1 Binary tails and Möbius sums

Erdős's argument for $\sum_n 1/(2^n - 1)$ chooses congruences so that divisor counts along a block are divisible by increasing powers of the base. A tail bound then forces a long run of zero digits [9, pp. 63–66]. It uses no finite automaton.

A different proposal would recover each later block from a fixed finite summary of the preceding history. The balanced-pulse counterexample [LEAN SOURCE](#) shows why that proposal fails for the family constructed there. For $m \geq 2$, its members have the same earlier history but different successors. A state that identifies all those histories cannot recover every successor. More generally, an exact label-and-decoder pair for that family needs at least $\lfloor (m + 1)/2 \rfloor + 1$ distinct labels. This is a lower bound on the number of labels, not a linear lower bound on the number of bits, and it is not a state-complexity theorem for the actual totient sequence. [Lean] [uniform] .

The cofinal certificate implication [LEAN SOURCE](#) is a separate theorem. The counterexample above neither proves its hypothesis nor makes it the only possible route to irrationality. Likewise, [LEAN SOURCE](#) shows that inverse-phase column weights can preserve a determinant, and [LEAN SOURCE](#) gives the exact growing ranks of totient sections. These are different statements with different hypotheses; none is a general prohibition on other representations of S .

The Möbius representation instead allows the identity [LEAN SOURCE](#) to be applied to μ , since $|\mu(d)| \leq d$ for $d \geq 1$. The polynomial identities from [LEAN SOURCE](#) to [LEAN SOURCE](#)

operate at each finite radical without an autonomous decoder. The exact reduced denominator and its lower bound are then given by [LEAN SOURCE](#) and [LEAN SOURCE](#). Those formulas concern a finite rational contribution. They do not exclude cancellation by the complementary tail and hence do not settle irrationality. The representations are equal expressions for S ; what differs is the information supplied by the particular arguments.

9 Index of unproved conditions

This section collects the unproved hypotheses used by the conditional results above. For each one, it states the additional arithmetic assertion and the implication to $S \notin \mathbb{Q}$. A proof of an implication is not a proof of its hypothesis. The record does not settle whether $S = \sum_{n \geq 0} \varphi(n)2^{-n}$ is irrational.

9.1 How to use the index

Each entry separates the proved implication from the additional hypothesis needed to apply it. The comparison concerns the displayed statements, not the prospects of an entire method.

The range tags describe quantification, not a ranking of mathematical strength. [fixed] records finitely many explicit instances; [bounded] records the displayed bounded parameter range; [uniform] ranges over every allowed value of the stated parameters; and [cofinal] includes a quantifier of the form $\forall a_0 \exists a \geq a_0$. All other hypotheses and quantifiers remain part of the statement. A cofinal assertion about a different quantity, or with a different order of quantifiers, need not supply the irrationality condition. [n/a] denotes the absence of a relevant growing parameter. Proof status is recorded separately by [Lean], [Cert], [Math], [Cited] and [Open], as in the reading guide; a uniform or cofinal statement may itself be proved or may remain an unproved hypothesis.

The quantified residue condition for #249, stated in Definition 1.2 and proved sufficient ([LEAN SOURCE](#)), is the following statement:

$$\forall h \geq 1, \forall N_0, \exists N \geq N_0, \exists L, C h N L.$$

The following catalogue compares fourteen proposed approaches to this condition. It starts with the exponential-sum estimate, then gives thirteen further approaches with their individual hypotheses and source references.

9.2 The exponential-sum criterion

The next three definitions recall the finite sum, its residue test and its phase. The following theorems explain which estimates for that phase would imply the quantified condition in Definition 1.2.

Definition 9.1 (The truncated difference). [LEAN SOURCE](#). For $h, N, L \in \mathbb{N}$,

$$D(h, N, L) = \sum_{j=0}^{L-1} (\varphi(N + h + 1 + j) - \varphi(N + 1 + j)) \cdot 2^{L-1-j} \in \mathbb{Z}.$$

This is the depth- L truncation of $2^L \cdot (R_{N+h} - R_N)$, where $R_N = \sum_{j \geq 1} \varphi(N + j)/2^j$ is the local totient tail ([LEAN SOURCE](#)).

Definition 9.2 (The central-residue condition). [LEAN SOURCE](#).

$$C h N L : \Leftrightarrow (N+h+L+2) < D(h, N, L) \bmod 2^L < 2^L - (N+h+L+2).$$

That is: the residue of $D(h, N, L)$ modulo 2^L avoids the radius- $(N+h+L+2)$ neighbourhood of 0. [LEAN SOURCE](#) records the necessary room condition this forces: $2(N+h+L+2) < 2^L$.

Definition 9.3 (The complex phase and its real part). [LEAN SOURCE](#) and [LEAN SOURCE](#):

$$\operatorname{Re} E(h, N, L) = \cos\left(2\pi \frac{D(h, N, L) \bmod 2^L}{2^L}\right), \quad E(h, N, L) = \exp\left(i 2\pi \frac{D(h, N, L) \bmod 2^L}{2^L}\right).$$

[LEAN SOURCE](#) records $\operatorname{Re}(E) = \operatorname{Re} E$ and [LEAN SOURCE](#) that $|E(h, N, L)| = 1$: this is literally the first additive character of the endpoint discrepancy modulo 2^L , a standard exponential-sum object.

Theorem 9.4 (A real-part bound gives a certificate). [LEAN SOURCE](#). [Lean]. [uniform]. . For all h, X, L with $0 < X$ and the room condition $16(2X+h+L+2) \leq 2^L$, if

$$\sum_{N=X}^{2X-1} \operatorname{Re} E(h, N, L) \leq \frac{9}{10} X,$$

then $\exists N \in [X, 2X)$ with $C h N L$.

Lean: [windowFirstCos_unfolded, exists_certifiedKill_of_block_real_part_bound](#).

Theorem 9.5 (The same implication for a nonempty subset). [LEAN SOURCE](#). [Lean]. [uniform]. . For any nonempty finite $T \subseteq \mathbb{N}$ with $T \subset [0, 2X)$ and the same room condition, if

$$\sum_{N \in T} \operatorname{Re} E(h, N, L) \leq \frac{9}{10} |T|,$$

then $\exists N \in T$ with $C h N L$. This generalises Theorem 9.4: T can be any explicitly chosen nonempty finite subset of the dyadic block, not the whole block, and the proof (an averaging pigeonhole, §below) never uses that T has positive density or comes from a partition.

Lean: [exists_certifiedKill_of_subset_real_part_bound, block_real_part_bound_of_subset_form](#).

Theorem 9.6 (A norm bound gives the real-part criterion). [LEAN SOURCE](#) and [LEAN SOURCE](#). [Lean] for both implications; the hypothesis `blocknormcondition` is [Open]. The complex norm bound implies the real-part bound ($|z| \geq \operatorname{Re}(z)$, and $21/25 < 9/10$ absorbs the slack), so it composes through Theorem 9.4 to the same certificate. Define

$$\text{block norm condition} : \Leftrightarrow \forall h > 0 \forall X_0 \exists X, L, \max(X_0, 1) \leq X \wedge 16(2X+h+L+2) \leq 2^L \wedge$$

$$\left\| \sum_{N=X}^{2X-1} E(h, N, L) \right\| \leq \frac{21}{25} X.$$

. [cofinal] (this is the open target itself). Then

$$\text{block norm condition} \implies \operatorname{Irrational}\left(\sum_{n \geq 0} \frac{\varphi(n)}{2^n}\right),$$

proved in full, with no further gap, by chaining Theorem 9.6 $\rightarrow$ Theorem 9.4 $\rightarrow$ [LEAN SOURCE](#).

Lean: [blockNormCondition_unfolded](#), [real_part_bound_of_norm_bound](#), [exists_certified_Kill_of_block_norm_bound](#), [irrational_of_blockNormCondition](#).

Observation 9.7 (What remains unproved). The implications above do not establish the cofinal block estimate. The following callers retain a separation or gap hypothesis: [LEAN SOURCE](#) uses the two-element set $\{N, M\}$, while [LEAN SOURCE](#) and [LEAN SOURCE](#) use, respectively, a supplied finite set T and the singleton $\{N\}$. They prove what follows from the relevant estimate; they do not provide that estimate at arbitrarily large scales. The finite numerical observations in Table 6 are a different kind of evidence and do not discharge this quantified hypothesis. [Open]

Remark 9.8 (What the reduction does and does not prove). The required savings are explicit: a real-part bound of $9X/10$, or the stronger norm bound of $21X/25$, on blocks at arbitrarily large scales for every positive shift h . An unspecified nonzero saving is not the stated hypothesis. A bound $o(X)$ would suffice, but it is more than the implication needs.

The inequality $16(2X + h + L + 2) \leq 2^L$ can be met by increasing L for fixed X, h . Doing so changes the exponential sum, and hence does not preserve an estimate proved at another depth. The arithmetic estimate must hold at a depth satisfying this inequality.

The subset theorem permits any *nonempty* finite sample inside $[0, 2X]$; no density assumption is required. A prime-defined sample is one possible choice. The decomposition in [LEAN SOURCE](#) isolates one prime factor and the remaining totient-dependent phase, but does not prove the required cancellation between them.

9.3 Proved results and the additional assumptions they need

Section 9.2 gives the exponential-sum condition, listed as **e1-companion** in the source catalogue. The remaining entries compare tail signs, endpoint residues, finite approximations, short windows, Farey bounds, divisibility constructions, denominator growth and carry rank. Each states the extra assumption needed for its proposed application. The entries are not a classification of all possible approaches to #249.

9.3.1 Positivity and the remaining residue inequality

Definition 9.9. The *actual LCM tail orbit* at exponent a is $\Omega_a = R_{2H} - R_H = R_{2H} - R_H$, where $H = H(2^a)$ ([LEAN SOURCE](#)).

Theorem 9.10. [LEAN SOURCE](#) and [LEAN SOURCE](#). [Lean]. [uniform]. For every $a \geq 8$ and every J with $J + (a+6) < 2 \cdot 2^a$,

$$0 < R(2H+J) - R(H+J), \quad H = H(2^a).$$

Hence $0 < \Omega_a$ for every $a \geq 8$. The proof is genuinely uniform in a : it rests on two facts proved for all $a \geq 8$ by a two-case structural split (divisor letter vs. foreign prime power), with absolute constants 4, 8, 32 and no lookup table; [LEAN SOURCE](#) and [LEAN SOURCE](#).

Lean: [actualLcm_corridor_pos](#), [corridor_letter_pos](#), [corridor_height_lt_letter](#).

Remark 9.11 (Which endpoint interval must be excluded). Under the hypotheses of Proposition 6.85, integrality would force the residue to be $2^K - e$, where $e = R_{2H+J+K} - R_{H+J+K}$ is a positive integer smaller than $2H+J+K+2$. Thus the residue lies near the upper endpoint, not near zero in its least nonnegative representation. The representative $-e$ is negative; the carry e is positive.

The one-sided test below excludes precisely that upper interval. This uses the bound $J + K + (a + 6) < 2 \cdot 2^a$, as well as the modulus bound. It does not follow from positivity that the residue already satisfies half of a nonintegrality certificate without those assumptions.

[Lean] [uniform] [LEAN SOURCE](#) [LEAN SOURCE](#)

9.3.2 An exact endpoint identity

The sign estimate permits a one-sided residue test under the bounds below. This is a sufficient criterion, with additional arithmetic hypotheses, rather than a claimed strict weakening of irrationality.

Theorem 9.12 (The proved implications). [LEAN SOURCE](#), [LEAN SOURCE](#), [LEAN SOURCE](#). [Lean]. [uniform]. For every $a \geq 8$ and every J, K, m inside the sign corridor ($J+K+(a+6) < 2 \cdot 2^a$), a one-sided residue gap at precision $m \leq K$; room $2H+J+K+2 < 2^m$ and $D(H, H+J, K) \bmod 2^m \leq 2^m - (2H+J+K+2)$; already forces $R(2H+J) - R(H+J) \notin \mathbb{Z}$. No lower margin at all is demanded. The proof chain is complete: the theorem holds for every $a \geq 8$, and cofinal upper-endpoint condition $\Rightarrow$ Irrational S is proved.

Lean: [topEdgeResidueGap_unfolded](#), [topEdgeResidueGap_forces_nonintegral](#), [topEdgeResidueGap_orbit_nonintegral](#), [irrational_of_topEdgeResidueGapSupply](#).

Definition 9.13 (The unproved quantified condition). [LEAN SOURCE](#). [Open]. [cofinal].

$$\forall a_0 \exists a, K, m, a_0 \leq a \wedge 8 \leq a \wedge K+(a+6) < 2 \cdot 2^a \wedge \mathcal{G}_a(0, K, m).$$

No unconditional statement anywhere in the record locates the residue of the diagonal word $D(H, H, K)$ (where $H = H(2^a)$) below the top strip at even one large a . [LEAN SOURCE](#) further reduces the condition to the last m arithmetic letters of the word alone.

Proposition 9.14 (Five sufficient conditions). Each of the five conditions below suffices for irrationality. The first four imply the upper-endpoint condition [LEAN SOURCE](#); the fifth gives nonintegrality directly by the endpoint identity. These are not asserted to form a linear hierarchy. Here $H = H(2^a)$, and every condition quantifies over arbitrarily large exponents a .

1. [LEAN SOURCE](#) ([Open], [cofinal]):

$$\forall a_0 \exists a, m, a_0 \leq a \wedge 8 \leq a \wedge m+1+(a+6) < 2 \cdot 2^a \wedge 2H+m+3 < 2^m \wedge$$

$$2H+m+2 \leq \text{suffix residue}(2^a) \ 0 \leq m \leq 2^m - (2H+m+2)$$

(a two-sided band on the adjacent-suffix residue directly, one candidate depth m).

2. **LEAN SOURCE** ([Open], [cofinal]): at the odd guarded depth $2q+1$, a two-sided band of half-width $H+q+2$ on the half-word residue modulo 4^q . The depth is fixed by the stated guard; it cannot be chosen independently of a .
3. **LEAN SOURCE** ([Open], [cofinal]), proved equivalent to the previous one via **LEAN SOURCE**:

$$\forall a_0 \exists a, q, \max(14, a_0) \leq a \wedge q = q_a \wedge H+q+2 \leq |u_{a,q}|.$$

4. **LEAN SOURCE** ([Open], [cofinal]): the same magnitude bound with the depth restriction relaxed from “canonical guarded” to any odd $2q+1$ satisfying the half-cell fit $2(H+q+2) \leq 4^q$ and the sign-corridor room $2q+2+(a+6) < 2 \cdot 2^a$.
5. **LEAN SOURCE**: under the same bounds as item 4, $\delta_{2^a}(2q+2) \leq 2u_{a,q}$. This is a different one-sided comparison, treated immediately below.

For the first four conditions, the relevant implications into **LEAN SOURCE** are as follows: midband $\rightarrow$ residue-gap at **LEAN SOURCE**; half-word-band $\rightarrow$ midband at **LEAN SOURCE**; final-magnitude $\Leftrightarrow$ half-word-band at **LEAN SOURCE**; flexible-magnitude $\rightarrow$ midband at **LEAN SOURCE**; final-magnitude $\rightarrow$ flexible-magnitude at **LEAN SOURCE**.

Lean: `paperTeChain_item_one_unfolded`, `teChain_item_two_unfolded`, `teChain_item_three_unfolded`, `teChain_item_four_unfolded`, and 11 further declarations in the **coverage** section.

9.3.3 A bound on the last totient difference

The next condition compares one totient difference with a centred residue. Its usefulness would depend on proving that comparison for arbitrarily large a within the required range of q .

Definition 9.15. **LEAN SOURCE**. [Open]. [cofinal]. Writing $H = H(2^a)$,

$$\forall a_0 \exists a, q, a_0 \leq a \wedge 8 \leq a \wedge 2q+2+(a+6) < 2 \cdot 2^a \wedge 2(H+q+2) \leq 4^q \wedge$$

$$\delta_{2^a}(2q+2) \leq 2 \cdot u_{a,q}.$$

Theorem 9.16 (The proved implication). **LEAN SOURCE** and **LEAN SOURCE**. [Lean]. [uniform]. The dominance hypothesis at a single odd rank already excludes integrality of Ω_a , and the supply predicate composes to $S \notin \mathbb{Q}$.

Lean: `terminalDominance_orbit_nonintegral`, `irrational_of_terminalDominanceSupply`.

Observation 9.17 (How the sign and size bounds enter the contradiction). Use the hypotheses and notation of Proposition 6.66. Under the assumption $z = \Omega_a \in \mathbb{Z}$, the exact identity is

$$2u_{a,q} = \delta_{2^a}(2q+2) - c_{H,H,z}(2q+1), \quad 0 < c_{H,H,z}(2q+1) < 2H + 2q + 3.$$

Terminal dominance $\delta_{2^a}(2q+2) \leq 2u_{a,q}$ would contradict the positive lower bound on the carry. The other sufficient inequality, $2u_{a,q} \leq \delta_{2^a}(2q+2) - (2H+2q+3)$, would contradict its upper bound. Both are valid ways to contradict integrality.

These bounds do not eliminate either approach: they hold under the integrality assumption that the approach is intended to refute. They neither prove nor disprove an unconditional estimate for $u_{a,q}$. Finite examples lying between the two boundaries do not establish an asymptotic exclusion of either inequality.

[Lean] (identities and conditional bounds) [LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#)

Proposition 9.18 (A sufficient extension). *Either (i) the lower-escape branch cofinally; as displayed just above; or (ii) the two-sided magnitude form [LEAN SOURCE](#) (item 4 of Proposition 9.14), which asks only $H+q+2 \leq |u_{a,q}|$ and which the file proves ([LEAN SOURCE](#)) implies corridor escape via a clean sign split (positive branch escapes above the terminal letter, negative branch escapes below the directed bound), without prescribing the sign of the centred representative. Neither branch is supplied by the conditional sign identity itself.*

Lean: [irrational_of_lower_escape_supply](#), [corridor_escape_and_irrational_of_magnitude](#), [teChain_item_four_unfolded](#).

9.3.4 Separation of a rational approximation

Theorem 9.19. [LEAN SOURCE](#) and [LEAN SOURCE](#). [Lean]. [uniform]. Unconditionally, for every a and q ,

$$|\Omega_a - \rho_{a,q}| < \frac{4H + 2(2q+1) + 4}{2^{2q+2}},$$

where $\rho_{a,q}$ is an explicit finite computable rational block. For fixed a , the error tends to zero as q grows. In the quantified condition below, however, the depth is prescribed, so increasing q is not a free way to satisfy the separation hypothesis.

Lean: [abs_actualLcmTailOrbit_sub_rawApprox_lt_paper_form](#), [two_pow_odd_eq](#), [actualLcmRawApprox_isRat](#), [actualLcmRawErrorRadius_tendsto_zero](#), and 1 further declaration in the [coverage section](#).

Remark 9.20 (Scope of the implication). A cofinal $1/32$ -separation of the raw approximants at the prescribed indices $q = q_a$ suffices for #249. Allowing arbitrary depths would be a different hypothesis. The cited short-window theorem gives nonintegrality of Ω_a at $a = 4, 6$ ([LEAN SOURCE](#), [Cert], [fixed], via [LEAN SOURCE](#) and [LEAN SOURCE](#)). It neither asserts the $1/32$ margin at those exponents nor says that these are the only successful exponents. Indeed, the diagonal table also gives short-window certificates $(a, L) = (2, 7)$ and $(3, 15)$, since $H_4 = 12$, $H_8 = 840$, and $7 < 8$, $15 < 16$. All these are finite nonintegrality examples, not the cofinal approximation-separation condition.

Definition 9.21 (The additional hypothesis). cofinal approximation-separation condition: [Open], [cofinal].

$$\begin{aligned} \forall a_0 \exists a \geq \max(2, a_0) \exists q, \quad q = q_a \wedge \\ \forall z \in \mathbb{Z}, \quad \frac{1}{32} + \varepsilon_{a,q} \leq |\Omega_a - z|. \end{aligned}$$

This hypothesis is about the real quantity Ω_a , not just a finite rational calculation. It implies $|\rho_{a,q_a} - z| > 1/32$ for every integer z , by the displayed error bound and the triangle inequality. A cofinal $1/32$ -separation condition on the finite approximants is another sufficient input, through the residue-band implication. These sufficient conditions are not asserted to be equivalent.

9.3.5 The short-window examples through exponent 6

Theorem 9.22. [LEAN SOURCE](#) and [LEAN SOURCE](#). *[Cert]for the witness, [Lean]for the implication. [bounded].*

$$\forall a_0 \leq 6, \exists a, L, a_0 \leq a \wedge L < 2 \cdot 2^a \wedge C(H_{2^a}, H_{2^a}, L)$$

The single witness $(a, L) = (6, 93)$ works for every threshold $a_0 \leq 6$. Its certificate is verified by exact integer arithmetic on the finite totient windows in [LEAN SOURCE](#). This proves the displayed bounded statement. It does not construct witnesses for unbounded thresholds; further finite checks would extend the verified range, not establish the universally quantified condition below.

Lean: [shortWindowSupply_through_six_paper](#), [shortWindowSupply_single_witness_six_ninetyThree](#), [shortWindowSupply_witness_eq_t64_certificate](#).

Definition 9.23 (The additional hypothesis). cofinal short-window condition: [Open], [cofinal].

$$\forall a_0 \exists a, L, a_0 \leq a \wedge L < 2 \cdot 2^a \wedge C(H_{2^a}, H_{2^a}, L).$$

The depth restriction gives a useful description of the offsets: if $1 \leq j < 2 \cdot 2^a$ and $j \nmid H_{2^a}$, then j is a prime power greater than 2^a . Indeed, some prime-power divisor p^r of j exceeds 2^a ; if $j/p^r \geq 2$, then $j > 2 \cdot 2^a$, a contradiction. This describes the possible offsets, not the size of their weighted sum.

The condition restricts the depth in the ordinary diagonal certificate criterion. It is not established by pointwise completeness, which permits arbitrarily large depths. Nor does a successful short-window certificate by itself yield the fixed $1/32$ separation required by the preceding approximation condition. These are distinct sufficient hypotheses; no implication from this condition to that fixed-margin condition is proved here.

9.3.6 The diagonal certificate table

Theorem 9.24. [LEAN SOURCE](#), with [LEAN SOURCE](#) and [LEAN SOURCE](#). *[Cert]. [fixed].*

$$\forall t \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17\}, \exists L, C(H_t)(H_t)L$$

at depths $\{6, 5, 7, 7, 9, 14, 15, 14, 21, 22, 23, 26\}$ respectively, extended by the separate T19...T64 modules to 28 historical values through $t = 64$ (the $t = 64$ endpoint is [LEAN SOURCE](#)). The later aggregate theorem closes every scale $t \leq 82$ with no holes ([LEAN SOURCE](#)). Each witness is checked by exact integer arithmetic on two finite totient windows. For example,

$H_{17} = 12252240$ and $L = 26$ use $\varphi(H_{17} + i)$ and $\varphi(2H_{17} + i)$ for $1 \leq i \leq 26$, not every totient between the endpoints 12252241 and 24504506. These computations verify the listed cases; they do not give a formula producing a certificate for every t .

Lean: `diagonalPincerCertificateScales_list`, `diagonalPincerKillDepth_list`, `certified-Kill_diagonal_table`, `exists_diagonalKill_on_table`, and 3 further declarations in the `coverage` section.

Remark 9.25 (The observed depths are finite data). `LEAN SOURCE` forces $2(2H_t + L + 2) < 2^L$, i.e. $L \gtrsim \log_2(4 \cdot H t)$ at any certified depth. Comparing that floor against `diagonalPincerKillDepth` shows every proved certificate fires within a small additive constant of the theoretical minimum depth, at every tested scale; a numerically supported ([Cert], not [Math]) separation observation, not a theorem.

Proposition 9.26 (A sufficient extension).

$$\exists C \forall t_0 \exists t \geq t_0 \exists L \leq \log_2(4 \cdot H t) + C, \quad C(H t) (H t) L.$$

This asks for a certificate within a fixed additive constant of the necessary logarithmic depth bound. The uniform constant is an additional requirement, not a consequence of the finite data.

Lean: `irrational_of_logarithmicDepth_diagonal_supply`, `irrational_of_restricted-Depth_diagonal_supply`.

9.3.7 One common certificate for sixteen shifts

Theorem 9.27. `LEAN SOURCE` and `LEAN SOURCE` (the earlier eight-shift computation is `LEAN SOURCE`). [Cert]. [fixed].

$$C(h, 14, 9) \quad (h \in \{1, \dots, 16\}).$$

The basepoint 14 and depth 9 are common to all sixteen shifts. Consequently, $S \neq r$ whenever $r \in \mathbb{Q}$ and $\text{den}(r) \mid 2^{14}(2^h - 1)$ for some integer h with $1 \leq h \leq 16$. The exact computation uses two nine-term windows for each shift and reduces their discrepancy modulo 2^9 . The earlier eight-shift computation uses basepoint 12 and depth 16.

Lean: `commonCertificate_sixteen_shifts_basepoint_fourteen`, `totientSeries_ne_rat_of_den_dvd_two_pow_fourteen_mul_mersenne`, `commonCertificate_eight_shifts_basepoint_twelve`.

Remark 9.28 (Finite simultaneity and the unbounded condition). The displayed computation concerns only $N = 14$, $L = 9$ and $1 \leq h \leq 16$. It does not provide certificates beyond every basepoint threshold. Simultaneous certification of finitely many shifts at a sufficiently large, unrestricted depth is not an additional obstacle: under irrationality it follows from the same truncation estimate, as shown next. The argument below does not prescribe the common depth in terms of the basepoint and number of shifts.

Proposition 9.29 (Simultaneous certificates with unrestricted depth). The existence of a function $f : \mathbb{N} \rightarrow \mathbb{N}$ with $f(N) \rightarrow \infty$ such that

$$\forall N_0 \exists N \geq N_0 \exists L \forall h \in \{1, \dots, f(N)\}, \quad C(h, N, L)$$

is equivalent to irrationality of S . Under irrationality, one can take $f(N) = N + 1$, and every N has a suitable depth. [Math]

Lean: `exists_growingShift_simultaneous_certificate_iff_irrational`, `exists_simultaneous_depth_succ_of_irrational`, `exists_simultaneous_depth_of_irrational`.

Proof. The displayed condition supplies Proposition 6.71's quantified certificate condition: for fixed $h \geq 1$, choose the basepoint threshold so large that $f(N) \geq h$ beyond it. That proposition gives irrationality.

Conversely, suppose S is irrational. For every $h \geq 1$ and $N \geq 0$,

$$R_{N+h} - R_N = 2^N(2^h - 1)S - (\Phi_{N+h} - \Phi_N)$$

is nonintegral. Fix N and a positive integer M . Choose L so large that

$$\frac{2(N + M + L + 2)}{2^L} < \min_{1 \leq h \leq M} \text{dist}(R_{N+h} - R_N, \mathbb{Z}).$$

The finite minimum is positive, whereas the left side tends to zero. For each $h \leq M$, the truncation error between $D(h, N, L)/2^L$ and $R_{N+h} - R_N$ is at most $(N + h + L + 2)/2^L$. The triangle inequality therefore puts $D(h, N, L)/2^L$ at distance greater than $(N + h + L + 2)/2^L$ from every integer, which is precisely $\mathcal{C}(h, N, L)$. Taking $M = N + 1$ proves the converse. $\square$

This proof gives no quantitative lower bound for the finite minimum as the basepoint or number of shifts grows. It selects a common depth after those parameters are fixed; it does not impose depth equal to the period.

9.3.8 The fixed Farey bound

Theorem 9.30. `LEAN SOURCE`, built from the classical mediant lemma `LEAN SOURCE` and the window sharpness certificate `LEAN SOURCE` / `LEAN SOURCE`. [Lean] for the general mediant lemma; [Cert] for the $K = 240$ window instantiation. [fixed]. If S is rational, its reduced denominator exceeds $7.9639646646701375323355774875831053 \times 10^{34}$ (the exact numeral in the declaration name). This is the finite denominator exclusion in the corpus, logically independent of the certificate-supply reduction.

Lean: `totientSeries_rational_den_gt_fareyBound`, `farey_gap_paper`, `gapCheck_window_1_240_paper`, `gapCheck_window_1_240_first_failure_paper`.

Remark 9.31 (What a larger Farey bound requires). Unbounded valid denominator exclusions would imply $S \notin \mathbb{Q}$. For the single window $K = 240$, however, `LEAN SOURCE` proves that the residue inequality fails at the mediant denominator $b + d = Q_0 + 1$. This exact first failure prevents a larger cutoff from that same inequality at that same window. It is not an obstruction to other windows or to other tests using the finite data.

The existing proofs for individual windows use explicit totient residues and rational endpoint calculations. Extending that implementation calls for new finite data; a uniform proof of a growing bound need not use hard-coded convergents. If S had reduced denominator q , every valid exclusion bound would be smaller than q . Thus unbounded exclusions suffice for irrationality; rationality would bound them, but would not force eventual constancy. No converse or comparison with the continued-fraction denominators of S is proved here.

Observation 9.32 (The limited gain from a unit-gap refinement). The finite condition [LEAN SOURCE](#) and its quantified version [LEAN SOURCE](#) are definitions; the latter is not asserted to hold. The separate results [LEAN SOURCE](#) and [LEAN SOURCE](#) ([Lean], [uniform]) show that, for p prime and $e \geq 1$, a window at denominator p^e that *passes the refined test* contains at most one candidate integer. A window with two or more candidates fails the test. This is a condition for successful certification, not a bound on all windows or on the improvement in the denominator cutoff. It does not apply to all odd denominators: [Observation 6.114](#) gives an actual totient window with two admissible nonunit candidates modulo 15.

Proposition 9.33 (A sufficient extension). *Suppose $g(K) \rightarrow \infty$ and, for every K , the $(N = 1, K)$ gap check excludes every rational of reduced denominator at most $g(K)$. Then S is irrational: any rational value of S would have a fixed finite denominator, contradicted at a sufficiently large K . Only unbounded exclusion bounds along a sequence of windows are needed for this argument. No equivalence with a bound for the continued-fraction denominators of S is asserted; such a comparison would require its own proof connecting those convergents to the certified intervals.*

Lean: [irrational_of_unbounded_window_one_gapCheck](#), [gapCheck_window_one_excludes](#).

9.3.9 A two-adic congruence that does not give a certificate

Theorem 9.34. [LEAN SOURCE](#), [LEAN SOURCE](#), [LEAN SOURCE](#), [LEAN SOURCE](#). [Lean]. [uniform] (the theorem is unconditional and holds for every K, H, B , not merely cofinally many). For every $K \geq 2$, every $H > K$, and every bound B , there are primes $p > B$ with a length- $(K-1)$ zero prefix and a terminal half-modulus, giving

$$D(H, p - K, K) \equiv 2^{K-1} \pmod{2^K}.$$

Under eventual integrality this transfers to an integer z with $(z : \mathbb{R}) = R(p+H) - R_p$ and $z \equiv 2^{K-1} \pmod{2^K}$.

Lean: [exists_prime_twoAdic_half_pulse_window](#), [eventual_integral_tailDiff_twoAdic_half_pulse](#).

Remark 9.35 (Why the central residue still fails the test). This lands the residue at the exact centre of the residue-certificate arc: 2^{K-1} is maximally far from 0 modulo 2^K , precisely the coordinate the obligation wants. Taking $N=p-K$, $h=H$, $L=K$ gives radius $N+h+L+2 = p+H+2$, so C requires $2^{K-1} > p+H+2$. But $z \equiv 2^{K-1} \pmod{2^K}$ forces $|z| \geq 2^{K-1}$, while the directed tail bound forces $|z| < p+H+2$; and the construction's own congruence $p \equiv 1+2^{K-1} \pmod{2^K}$ (equivalently $v_2(p-1) = K-1$) forces $p \geq 1+2^{K-1}$. So $p+H+2 > 2^{K-1}$ always, and no choice of p rescues it. The quantifier order is inverted: the theorem gives, for fixed K , cofinally many *large* p ; the certificate needs a *small* relative to 2^{K-1} , impossible for a single letter since $|\varphi(N+H) - \varphi(N)| < N+H$. Depth-promotion is not the issue: it has already been performed, generalizing an exponent-2 construction to every K uniformly by CRT-gluing over a $\text{Unit} \oplus \text{Fin}(K-1) \oplus \text{Fin}(K-1)$ family.

Proposition 9.36 (A sufficient accumulated-residue condition). *Suppose that for every integer $h \geq 1$ and every $N_0 \in \mathbb{N}$ there are $N \geq N_0$ and $L \geq 1$ such that*

$$D(h, N, L) \equiv 2^{L-1} \pmod{2^L}, \quad N + h + L + 2 < 2^{L-1}.$$

Then $S \notin \mathbb{Q}$: these two inequalities imply $C(h, N, L)$, and the full quantified certificate condition applies. The half-modulus condition here is imposed on the accumulated weighted sum $D(h, N, L)$, not on a single totient difference. An unbounded set of triples with no control of the shift does not supply the displayed quantifiers. The required family remains unproved.

Lean: [irrational_of_accumulated_halfModulus_supply](#), [certifiedKill_of_halfModulus_residue](#).

9.3.10 The residue margin at an LCM jump

Theorem 9.37 (Arbitrarily large prime-power LCM jumps). [LEAN SOURCE](#). [Lean]. [cofinal] (proved): for every $t_0 \in \mathbb{N}$ there is $t \geq t_0$ with $H(t) < H(t+1)$. One may take $t = p - 1$ for any prime $p > t_0$. The positions $t = 2^a - 1$ are also strict LCM jumps for $a \geq 1$: 2^a is the next required power of 2. The restriction excludes $a = 0$, since $H(0) = H(1) = 1$. The condition [LEAN SOURCE](#) asks for the additional residue margin along this explicit sequence. Knowing these jump positions does not establish that margin.

Lean: [exists_periodLcm_strict_jump_ge_paper](#), [periodLcm_strict_jump_at_prime_pred](#), [periodLcm_strict_jump_at_powerTwo_pred](#), [periodLcm_zero_and_one_eq_one](#).

Definition 9.38 (The remaining residue margin). For $t \geq 3$, put $H = H_t$ and define

$$\begin{aligned} m &= \lfloor \log_2 H \rfloor + 10, \\ d &= (D(H, H+1, m) - D(H, H, m)) \bmod 2^m, \\ \sigma_t &= \min\{d - 2^{m-5}, 2^m - 2^{m-5} - d\}. \end{aligned}$$

Thus d is the difference of two adjacent window numerators, reduced modulo 2^m , not the residue of $D(H, H, m)$ alone. The integer σ_t is nonnegative exactly when $2^{m-5} \leq d \leq 2^m - 2^{m-5}$. The choice of m makes the error bound $3H + 2m + 5$ smaller than 2^{m-5} ; nonnegativity is the additional arithmetic requirement, not a consequence of that choice.

[LEAN SOURCE](#) [LEAN SOURCE](#) [LEAN SOURCE](#)

The first unproved condition asks for every t_0 for an index $t \geq \max(3, t_0)$ with $H_t < H_{t+1}$ and $\sigma_{t+1} \geq 0$. The second restricts these endpoints to powers of two: for every a_0 , there must be $a \geq \max(2, a_0)$ with $\sigma_{2^a} \geq 0$. These are respectively [LEAN SOURCE](#) and [LEAN SOURCE](#). Each condition implies $S \notin \mathbb{Q}$, by [LEAN SOURCE](#) and [LEAN SOURCE](#). [Open] [cofinal] (hypotheses; the implications are proved).

Remark 9.39 (Finite numerical evidence). [Cert], [fixed]. All 40 strict jumps up to endpoint 113 pass, closest margin ≈ 0.000221 of the modulus at $t = 100$; the power-of-two endpoints 4, 8, 16, 32 pass with slack fractions 0.41, 0.036, 0.40, 0.19. This is a finite computation, not a cofinal theorem. The identity [periodLcm_succ_eq_prime_mul_of_strict_jump](#) ([LEAN SOURCE](#)) says a strict jump has $t+1 = p^k$ for a prime p and $k \geq 1$, with $H_{t+1} = pH_t$. Thus σ_{t+1} uses the new height pH_t , not the pre-jump height H_t . Estimating its change would require information about the adjacent-window residue, not just the LCM ratio. [LEAN SOURCE](#) proves $\sigma_t = \sigma_u$ whenever $H_t = H_u$. Thus the value changes only when the LCM height changes. This reduces the number of distinct values to study; it does not determine their signs.

Proposition 9.40 (A sufficient inequality). *If, for every $a_0 \in \mathbb{N}$, there is $a \geq \max(2, a_0)$ with $\sigma_{2^a} \geq 0$, then $S \notin \mathbb{Q}$. This is the sufficient condition defined above, not a claim that the inequality holds at arbitrarily large indices. Equality at the edge is allowed: the condition is nonnegativity, not strict positivity.*

Lean: [irrational_of_powerTwo_postJump_slack_supply](#), [canonicalAdjacentSuffixCentral-Slack_paper_formula](#).

9.3.11 A certificate for a four-tail combination

Theorem 9.41. [LEAN SOURCE](#) and [LEAN SOURCE](#). [Lean]. [uniform]. Use $J(H, p)$, $W(H, p, L)$ and $B(H, p, L)$ from Theorem 6.60. For all $H, p, L \in \mathbb{N}$,

$$B(H, p, L) < W(H, p, L) \bmod 2^L < 2^L - B(H, p, L) \implies J(H, p) \notin \mathbb{Z}.$$

The bound $B(H, p, L) = 3pH + (p+1)(L+2)$ follows by grouping the nonnegative remainders as in that theorem. Compared with the $4pH + (p+1)(L+2)$ bound obtained by treating the two diagonal differences separately, it saves pH . This is a strict improvement when $pH > 0$; optimality for totient tails is not asserted.

Lean: [primeJumpTailCommutator_notMem_int_of_central_window](#), [primeJumpSharpRadius_formula](#), [primeJumpSharpRadius_saves_pH](#), [primeJumpSharpRadius_lt_twoCellRadius](#).

Theorem 9.42 (One explicit witness). [LEAN SOURCE](#). [Cert]. [fixed]. For $H = H(4) = 12$, $p = 5$ and $L = 15$, direct integer evaluation gives

$$W(12, 5, 15) = 149906, \quad W(12, 5, 15) \bmod 32768 = 18834, \quad B(12, 5, 15) = 282.$$

Thus $282 < 18834 < 32486$, and $J(12, 5) \notin \mathbb{Z}$. This is the one instance singled out by the cited declaration, not a claim of uniqueness or of the smallest possible height. The implication from a quantified family is [LEAN SOURCE](#).

Lean: [periodLcm_four_eq_twelve](#), [primeJump_witness_twelve_five_values](#), [primeJumpTail-Commutator_twelve_five_notMem_int](#), [irrational_of_primeJumpSharp_supply](#).

Remark 9.43 (Scope of the witness). The quantified hypothesis asks, for every $t_0 \in \mathbb{N}$, for $t \geq t_0$, $p \geq 1$ and $L \geq 0$ satisfying the displayed residue condition with $H = H(t)$. The multiplier p need not be prime, and need not be tied to the next LCM jump. No comparison of the difficulty of this condition with the prescribed-depth separation condition is proved.

At a strict LCM jump, $H(t+1) = pH(t)$ for a prime p . This gives one natural choice of multiplier, but the prime need not be a new divisor: $H(4) = 2H(3)$ and 2 already divides $H(3) = 6$. The identity in [LEAN SOURCE](#) cancels contributions affine in the multiplier. It does not bound the remaining terms of the four-tail combination. The finite example above therefore does not establish the quantified condition.

9.3.12 Denominators of Möbius sums

Theorem 9.44. [LEAN SOURCE](#), [LEAN SOURCE](#), [LEAN SOURCE](#). [Lean]. [uniform]. For every integer $t \geq 5$, with $\mathcal{P}_t = \{p \text{ prime} : t/2 < p \leq t\}$ as above,

$$2^{\lfloor t/2 \rfloor} \leq \prod_{p \in \mathcal{P}_t} (2^p - 1) \leq \text{den}(H_t \beta_{H_t}).$$

Moreover, some $p \in \mathcal{P}_t$ satisfies

$$2^{\lfloor t/2 \rfloor} \leq 2^p - 1 < 2^t, \quad 2^p - 1 \mid \text{den}(H_t \beta_{H_t}).$$

Indeed, Bertrand's postulate makes $\mathcal{P}_t$ nonempty. Its members satisfy $p - 1 \geq \lfloor t/2 \rfloor$ and $p \leq t$, giving the size bounds. The divisibility follows from Proposition 6.6; divisibility of a positive reduced denominator also gives the product inequality.

Lean: `upperHalfMersenneProduct_between_bounds`, `exists_upperHalf_channel_paper`, `upperHalfPrimes_nonempty_paper`, `upperHalfPrimes_member_bounds`, and 2 further declarations in the [coverage section](#).

Remark 9.45 (Growth of these denominators does not give approximation of S). The number $H_t \beta_{H_t}$ is the finite rational contribution to a totient-tail decomposition. Its complementary term can still cancel its fractional part. The stated denominator bound supplies neither an error estimate for approximation of S nor nonintegrality of $R_{2H_t} - R_{H_t}$.

To apply the diagonal criterion in Theorem 6.49, one must establish this tail nonintegrality at arbitrarily large LCM scales. Denominator growth alone does not supply that condition.

Proposition 9.46 (The additional approximation hypothesis). *A sequence of rationals u_t proves irrationality of S if*

$$u_t \neq S \quad \text{for all sufficiently large } t, \quad \text{den}(u_t)|S - u_t| \rightarrow 0.$$

This is the rational-separation criterion in [LEAN SOURCE](#): if $S = a/b$ were reduced, every unequal u_t would instead satisfy $\text{den}(u_t)|S - u_t| \geq 1/b$. An upper bound tending to zero for this product is needed. A lower bound for the denominator cannot establish it; improving a lower bound does not change the actual approximation error.

A different argument could deduce a residue certificate from the surviving divisor and further information about the complementary term. The denominator theorem alone contains no such separation statement. Neither sufficient argument is established by its lower bound, and these are not claimed to exhaust possible approaches.

Lean: `irrational_totientSeries_of_rational_separation`, `den_mul_abs_sub_ge_one_div_den`.

9.3.13 Rationality and carry rank

The next two theorems concern the totient sections and, under rationality, the sections of an integral carry. Both give lower bounds for dimension. A rank argument for irrationality would also need an incompatible upper bound. Rationality alone does not provide one.

Theorem 9.47. [LEAN SOURCE](#). [Lean]. [uniform]. *For every $e \geq 1$, the retained dyadic totient sections form a basis for all sections through level e , of dimension $2^e + 1$. The proof uses the Chinese remainder theorem and Dirichlet's theorem. The complete level-zero truncation, consisting only of φ , has dimension one.*

Lean: [retainedSections_basis_and_rank](#), [completeLevelZeroTruncation_is_totient_and_rank_one](#), [canonicalTotientKernelFamily_entries](#).

Theorem 9.48. [LEAN SOURCE](#). [Lean]. [uniform]. If $S \in \mathbb{Q}$, choose an integer $v > 0$ with $vS \in \mathbb{Z}$ and put $u_N = vR_N$. Then $u_N \in \mathbb{Z}$, $u_{N+1} = 2u_N - v\varphi(N+1)$, and $0 \leq u_N \leq v(N+2)$. For every $e \geq 0$,

$$\dim_{\mathbb{Q}} \text{span}\{n \mapsto u_{2^j n+r} : 1 \leq j \leq e, 0 \leq r < 2^j\} \geq 2^e - 1.$$

The proof is given in Section 10.8.

Lean: [rationalValue_integral_carry_and_rank_floor](#).

Remark 9.49 (The generic upper-bound inference is false). The bound $2^e - 1$ holds at every level. An upper bound contradicting it would need additional information about the totient, since the rational $5/4$ comparison has the same lower bound. One proposed system of integer equalities and strict inequalities is inconsistent: [LEAN SOURCE](#) ([Lean], [n/a]) shows no [LEAN SOURCE](#) with $Q, v > 0$, $A \neq 0$, $QvA = \text{boundary}$ and $|\text{boundary}| < Qv$ can exist: a nonzero integer multiple of Qv has absolute value at least Qv . This excludes those simultaneous requirements, not every argument using a matrix or its adjugate. Separately, infinite-dimensionality of the totient kernel span is the conclusion of the source theorem immediately preceding it.

Proposition 9.50 (An additional hypothesis for a totient-specific rank argument). *A rank bound that contradicts the lower bound $2^e - 1$ would have to use additional arithmetic of the actual totient coefficients. The generic proposal that every rational coefficient series has bounded tempered-carry rank is ruled out by the rational control recorded in the short paper: its carry rank is at least $2^e - 1$ at every level. It is therefore not a remaining general lemma from which totient irrationality follows. The conditional theorem in the totient tail carry period source, line 224, records eventual periodicity modulo v together with unbounded rational rank. Periodicity of the residue sequences concerns values in a finite quotient; it does not bound the rational span of the integer-valued sections. The $5/4$ comparison shows that this distinction persists for a rational coefficient series.*

Lean: [rationalControl_periodic_with_unbounded_carry_rank](#), [rationality_gives_mod_period_and_unbounded_rank](#).

9.4 What the preceding comparisons show

The comparison separates three issues. A block estimate with a fixed saving in the real part or norm of an exponential sum would provide certificates. The upper-endpoint conditions instead ask for a particular residue or size estimate along an LCM sequence. Their sufficient forms are related by Proposition 9.14, but those implications do not prove any of the arithmetic hypotheses.

The generic bounded-rank argument fails for a separate reason: the rational comparison sequence has unbounded section rank. A totient-specific upper bound would require hypotheses not shared by that sequence.

These are statements about the displayed conditions and comparisons, not an exhaustive classification of possible proofs. Some conditions are equivalent to irrationality; others are only sufficient. No relative difficulty, truth probability, or independent formal verification follows from their position in the table.

10 Counterexamples to proposed deductions

The following examples disprove particular deductions proposed for $S = \sum_{n \geq 1} \varphi(n)2^{-n}$. Each example specifies the assumptions that it satisfies and the conclusion that fails. Some concern other coefficient sequences; others concern a particular way of estimating a totient sum. Neither kind decides the irrationality of S , and neither excludes arguments using additional properties of φ .

10.1 Changing the coefficients after a finite prefix

(a) **Proposed argument.** The quantified residue condition requires certificates beyond every starting index. A finite computation can check many instances, but one fixed inspection bound cannot establish these quantifiers. The following comparison explains this limitation without making any claim about all possible irrationality arguments.

[LEAN SOURCE](#)

(b) **Counterexample.** Theorem 2.4 gives an explicit splice construction; its series value, exact reduced denominator and failure of the separation condition are checked in Lean ([Lean]). Choose an integer $P > B$, set $\gamma(n) = \varphi(n)$ for $n \leq B$, and $\gamma(n) = n - \mathbf{1}_{P|n}$ for $n > B$. Then $0 \leq \gamma(n) \leq n$, and the binary series $\sum \gamma(n)/2^n$ is *rational*: it is a finite rational prefix, plus the tail of $\sum_n n2^{-n}$, minus a geometric sum over the multiples of P beyond B . Because γ agrees with φ on every index $\leq B$, γ passes every finite certificate that only inspects indices $\leq B$; in particular it passes or fails every instance of $\mathcal{C}(h, N, L)$ with $N + h + L \leq B$ exactly as φ does. Yet $\sum \gamma(n)/2^n \in \mathbb{Q}$ by construction.

(c) **Scope.** For each fixed bound B , a different rational comparison sequence can agree with φ on all inspected coefficients. This rules out a deduction from that finite prefix alone. It does not rule out a proof that establishes the required instances for arbitrarily large B with the correct quantifiers. Such a proof would no longer be a single finite inspection. The comparison also does not preserve multiplicativity or every other arithmetic property of φ .

(d) **Consequence.** The finite-prefix example and the parity example below are distinct constructions. The former preserves a prescribed finite prefix; the latter preserves parity and has separated nonperiodic blocks. Each shows why its own list of assumptions is insufficient. The recorded finite diagonal examples remain useful individual certificates, but their number alone gives no assertion at arbitrarily large scales. [LEAN SOURCE](#)

10.2 A rational series with matching totient parity

(a) **Proposed argument.** Consider the claim that a uniformly bounded sequence $c : \mathbb{N} \rightarrow \mathbb{N}$, with $c(n) \leq n$, exact totient parity and no eventual period, must have an irrational binary sum. The next example refutes this claim. It also has arbitrarily many paired changes beyond every threshold, with an arbitrarily prescribed minimum separation.

(b) **Construction.** Start with the sequence $0, 1, 1, 2, 4, 4, 4, \dots$. At each position $m = 2^{k+3}$, $k \geq 0$, replace the pair $(4, 4)$ at $m, m + 1$ by $(6, 0)$. Equivalently,

$$c(n) = \begin{cases} 0, & n = 0, \\ 1, & n = 1, 2, \\ 2, & n = 3, \\ 6, & n = 2^{k+3} \text{ for some } k \geq 0, \\ 0, & n = 2^{k+3} + 1 \text{ for some } k \geq 0, \\ 4, & \text{otherwise.} \end{cases}$$

Each replacement leaves the binary sum unchanged because $2 \cdot 2^{-m} - 4 \cdot 2^{-(m+1)} = 0$. The increasingly large gaps between replacements prevent eventual periodicity. The strongest recorded result also quantifies their separation:

Proposition 10.1 (A rational series preserving totient parity and the stated separation properties). *There exists $c : \mathbb{N} \rightarrow \mathbb{N}$ such that: $c(n) \leq 6$ for all n ; $c(n) \leq n$; $c(n) \equiv \varphi(n) \pmod{2}$ for every n ; for every N, G, K there is a block of K explicit $(6, 0)$ carry-pulse pairs beyond N , each pair separated by more than G ; and $\sum_n c(n)/2^n = 3/2$.*

Lean: `parityCoboundaryWeight_le_six`, `parityCoboundaryWeight_le_self`, `parityCoboundaryWeight_mod_two_eq_totient`, `exists_later_arbitrarily_many_separated_parityCoboundaryWeight_carry_pairs`, and 1 further declaration in the [coverage section](#).

LEAN SOURCE (sum computation [LEAN SOURCE](#); non-periodicity [LEAN SOURCE](#); exact parity match [LEAN SOURCE](#)). [Lean] [cofinal] .

(c) **Scope.** The displayed bounds, parity agreement and aperiodicity, including the stated number and separation of the edits, are jointly compatible with a rational binary sum. They therefore do not imply irrationality. The sequence c differs from φ : the example preserves its parity, not all its arithmetic properties. It excludes an argument using only the listed assumptions, not every strengthening of an aperiodicity condition or every use of totient coefficients.

(d) **Consequence.** The paired edits have zero total contribution, since $2 \cdot 2^{-m} - 4 \cdot 2^{-(m+1)} = 0$. Applied at the specified sparse positions of this particular base sequence, they preserve the rational sum and the required bounds while preventing eventual periodicity. This does not construct a rational series for every prescribed parity pattern, and it is not the finite-prefix construction of Theorem 2.4. A proposed criterion using only the assumptions of this example must fail; a criterion using further arithmetic information is not addressed.

10.3 What the square-CRT construction proves

For a prime p and $m \geq 1$,

$$\varphi(pm) = (p-1)\varphi(m) + \mathbf{1}_{p|m}\varphi(m).$$

A congruence modulo p^2 can prescribe the cofactor modulo p and hence remove the second term over a chosen finite range. This makes each product formula simpler; it does not imply that differences of the resulting values are nonzero.

The examples in Observation 6.117 show both possibilities. At $n = 52$, the offset pairs $(13, 13)$ and $(26, 18)$ give respectively $\varphi(65) - \varphi(65) = 0$ and $\varphi(78) - \varphi(70) = 0$

([LEAN SOURCE](#)). All four prime-cofactor products are coprime ([LEAN SOURCE](#)). At $n = 27$, the offset pairs $(3, 3)$ and $(6, 8)$ instead give $\varphi(30) - \varphi(30) = 0$ and $\varphi(33) - \varphi(35) = -4$ ([LEAN SOURCE](#)), again with coprime factors ([LEAN SOURCE](#)). These finite equalities are checked by exact arithmetic. No minimality claim about either witness is needed. The general existence theorem [LEAN SOURCE](#) supplies a common base for the stated finite congruence requirements, not a nonvanishing assertion. [Cert] [fixed] .

The vanishing example disproves the implication from correction-free factorisation to nonvanishing; the other example shows that this factorisation does not force vanishing either. This does not establish formal independence of two propositions, or exclude an argument using additional hypotheses. The prime-dilation identity [LEAN SOURCE](#) and the CRT constructions [LEAN SOURCE](#), [LEAN SOURCE](#) remain available with their own prime and residue hypotheses.

A separate elementary identity in the same file concerns alternating sums. Let I be finite, let $i \notin I$, and suppose an integer-valued function f on finite subsets satisfies $f(T \cup \{i\}) = f(T)$ for every $T \subseteq I$. Pairing T with $T \cup \{i\}$ gives

$$\sum_{T \subseteq I \cup \{i\}} (-1)^{|T|} f(T) = 0.$$

This is the content of [LEAN SOURCE](#). It has no totient hypothesis, but an application must still verify invariance under the indicated insertion. Neither it nor the congruence construction supplies the missing nonzero totient difference at a prescribed LCM height.

10.4 The cost of clearing a rational denominator

For example, multiplying $5/12$ by 4 gives $5/3$: the factor 4 removed from the denominator appears in the multiplier. The following lemma gives the exact divisibility behind this calculation. It concerns ordinary reduced fractions, without assumptions on totients.

Lemma 10.2 (Complement divisibility after multiplication). *Write $x = a/b$ in lowest terms, where $a \in \mathbb{Z}$ and $b \geq 1$. Let $c \in \mathbb{Z}$ and let H be a positive divisor of b . If the reduced denominator of cx divides H , then*

$$b/H \mid c.$$

Lean: [complementDenominator_dvd_scalar](#).

[LEAN SOURCE](#). Indeed, the hypothesis makes $Hcx = Hca/b$ integral. Since $\gcd(a, b) = 1$, it follows that $b \mid Hc$, hence $b/H \mid c$. Writing $c = (b/H)t$ also gives $Hcx = ta$; this equality is recorded in [LEAN SOURCE](#). If $c \neq 0$, the divisibility implies $|c| \geq b/H$. The nonzero hypothesis is essential for this size bound: $c = 0$ has reduced denominator one and satisfies the divisibility, but has no positive size cost. The lemma is about multiplication of one rational value, not arbitrary cancellations among several values.

A related elementary inequality compares a positive numerator with its denominator. If $a > 0$, $r \geq 0$, $n \geq 1$, $2^r \mid a$, and $a/b < 2/(2^n - 1)$, then

$$2^r(2^n - 1) \leq a(2^n - 1) < 2b.$$

The source gives this Mersenne instance [LEAN SOURCE](#) and the version with general positive denominator parameter [LEAN SOURCE](#). Positivity of a is what turns numerator divisibility into a lower bound; it cannot be omitted.

The accompanying linear-algebra statement has a one-dimensional quotient. Let V, W be vector spaces over $\mathbb{Q}$, let $\text{ev} : V \rightarrow \mathbb{Q}$ and $\Lambda : V \rightarrow W$ be linear, and suppose $\ker \text{ev} \subseteq \ker \Lambda$. Choose $e \in V$ with $\text{ev}(e) = 1$. Then, for every $v \in V$,

$$v - \text{ev}(v)e \in \ker \text{ev}, \quad \Lambda(v) = \text{ev}(v)\Lambda(e).$$

This is factorisation through the scalar quotient $V / \ker \text{ev}$ ([LEAN SOURCE](#)). A surjection onto a higher-dimensional space would imply only a linear factorisation through that space, not multiplication by one fixed vector.

The denominator calculations use [LEAN SOURCE](#) and [LEAN SOURCE](#). For integers a and $D \geq 1$, the reduced denominator of a/D is $D / \gcd(|a|, D)$. Hence a divisor of D coprime to a divides the reduced denominator. Coprimality is sufficient, not necessary: $2/8 = 1/4$ retains the divisor 2. More generally, if C is a positive divisor of D , $\gcd(C, a) = 1$, and $h \geq 0$ is an integer, then $C / \gcd(C, h)$ divides the reduced denominator of ha/D . At $h = 0$ this assertion retains only the divisor one. [Lean] [uniform] .

For a nonzero multiplier, a lower bound for b/H therefore gives a lower bound for its absolute value. For an integer $t \geq 5$, the product estimate $2^{\lfloor t/2 \rfloor} \leq \prod_{p \in \mathcal{P}_t} (2^p - 1)$ ([LEAN SOURCE](#)) can be used when that product is shown to divide b/H . Neither the product estimate nor the scalar lemma proves a quadratic growth law for arbitrary approximation schemes.

The finite Farey exclusion for S is a separate use of reduced denominators. Its certified interval contains no rational whose reduced denominator is at most

$$79,639,646,646,701,375,323,355,774,875,831,053$$

([LEAN SOURCE](#)). That numerical exclusion requires the interval and Farey calculation; it is not a consequence of the scalar lemma alone.

10.5 Signed moments and parity

A determinant construction needs a nonzero numerator as well as a small approximation error. The following parity test can establish the first requirement for a specified finite sum.

Lemma 10.3 (Nonvanishing from a unique largest denominator exponent). *Let I be a finite set, let $u_i \in \mathbb{Z}$ and $e_i \in \mathbb{N}$ for $i \in I$, and suppose that $m \in I$ satisfies $e_i < e_m$ for every $i \neq m$. If u_m is odd, then*

$$2^{e_m} \sum_{i \in I} \frac{u_i}{2^{e_i}} = \sum_{i \in I} u_i 2^{e_m - e_i} \equiv 1 \pmod{2}.$$

In particular, the sum is nonzero.

Lean: [signed_dyadic_clearing](#), [signed_dyadic_sum_ne_zero](#), [scaled_dyadic_sum_odd](#).

Every term except u_m is even, which proves the assertion. For example, $1/2 - 1/8 = 3/8$ has odd cleared numerator. Uniqueness is necessary for this argument: two odd

contributions with the same largest exponent can cancel. The formal statement is [LEAN SOURCE](#).

The same file gives a separate determinant expansion from the Leibniz formula [LEAN SOURCE](#), together with the truncated signed moments $\sum_{N=1}^Y a_N 2^{-N(k+1)}$ and their Hankel determinants ([LEAN SOURCE](#) and [LEAN SOURCE](#)). The parity lemma is not derived from that determinant identity. A determinant application must first identify a sum to which the parity hypotheses apply. [Lean] [uniform] .

For the totient series, one would still need suitable configurations at arbitrarily large scales and a denominator-weighted error tending to zero. Nonvanishing alone provides neither. No such family is supplied by these two algebraic statements; compare the separate rank discussion in §10.8. In another base, the analogous argument would require divisibility by a prime factor of that base rather than an unchanged appeal to parity.

10.6 Column rescaling does not certify cancellation

A nonzero determinant need not prevent one row from being constant. For example,

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

The first and last matrices have determinants -2 and 2 , respectively; the second row of the resulting matrix is constant. This simple column rescaling matters when a determinant is proposed as evidence for cancellation in the first-harmonic sums ([LEAN SOURCE](#)).

Proposition 10.4 (A nonzero minor survives inverse-phase column weights). *Let $d \geq 1$ be an integer, let $e_0, \dots, e_{d-1}$ be nonnegative integers, and let $z_0, \dots, z_{d-1}$ be nonzero complex numbers. Write $P_{ij} = z_j^{e_i}$, and suppose $e_{i_0} = 1$ for some i_0 . Multiplying column j by $W_j = z_j^{-1}$ gives*

$$(P \operatorname{diag}(W))_{i_0 j} = 1, \quad \det(P \operatorname{diag}(W)) = \det(P) \prod_{j=0}^{d-1} z_j^{-1}.$$

Thus a nonzero determinant remains nonzero while row i_0 becomes constant. If all $|z_j| = 1$, its absolute value is also unchanged.

Lean: [inversePhaseGauge_locks_row_and_preserves_minor](#).

The entry formula and the determinant of a product prove the statement ([LEAN SOURCE](#), [LEAN SOURCE](#)). If row-dependent weights are allowed, the choice $W_{ij} = z_j^i / z_j^{e_i}$ gives $W_{ij} P_{ij} = z_j^i$ and makes row zero constant. It produces the ordinary Vandermonde matrix, which is nonsingular when the z_j are pairwise distinct ([LEAN SOURCE](#), [LEAN SOURCE](#)). [Lean] [uniform] .

These are statements about complex matrices, not arithmetic properties of the totient sequence. In particular, determinant nonvanishing, even with unit-modulus weights, does not by itself exclude complete alignment in the indicated row. An application of the first-harmonic criterion ([LEAN SOURCE](#)) needs additional information that excludes such weights or controls the resulting weighted sum. The example does not require that information to take the form of an identity coupling different rows, and it does not rule out determinant arguments that use appropriate additional hypotheses.

10.7 Tail integrality on an LCM grid

(a) **The comparison.** The diagonal condition ([LEAN SOURCE](#)) uses the pair $(H_t, 2H_t)$. Under rationality, the same fractional part occurs at every positive multiple of H_t once t is sufficiently large. A finite set of these multiples therefore gives another possible place to test for a nonintegral difference. The criterion below does not assert a smaller successful depth than the diagonal test.

(b) **Eventual equality of fractional parts.**

Proposition 10.5 (Tail integrality on an LCM grid). $S \in \mathbb{Q} \implies \exists t_1, \forall t \geq t_1, \forall q, m : \mathbb{N}, 0 < q \implies R(q \cdot H(t) + m \cdot H(t)) - R(q \cdot H(t)) \in \mathbb{Z}$.

Lean: [lcm_grid_flatness](#).

[LEAN SOURCE](#) (body lcm cone flatness source). [Lean] [uniform] (for all sufficiently large t , given rationality). Rationality makes the fractional parts of R_{kH_t} equal for all $k \geq 1$ at each scale $t \geq t_1$. The corresponding implication [LEAN SOURCE](#) shows that a certificate $\mathcal{C}(mH_t, qH_t, L)$ with $q \geq 1$ at arbitrarily large t implies $S \notin \mathbb{Q}$.

(b') **A finite test on the grid.**

Proposition 10.6 (A finite-grid certificate gives a nonintegral pair). Let $Q \subseteq \mathbb{N}_{>0}$ be finite and nonempty. For each $q \in Q$, put $A_q = \sum_{j=1}^L \varphi(qH + j)2^{L-j}$ and $B_q = qH + L + 2$. Suppose $B_q < 2^L$ for every $q \in Q$ and

$$\forall q_i \in Q \exists q_j \in Q, \quad B_{q_j} < (A_{q_i} - A_{q_j}) \bmod 2^L.$$

Then $R_{q_j H} - R_{q_i H} \notin \mathbb{Z}$ for some $q_i, q_j \in Q$. The proof and a four-point example are given in Theorem 6.56.

Lean: [paperGridNumerator_eq](#), [finite_grid_nonintegral_pair](#).

[LEAN SOURCE](#) (quantified assertion [LEAN SOURCE](#)). [Lean] [bounded] (proved for each finite set satisfying the hypotheses; the required family at arbitrarily large t is not established).

(c) **Scope.** Eventual integrality on the LCM grid is a necessary consequence of rationality, not a claim about the actual series made without that hypothesis. The finite-grid test uses the individual one-sided bounds $B_q < 2^L$, rather than charging the larger radius to both ends of a pairwise interval. This compares modulus bounds, not binary depths. A finite nonempty set and every stated size inequality are needed; three or more points and pairwise compatibility are not.

Neither statement supplies the unbounded family required to decide #249. Exhibiting a nonintegral tail difference on the LCM grid at arbitrarily large t would refute rationality. The finite-grid test is a way to certify such a difference, not a proof that these tests succeed at arbitrarily large scales.

The implication from rationality to grid flatness ([LEAN SOURCE](#)) should not be confused with an assertion of flatness for the actual series. The exact diagonal integrality reformulation and its quantified converse ([LEAN SOURCE](#), [LEAN SOURCE](#)) give other equivalent forms of the problem. They are not the only such forms; Proposition 1.8 lists further equivalent certificate conditions.

(d) **Consequence.** Choosing the least deep tail is the step that turns a common-fractional-part assumption into an intersection of arcs. A related finite comparison concerns first and second differences: at $(h, N) = (1, 8)$ the first-difference test holds at depth 8, while no second-difference certificate exists at depth at most 8 ([LEAN SOURCE](#)). This disproves a universal claim of smaller depth for second differences; it does not show that they can never improve a certificate.

10.8 Exact dyadic rank and the limits of a rank argument

(a) **The question.** Which linear relations hold among the sections $n \mapsto \varphi(2^j n + r)$, and how does their span grow as the level increases? Coons had already proved nonregularity in every base $k \geq 2$ [2, Theorem 3.2] [Cited]. The result below identifies a basis and gives the exact rank at each dyadic level. Martin's Theorem 1 implies the independence of the positive-residue affine forms [4, Theorem 1]; the two sections with residue zero need a separate argument. The short note gives the all-base proof and the integral relations.

(b) **Construction.**

Theorem 10.7 (Exact dyadic rank and infinite-dimensionality). *For every $e \geq 1$, the family*

$$\{n \mapsto \varphi(n), n \mapsto \varphi(2n)\} \cup \{n \mapsto \varphi(2^j n + r) : 1 \leq j \leq e, 0 < r < 2^j, r \text{ odd}\}$$

is linearly independent over $\mathbb{Q}$. It contains $2^e + 1$ sequences ([LEAN SOURCE](#)) and spans the sections through level e by the zero- and even-residue reductions.

Lean: `canonicalTotientKernelFamily_entries`, `canonicalTotientKernelFamily_independent_card_and_span`.

The forms n and $2n$ are proportional, so their coefficients need a separate argument. Restrict a proposed relation to $n = 2m + 1$, where $\varphi(2n) = \varphi(n)$. The odd-residue sections and $\varphi(n)$ then correspond to pairwise nonproportional forms. CRT and Dirichlet's theorem separate them by an evaluation matrix: make the primitive part of one form prime and force suitable prime divisors into the others. This kills the positive-residue coefficients and gives $A + B = 0$ for the two remaining coefficients. Evaluating their relation at $n = 2$ gives $A + 2B = 0$. Hence both vanish. This gives the dyadic specialization of the proof in the short note. The formal independence statement is [LEAN SOURCE](#); consequently the full infinite family spans an infinite-dimensional $\mathbb{Q}$ -space, [LEAN SOURCE](#). A separate elementary observation concerns a proposed integer identity. There are no $Q, v \in \mathbb{N}_{>0}$ and $A, b \in \mathbb{Z}$ with $QvA = b$, $A \neq 0$, and $|b| < Qv$: the identity forces $|b| = Qv|A| \geq Qv$. The source packages these incompatible requirements as one structure, [LEAN SOURCE](#). [Lean] [uniform] (holds for every $e \geq 1$; existence side is unconditional, via Mathlib's CRT + primes-in-AP machinery) (dyadic totient-kernel; *not* the Möbius coordinate).

The Lean independence theorem is also stated for $e = 0$, but its auxiliary canonical index then still contains the two zero-residue sections $\varphi(n)$ and $\varphi(2n)$. It is therefore not the actual truncation through level zero, which consists only of $\varphi(n)$ and has dimension one.

Integral relations. The independence statement also determines the arithmetic of the section lattice. Fix $e \geq 1$, write $F_{j,r}(n) = \varphi(2^j n + r)$, and let $I_e = \{(j, r) : 0 \leq j \leq e, 0 \leq r < 2^j\}$. Retain $F_{0,0}, F_{1,0}$, and the odd-residue sections at levels $1, \dots, e$; call this retained index set J_e . The totient identities reduce every omitted section to $F_i = a_i F_{j(i)}$ for a retained section $j(i)$ and an integer $a_i \geq 0$. Thus the retained family generates the lattice $L_e = \sum_{i \in I_e} \mathbb{Z} F_i$. Its rational independence gives integer independence as well: every element of L_e has unique integer coordinates in these $2^e + 1$ retained sections.

To describe *all* integral relations, let E_i denote the formal coordinate vector in $\mathbb{Z}^{I_e}$ and use the evaluation map

$$\text{ev}_e : \mathbb{Z}^{I_e} \longrightarrow L_e, \quad (c_i) \longmapsto \sum_{i \in I_e} c_i F_i.$$

For each omitted index put $R_i = E_i - a_i E_{j(i)}$. Then

$$\ker(\text{ev}_e) = \bigoplus_{i \in I_e \setminus J_e} \mathbb{Z} R_i, \quad \text{rank}_{\mathbb{Z}} \ker(\text{ev}_e) = 2^e - 2.$$

Indeed, subtracting $\sum_{i \notin J_e} c_i R_i$ from a relation removes all omitted coordinates. The remainder is a relation among retained sections, so independence makes it zero. Conversely, the coefficient of E_i in $\sum_{h \in J_e} b_h R_h$ is exactly b_i , proving uniqueness. The decisive feature is the unit pivot 1 in each omitted coordinate (or -1 if a relation is oriented oppositely). Elimination uses no division; it therefore identifies the integral relation lattice, not merely its rational span. The depth-two example in the short note illustrates these two-term rows. The distinction between the retained basis and the redundant spanning family of all sections already matters for k -regular sequences that are not eventually zero: their growth exponent is the base- k logarithm of the joint spectral radius of their section matrices in a basis, whereas a redundant spanning family gives only an upper bound (Coons [3, Theorem 1, Proposition 4 and Corollary 7]) [Cited]. The totient is not regular, so that theorem does not apply; the short note records how the section maps act on the retained bases from one level to the next.

The [integral normal form](#) and the [full dyadic rational basis](#) are formally checked in Lean. The formal statement gives the same construction for every integer base $k \geq 2$, with relation rank $\sum_{j=1}^{e-1} k^j$. The full dyadic assembly separately gives a basis for the infinite rational span and for its finitely supported rational relations. These finite and infinite statements should not be confused with a claim about infinite sums of relations.

(c) What the rank theorem does not prove. Infinite section rank does not imply irrationality of S . The result quantifies the nonregularity already cited above: it gives the rank, the retained sections and their integral relations. Martin's simultaneous inequalities supply another proof for pairwise nonproportional affine forms with positive slopes [4, Theorem 1] [Cited]. His theorem does not separate n from $2n$; the two evaluations above do that. The linked Lean proof is independent of Martin's theorem.

The same independence argument also bounds the rank of an integral carry. If $S \in \mathbb{Q}$, choose an integer $v > 0$ with $vS \in \mathbb{Z}$ and put $u_N = vR_N$, where $R_N = \sum_{j \geq 1} \varphi(N + j)2^{-j}$. Then $u_N \in \mathbb{Z}$,

$$u_{N+1} = 2u_N - v\varphi(N + 1), \quad 0 \leq u_N \leq v(N + 2), \quad u_N/2^N \longrightarrow 0.$$

For every $e \geq 0$,

$$\dim_{\mathbb{Q}} \text{span}\{n \mapsto u_{2^j n+r} : 1 \leq j \leq e, 0 \leq r < 2^j\} \geq 2^e - 1.$$

This is the finite family in the linked statement ([LEAN SOURCE](#)); it contains all carry residues at levels $1, \dots, e$, not just odd residues or an unspecified retained subfamily. Indeed, for each odd r with $0 < r < 2^j$, the recurrence gives

$$v\varphi(2^j n + r) = 2u_{2^j n+r-1} - u_{2^j n+r}.$$

These $2^e - 1$ totient sections are independent and belong to the span of the displayed carry sections after multiplication by $v \neq 0$. The rank bound follows. The boundary condition $u_N/2^N \rightarrow 0$ identifies the canonical carry, but is not used in this rank-transfer argument.

To contradict rationality by rank, one would also need an incompatible upper bound. The comparison series with value $5/4$ in Section 12.6 shows why rationality, a binary carry recurrence and linear coefficient growth alone do not supply such a bound. Its coefficients agree with φ at odd indices but differ at even indices; they are not the totient sequence. Thus this comparison does not exclude an argument using additional totient identities.

A separate finite-dimensional calculation concerns the matrix $U_N(i, j) = \mu((i+1)/(j+1))$ when $(j+1) \mid (i+1)$ and zero otherwise, for $0 \leq i, j < N$. It is lower triangular with diagonal entries one. Hence $\det U_N = 1$ and the map $x \mapsto U_N x$ is injective ([LEAN SOURCE](#)). This proves injectivity of that particular map, not the impossibility of every finite-dimensional argument for irrationality.

(d) The evaluation-matrix argument. For functions $f_1, \dots, f_s$ with values in a field, points $n_1, \dots, n_s$ for which $\det(f_j(n_i)) \neq 0$ prove linear independence: evaluating a relation gives an invertible linear system for its coefficients. This elementary criterion is the content of [LEAN SOURCE](#) and [LEAN SOURCE](#). It does not, by itself, construct the points. Here the totient formula, CRT and Dirichlet's theorem provide them for non-proportional affine forms, followed by the separate zero-residue argument. Another arithmetic function would require its own argument for nonvanishing of the evaluation determinant.

10.9 Further counterexamples and sharp bounds

The following table records further counterexamples and bounds. Each row states the argument, identifies its source and specifies what remains possible beyond the assumptions it tests.

Approach	Argument and source	Scope and remaining question
Complete terminal divisibility pattern	Under $a \geq 8$, $m \leq K$, $J + K + a + 6 < 2 \cdot 2^a$ and $2H + J + K + 2 < 2^m$, the full set of divisibilities in Observation 6.107 is impossible: the last difference would be a positive multiple of 2^m smaller than 2^m . LEAN SOURCE . [Lean] [bounded]	The obstruction is specific to that pattern and those bounds. If the last divisibility is omitted, the additional terminal bound of Proposition 6.86 forces the penultimate difference to be 2^{m-1} . It does not prove that such partial patterns exist. LEAN SOURCE .
Finite linear combinations of shifted carry recurrences	The explicit sequence in Observation 6.115 has $a_{(q-1)H-1} = \varphi(H)$ for $2 \leq q < t$ and a bounded state satisfying $a_i = 2c_i - c_{i+1}$. Every finite integer combination of its shifts retains a recurrence of this form, including products of differences along multiples of H . Its bounds acquire the factor $\sum_j \lambda_j $; they are not uniform in the coefficients of the linear combination. The original module introduction is at LEAN SOURCE ; the explicit theorem and ordinary proof are given in Observation 6.115. [Math; formal source cited] [uniform]	The constructed coefficients are not asserted to be actual totient differences. Preserving the displayed recurrence under shifts therefore does not distinguish them from the coefficients required in the proposed argument. The full totient divisor identity LEAN SOURCE retains information absent from the recurrence alone. The counterexample does not require every future proof to use that particular identity.
Adding three multiplicative scales to an integrality test	For $H \geq 0$ and positive integers p, q , requiring $R_{2kH} - R_{kH} \in \mathbb{Z}$ for every $k \in \{1, p, q, pq\}$ is equivalent to requiring only $R_{2H} - R_H \in \mathbb{Z}$. Neither multiplier need be prime. The affine transport formula along $H \mapsto kH$ carries integrality at the first scale to each of the other three. LEAN SOURCE . [Lean] [uniform]	Adding these three integrality assertions does not strengthen the first one. A different test involving part of the tail would need an estimate for the part it discards; this equivalence provides no such estimate. The result concerns the displayed four conditions, not every test using several scales.
Prescribing valuations and finite unit residues	For the valuations v_j and odd residues a_j of Proposition 2.13(iii), the free integers z_j can always be chosen so that $e_{j+1} = 2e_j + 2^{v_j}(a_j + 2^u z_j)$ lies in the stated centred interval at every step. LEAN SOURCE . [Lean] [bounded]	The specified residues do not exclude all finite carry sequences while the higher quotients z_j remain unrestricted. An argument would need further constraints on those quotients or on the actual coefficients. The lemma does not assert that its freely chosen coefficients are totient differences.

Approach	Argument and source	Scope and remaining question
The information retained by an endpoint residue	For an integer sequence a and integer initial values u_0, v_0 , let $u_{n+1} = 2u_n - a(n+1)$ and $v_{n+1} = 2v_n - a(n+1)$. Induction gives $u_L - v_L = 2^L(u_0 - v_0)$ for every $L \geq 0$. LEAN SOURCE . [Lean] [uniform]	The endpoints agree modulo 2^L, but they are distinct integers whenever $u_0 \neq v_0$. Indeed, if the forcing terms are known, then $u_0 = \frac{u_L + \sum_{j=1}^L 2^{L-j} a(j)}{2^L}.$ Only reduction modulo 2^L discards the initial value. The identity does not exclude an argument that retains the full endpoint or other information about the sequence.
Bounded finite-state / autonomous successor decoder for the binary carry	For the family in Proposition 2.13(i), with $m \geq 2$, all histories before m agree but the next tail takes $\lfloor (m+1)/2 \rfloor + 1$ distinct values. A common state cannot determine that tail; a set of labels permitting exact recovery must have at least that many elements. LEAN SOURCE . [Lean] [uniform]	This counterexample concerns states determined only by the common history of the displayed family. It does not exclude a state using additional information specific to the totient sequence. The bound counts labels, not bits; it is not a complexity theorem about the binary digits of S .
Zeros on a progression versus nonzero coefficients arbitrarily far out	Let $\chi_4(n)$ be 0 for even n , 1 for $n \equiv 1 \pmod{4}$, and -1 for $n \equiv 3 \pmod{4}$, and put $c(n) = \sum_{d n} \chi_4(d)$ for $n \geq 1$. Divisor pairing gives $c(n) = 0$ when $n \equiv 3 \pmod{4}$. LEAN SOURCE . Nevertheless $c(2^k) = 1$ for every $k \geq 0$, since 1 is the only odd divisor of 2^k . [Lean (progression identity)] [uniform]	This rules out nonvanishing on an arbitrarily prescribed progression, not nonzero coefficients arbitrarily far out. In fact c is multiplicative. Its prime-power values are 1 at powers of 2, $a+1$ at p^a with $p \equiv 1 \pmod{4}$, and 1 or 0 at p^a with $p \equiv 3 \pmod{4}$ according as a is even or odd. Thus $c(n) \geq 0$ for every $n \geq 1$. These elementary calculations verify both coefficient hypotheses of LEAN SOURCE : $\sum_{a \geq 1} \chi_4(a) / (b^a - 1)$ is irrational for every integer $b \geq 2$. This application concerns the displayed weighted Lambert series, not the totient series S .

Approach	Argument and source	Scope and remaining question
Using a finite exclusion test to prove membership	finite Mersenne-weight exclusion(x , level, lookahead) is a decidable, finite-depth certificate proving non-membership only (LEAN SOURCE ; witness LEAN SOURCE , 3/4 excluded at level 1). [Cert] [bounded]	The cited test has a sound exclusion implication. The analogous soundness statements in LEAN SOURCE and LEAN SOURCE also do not make a finite unsuccessful search a proof of integrality. This is a limit of what follows from those tests, not an impossibility theorem for rationality or membership proofs using additional identities.
Reconstructing one totient value from a two-tail absolute-value linear combination	For any finite $w : \iota \rightarrow \mathbb{Q}$, $x : \iota \rightarrow \mathbb{N}$ with $\sum w_i \varphi(x_i) = 1$ (exact isolation), the crude two-tail cost $\sum w_i (2(x_i + 1) + (x_i + 2)) \geq 3$, using only $\varphi(x) \leq x$. LEAN SOURCE (closure LEAN SOURCE). [Lean] [uniform]	The displayed termwise absolute-value estimate cannot be less than 1. The same inequality holds for any coefficients $0 \leq c(n) \leq n$ with $\sum_i w_i c(x_i) = 1$. It does not rule out a smaller error bound that uses cancellation between the tails, or a different approximation identity.
Homogeneous Mersenne-primitive prime factor alone forces a contradiction	For the completely multiplicative sequence $c(n) = n$, if $0 < K < q$ and $q \mid 2^K - 1$, then the scaled-tail difference is the integer K at every starting index, but $q \nmid K$. LEAN SOURCE . [Lean] [n/a]	A factor of $2^K - 1$ does not, from the recurrence alone, have to divide the inhomogeneous tail difference. The example $K = 3$, $q = 7$ already has a primitive prime factor. An application to another series would need a correspondence between its coefficients and this recurrence; Mersenne denominators alone do not provide one.
The unit-gap test at a prime-power denominator	For p prime and $e \geq 1$, if every candidate integer in the interval at denominator p^e is a nonunit modulo p^e , then there is at most one candidate. Equivalently, the notation of Observation 6.114 satisfies $L = 0$, or $L = 1$ with $p \mid J + 1$. LEAN SOURCE (iff-characterisation LEAN SOURCE). [Lean] [uniform]	This candidate-count bound does not apply to every odd denominator; the example modulo 15 in Observation 6.114 has two candidates. It also does not bound the denominator cutoff obtainable by changing the window or approximation. It supplies no proof that the fixed exclusion in Section 10.4 extends to arbitrarily large denominators.

Approach	Argument and source	Scope and remaining question
Extra divisibility of a three-term totient difference	At $H = 2^{n+2}$, $j = 3 \cdot 2^{n+1}$, $\varphi(3H + j) - 2\varphi(2H + j) + \varphi(H + j) = -2^{n+1}$ for every $n \geq 0$. LEAN SOURCE . [Lean] [uniform]	The exact valuation concerns this scaled family, not the LCM family with $j^2 \leq 2^a$; see Observation 6.108. The independent coefficient calculation says that integer solutions of $c_1 + c_2 + c_3 = c_1 + 2c_2 + 3c_3 = 0$ are exactly the multiples of $(1, -2, 1)$ (LEAN SOURCE). It classifies these three coefficients; it does not determine the valuation of their totient combination on every restricted family.

11 Why the stated hypotheses cannot be omitted

A conditional theorem can hold at every scale while its arithmetic hypothesis is known at only a few scales, or at none. The distinctions below concern the specific implications already stated. They are not claims that every argument using the same objects must fail.

Implications concerning the totient series

Cancellation versus a certificate. The first-harmonic implication converts the stated cosine bound on one block into a finite residue certificate ([LEAN SOURCE](#)). To prove irrationality, the bound must hold with the quantifiers in Section 3: for every shift and beyond every block threshold. A finite identity for the sum, or a computation at one block, does not establish this assertion.

Sign versus distance from an integer. The positivity theorem for the LCM tail difference ([LEAN SOURCE](#)) identifies the relevant endpoint under the stated room and modulus bounds. Nonintegrality follows only after excluding the corresponding residue interval ([LEAN SOURCE](#)). Likewise, the terminal-dominance condition is a sufficient hypothesis, not a construction of the required parameters ([LEAN SOURCE](#)). Under integrality its inequality contradicts positivity of the terminal carry; the contradiction establishes sufficiency, not the inequality itself.

An error bound versus a separated approximation. The finite-tail approximation has a uniform error radius ([LEAN SOURCE](#)). The cited exact check supplies nonintegrality at the two exponents $a = 4, 6$ ([LEAN SOURCE](#)); it is not an assertion that no other exponent works. To obtain irrationality by this criterion, separation must occur at arbitrarily large LCM scales. In the nondivisor approximation of Section 6.6.8, the error bound is also proved when $D \geq 2H$. Its remaining requirement is finite rational separation, not an additional convergence hypothesis.

Denominator growth versus approximation quality. The lower bound for the product of the upper-half Mersenne factors ([LEAN SOURCE](#)) concerns a denominator expression. An irrationality argument needs a comparison with the error of the corresponding approximation. Growth of that denominator expression alone does not give either a successful residue test or a nonintegral totient tail.

A rank lower bound versus a contradiction. The coefficient independence and hypothetical-carry rank bounds hold uniformly in the level ([LEAN SOURCE](#), [LEAN SOURCE](#)). A contradiction would require an upper bound for the actual carry that is strictly smaller than the lower bound at some level. The rational $5/4$ comparison disproves the proposed generic upper bound. The separate integer boundary conditions in the cited adjugate calculation are inconsistent ([LEAN SOURCE](#)), but that calculation does not exclude matrix arguments with different boundary conditions.

Related source comparisons for Problem #257

The following related statements concern Problem #257. They are not inputs to any proof about S ; the comparisons below do not extend the cited statements beyond their hypotheses.

The comparison of the middle-carry estimate ([LEAN SOURCE](#)) with the remainder estimate ([LEAN SOURCE](#)) distinguishes the direction of an inequality from control of the required deviation. The associated sign claim in the earlier working notes had no unconditional proof supplied. A sign alone would not give a lower bound for the magnitude in any event.

The row-capacity and later-skip implications retain their respective local deficit and skip hypotheses ([LEAN SOURCE](#), [LEAN SOURCE](#)). The finite-shadow comparison distinguished an existential horizon, obtained from convergence, from the particular horizon $J = k - 2$ needed in that application ([LEAN SOURCE](#)). The terminal-strip formulation also imposes its own support and carry restrictions ([LEAN SOURCE](#)); a bound on carry size does not automatically replace a restriction on the positions of the nonzero coefficients.

For finite sums of $1/(2^n - 1)$, the common denominator is odd. Their reduced denominators are therefore odd, so they cannot equal a rational number with even reduced denominator. This is the elementary reason behind the finite-support exclusion and its denominator source ([LEAN SOURCE](#), [LEAN SOURCE](#)). The cited infinite-skipping theorem is stated for the target $1/2$ ([LEAN SOURCE](#)). A proposed extension to other targets requires checking that theorem's hypotheses and proof; the finite-denominator observation alone does not prove such an extension or the needed infinite-orbit estimate.

Finally, the support-fraction statement and the more general collision bound have distinct hypotheses ([LEAN SOURCE](#), [LEAN SOURCE](#)). The earlier review proposed a broader use but did not supply its additional mean-residue estimate. It also left the endpoint proof bodies in the following dichotomy unverified for a changed target ([LEAN SOURCE](#)). No such extension is used here.

Quantifiers must be checked independently. For example, a statement of the form $\exists N \exists L \forall h$ does not provide the required $\forall h \forall N_0 \exists N \geq N_0 \exists L$. The first assertion controls many shifts at one location; the second requires arbitrarily late locations for each fixed shift. Relabelling a source statement does not bridge this difference.

12 Additional mathematics required by the sufficient conditions

The preceding sections contain identities, conditional implications, comparison sequences and finite audits. This section develops the arithmetic estimates needed by the sufficient conditions in Section 3. It separates proved reductions from proposed applications of analytic methods; a limitation of one proposed application is not a theorem excluding the method.

The section contains a doubling-orbit formulation with explicit thresholds, a lacunary comparison sequence for the strength of a block estimate, and sufficient estimates for the prime-factor decomposition. Each result retains its stated evidence label. The

discussion of a rank upper bound must also be read with the rational 5/4 comparison sequence: rationality alone does not give bounded rank. These are the scope distinctions needed to interpret the individual statements, not priority claims.

Where a claim is proved it is marked [Math]; where it is checked by exact or floating-point computation, [Cert]; where it is quoted, [Cited]; where it is a proposal, [Open]. The proved reductions identify explicit sufficient hypotheses. Those hypotheses remain to be established; the reductions do not prove them.

Remark 12.1. Declarations in the shared tree are cited as Erdos249257.name with file and line. Declarations in the newer per-problem tree are cited with their path from the repository root, erdos problems/erdos249/file source:line; the hyperlink target for those is the per-problem directory, not Erdos249257/.

12.1 The doubling identity

Write $S = \sum_{n \geq 1} \varphi(n)/2^n$, $R_N = \sum_{m \geq 1} \varphi(N+m)/2^m$ for the local tail ([LEAN SOURCE](#)), and $\Phi_N = \sum_{n \leq N} \varphi(n)2^{N-n} \in \mathbb{Z}$ for the integer prefix, so that $2^N S = \Phi_N + R_N$. For $h \geq 1$ set

$$\alpha_h := (2^h - 1) S.$$

Lemma 12.2 (The doubling identity). *For all $N \geq 0$ and $h \geq 1$,*

$$R_{N+1} = 2R_N - \varphi(N+1), \quad R_{N+h} - R_N = 2^N \alpha_h - (\Phi_{N+h} - \Phi_N),$$

so $R_{N+h} - R_N \equiv 2^N \alpha_h \pmod{1}$, and consequently the exact first character of the tail difference is the $\times 2$ orbit of one real number:

$$e(R_{N+h} - R_N) = e(2^N \alpha_h), \quad e(x) := \exp(2\pi i x).$$

Hence, for fixed h , the phases $\{R_{N+h} - R_N \pmod{1}\}_{N \geq 0}$ are the forward orbit of $\alpha_h \pmod{1}$ under $x \mapsto 2x$.

Lean: [orbit_tail_recurrence](#), [orbit_tail_diff_eq](#), [orbit_tail_diff_sub_scaled_is_int](#), [orbit_tail_diff_firstChar_eq](#), and 2 further declarations in the [coverage section](#).

The second identity and the character form are proved: [LEAN SOURCE](#) and [LEAN SOURCE](#) [Lean] [uniform]. The recurrence $R_{N+1} = 2R_N - \varphi(N+1)$ is one line from the definition and is used below only for intuition [Math].

These identities express the phases as a lacunary exponential sum in one real variable. The next proposition transfers an estimate for that orbit to the truncated phases, accounting for the truncation error. The identities themselves are already in the cited sources.

Proposition 12.3 (Transfer to the doubling orbit). *Suppose that*

$$\forall h \geq 1 \forall X_0 \exists X \geq \max(X_0, 1) : \sum_{N=X}^{2X-1} \cos(2\pi 2^N \alpha_h) \leq \frac{89}{100} X.$$

This hypothesis implies that S is irrational.

Lean: `irrational_totientSeries_of_block_cosine_gap`.

Proof. Fix $h \geq 1$ and X_0 , and take X as supplied. Choose $L \geq h$ with $2^L \geq 1024(2X + h + L + 2)$; such L exists because $2^L/L \rightarrow \infty$, and it satisfies the room inequality $16(2X + h + L + 2) \leq 2^L$ a fortiori. For every $N < 2X$ the proved truncation bound `LEAN SOURCE` gives $\|E(h, N, L) - e(2^N \alpha_h)\| < 2\pi(N + L + h + 2)/2^L \leq 2\pi/1024 < 1/100$, so by `LEAN SOURCE`, $\operatorname{Re} E(h, N, L) \leq \cos(2\pi 2^N \alpha_h) + 1/100$. Summing over the X values $N \in [X, 2X)$ gives $\sum_N \operatorname{Re} E(h, N, L) \leq \frac{89}{100}X + \frac{1}{100}X = \frac{9}{10}X$, so `LEAN SOURCE` produces $N \in [X, 2X)$ with $C(h, N, L)$, and $N \geq X \geq X_0$. This is exactly the quantified certificate condition consumed by `LEAN SOURCE`. $\square$

[Math] [cofinal] . The proof above makes the truncation margin explicit. The full implication is also recorded in the formal source `LEAN SOURCE`. It uses a real-part estimate for the exact phases and spends $1/100$ on truncation. Compare the sufficient norm condition `LEAN SOURCE`. This is a separately sufficient condition. It is not asserted to be equivalent to the $21/25$ norm bound or to the four separate estimates stated above. The constants account for different quantities and different error margins.

Corollary 12.4 (A digit version of the analytic condition). *Let $\rho_h(X)$ denote the proportion of $N \in [X, 2X)$ with $\|2^N \alpha_h\|_{\mathbb{R}/\mathbb{Z}} \geq 1/4$. If for every $h \geq 1$ there are cofinally many X with $\rho_h(X) \geq 11/100$, then S is irrational. For nondyadic α_h , the condition counted by $\rho_h(X)$ is exactly a change between binary digits $N + 1$ and $N + 2$. Thus the same sufficient hypothesis asks for at least $11X/100$ such changes, counted with $X \leq N < 2X$, on arbitrarily large blocks for every h .*

Lean: `irrational_totientSeries_of_quarterFarPhase_proportion`, `quarterFarFromInt_iff_binaryDigitAt_change`, `irrational_totientSeries_of_digitChange_count`, `irrational_totientSeries_of_quarterFarPhase_count`, and 3 further declarations in the `coverage` section.

Proof. The inequality $\|x\|_{\mathbb{R}/\mathbb{Z}} \geq 1/4$ gives $\cos(2\pi x) \leq 0$. Every remaining term is at most 1, so

$$\sum_{N=X}^{2X-1} \cos(2\pi 2^N \alpha_h) \leq (1 - \rho_h(X))X \leq \frac{89}{100}X.$$

Proposition 12.3 now applies. For the digit reading, for a nondyadic α_h the fractional part of $2^N \alpha_h$ is in $[1/4, 3/4]$ exactly when those two binary digits differ. The nondyadic qualification excludes the endpoint ambiguity at $3/4$. $\square$

[Math]. The exact hypothesis is the digit-change inequality in the corollary. It is not a claim about an average run length: the two boundary runs of a finite block need not be complete, and their lengths are not specified by the number of interior digit changes. The finite observations below satisfy the displayed bound at their sampled parameters. They do not establish it on arbitrarily large blocks.

For the numerical enclosure, compute $P = \sum_{n=1}^{6080} \varphi(n) 2^{6080-n}$. The elementary tail bound gives

h	X	$\rho_h(X)$	$X^{-1} \sum_{N \in [X, 2X)} \cos(2\pi 2^N \alpha_h)$	required
1	256	0.4766	0.0037	≤ 0.89
1	2048	0.5073	-0.0043	≤ 0.89
3	256	0.4648	0.0296	≤ 0.89
3	2048	0.5020	0.0027	≤ 0.89
8	2048	0.5112	-0.0088	≤ 0.89
13	2048	0.5049	-0.0195	≤ 0.89

Table 6: Finite block statistics for $\alpha_h = (2^h - 1)S$. The exact digit-change counts in the six rows are 122, 1039, 119, 1028, 1047 and 1034; the proportions and cosine averages are rounded to four decimal places. The interval calculation described below certifies these counts and displayed roundings. [Cert]

$$P \leq 2^{6080}S \leq P + 6082.$$

The endpoints have the same quotient on division by 2^{80} , which certifies $\lfloor 2^{6000}S \rfloor$; an error bound smaller than one would not alone certify a floor near an integer boundary. Multiplying this interval by $(2^h - 1)2^{N-6080}$ encloses each phase. None of the resulting intervals crosses a digit-test boundary, and interval evaluation of the cosine certifies the rounded averages. Among the first 5800 fractional digits, the numbers of runs for $h = 1, 3, 8$ are 2864, 2964, 2927, and the longest runs have lengths 13, 12, 14. These count the two boundary runs as observed, whether or not they continue outside the window. No randomness assumption or asymptotic conclusion is part of these finite calculations.

Observation 12.5. Lemma 12.2 expresses each tail phase in terms of the doubling orbit of $\alpha_h = (2^h - 1)S$. The block conditions average those phases, the prime condition samples prescribed prime-indexed positions, and the full-depth condition requires a finite residue certificate for each (d, N) at some multiple of d . Corollary 12.4 gives one sufficient digit-change test for the real-part estimate; it is not an equivalence with the norm bound or with the four separate budgets. Likewise, a bound on a binary run length is not the exact prime-gap condition. Section 12.7 compares the original inequalities.

12.2 The full-block exponential-sum estimate

The required inequality.

$$\forall h \geq 1 \forall X_0 \exists X, L : \quad \max(X_0, 1) \leq X, \quad 16(2X + h + L + 2) \leq 2^L, \quad \left\| \sum_{N=X}^{2X-1} e(D(h, N, L)/2^L) \right\| \leq \frac{21}{25}X,$$

where $D(h, N, L) = \sum_{j < L} (\varphi(N + h + 1 + j) - \varphi(N + 1 + j))2^{L-1-j}$. [LEAN SOURCE](#), implication [LEAN SOURCE](#) [Open] [cofinal]. By Proposition 12.3 the real-part bound with constant 89/100 for the *exact doubling phases* suffices; its transfer to the truncated phases uses the separate error estimate in that proposition.

The discrepancy in terms of the doubling map. For $L \geq h$, dividing by 2^L and reindexing the two ranges gives

$$\frac{D(h, N, L)}{2^L} = \underbrace{-\sum_{t=1}^h \frac{\varphi(N+t)}{2^t}}_{\text{head}} + \underbrace{(2^h - 1) \sum_{t=h+1}^L \frac{\varphi(N+t)}{2^t}}_{\text{body}} + \underbrace{\sum_{t=L+1}^{L+h} \frac{\varphi(N+t)}{2^{t-h}}}_{\text{tail}}. \quad (1)$$

The phase is a geometrically weighted sum of totients at $L+h$ consecutive arguments. In the first two ranges, $1 \leq t \leq L$, the coefficient has denominator 2^t and odd numerator. Its contribution is therefore integral exactly when $v_2(\varphi(N+t)) \geq t$. In the last range, $L < t \leq L+h$, the denominator is instead 2^{t-h} , so the corresponding threshold is $t-h$. Here $v_2(a)$ denotes the exponent of 2 in the positive integer a . The selected offset in the four-term decomposition lies in the middle range, where the odd multiplier is $2^h - 1$.

Relevant comparison estimates. The following comparisons identify the estimates still needed; they do not rule out these methods.

Weyl differencing and van der Corput. The van der Corput inequality applies to arbitrary bounded complex sequences [21, Theorem 2.1], but its use here requires estimates for shifted correlations. For the exact phases $x_N = 2^N \alpha_h$ of Lemma 12.2,

$$x_{N+q} - x_N = (2^q - 1)x_N.$$

Thus differencing gives another doubling orbit, not a polynomial phase of lower degree. The finite phase $D(h, N, L)/2^L$ approximates this orbit modulo 1; it is not the exact orbit itself. No bound for the resulting correlations is established here.

The large sieve. A proposed application would need a suitable family of separated frequencies and a way to recover a bound for each fixed h . Averaging only over h would not suffice: the condition asks for a block estimate for every h , not for most of them.

Bilinear sums over prime factors. Factoring $N+t = mp$ and using $\varphi(mp) = \varphi(m)(p-1)$ makes the selected totient contribution linear in the prime p . Section 12.3 gives this decomposition and identifies the remaining weight. A bound for that weight is still needed before a Vaughan- or Vinogradov-type argument can be applied.

Erdős's 1948 digit method. The comparison series is $E = \sum_{n \geq 1} 1/(2^n - 1) = \sum_{N \geq 1} d(N)/2^N$, whose irrationality Erdős proved [9]. For the totient series, the corresponding Möbius expansion is exact: $\varphi = \mu * \text{id}$ gives

$$S = \sum_{d \geq 1} \mu(d) \frac{2^d}{(2^d - 1)^2} \quad (2)$$

The identity is labelled [Math]; the recorded numerical check gives $S = 1.3676308019850223 \dots$ to sixteen digits [Cert].

Erdős combines congruences forcing increasing powers of 2 to divide a block of divisor counts with an averaged tail bound, producing long zero blocks in the binary expansion [9, pp. 63–66]. To adapt that argument here, one would need congruences for the actual totient coefficients together with a compatible tail estimate. The identities alone provide neither combination. In (2), the coefficients $\mu(d)$ can have either sign,

so positivity of the original totient series does not give positivity term by term in this expansion. This is a limitation of the proposed direct transfer, not a proof that a digit argument or a different use of the Möbius expansion cannot work.

The unresolved estimate. For general coefficients $0 \leq c(n) \leq n$, irrationality alone does not imply the block norm condition. The following example has very sparse nonzero coefficients. It is irrational, but almost all phases in each sufficiently large block lie close to 1.

Theorem 12.6 (The block norm condition is stronger in the class $0 \leq c(n) \leq n$). *Let $c(n) = 1$ if $n = k!$ for some $k \geq 1$ and $c(n) = 0$ otherwise, so $0 \leq c(n) \leq n$ for all $n \geq 1$, and let $\beta = \sum_{n \geq 1} c(n)/2^n = \sum_{k \geq 1} 2^{-k!}$. Then β is irrational, and for every $h \geq 1$ and every $X \geq 81(h + 5)$,*

$$\sum_{N=X}^{2X-1} \cos(2\pi 2^N (2^h - 1)\beta) > \frac{9}{10}X.$$

Consequently the block norm condition fails for β at every sufficiently large scale, for every admissible truncation depth, even though β is irrational. Thus irrationality, nonnegative integer coefficients and the bound $c(n) \leq n$ do not imply the block gap. A proof for S needs an additional property not shared by this example; the comparison does not specify which additional property will suffice.

Lean: `lacCoef_bounds`, `lacBeta_eq_factorial_series`, `irrational_lacBeta`, `lacunary_block_cos_gap`, and 2 further declarations in the [coverage section](#).

Proof. The binary expansion of β has infinitely many ones separated by unbounded strings of zeros. It is not eventually periodic, so β is irrational. Write $\gamma = (2^h - 1)\beta$, and let $k!$ be the least factorial greater than N . The earlier terms become integers after multiplication by 2^N , so

$$2^N \gamma \equiv (2^h - 1) \sum_{j \geq k} 2^{N-j!} \pmod{\mathbb{Z}}.$$

If $N \leq k! - h - 5$, the expression on the right lies between 0 and $2 \cdot 2^{N+h-k!} \leq 1/16$. Thus $\|2^N \gamma\|_{\mathbb{R}/\mathbb{Z}} \leq 1/16$ and $\cos(2\pi 2^N \gamma) \geq \cos(\pi/8) > 9238/10000$. The remaining N lie in $(k! - h - 5, k!]$, at most $h + 5$ values per factorial. The relevant factorials have ratio at least 3, and $h + 5 < X$, so $[X, 2X)$ meets at most one such interval. Hence the sum is greater than

$$\frac{9238}{10000}(X - h - 5) - (h + 5) \geq \frac{9}{10}X + \frac{h + 5}{250}$$

when $X \geq 81(h + 5)$.

To pass from these exact phases to the finite residue test, note that $0 \leq c(n) \leq 1$ puts each scaled tail in $[0, 1]$. The error in the truncated tail difference is therefore at most 2^{-L} , so each complex phase changes by at most $2\pi 2^{-L}$. At an admissible depth, $2^L \geq 16(2X + h + L + 2) > 32X$, and the total change in the block sum is less than $\pi/16$. Its real part is consequently greater than $9X/10 - \pi/16 > 21X/25$ for $X \geq 81(h + 5)$. Its norm cannot satisfy the $21X/25$ bound either. $\square$

[Math] [cofinal] . This example rules out the implication from irrationality and the stated coefficient bound to the block norm condition. It does not say that the totient sum is Liouville-like, or that every argument based on its digits fails. A proof of the block estimate must use information that distinguishes the actual totient phases from this comparison sequence.

The size of the required saving. Under Lebesgue measure on the phase circle, the binary digit maps are independent fair bits. The digit-change proportion therefore tends to $1/2$ for almost every phase. For almost every phase, the real-part block sum is also $O(\sqrt{X \log \log X})$, by the law of the iterated logarithm of Erdős and Gál for lacunary series [27, p. 65]. The block bound follows by subtracting two initial partial sums. An almost-everywhere theorem supplies no conclusion at the particular phase $\alpha_h = (2^h - 1)S$ [Cited]. The requirement is $\rho_h \geq 0.11$ and a saving of a constant factor. The finite values in Table 6 have ρ_h near 0.50 and small block averages. They are compatible with square-root cancellation, but do not establish its asymptotic order. The sufficient condition asks only for the stated fixed saving on arbitrarily large blocks. No estimate in this record supplies it for the totient phases.

12.3 The four-term decomposition

The simultaneous inequalities. Use the four finite sums defined in Section 3.2; in particular, each group mean averages the selected prime phase, not the residual weight. For every $h \geq 1$ there must exist $s \geq 1$ and $\eta \in (0, 1)$ such that for every X_0 there are $X \geq \max(X_0, 1)$ and L with $h \leq L - s$, $16(2X + h + L + 2) \leq 2^L$, and all four of

$$\begin{aligned} \operatorname{Re} \sum_{N \in \mathcal{G}} w_N (z_N - \bar{z}_{m_N}) &\leq \frac{14}{25} X, \\ \left\| \sum_{N \in \mathcal{G}} w_N \bar{z}_{m_N} \right\| &\leq \frac{1}{100} X, \\ \left\| \sum_{N \in A \setminus \mathcal{G}} E(h, N, L) \right\| &\leq \frac{1}{100} X, \\ \left\| \sum_{\substack{X \leq N < 2X \\ N \notin A}} E(h, N, L) \right\| &\leq \frac{8}{25} X. \end{aligned}$$

LEAN SOURCE, implication LEAN SOURCE [Open] [cofinal]. The decomposition is an exact identity, LEAN SOURCE [Lean]. For $X \geq 4$ and $s = L - h$, the single cofactor group $m_N = 1$, not the full assigned set, consists of the indices for which $N + h + 1$ is prime, LEAN SOURCE [Lean].

An upper bound for the unassigned terms. An unassigned index has a smooth shifted argument, but the converse is not asserted. The factorisation gives the containment needed for an upper bound on the number of unassigned indices.

Proposition 12.7 (A one-sided bound for the unassigned terms). *Fix h, s and choose the admissible depth L minimally for each large X . Put $t = L - s + 1 = O_{h,s}(\log X)$ and $y_X = 4\sqrt{X} + 2t/\sqrt{X}$. If $n = N + t$ is unassigned, then its largest prime factor satisfies $P(n) \leq y_X$. Consequently*

$$\begin{aligned} \#\{N \in [X, 2X) : N \notin A\} &\leq \Psi(2X + t - 1, y_X) - \Psi(X + t - 1, y_X) \\ &= (1 - \log 2 + o(1))X < \frac{8}{25}X \end{aligned}$$

for all sufficiently large X . Here $\Psi(x, y)$ counts the positive integers at most x whose prime factors are all at most y . This is an upper bound for the unassigned count, not an asymptotic equality for that count.

Lean: [prop_dickman](#).

Proof. Let $p = P(n)$ and $m = n/p$. Since $n > 1$, failure of the assignment condition means either $p \leq 2\lfloor\sqrt{X}\rfloor$ or $m > \lfloor\sqrt{X}/2\rfloor$. In the second case $m > \sqrt{X}/2$, so

$$p = \frac{n}{m} < \frac{2(2X + t)}{\sqrt{X}} = y_X.$$

The first case also gives $p \leq y_X$. Counting the resulting smooth integers in $[X + t, 2X + t)$ proves the first inequality.

For completeness, the classical fixed-parameter asymptotic $\Psi(x, x^{1/u}) \sim x\rho(u)$ suffices; no additional short-interval estimate is needed. Indeed, for either $x = X + t - 1$ or $x = 2X + t - 1$, $\log y_X / \log x \rightarrow 1/2$. For every fixed $0 < \epsilon < 1$, eventually

$$x^{1/(2+\epsilon)} \leq y_X \leq x^{1/(2-\epsilon)}.$$

Monotonicity in y bounds $\Psi(x, y_X)/x$ between quantities tending to $\rho(2 + \epsilon)$ and $\rho(2 - \epsilon)$. Letting $\epsilon \rightarrow 0$ and using continuity gives the limit $\rho(2)$. Subtracting the two endpoint asymptotics yields $(\rho(2) + o(1))X$. Since $\rho(2) = 1 - \log 2 < 8/25$, the claimed bound follows. $\square$

[Lean] for the whole proposition, [LEAN SOURCE](#). The Lean proof uses no smooth-number asymptotic. For large X , $y_X^2 > 2X + t - 1$, so an integer in $(X + t - 1, 2X + t - 1]$ that is not y_X -smooth has exactly one prime factor above y_X , and

$$\Psi(2X + t - 1, y_X) - \Psi(X + t - 1, y_X) = X - \sum_{y_X < p \leq 2X + t - 1} \left(\left\lfloor \frac{2X + t - 1}{p} \right\rfloor - \left\lfloor \frac{X + t - 1}{p} \right\rfloor \right).$$

Mertens' theorems, proved in the development with explicit constants, then give $(1 - \log 2)X + O(X/\log X)$. The written proof above cites instead the fixed- u asymptotic [18, (1.1), p. 268] and the value $\rho(u) = 1 - \log u$ on $[1, 2]$ [18, (1.3), p. 268]. No effective crossover scale or exact density for the unassigned indices is asserted. The estimate uses the minimal admissible depth; it does not supply either of the two remaining correlation bounds at that depth.

Proposition 12.8 (The excluded-cofactor estimate). *Fix h, s and use the minimal admissible depth L , as in Proposition 12.7; thus $t = L - s + 1 = O_{h,s}(\log X)$. For $\eta \in (0, 1)$ let $B(\eta) = \{m \geq 1 : \varphi(m) < \eta m\}$, with natural density $D(\eta)$; by Schoenberg's theorem D exists, is continuous, and $D(0+) = 0$ [19, §17, p. 193], in the framework of [20, Theorem 1, pp. 318–319, and §8, p. 323] [Cited]. Then*

$$\#\{N \in A : m_N \in B(\eta)\} \leq (D(\eta) + o(1))X,$$

so a single choice of η with $D(\eta) < 1/200$ meets the $\frac{1}{100}X$ budget for all large X . This choice fixes η before X_0 , as required. It supplies only the excluded-cofactor bound: the mean and mean-subtracted estimates must still hold for this same η , and do not follow from making η smaller.

Lean: `prop_badcof`. Conditional on the prime number theorem; see the [coverage section](#).

Proof. For a fixed cofactor $m \leq \sqrt{X}/2$, each assigned index gives a prime p with $X + t \leq mp < 2X + t$. Dropping the other assignment restrictions bounds the number of indices by the number of primes in this interval. The prime number theorem, uniformly for $X/m \geq 2\sqrt{X}$, and $t = O_{h,s}(\log X)$ give the upper bound

$$(2 + o(1)) \frac{X}{m \log X}.$$

The possible endpoint contribution is $O(1)$ for each m , hence $o(X)$ after summing over $m \leq \sqrt{X}/2$. Natural density and partial summation give

$$\sum_{\substack{m \leq \sqrt{X}/2 \\ m \in B(\eta)}} \frac{1}{m} = \left(\frac{D(\eta)}{2} + o(1) \right) \log X.$$

Multiplying the two estimates proves the stated bound. If $D(\eta) < 1/200$, its right side is less than $X/100$ for all sufficiently large X . $\square$

The existence of a usable cutoff can also be made explicit by an elementary estimate. For $x \geq 1$,

$$\begin{aligned} \frac{1}{x} \sum_{n \leq x} \frac{n}{\varphi(n)} &\leq \sum_{d \geq 1} \frac{\mu(d)^2}{d \varphi(d)} = \prod_p \left(1 + \frac{1}{p(p-1)} \right) \\ &\leq \exp \left(\sum_{n \geq 2} \frac{1}{n(n-1)} \right) = \exp(1) < 3. \end{aligned}$$

The first step uses $n/\varphi(n) = \sum_{d|n} \mu(d)^2/\varphi(d)$ and $\lfloor x/d \rfloor \leq x/d$; the Euler product has nonnegative terms and converges by the displayed majorant. Markov's inequality gives $\#(B(\eta) \cap [1, x]) < 3\eta x$, hence $D(\eta) \leq 3\eta$. In particular, $\eta = 1/1000$ gives $D(\eta) \leq 3/1000 < 1/200$, an explicit sufficient choice for the excluded-cofactor budget. This does not establish either weighted phase estimate for that choice, and it is not claimed to be an optimal cutoff. [Math] Lean checks the estimate $\#(B(\eta) \cap [1, x]) < 3\eta x$ with no further input (`card_excludedCofactorSet_lt`) and, with the prime number theorem as its only

input, the excluded-cofactor budget at $\eta = 1/1000$ (`excluded_budget_one_thousand` `th`).

The first element of $B(1/5)$ is $30030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$, and $\varphi(30030)/30030 = 5760/30030 \approx 0.191808$. Enumeration through $3 \cdot 10^5$ gives 38 elements of $B(1/5)$ and

$$\frac{1}{\log(3 \cdot 10^5)} \sum_{\substack{m \leq 3 \cdot 10^5 \\ m \in B(1/5)}} \frac{1}{m} \approx 2.82 \cdot 10^{-5}.$$

This finite weighted proportion is not the limiting density $D(1/5)$ [Cert]. Using $\eta = 1/5$ in the asymptotic bound would require a rigorous bound for $D(1/5)$, including the integers beyond the enumeration; the finite count does not supply one.

The growing dyadic modulus. The selected offset is $t = L - s + 1$ in (1). The four-term condition fixes s after h but before the threshold X_0 , whereas

$$2^L \geq 16(2X + h + L + 2) > 32X, \quad 2^t > 2^{6-s}X.$$

On a group with cofactor m , the coefficient of $p - 1$ in the phase has reduced denominator $2^{\max(t - v_2(\varphi(m)), 0)}$, because $2^h - 1$ is odd. Assignment gives $\varphi(m) \leq m \leq \sqrt{X}/2$. Thus, for fixed s and all sufficiently large X , the reduced denominator is greater than $2^{7-s}\sqrt{X}$. In particular, the assignment conditions do not reduce these moduli to a fixed power of $\log X$. This is the modulus for variation in p , not a lower bound for the denominator of the phase value at each individual prime.

Choosing $t = O(\log \log X)$ would require s to grow with X , contrary to the stated quantifiers. Likewise, the formal observation that the phase is 1 when $v_2(\varphi(m)) \geq t$ cannot provide such groups at large X under these size bounds. The proposed small-modulus prime-distribution argument therefore does not establish the mean estimate. This does not disprove the four-term condition or exclude other estimates for the same prime sums. [Math] [cofinal].

The remaining prime-weighted correlation. Propositions 12.7 and 12.8 give the unsigned and excluded-cofactor bounds at minimal admissible depth. They must be combined with mean and correlation estimates at the same X, L, s, η ; bounds obtained at unrelated depths cannot be substituted.

With the notation of Section 3.2, the correlation estimate still required is

$$\operatorname{Re} \sum_{N \in \mathcal{G}} w_N \left(e \left(\frac{(2^h - 1)\varphi(m_N)(p_N - 1)}{2^t} \right) - \bar{z}_{m_N} \right) \leq \frac{14}{25}X.$$

The sum is over the assigned indices with good cofactors, not all primes and cofactors in the interval. By (1), w_N is the phase with the selected offset t removed.

Regrouping by $m_N = m$ gives weighted sums over the primes $p_N = p$. A bilinear argument would need to separate the m - and p -dependence or estimate the resulting two-variable weight directly. For $N = mp - t$, that weight depends on $\varphi(mp + j - t)$ at the other offsets $1 \leq j \leq L + h, j \neq t$. The identity $\varphi(mp) = \varphi(m)(p - 1)$ controls the

selected value, not these neighbouring totients. No bound for the resulting correlation is established here; this is the missing step, not a limitation of every bilinear method.

A fixed-offset example illustrates the correlation at issue. At $h = 1$ and offset $t = 2$, a prime $p \geq 5$ satisfies $\varphi(p-1) \equiv 0 \pmod{2}$ and $\varphi(p) = p-1$. The phase of R_{p-2} is therefore $\chi_{-4}(p)$ times the phase of the following weighted totients, since $e((p-1)/4) = i^{p-1} = \chi_{-4}(p)$. One possible fixed-saving estimate, with a fixed $\delta > 0$, would be

$$\operatorname{Re} \sum_{X < p \leq 2X} \chi_{-4}(p) e\left(\frac{\varphi(p+1)}{8} + \frac{\varphi(p+2)}{16} + \dots\right) \leq \left(\frac{9}{10} - \delta\right) (\pi(2X) - \pi(X)).$$

This fixed-offset example is not Definition 3.4: there s is fixed while $t = L - s + 1$ grows. The example isolates a correlation between a quadratic character at p and the binary phases of totients at $p+1, p+2, \dots$. It is analogous in this respect to shifted correlations studied in the Chowla and Elliott problems, but the analogy does not supply an applicable theorem. The cited theorems have different hypotheses and types of average: Matomäki and Radziwiłł compare short and long averages of real multiplicative functions bounded by 1, outside an exceptional set of short intervals [22, Theorem 1]; Tao proves logarithmically averaged two-point Chowla and a corrected Elliott statement, the latter under an explicit non-pretentiousness hypothesis on one of the bounded multiplicative functions [23, Theorems 1.2–1.3]; and Tao and Teräväinen prove the odd-order logarithmically averaged Chowla conjecture [24, Theorem 1.1] [Cited]. The phase here depends on φ at a growing window of shifts, so none of these applies directly.

Averaging over the block parameter. For each fixed h , an average of the nonnegative, normalised block norms can establish the full-block condition. It is enough to prove

$$\sum_{k=1}^K 2^{-k} \left| \sum_{N=2^k}^{2^{k+1}-1} e(D(h, N, L_k)/2^{L_k}) \right| = o(K),$$

where every L_k satisfies the room inequality for $X = 2^k$. Indeed, the number of $k \leq K$ for which the normalised norm exceeds $21/25$ is at most $25/21$ times the displayed sum, hence is $o(K)$. There are therefore arbitrarily large admissible blocks satisfying Definition 3.1. This argument does not establish the four-term condition: that condition requires all four separate budgets with common X, L, s, η .

The displayed average is a Cesàro average of norms on dyadic blocks. It is not the signed logarithmically weighted correlation appearing in the cited Chowla–Elliott theorems. The position of the absolute value matters: cancellation between different blocks would not bound this sum of norms. Invoking entropy decrement would therefore require an additional argument yielding this particular estimate.

The finite phase depends on the $L + h$ totient values at offsets $1, \dots, L + h$, with the coefficients in (1). Although its absolute value is 1, it is not a multiplicative function of N . Both the number of shifts and their dyadic denominators grow with L . The cited fixed-shift correlation theorems do not supply the uniform bounds for this growing family. No such extension is proved here. [Open]

12.4 Separation along prime positions

The quantified prime condition.

$$\forall h \geq 1 \forall N_0 \exists p \text{ prime} : \max(N_0 + h + 1, h + 5) \leq p \wedge \operatorname{Re}(e(R_{p-1} - R_{p-h-1})) < \frac{9}{10}.$$

[LEAN SOURCE](#), implication [LEAN SOURCE](#) [Open] [cofinal].

Its meaning for the doubling map. $\operatorname{Re} e(2^{p-h-1}\alpha_h) < 9/10$ says $\|2^{p-h-1}\alpha_h\|_{\mathbb{R}/\mathbb{Z}} > \arccos(9/10)/2\pi = 0.0717831\dots$, so the requirement is exactly: *for each h , the indicated prime-indexed $\times 2$ orbit point stays more than $0.0717831\dots$ from the nearest integer, for cofinally many primes p .* A short-run condition on the binary digits can be a sufficient proxy for this circle-distance inequality, but it is not equivalent to it; a run-length description alone does not determine the residual position inside the dyadic cylinder.

What the prime condition adds. The preceding condition asks for a separated point at a prime-indexed position. The full-block average does not by itself locate such a point on a prescribed sparse set. Conversely, the prime condition does not assert cancellation over the whole interval. They are different sufficient hypotheses; no general comparison of their difficulty is proved here.

Finite comparison. Among the 501 primes in $[1000, 5000)$, the strict inequality holds at 432, 426 and 430 primes for $h = 1, 2, 5$, respectively. The corresponding proportions are 0.8623, 0.8503 and 0.8583 to four decimal places, and each sample has a cosine value below -0.9999 . These counts were recomputed using the same interval for S as in Table 6, with no interval meeting the threshold $9/10$. They do not establish separation at arbitrarily large primes.

An average over a prescribed prime sample. The subset implication [LEAN SOURCE](#) allows a nonempty finite set T . If T is chosen after a separated point is known, the subset formulation may add no arithmetic content: the equivalence proved using selected pairs illustrates this issue [LEAN SOURCE](#). To avoid that particular choice, fix the sample in advance by

$$T_{h,X} = \{N \in [X, 2X) : N + h + 1 \text{ is prime}\}.$$

For $X \geq 4$ and $L \geq h$, this is precisely the prime-factor sample $\text{pivotFiber}(X, L, L - h, 1)$ in the supplied source [LEAN SOURCE](#).

A sufficient, unproved condition is

$$\begin{aligned} \forall h \geq 1 \forall X_0 \exists X \geq \max(X_0, 4) \exists L \geq h : \\ T_{h,X} \neq \emptyset, \quad 16(2X + h + L + 2) \leq 2^L, \\ \sum_{N \in T_{h,X}} \operatorname{Re} E(h, N, L) \leq \frac{9}{10} |T_{h,X}|. \end{aligned} \tag{3}$$

The subset theorem then gives a certificate at some $N \geq X \geq X_0$. Nonemptiness matters: for the empty set the displayed sum inequality would hold automatically and would

produce no certificate. The bounds $X \geq 4$ and $L \geq h$ are also the hypotheses of the identification with the source's prime-factor sample. [Open]

On this sample the selected cofactor is 1, so its coefficient is the odd integer $2^h - 1$ and its modulus is 2^{h+1} , fixed once h is fixed. The unweighted prime average has zero Ramanujan main term, since $c_{2^{h+1}}(2^h - 1) = \mu(2^{h+1}) = 0$. That observation does not establish (3): the prime phase is multiplied by a residual phase built from the other totient values. Their correlation is the missing estimate. Restricting the sample removes the complement and unwanted cofactor terms, but not this residual correlation.

Remark 12.9. Fixing the sample rules out the particular argument that selects a pair only after locating separation. It does not prove that (3) is inequivalent to irrationality, or that it is a strictly stronger condition. No such comparison is established here.

12.5 Depth equal to a period multiple

The quantified residue condition.

$$\text{full-depth condition} := \forall d \geq 1 \forall N \exists t \geq 1 : C(td, N, td),$$

unpacked, $(N+2td+2 : \mathbb{Z}) < D(td, N, td) \bmod 2^{td} < 2^{td} - (N+2td+2)$. [LEAN SOURCE](#), implication [LEAN SOURCE](#), ambient equivalence [LEAN SOURCE](#), equivalence [LEAN SOURCE](#) [Open] [cofinal]. The depth-locked condition is equivalent to irrationality; the remaining gap is a nonintegral seed for every ray and basepoint, not an unknown converse.

The condition in terms of binary digits. Writing $h = td$ and using $D(h, N, L)/2^L = (R_{N+h} - R_N) - (R_{N+L+h} - R_{N+L})/2^L$ at $L = h$, together with the proved strip $|R_{M+h} - R_M| < M + h + 2$ ([LEAN SOURCE](#), [Lean]), applied at $M = N + h$, one gets:

Proposition 12.10. $C(h, N, h)$ holds whenever $\|2^{N+h}S - 2^N S\|_{\mathbb{R}/\mathbb{Z}} > 2(N + 2h + 2)/2^h$.

Lean: [certifiedKill_of_fullDepth_phase_separation](#), [bracket_of_two_sided_separation](#).

Proof. $\|D(h, N, h)/2^h\| \geq \|R_{N+h} - R_N\| - |R_{N+2h} - R_{N+h}|/2^h > 2(N + 2h + 2)/2^h - (N + 2h + 2)/2^h = (N + 2h + 2)/2^h$, and $R_{N+h} - R_N \equiv 2^N(2^h - 1)S = 2^{N+h}S - 2^N S \pmod{1}$ by Lemma 12.2. A residue at distance more than $(N + 2h + 2)/2^h$ from $\mathbb{Z}$ is exactly a residue certificate at depth h . $\square$

[Math]. Proposition 12.10 gives a sufficient analytic criterion for Route 4: for each $d \geq 1$ and $N \geq 0$, it is enough to find $h = td$, $t \geq 1$, for which the displayed distance exceeds the stated threshold. The criterion is not asserted to be equivalent to each individual finite certificate. It involves the phase of S itself, and the threshold tends to zero exponentially along a fixed ray. The earlier full-depth amplifier explains why a nonintegral seed produces late certificates; it does not supply seeds for all d, N .

Comparison with digit complexity. The phase inequality above is a quantitative statement at specified positions and period multiples. A general block-complexity assertion is a different quantified statement, of the kind treated by the Adamczewski–Bugeaud method for algebraic numbers [25, Theorem 1, p. 549]. No implication from such an assertion to the required inequality is established here. In particular, this record does not

justify describing one existing method as the only possible route, or algebraicity as the sole missing input. The arithmetic nonintegrality obligation must be stated and proved in its own right.

Finite examples. At $N = 300$, the least t with $\mathcal{C}(td, 300, td)$ exists and is small for every ray $d \leq 24$: $t = 10$ for $d = 1$, $t = 5$ for $d = 2$, $t = 4$ for $d = 3$, $t = 3$ for $d = 4$, $t = 2$ for $5 \leq d \leq 9$, and $t = 1$ for $10 \leq d \leq 24$; in every case at or near the first depth admitted by the room floor. The residues are not marginal: $r/2^h = 0.3594$ at $d = 1$, 0.5977 at $d = 3$, 0.6853 at $d = 7$, 0.2620 at $d = 17$, 0.5887 at $d = 23$, against edge radii $3.1 \cdot 10^{-1}$, $8.0 \cdot 10^{-2}$, $2.0 \cdot 10^{-2}$, $2.6 \cdot 10^{-3}$, $4.1 \cdot 10^{-5}$ [Cert]. This is a finite verification at the displayed basepoint and shifts. It neither bounds the required multiplier for arbitrary d, N nor establishes the universally quantified condition.

A possible heuristic, not an estimate. If a phase U were uniform on $\mathbb{R}/\mathbb{Z}$, then $\Pr(\|U\| \leq \epsilon) = \min(1, 2\epsilon)$ for $\epsilon \geq 0$. At $\epsilon = 2(N + 2h + 2)2^{-h}$ this gives the marginal model $\min(1, 4(N + 2h + 2)2^{-h})$. The phases $2^N(2^{td} - 1)S$ on a fixed ray are arithmetically dependent; no independence model or uniform distribution theorem for this orbit is proved here. Consequently the marginal estimate cannot be multiplied across depths, and no summability conclusion over all basepoints follows from it. The finite experiments remain finite evidence, not a proof of the all-ray supply. [Open] (heuristic only).

12.6 Why rationality does not bound carry rank

For an integer sequence u , write $\text{rk}_e(u)$ for the rational dimension of the span of all sequences $n \mapsto u_{2^j n + r}$ with $0 \leq j \leq e$ and $0 \leq r < 2^j$. Rationality does not force these ranks to be bounded, even for one fixed coefficient sequence. More precisely, the assumptions

$$0 \leq c(n) \leq n, \quad \sum_{n \geq 1} c(n)2^{-n} \in \mathbb{Q}, \quad v \in \mathbb{N}_{>0}, \quad u_n \in \mathbb{Z},$$

$$u_{n+1} = 2u_n - vc(n+1), \quad u_n/2^n \rightarrow 0$$

do not imply $\sup_e \text{rk}_e(u) < \infty$. Nor do they imply an upper bound that is $o(2^e)$. Such a bound for the actual totient carry would contradict the lower bound [LEAN SOURCE](#) [Lean], but the following rational comparison already has that lower bound.

The comparison with value $5/4$. Use a centred base-four expansion. The elementary bounds $9/8 \leq S \leq 3/2$ follow respectively from the first four terms and from $\varphi(n) \leq n-1$ for $n \geq 2$. Put $x_0 = 5/4 - S$ and, for $m \geq 1$, define

$$d_m = \left\lfloor 4x_{m-1} + \frac{2}{3} \right\rfloor, \quad x_m = 4x_{m-1} - d_m.$$

The interval $[-2/3, 1/3)$ contains x_0 and is preserved by this recurrence, so $d_m \in \{-2, -1, 0, 1\}$. Iteration gives

$$x_0 = \sum_{m=1}^M d_m 4^{-m} + 4^{-M} x_M, \quad |x_M| \leq \frac{2}{3}.$$

Consequently $x_0 = \sum_{m \geq 1} d_m 4^{-m}$. Define

$$c(0) = 0, \quad c(2m-1) = \varphi(2m-1), \quad c(2m) = \varphi(2m) + d_m \quad (m \geq 1).$$

These are nonnegative integers: for $m \geq 2$, $\varphi(2m) \geq 2$; for $m = 1$, the sharper bound $x_0 \geq -1/4$ gives $d_1 \geq -1$. The inequalities $d_m \leq 1$ and $\varphi(2m) \leq 2m-1$ also give $c(n) \leq n$. Thus $|c(n) - \varphi(n)| \leq 2$ and

$$\sum_{n \geq 1} c(n) 2^{-n} = S + \sum_{m \geq 1} d_m 4^{-m} = \frac{5}{4}.$$

This is the centred base-four construction in the supplied [comparison-series theorem](#).

Take $v = 4$ and $u_N = 4 \sum_{j \geq 1} c(N+j) 2^{-j}$. The rational sum makes u_N integral; the coefficient bound gives $0 \leq u_N \leq 4(N+2)$. Since c agrees with φ at every odd index, for odd r ,

$$4\varphi(2^j n + r) = 2u_{2^j n + r - 1} - u_{2^j n + r}.$$

The same $2^e - 1$ independent totient sections therefore lie in the span of the carry sections at levels $1, \dots, e$, proving $\text{rk}_e(u) \geq 2^e - 1$ for every $e \geq 0$. The corresponding formal statement is the [unbounded-rank theorem for this comparison](#). This example is distinct from the bounded $3/2$ parity comparison in Proposition 1.9. It disproves an inference from the listed assumptions alone, not one that also uses identities of the actual totient coefficients. [Math]

The rank quantities. Finite-dimensional span of all dyadic sections is the definition of a 2-regular sequence in the sense of Allouche and Shallit [1, Definition 2.1] [Cited]. For integer-valued sequences this is equivalent to finite generation of the $\mathbb{Z}$ -module generated by the sections: if their span has dimension d , some d evaluations embed it in $\mathbb{Q}^d$ and that module in $\mathbb{Z}^d$ [Math]. Coons [2, Theorem 3.2] proved that φ is not k -regular for any $k \geq 2$ [Cited]. The unconditional theorem [LEAN SOURCE](#) proves the case $k = 2$ independently. For integer coefficient and carry sequences, the recurrence transfers non-2-regularity of the coefficients to non-2-regularity of the carry. A proof must include the zero-residue sections, not just the positive residues. The following argument gives an explicit dimension bound.

For a rational-valued sequence f , let $V_e(f)$ be the span of all its sections through levels $0, \dots, e$. Define the backward shift by $(Bf)(0) = 0$ and $(Bf)(n) = f(n-1)$ for $n \geq 1$, and let δ_0 be 1 at 0 and 0 elsewhere. A level- j section of Bf is a section of f if its residue is positive. At residue zero it is the backward shift of $n \mapsto f(2^j n + 2^j - 1)$. Hence

$$V_e(Bf) \subseteq V_e(f) + B(V_e(f)).$$

For integer sequences c, u and a positive integer v , the recurrence $vc(n+1) = 2u(n) - u(n+1)$ gives

$$c = \frac{2Bu - u}{v} + \left(c(0) + \frac{u(0)}{v} \right) \delta_0.$$

Since every section of δ_0 is 0 or δ_0 , it follows that

$$V_e(c) \subseteq V_e(u) + B(V_e(u)) + \mathbb{Q}\delta_0, \quad \dim_{\mathbb{Q}} V_e(c) \leq 2 \dim_{\mathbb{Q}} V_e(u) + 1.$$

In particular, bounded carry ranks would force bounded coefficient ranks. Non-2-regularity of c therefore forces unbounded carry ranks. This is an ordinary proof, not a claim of a new Lean declaration. It also gives the generic bound $\dim_{\mathbb{Q}} V_e(u) \geq 2^{e-1}$ for $c = \varphi$ and $e \geq 1$, using the coefficient rank $2^e + 1$. The factor 2 is not an additive $O(1)$ loss, and this bound concerns all sections, including level zero. [Math]

The source theorem for φ is sharper: the carry sections at all residues of levels $1, \dots, e$ have rank at least $2^e - 1$. It uses the independence of the odd-residue totient sections. Neither lower bound is an irrationality proof, since rational coefficient series can have nonregular carries. The separate 5/4 comparison verifies this directly. Any proposed rank upper bound must use additional totient-specific hypotheses not satisfied by the 5/4 comparison sequence.

12.7 What the available results suggest

This subsection compares the stated conditions and the recorded finite observations. It does not infer an asymptotic digit distribution or rank the prospects of different proof methods.

Comparison of conditions. The following table records the conditions themselves, rather than sufficient digit proxies. Lemma 12.2 explains how their phases are related.

Condition	Required assertion	Indices and quantifiers
Block norm	Norm at most $21X/25$ for the truncated phase sum	All $N \in [X, 2X]$ at one depth; arbitrarily large X for each h
Four terms	The four separate budgets of Definition 3.4	One common $X, L, s, \eta; s, \eta$ fixed before X_0
Prime gap	Distance from $\mathbb{Z}$ greater than $\arccos(9/10)/(2\pi)$	The phase $2^{p-h-1}\alpha_h$ at arbitrarily large primes for each h
Full depth	$\mathcal{C}(td, N, td)$ for some $t \geq 1$	Every $d \geq 1$ and every $N \geq 0$
Generic rank bound (false)	A uniform or subexponential upper bound for carries of rational coefficient series	All finite dyadic levels; refuted by the 5/4 comparison

The first four conditions concern the same constant but impose different averaging, sampling and separation requirements. The 11/100 digit-change test in Corollary 12.4 is sufficient for a real-part bound; it is not a replacement for the norm or four-term rows. Only the full-depth row is proved here to be equivalent to irrationality. The last row records a false generic upper bound, not an additional open hypothesis.

Quantitative estimates. The counterexamples identify missing assumptions in the specific arguments tested here. They do not imply that the binary digits of S are random, normal, or unusually structured. A constructed counterexample can disprove an implication without resembling S in other respects. Likewise, failure of one congruence construction does not exclude every use of the Chinese remainder theorem.

Additional arithmetic information. Table 6 records finite observations: the sampled digit-change proportions are near 0.50, above the sufficient threshold 0.11, and the displayed normalized cosine sums are below 0.89. These computations are compatible with the proposed conditions. They do not establish those conditions for every shift and arbitrarily large blocks, or prove a fair-coin model for the digits. The relevant quantified estimates remain unproved here.

The recorded small margins near a forbidden endpoint also require careful interpretation. One small margin disproves a proposed lower bound larger than that margin at that index. It does not rule out every positive margin on a suitable subsequence, or every estimate with scale-dependent bounds. No numerical census in this record determines which of these possibilities holds asymptotically.

Scope of the comparison. Corollary 12.4 specifies one sufficient digit statistic: for every fixed h , the digit-change proportion must reach at least 11/100 on arbitrarily large blocks. A statement for one shift, a smaller positive proportion, or finitely many blocks is not this hypothesis. An upper bound $o(N)$ on the longest run alone would not imply irrationality; a periodic alternating expansion has bounded runs. The prime-index condition in (3) is another proposed sufficient estimate. Its exact algebraic decomposition does not establish the required correlation bound.

13 Remaining arithmetic questions

Erdős #249 asks whether $S = \sum_{n \geq 0} \varphi(n)/2^n$ is irrational. It is [Open]: nothing in this corpus decides it.

This section collects quantified conditions on the scaled tail $R_N = \sum_{j \geq 0} \varphi(N+1+j)2^{-(j+1)}$ that suffice for irrationality. The implication theorems and the arithmetic assumptions have different status: the former are proved in the indicated sources, while the latter are not supplied for the actual totient sequence. [LEAN SOURCE](#)

Each condition is stated with its quantifiers and linked to its formal source. The comparison distinguishes proved equivalences from one-way sufficient conditions. It is not an exhaustive list of possible methods. A finite list of certificates does not stand in for a condition beyond every threshold, and an implication does not establish its antecedent.

Recall that $H_t = H(t) = \text{lcm}(1, \dots, t)$ ([LEAN SOURCE](#)). For nonnegative integers h, N, L , put

$$D(h, N, L) = \sum_{j=0}^{L-1} (\varphi(N+h+1+j) - \varphi(N+1+j))2^{L-1-j}.$$

[LEAN SOURCE](#) The certificate $C(h, N, L)$ is the pair of strict inequalities

$$N + h + L + 2 < D(h, N, L) \bmod 2^L < 2^L - (N + h + L + 2).$$

LEAN SOURCE [Lean] (the definition; decidable, instance at line 84) [n/a] . Thus the least nonnegative residue must lie more than $N + h + L + 2$ from either endpoint. For fixed h, N , this excluded radius grows linearly with L , while its proportion of the modulus tends to zero.

13.1 Certificate completeness

For each fixed $h, N \geq 0$, a certificate at some *unrestricted* depth exists exactly when the tail difference is nonintegral:

$$(\exists L \geq 0, C(h, N, L)) \iff R_{N+h} - R_N \notin \mathbb{Z}.$$

LEAN SOURCE (The same equivalence is also stated at **LEAN SOURCE**.) [Lean] [uniform] (holds for all h, N) .

This equivalence removes the existential depth from the unrestricted certificate conditions below, including the diagonal and LCM-grid forms. It does not by itself supply a depth bound, a prescribed separation margin, a common depth for a family, or an exponential-sum estimate. Such requirements must be retained unless a separate argument removes them. In particular, completeness does not establish any of the quantified nonintegrality assertions needed for irrationality.

13.2 The original quantified residue condition

The quantified residue condition.

$$S \notin \mathbb{Q} \iff \forall h \geq 1 \forall N_0 \exists N \geq N_0 \exists L, C(h, N, L).$$

Here $C(h, N, L)$ is the finite residue test defined earlier; the universal quantifiers over h and N_0 belong to the displayed assertion, not to that finite test. **LEAN SOURCE** [Lean] (the equivalence is proved; the quantified existence assertion is not proved) [cofinal] .

The reverse direction is by contradiction against the tail-period law: if S were rational, the eventual-period theorem (**LEAN SOURCE**, [Lean], unconditional) supplies a period $h > 0$ and pre-period N_0 with $R_{N+h} - R_N \in \mathbb{Z}$ for all $N \geq N_0$; the certificate hypothesis instantiated at that (h, N_0) produces an $N \geq N_0$ where $R_{N+h} - R_N \notin \mathbb{Z}$, a contradiction via nonintegrality from a certificate (**LEAN SOURCE**).

For the forward direction, irrationality gives pointwise non-integrality of every positive tail shift and certificate completeness supplies a witness already at $N = N_0$.

For fixed h , the required certificate may choose both $N \geq N_0$ and L . It is not required at every N or every depth. Its error radius is $N + h + L + 2$, so for fixed h, N it grows only linearly in L , whereas the modulus is 2^L . The missing assertion is that an admissible residue lies outside both endpoint intervals beyond every threshold, for every positive h . The reformulations below specify other ways to choose such witnesses; none establishes this assertion for the actual totient sequence.

13.3 Multiples, diagonals, and LCM grids

The next forms change which tail differences may provide a witness. Each follows from irrationality and each is sufficient for irrationality, by the displayed implications and the preceding equivalence. Their different quantifiers may make different arithmetic estimates applicable, but logical equivalence alone does not provide any of those estimates.

Multiples of a period.

$$\forall h_0 > 0, \forall N_0, \exists m > 0, \exists N \geq N_0, \exists L, C(m \cdot h_0, N, L) \implies S \notin \mathbb{Q}.$$

[LEAN SOURCE](#) [Lean] [cofinal] .

The preceding residue condition implies this one by taking $m = 1$. Conversely, a hypothetical period h_0 also makes every positive multiple mh_0 a period of the tail fractional parts. A certificate for one such multiple contradicts that consequence. The freedom to choose m changes the allowed witnesses, while the quantified assertion remains equivalent to irrationality.

The diagonal.

$$S \notin \mathbb{Q} \iff \forall t_0, \exists t \geq t_0, \exists L, C(H_t, H_t, L).$$

[LEAN SOURCE](#) [Lean] [cofinal] .

The two parameters in a hypothetical rational tail period are absorbed by one LCM index. For $t \geq \max(h_0, N_0)$, both $h_0 \mid H_t$ and $H_t \geq t \geq N_0$ hold. A diagonal certificate then contradicts periodicity through the multiple-period argument; the precise intermediate implication is [LEAN SOURCE](#). Conversely, irrationality and pointwise completeness give a diagonal certificate already at $t = t_0$.

The finite assertion $P(t) := \exists L C(H_t, H_t, L)$ is verified for every $t \leq 82$ ([LEAN SOURCE](#)). The historical bank contained 28 explicit values through $t = 64$ ([LEAN SOURCE](#), depths [6, 5, 7, 7, 9, 14, 15, 14, 21, 22, 23, 26, ...] for $t \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, \dots\}$; end-point [LEAN SOURCE](#) [Cert] [fixed]). The current contiguous

range is still finite and does not establish witnesses at arbitrarily large indices; no certificate at $t = 83$ is claimed here.

The LCM grid.

$$\forall t_0, \exists t \geq t_0, \exists q, m, L, 0 < q \wedge C(m \cdot H_t, q \cdot H_t, L) \implies S \notin \mathbb{Q}.$$

[LEAN SOURCE](#) [Lean] [cofinal] . The

diagonal choice is $q = m = 1$. Allowing other positive grid points therefore enlarges the choice of finite witnesses. To prove sufficiency, suppose rationality supplies a period h and pre-period N_0 . For $t \geq \max(h, N_0)$, all positive multiples of H_t are past the pre-period and their differences are multiples of h . Consequently their tails have the same fractional part: $R_{qH_t+mH_t} - R_{qH_t} \in \mathbb{Z}$ for every $q > 0$ and $m \geq 0$ ([LEAN SOURCE](#), [Lean]). A certificate for any such difference is a contradiction.

The finite-grid condition of Theorem 6.56 offers a second test. At $H = H_t$, choose a finite nonempty $Q \subseteq \mathbb{N}_{>0}$ with $B_q = qH_t + L + 2 < 2^L$ for every q . It requires

$$\forall q_i \in Q \exists q_j \in Q, \quad B_{q_j} < (A_{q_i} - A_{q_j}) \bmod 2^L, \quad A_q = \sum_{j=1}^L \varphi(qH_t + j)2^{L-j}.$$

For every threshold t_0 , such data at some $t \geq t_0$ would imply irrationality. This quantified condition, including the nonemptiness and size assumptions, remains [Open], [cofinal]; [LEAN SOURCE](#). The modulus bound is one-sided at each point; it does not imply a factor-of-two reduction in the number of binary digits.

Equivalent nonintegrality statements Via the completeness iff above, the diagonal and cone forms restate, *exactly*, with no reference to C at all:

$$\forall t_0, \exists t \geq t_0, R_{2H_t} - R_{H_t} \notin \mathbb{Z} \iff \text{diagonal certificate condition} \implies S \notin \mathbb{Q},$$

[LEAN SOURCE](#) (cone analogue at [LEAN SOURCE](#)) [Lean] [cofinal] .

Thus one can ask directly whether $R_{2H_t} - R_{H_t} \notin \mathbb{Z}$ for arbitrarily large t . Finite instances are known in this record; the universally quantified assertion is not proved here.

13.4 The block exponential-sum condition

A bound for an exponential sum is sufficient for the quantified residue condition. With $e(x) = \exp(2\pi ix)$, the assertion is

$$\text{block norm condition} := \forall h > 0, \forall X_0, \exists X, L, \quad \max(X_0, 1) \leq X \wedge 16(2X + h + L + 2) \leq 2^L \wedge \left\| \sum_{N \in [X, 2X)} e(D(h, N, L)/2^L) \right\| \leq \frac{21}{25}X.$$

[LEAN SOURCE](#)

[Open] (the universally quantified assertion is not proved) [cofinal] .

$$\text{block norm condition} \implies S \notin \mathbb{Q}.$$

[LEAN SOURCE](#) [Lean] [cofinal] . The

threshold is X_0 , while X and L are chosen witnesses and may depend on both h and X_0 . Taking $X_0 = N_0$ gives a certificate at some $N \in [X, 2X)$, hence at $N \geq N_0$. The elementary reason is that, if every residue failed the test, each would be within $(2X + h + L + 2)/2^L \leq 1/16$ of an integer. Every summand would then have real part at least $\cos(\pi/8) > 21/25$, contradicting the norm bound. The implication is proved; the bound on suitable blocks beyond every threshold remains to be established.

The underlying finite lemma is more general: it applies to any nonempty finite $T \subseteq [0, 2X)$ with the stated size bound and $\left\| \sum_{N \in T} e(D(h, N, L)/2^L) \right\| \leq (21/25)|T|$. Taking $T = [X, 2X)$ gives the full-block implication. Allowing a chosen subset weakens the hypothesis; it does not strengthen it. The source [LEAN SOURCE](#) uses the elementary block lemma [LEAN SOURCE](#) ([Lean], unconditional as an implication). The real-part version uses $\text{Re} \sum_{N \in T} e(D(h, N, L)/2^L) \leq (9/10)|T|$. Its proof uses $\cos(\pi/8) > 9/10$ and averaging. An unspecified saving below $|T|$ is not the stated hypothesis.

What supplying this form would take: genuine cancellation for the additive character $e(D(h, N, L)/2^L)$ at the growing dyadic moduli permitted by the size bound. The quantifiers require suitable blocks beyond every threshold for each fixed $h > 0$; they do not

require one threshold that works simultaneously for all h . [LEAN SOURCE](#) gives a comparison in which every weighted phase equals 1 while the specified determinant remains nonzero. Thus nonvanishing of that determinant alone cannot prove cancellation. A successful use of a determinant would need further information connecting its entries to the phase sum; the comparison does not exclude every such argument.

13.5 The actual tail difference on the LCM diagonal

The same irrationality question can be expressed on the sparse sequence $H = H_{2^a}$. The equivalence below is useful because its right-hand side contains only the actual tail differences Ω_a , not an auxiliary approximation.

The exact equivalence.

$$S \notin \mathbb{Q} \iff \forall a_0, \exists a \geq a_0, \Omega_a \notin \mathbb{Z}.$$

[LEAN SOURCE](#) [Lean] [n/a] (an unconditional equivalence, not a supply) , where $\Omega_a := 2^H(2^H - 1)S - (\Phi_{2H} - \Phi_H)$, $H = H_{2^a}$ ([LEAN SOURCE](#)). The right-hand side is an equivalent reformulation of irrationality, not a weaker hypothesis already established by the finite examples. The following sufficient conditions are attempts to prove it.

An impossible complete divisibility pattern Let $a \geq 8$, $H = H_{2^a}$ and $J, K, m \in \mathbb{N}$, with

$$m \leq K, \quad J + K + a + 6 < 2 \cdot 2^a, \quad B := 2H + J + K + 2 < 2^m.$$

It is impossible that

$$2^{r+1} \mid \delta_{2^a}(J + K - m + r + 1) \quad \text{for every } 0 \leq r < m.$$

[LEAN SOURCE](#) [Lean] [bounded] . Indeed, the last requirement would make $\delta_{2^a}(J + K)$ a multiple of 2^m , whereas the short-window bounds give $0 < \delta_{2^a}(J + K) < B < 2^m$. The positivity input is

$$\delta_{2^a}(j) > 0 \quad (a \geq 8, 0 < j < 2 \cdot 2^a),$$

[LEAN SOURCE](#), without a rationality hypothesis.

Omitting only the last divisibility requirement leaves the partial pattern in Proposition 6.86. It additionally requires $\delta_{2^a}(J + K) \leq 2^m - B$. The resulting necessary equality is *penultimate*:

$$\delta_{2^a}(J + K - 1) = 2^{m-1}, \quad B < 2^m < 2B.$$

[LEAN SOURCE](#) [Lean] [bounded]. The terminal difference is not pinned to 2^{m-1} . Neither this necessary restriction nor the full-pattern impossibility supplies an unbounded family of successful partial patterns.

The remaining upper-endpoint residue

$\mathcal{G}_a(J, K, m) := m \leq K \wedge 2H + J + K + 2 < 2^m \wedge D(H, H + J, K) \bmod 2^m \leq 2^m - (2H + J + K + 2)$.

LEAN SOURCE (definition; an unbounded family satisfying it is [Open]) [bounded]

The positivity theorem gives

$$R_{2H+J} - R_{H+J} > 0 \quad \text{if } a \geq 8 \text{ and } J + a + 6 < 2 \cdot 2^a.$$

LEAN SOURCE For $a \geq 8$, the residue test at depth K uses the stronger bound $J + K + a + 6 < 2 \cdot 2^a$. Under these hypotheses, an integral initial tail difference would give an integer terminal carry $e = R_{2H+J+K} - R_{H+J+K}$ with $0 < e < 2H + J + K + 2$. The tail identity is

$$D(H, H + J, K) = 2^K(R_{2H+J} - R_{H+J}) - e.$$

Hence, when $2H + J + K + 2 < 2^K$, it is the residue of $D(H, H + J, K)$, not of e , that equals $2^K - e$: **LEAN SOURCE**. More generally, the bounds $m \leq K$ and $2H + J + K + 2 < 2^m$ give

$$D(H, H + J, K) \bmod 2^m = 2^m - e > 2^m - (2H + J + K + 2),$$

because 2^m divides 2^K and $0 < e < 2^m$. This contradicts $\mathcal{G}_a(J, K, m)$. Consequently, for $a \geq 8$ and $J + K + a + 6 < 2 \cdot 2^a$,

$$\mathcal{G}_a(J, K, m) \implies R_{2H+J} - R_{H+J} \notin \mathbb{Z}.$$

LEAN SOURCE The one-sided inequality excludes the upper-endpoint interval forced by integrality. Positivity alone does not establish that inequality.

The cofinal target built from it:

cofinal upper-endpoint condition $:= \forall a_0, \exists a, K, m, \quad a_0 \leq a \wedge 8 \leq a \wedge K + (a+6) < 2 \cdot 2^a \wedge \mathcal{G}_a(0, K, m)$.

LEAN SOURCE [Open] [cofinal] .

$$\text{cofinal upper-endpoint condition} \implies S \notin \mathbb{Q}.$$

LEAN SOURCE [Lean] [cofinal] .

Five sufficient conditions for that residue inequality The next five quantified conditions each imply irrationality. Conditions (i)–(iv) reach the upper-endpoint test; (v) gives nonintegrality directly. Their proven relations are summarised in Proposition 6.65.

(i) For every a_0 , there are $a \geq \max(a_0, 8)$ and $m \geq 0$ with $H = H_{2^a}$ such that

$$\begin{aligned} m + 1 + a + 6 &< 2 \cdot 2^a, & 2H + m + 3 &< 2^m, \\ 2H + m + 2 &\leq (D(H, H, m) + \delta_{2^a}(m + 1)) \bmod 2^m \leq 2^m - (2H + m + 3), \end{aligned}$$

where $\delta_{2^a}(j) = \varphi(2H+j) - \varphi(H+j)$. The size condition includes the larger depth $m+1$, so the conclusion remains valid whichever of the two adjacent depths gives the certificate. **LEAN SOURCE** [Open] [cofinal] . Sufficiency: **LEAN SOURCE**, direct endpoint **LEAN SOURCE**.

- (ii) For every a_0 , there is $a \geq \max(a_0, 14)$ with the following property. Put $H = H_{2^a}$ and choose q so that $2q + 1$ is the least odd integer not smaller than $\lfloor \log_2 H \rfloor + 10$. Define integers $W_{a,j}$ recursively by

$$W_{a,0} = 0,$$

$$W_{a,j+1} = 4W_{a,j} + \frac{\delta_{2^a}(2j+3) - 2\delta_{2^a}(2j+2) + \delta_{2^a}(2j+4)}{2}.$$

The increments $\delta_{2^a}(s)$ are even, so the recurrence is integral. The required inequality is

$$H + q + 2 \leq W_{a,q} \bmod 4^q \leq 4^q - (H + q + 2).$$

At depth $m = 2q + 1$, the residue in (i) equals $2(W_{a,q} \bmod 4^q)$. Dividing its two endpoint bounds by 2 and using integrality gives exactly the threshold $H + q + 2$. [LEAN SOURCE](#) [Open] [cofinal]. The width is $H + q + 2$ and the depth is the prescribed guarded depth; both must be retained when comparing this with a fixed-fraction band. Sufficiency: [LEAN SOURCE](#).

- (iii) Use exactly the quantifiers and prescribed depth of (ii). Let $r = W_{a,q} \bmod 4^q$ and define its centred representative by

$$u_{a,q} = \begin{cases} r, & 2r \leq 4^q, \\ r - 4^q, & 2r > 4^q. \end{cases}$$

Thus $-4^q/2 < u_{a,q} \leq 4^q/2$; when a midpoint occurs, the positive representative is used. The equivalent inequality is

$$H + q + 2 \leq |u_{a,q}|.$$

[LEAN SOURCE](#) [Open] [cofinal]. At the prescribed depth the size bound $2(H + q + 2) \leq 4^q$ holds. For a residue in $[0, 4^q)$, the band in (ii) is therefore exactly this absolute-value condition. The equivalence, including its quantifiers, is proved: [LEAN SOURCE](#) [Lean].

- (iv) Allow the odd depth to vary, but retain both size bounds. For every a_0 , require $a \geq \max(a_0, 8)$ and $q \in \mathbb{N}$ with

$$2q + 2 + a + 6 < 2 \cdot 2^a, \quad 2(H + q + 2) \leq 4^q,$$

$$H + q + 2 \leq |u_{a,q}|, \quad H = H_{2^a}.$$

[LEAN SOURCE](#) [Open] [cofinal]. The first bound keeps the adjacent depth inside the short window; the second makes the required residue band nonempty. One implication passes through (i), and another gives the two-sided carry exclusion below: [LEAN SOURCE](#), direct endpoint [LEAN SOURCE](#).

- (v) Retain the full quantifiers and the two size bounds in (iv), but replace its absolute-value inequality by

$$\delta_{2^a}(2q+2) \leq 2u_{a,q}.$$

This compares the signed centred residue with the terminal totient difference. It is not an unrestricted assertion at an arbitrary q . [LEAN SOURCE](#) [Open] [cofinal] . Direct endpoint [LEAN SOURCE](#).

A further sufficient terminal-carry condition With $H = H_{2^a}$, the unproved quantified condition is

$$\begin{aligned} \forall a_0 \exists a \geq \max(a_0, 8) \exists q : \\ 2q+2+(a+6) < 2 \cdot 2^a, \quad 2(H+q+2) \leq 4^q, \\ 2u_{a,q} \leq \delta_{2^a}(2q+2) - (2H+2q+3) \quad \text{or} \quad \delta_{2^a}(2q+2) \leq 2u_{a,q}. \end{aligned}$$

[LEAN SOURCE](#) Either alternative gives nonintegrality at that scale, so the quantified condition implies irrationality [LEAN SOURCE](#).

The proof assumes an integer $z = \Omega_a$ and uses

$$2u_{a,q} = \delta_{2^a}(2q+2) - c_{H,H,z}(2q+1).$$

[LEAN SOURCE](#) The carry is then a positive tail difference, strictly smaller than $2H+2q+3$. The source name referring to a negative endpoint concerns the representative *minus* this carry [LEAN SOURCE](#). The first alternative contradicts the upper bound; the second contradicts positivity, which is the terminal-dominance argument [LEAN SOURCE](#).

Both arguments are conditional contradictions, not evidence against either unproved estimate. No converse from nonintegrality to either finite inequality is asserted. In particular, the exact identity must not be described as making this sufficient condition necessary.

Certificates in short windows For every $t, j \in \mathbb{N}$, the arithmetic difference on the LCM progression is exactly

$$\delta_t(j) = \varphi(2H_t + j) - \varphi(H_t + j).$$

No coprimality restriction on j is needed ([LEAN SOURCE](#), [Lean]). Consequently the corresponding window numerator is

$$D(H_t, H_t, L) = \sum_{j=1}^L \delta_t(j) 2^{L-j},$$

and applying the same residue and error bounds gives precisely the test $C(H_t, H_t, L)$ ([LEAN SOURCE](#), [Lean]). Thus the quantified assertion

$$\forall a_0 \exists a \geq a_0 \exists L < 2 \cdot 2^a, \quad C(H_{2^a}, H_{2^a}, L)$$

LEAN SOURCE, [Open], [cofinal], , implies irrationality through **LEAN SOURCE**. It restricts the diagonal indices to powers of two and imposes the additional bound $L < 2 \cdot 2^a$. Pointwise completeness allows an unspecified finite depth; it does not imply a witness within this short window.

The displayed certified examples have $a = 4, L = 23$ and $a = 6, L = 93$: **LEAN SOURCE** [Cert] [fixed]. The bounded statement **LEAN SOURCE** ([Lean], [bounded]) uses the single witness $(a, L) = (6, 93)$ to cover all thresholds $a_0 \leq 6$. It makes no assertion for larger thresholds, or that these are the only successful exponents.

A sufficient approximation bound Put $H = H_{2^a}$ and define the finite rational and its error bound by

$$\rho_{a,q} = \frac{D(H, H, 2q+1) + \delta_{2^a}(2q+2)}{2^{2q+1}}, \quad \varepsilon_{a,q} = \frac{2H + 2q + 3}{2^{2q+1}}.$$

For every $a, q \in \mathbb{N}$,

$$|\Omega_a - \rho_{a,q}| < \varepsilon_{a,q}.$$

LEAN SOURCE [Lean] [uniform] . Let $2q_a + 1$ be the least odd integer not smaller than $\lfloor \log_2 H \rfloor + 10$, as in condition (ii). The sufficient quantified separation condition is

$$\forall a_0 \exists a \geq \max(2, a_0) \forall z \in \mathbb{Z}, \quad \frac{1}{32} + \varepsilon_{a,q_a} \leq |\Omega_a - z|.$$

[Open] [cofinal] ; sufficiency is proved in **LEAN SOURCE** ([Lean]). The triangle inequality then gives $|\rho_{a,q_a} - z| > 1/32$ for every integer z . The hypothesis concerns the real tail Ω_a , not merely its computable approximant. The truncation estimate transfers the stated separation, but does not itself establish it. Also, the depth is prescribed by a ; it is not a free parameter in this hypothesis.

13.6 An equivalent test using two leading residue bits

Proposition 6.87 gives an equivalent certificate condition: there are $s, b \in \mathbb{N}$ such that either $\mathcal{C}(h, N + s, b + 1)$ holds, or both

$$N + s + h + b + 4 < 2^b, \quad 2^b \leq D(h, N + s, b + 2) \bmod 2^{b+2} < 3 \cdot 2^b.$$

LEAN SOURCE [Lean] [uniform] . The forward proof selects b using the depth L of the first certificate, so this is an encoding of a witness rather than a bound on how far one must search. Existence for the actual totient sequence remains the missing assertion. A use for another coefficient sequence would require its own recurrence and tail bounds.

13.7 Farey growth and the failed generic rank argument

The Farey growth law below is a separate unproved assertion. The rank statement is retained only to record a counterfactual sufficient input and the reason that the generic compression route to it is closed; it is not a second open problem supported by the present argument.

The hypothetical rank bound. Unconditionally, for every level $e \geq 1$, the dyadic totient basis of $2^e + 1$ sequences is linearly independent over $\mathbb{Q}$ (via CRT and Dirichlet's theorem on primes in arithmetic progression): [LEAN SOURCE](#) [Lean] [uniform] . Consequently, if $S \in \mathbb{Q}$, there is an integer $v > 0$ with $vS \in \mathbb{Z}$. Its integral carry $u_N = vR_N$ satisfies, for every $e \geq 0$,

$$\dim_{\mathbb{Q}} \text{span}\{n \mapsto u_{2^j n + r} : 1 \leq j \leq e, 0 \leq r < 2^j\} \geq 2^e - 1.$$

It obeys $u_{N+1} = 2u_N - v\varphi(N+1)$ and $0 \leq u_N \leq v(N+2)$. [LEAN SOURCE](#) [Lean] [uniform] .

A bound independent of e for these same carry-section ranks would contradict the displayed lower bound; so would an upper bound that is $o(2^e)$. Rationality and linear coefficient growth alone do not give either conclusion: the fixed sequence constructed in Section 12.6 has sum $5/4$ and the same rank lower bound. An incompatible upper bound for the actual totient carry would have to use information not shared by that comparison sequence.

The most direct generic construction is also impossible: no integer certificate satisfying the following conditions ($Q \cdot v \cdot A = \text{boundary}$ with $|\text{boundary}| < Q \cdot v$ and $A \neq 0$) can exist; [LEAN SOURCE](#) [Lean]; and the full family is proved infinite-dimensional in span, the case $k = 2$ of Coons's non-regularity theorem (§10.8): [LEAN SOURCE](#) [Lean].

The conditional source statement gives a common eventual period modulo v for the sections of this same u , together with their unbounded rational ranks: [LEAN SOURCE](#) [Lean]. These conclusions concern different properties of the same carry; they supply no rank upper bound.

Growth of the Farey denominator bound. The finite denominator exclusion for S comes from a Farey-gap denominator exclusion at a single fixed window $K = 240$:

$$S \neq a/b \quad \text{for all } a \in \mathbb{Z}, \quad 1 \leq b \leq 79639646646701375323355774875831053, \quad \gcd(a, b) = 1.$$

[LEAN SOURCE](#) [Lean] [bounded] . This bound is *sharp* at $K = 240$; the mediant denominator $q = 79639646646701375323355774875831054$ is the exact first failing denominator ([LEAN SOURCE](#), [Lean]); so the same interval cannot exclude every rational at the next denominator. A different window requires a new certificate or a uniform argument; the fixed-window calculation supplies neither.

A sufficient extension is a function $g(K) \rightarrow \infty$ for which every $(N = 1, K)$ gap check excludes all rationals of reduced denominator at most $g(K)$. More generally, unbounded valid exclusion bounds along a sequence of windows suffice. This is a statement about the certified intervals. The record does not prove it equivalent to a lower bound on the continued-fraction denominators of S . [Open] [cofinal] .

For p prime and $e \geq 1$, a successful unit-gap certificate at denominator p^e has zero candidates or a single candidate divisible by p ; [LEAN SOURCE](#) [Lean]. The success hypothesis is essential: the bound is not a statement about every window at that denominator. Nor does it bound the eventual denominator cutoff. The prime-power restriction cannot be omitted, as the modulo-15 example in Observation 6.114 shows. No obstruction to a uniform estimate as K varies is proved here.

13.8 Summary of implications

The second column gives the relation to the quantified residue condition or to the explicitly named intermediate condition.

Condition	Proved relation	Source
Quantified $\mathcal{C}(h, N, L)$ condition (the base condition)	equivalent to $S \notin \mathbb{Q}$	LcmConeFlatness.lean:412
Certificates for multiples of a period	equivalent via the base condition: take multiplier one in one direction and use the cited implication in the other	CarrySurvivorExtinction.lean:502
Diagonal certificates beyond every threshold	equivalent to $S \notin \mathbb{Q}$; one universally quantified threshold	LcmConeFlatness.lean:426
Certificates on LCM grids beyond every threshold	equivalent via the base condition; diagonal certificates have $q = m = 1$	CertificateKernel.lean:19014
Quantified finite-grid condition	sufficient; requires each one-sided size bound	CertificateKernel.lean:19140
Nonintegral diagonal or LCM-grid tail differences	equivalent to the corresponding unrestricted-depth certificate condition	CertificateKernel.lean:19034, CertificateKernel.lean:19046
Block norm condition	sufficient for the quantified residue condition; no converse proved	FirstHarmonicPivot.lean:83
$\Omega_a \notin \mathbb{Z}$ beyond every threshold	equivalent to $S \notin \mathbb{Q}$ itself	TotientActualLcmOrbitNonintegrality.lean:37
Short-window certificate condition	sufficient; imposes an additional short-depth restriction at powers of two	TotientActualLcmOrbitArithmetic.lean:2107
Upper-endpoint residue condition beyond every threshold	sufficient for $\Omega_a \notin \mathbb{Z}$ beyond every threshold	TotientActualLcmTopEdgesTaircase.lean:1325
Adjacent-depth residue band, condition (i)	sufficient for the upper-endpoint residue gap	TotientActualLcmTopEdgesTaircase.lean:1334
Prescribed-depth residue band, condition (ii)	equivalent to condition (iii)	TotientActualLcmTopEdgesTaircase.lean:1349

Condition	Proved relation	Source
Prescribed-depth centred magnitude, condition (iii)	equivalent to condition (ii)	TotientActualLcmTopEdgeS taircase.lean:1471
Variable-depth centred magnitude, condition (iv)	sufficient for both the midband and the carry exclusion	TotientActualLcmTopEdgeS taircase.lean:1927
Terminal comparison, condition (v)	sufficient for carry exclusion; one-sided	TotientActualLcmTopEdgeS taircase.lean:1908
Either terminal-carry exclusion inequality	sufficient; no converse asserted	TotientActualLcmTopEdgeS taircase.lean:1895
Real-tail separation with approximation margin	sufficient; fixes the depth and approximation margin	TotientActualLcmOrbitSep aration.lean:305
Carry-rank upper bound	the rational 5/4 comparison refutes a bound from rationality and the stated coefficient hypotheses alone	ParityPerturbedRationalC ontrol.lean:478
Unbounded certified denominator exclusions	sufficient; the cited result is one finite exclusion, not an unbounded family	GapFareyBound.lean, CertificateKernel.lean: 18384
Shifted short certificate or two-bit residue test	equivalent existence test with unrestricted depth; no a priori search-depth bound follows	TotientActualLcmTopEdgeS taircase.lean:844

Reading the table. “Sufficient” records only the indicated implication. It does not assert a converse or a strict comparison of strength. “Equivalent” records two proved directions, either directly or through the base equivalence with $S \notin \mathbb{Q}$. Logical equivalence does not by itself provide a direct construction from the finite witnesses in one formulation to those in another.

13.9 What remains to be proved

The implications above leave explicit arithmetic assertions to prove for φ . Two types of estimate can be stated directly, without assuming either that they follow from known methods or that they exhaust the possible approaches.

Exponential sums. For each fixed $h > 0$ and every threshold X_0 , the block norm condition asks for some $X \geq \max(X_0, 1)$ and L satisfying both the modulus bound and

$$\left| \sum_{N \in [X, 2X)} e(D(h, N, L)/2^L) \right| \leq \frac{21}{25} X.$$

The saving $21/25$ is fixed, but the chosen block and depth may depend on h and X_0 ; uniformity of these choices in h is not required. The preceding four-term decomposition and the proved counting bounds identify contributions to this sum. The two remaining weighted estimates must still be obtained with the same parameters. The record therefore gives an exact reduction and some of its estimates, not a proof of the displayed cancellation bound.

The last totient difference. The endpoint criteria compare

$$\delta_{2^a}(2q+2) = \varphi(2H+2q+2) - \varphi(H+2q+2), \quad H = H_{2^a},$$

with twice the centred residue $u_{a,q}$. An appropriate inequality, proved for arbitrarily large a with the stated bounds on q , would contradict the carry interval required by integrality. The exact identity explains why such an inequality would suffice; it does not supply the inequality. The finite examples through $t = 82$, and the short-window examples at $a = 4, 6$, do not establish an asymptotic pattern.

Declarations

Further background. Erick Wong posted a proof that $\sum_{n \geq 1} (\varphi(n) \bmod k) k^{-n}$ is irrational for every integer $k > 2$ on Mathematics Stack Exchange on 29 March 2015 [5]; there the base equals the modulus. The Möbius-square form of S and its coprimality interpretation were posted publicly before this record, as credited in §5.4. Kovač and Tao show that suitable subsums of Lambert series in distinct integer bases can be rational [16, Theorem 2.3]; the freedom to choose those subsums is absent for S . The Lambert-series factorisations of Merca [15, Theorem 1.2] and of Merca and Schmidt [14] are transform background for the Möbius–Mersenne coordinate. Balasubramanian, Giri and Srivastav prove asymptotics for finite shifted convolutions of divisor-convolution functions, including $\varphi(n)/n$, uniformly for $|h| \leq x/2$ at cutoff x [17, Theorem 2.2]. Their partial-summation extension uses differentiable weights with bounded derivative on the summation interval [17, Rem. 2.9]. Applying that observation here would require control of the weighted error for the present phases; no such estimate is established in this record. The cited results are not themselves the dyadic residue separation required by Sep. The remaining sources are cited where their hypotheses are used.

At the generating-function level, the published height criterion of Adamczewski, Bell and Smertnig gives k -regularity for a k -Mahler series with integer coefficients of polynomial growth [33, Thm. 1.2(b), p. 2528]. For the totient coefficients the logarithmic heights are $\log \max(1, \varphi(n)) \leq \log n$ for $n \geq 1$, so Coons’s nonregularity already rules out k -Mahler equations in every base. The later multiplicative classification of Bell and Smertnig provides another route [30, Thm. 1.3 and p. 3]. The algebraic and D -finite classifications of Bell, Bruin and Coons belong to the same function-level context [31, Thms. 1.5–1.6]. They do not by themselves decide the value at $z = 1/2$.

Automatic prime-sampling theorems address a different issue. Adamczewski, Drmota and Müllner identify a computable local coprimality condition governing the relevant logarithmic densities, with a conditional ordinary-density transfer [32, Thm. 1.4 and Rem. 1.5]. Their theorem does not assert that every output state has positive density on a prescribed prime progression, nor does it prove independence with automatic coefficient weights. Any use for that purpose needs its own compatibility argument. [Cited]

Sources and data. The linked Lean declarations, fixed toolchain, and library manifest are published in the companion repository. The ordinary arguments in these papers explain the mathematics; the pinned declarations record the corresponding formal statements. Their hypotheses must be checked separately when an implication is applied.

Formalisation coverage and remaining dependencies

Every theorem, lemma, proposition and corollary of this record other than the five listed below has a Lean statement of the same assertion, with the same hypotheses, checked by the Lean kernel using only the axioms `propext`, `Classical.choice` and `Quot.sound`, in the development at revision 181078b6b009.

Proposition 12.7, one of the five, was proved later. It has a Lean statement of the same assertion, checked by the Lean kernel using only those three axioms, at revision e4ba6841dfdb; the paragraph after its proof describes the Lean argument.

Propositions 5.17 and 6.30 share one input. The displayed three-term identity is unconditional and checked in both. The development verifies the reduction of the irrationality clause to the theorem of Postelmans and Van Assche that $1, \zeta_q(1)$ and $\zeta_q(2)$ are linearly independent over $\mathbb{Q}$ for $q = 1/p$ with an integer $p \geq 2$ [10, Theorem 1.3, p. 3]. A formal proof of that input is not included in the verified development, so those two endpoints remain conditional on it. Proposition 6.102 is the same situation with a different input. Its four Lambert identities and the identification of $L(\text{Id})$ with $\sum_{m \geq 1} \sigma(m)/2^m$ are checked, and the development verifies the reduction of the transcendence of that value to Nesterenko's theorem on the algebraic independence of the three Eisenstein values at an algebraic point of the punctured unit disc [11, Cor. 2, p. 1320].

Proposition 12.8 is checked in Lean with the prime number theorem as its one input (`prop_badcof`, revision 76f44c5329c0). Its density $D(\eta)$ enters as the premise that $B(\eta)$ has a natural density, which the statement takes from Schoenberg's theorem.

A theorem whose own statement is conditional is formalised exactly as stated. Propositions 5.17, 6.30, 6.102 and 12.8 are asserted unconditionally in this record, and all four have Lean counterparts carrying a named input.

Concordance of statements and Lean declarations. Each result of this record that has a kernel-checked Lean statement of the same assertion is listed below with the declarations that jointly state it. Each name links to its declaration at revision a25cb360bef8. Where the Lean statement is stronger than the printed one and implies it by an immediate specialisation, the entry says so. An entry marked *compared* was also checked independently: a restatement of the same declarations against Mathlib alone, together with its proof, was verified by Comparator (`leanprover/comparator`) in a clean continuous integration environment, in the run named by its number. Comparator trusts the restated statement, so the correspondence between the printed result and that statement is the one this concordance records.

Theorem 1.3: `tsum_totient_div_pow_two_ne_ratCast_of_den_le_79639646646701375323355774875831053, gap_check_window_1_240_first_failure`. *Compared*, runs 35544127144, 35715779405.

Proposition 1.4: `certifiedKill_diagonal_all_imported_through_t64, certifiedKill_diagonal_t64, certifiedKill_all_upto_sixteen, certifiedKill_67_300, certifiedKill_81_300, certifiedKill_97_300, certifiedKill_101_300, certifiedKill_121_300, certifiedKill_125_300, certifiedKill_127_300, certifiedKill_128_300, exists_diagonalKill_le_82`. *Compared*, runs 35643815458, 35674034595, 35682858662.

Proposition 1.5: `actualLcmTailDiff_shift_pos, actualLcm_integral_forces_topEdgeResidue, carryOrbit_eq_tail_diff`. *Compared*, run 35674034595.

Proposition 1.6: `not_irrational_totientSeries_implies_unbounded_carryRank_unconditional`. *Compared*, run 35544127144.

Proposition 1.7: `not_irrational_totientSeries_implies_mod_period_and_unbounded_rank`. *Compared*, run 35544127144.

Proposition 1.8: `periodMultipleKillSupply_iff_irrational`,
`irrational_totient_series_iff_certificate_supply`,
`irrational_totient_series_iff_lcm_diagonal_certificate_supply`,
`irrational_totientSeries_iff_cofinalDirectedLcmCertificateSupply`,
`dtwWindowSeparatedPairs_iff_irrational_totient_series`. *Compared*, runs
35643815458, 35674034595.

Proposition 1.9:
`exists_totientParity_arbitrarilyManySeparatedCarry_rational_countermodel`,
`tsum_parityCoboundaryWeight_eq_three_halves`. *Compared*, run 35624228171.

Lemma 2.1: `certificate_sound`, `scaled_difference`, `truncation_error_bound`. *Compared*,
run 35643815458.

Lemma 2.2: `generic_tail_period`, `scaled_tail_split`. *Compared*, run 35643815458.

Theorem 2.4: `gamma_le`, `gamma_prefix`, `discrepancy_prefix`, `exact_series`, `value_cast`,
`exact_denominator`, `no_certificate_after_prefix`, `gamma_not_separation`. *Compared*,
run 35643815458.

Corollary 2.5: `no_uniform_prefix_rule`, `gamma_not_separation`. *Compared*, run 35643815458.

Proposition 2.7: `three_particular_equivalences`,
`two_point_sample_numerical_requirement`. *Compared*, runs 35643815458, 35674034595.

Proposition 2.11: `twoAdic_pulse_construction_never_certifies`,
`twoAdic_pulse_defining_congruence`, `twoAdic_pulse_error_bound`. *Compared*, runs
35643815458, 35674034595.

Proposition 2.13: `balancedPulse_common_history`, `balancedPulse_tail_at`,
`balancedPulse_series`, `balancedPulse_label_lower_bound`,
`balancedPulse_no_decoder_from_common_state`,
`affineBinaryOrbit_difference_and_reset`, `fixed_precision_carry_completion`.

Proposition 2.14 (the Lean statement is stronger):
`b6_retained_dyadic_sections_independent`,
`b6_compressed_adjoint_identity_impossible`,
`b6_mobius_incidence_unimodular_and_injective`, `b6_adjugate_tail_cost_floor`,
`b6_synthetic_sequence_prescribed_differences`,
`b6_synthetic_shift_combinations_same_form`, `b6_mobiusMersennePrefix_eq_icc_sum`,
`b6_mobiusMersenne_rung_estimates`,
`b6_mobiusMersenneTheta_two_eq_totientSeries_sub_half`,
`b6_rankOneSubrankQuotient_sub_totientSeries_offset_gt`. *Compared*, run 35674034595.

Proposition 2.15: `exists_rational_parityComparison`,
`parityComparisonProperties_do_not_imply_irrational`. *Compared*, run 35643815458.

Proposition 5.3: `two_pow_mul_totient_series_eq`. *Compared*, run 35643815458.

Lemma 5.4: `farey_gap`. *Compared*, run 35715779405.

Proposition 5.5: `totient_carry_residue_window_1_240_eq`,
`gap_check_window_1_240_le_79639646646701375323355774875831053`,
`gap_check_window_1_240_first_failure`. *Compared*, runs 35624228171, 35715779405.

Theorem 5.6: `tsum_totient_div_pow_two_ne_ratCast_of_den_le_7963964664670137532335`
5774875831053. *Compared*, run 35544127144.

Proposition 5.9: `tsum_visible_coprime_pairs_eq_totient_series`,
`totient_series_eq_half_add_visible_coprime_pairs`. *Compared*, run 35624228171.

Proposition 5.10: `tsum_gcd_layer_pos_coprime_half_eq_one`,
`tsum_pos_pair_both_dvd_half_eq_inv_mersenne_sq`. *Compared*, run 35715779405.

Theorem 5.11: `tsum_visible_coprime_pairs_ne_int_div_of_den_le_39819823323350687661`
677887437915526. *Compared*, run 35624228171.

Theorem 5.12: `tsum_moebius_div_two_pow_sub_one_sq_ne_int_div_of_den_le_39819823323350687661677887437915526`. Compared, run 35624228171.

Proposition 5.14: `mobius_square_reduction, irrational_totient_iff_mobius_square, moebius_three_values`. Compared, runs 35624228171, 35643815458.

Proposition 5.16: `tsum_lambert_linear_weight_sq_pure`. Compared, run 35715779405.

Proposition 5.18: `gcd_moment_identity_three_members, tsum_pos_pair_gcd_half_eq_totient_div_mersenne_sq, sum_gcd_Icc_eq_pillaiP, pillaiP_eq_totient_mul_id, tsum_totient_div_mersenne_sq_eq_gcd_moment_series`. Compared, runs 35643815458, 35715779405.

Theorem 6.1: `specified_euler_tail_period`. Compared, run 35674034595.

Theorem 6.2:

`not_irrational_totientSeries_implies_unbounded_carryRank_unconditional`. Compared, run 35544127144.

Theorem 6.3: `displayed_canonical_and_full_dyadic`. Compared, run 35605146048.

Proposition 6.4: `tail_diff_int_of_den_dvd`. Compared, run 35643815458.

Proposition 6.5: `mobiusNumeratorPolynomial_eval_two, mobiusNumerator_gcd_cyclotomicValue, cyclotomicValue_dvd_baseMobiusShadow_den`. Compared, run 35624228171.

Proposition 6.6: `lcmHeight_upperHalf_product_dvd_den, upperHalfChannel_product_dvd_den_of_coprime_scale, upperHalfChannel_product_dvd_den_of_scale_primeFactors_le`. Compared, run 35643815458.

Proposition 6.7: `upper_half_product_denominator_bounds`. Compared, run 35643815458.

Proposition 6.8: `lcmHeight_scaledMobiusShadow_den_exact, lcmHeight_five_scaledMobiusShadow_den_exact`. Compared, run 35643815458.

Proposition 6.9: `signed_dyadic_clearing, signed_dyadic_sum_ne_zero, scaled_dyadic_sum_odd`. Compared, runs 35624228171, 35643815458.

Proposition 6.10: `small_certificate_windows, small_certificates_and_exclusions`. Compared, run 35674034595.

Proposition 6.11: `sixteen_certificate_windows, sixteen_certificates_and_exclusions`. Compared, run 35674034595.

Proposition 6.12: `historical_table_size_and_initial_depths, historical_table_and_complete_band`. Compared, run 35674034595.

Proposition 6.16: `prefix_fractional_part, two_pow_mul_totient_series_eq, tail_diff_mem_int_iff_scaled_series_mem_int`. Compared, runs 35643815458, 35674034595.

Lemma 6.17: `certificate_logarithmic_depth, fixed_depth_bounds_indices, certifiedKill_depth_floor`. Compared, runs 35643815458, 35674034595.

Proposition 6.18: `totient_scaled_truncation_error, tail_diff_notMem_int_of_certifiedKill`. Compared, runs 35643815458, 35674034595.

Theorem 6.19: `exists_certifiedKill_iff_tail_diff_notMem_int`. Compared, run 35643815458.

Proposition 6.21: `farey_gap`. Compared, run 35715779405.

Proposition 6.22: `irrational_of_den_mul_abs_sub_tendsto_zero`. Compared, run 35624228171.

Proposition 6.23 (the Lean statement is stronger): `irrational_of_int_mul_near_int, irrational_of_pow_mul_near_int, irrational_of_near_integer_multiples, irrational_of_near_integer_base_powers, one_div_den_le_abs_int_combination, digit_block, not_eventually_periodic, irrational_xi, no_three_consecutive_equal, fract_two_pow_mul_xi_mem_Icc, fract_mem_Icc_of_no_three_equal`.

- one_div_eight_le_dist_xi, not_approaches_int,
not_near_integer_along_powers_of_two, exists_irrational_basePower_bounded_away,
strict_lower_bound_needed, upper_bound_needed_for_every_q, tail_mem_Icc,
irrational_tail_zero, one_div_eight_le_abs_sub_int. *Compared, runs 35624228171, 35643815458.*
- Proposition 6.24: linearIndependent_of_separatedMinorCertificate. *Compared, run 35682858662.*
- Proposition 6.25: rational_difference_exact, rational_cross_numerator_positive,
rational_error_denominator_bound, positive_rational_difference_lower_bound.
Compared, run 35643815458.
- Proposition 6.26: tsum_moebius_lambert_sq,
totient_series_eq_half_add_moebius_mersenne_square. *Compared, runs 35624228171, 35715779405.*
- Corollary 6.27: irrational_totient_iff_moebius_square. *Compared, run 35682858662.*
- Proposition 6.28: tsum_lambert_linear_weight_sq_pure. *Compared, run 35715779405.*
- Proposition 6.31: gcd_moment_identity_three_members,
tsum_pos_pair_gcd_half_eq_totient_div_mersenne_sq, pillaiP_eq_totient_mul_id,
sum_gcd_Icc_eq_pillaiP, tsum_totient_div_mersenne_sq_eq_gcd_moment_series.
Compared, runs 35643815458, 35715779405.
- Proposition 6.33: pair_divisibility_mass, tsum_geometric_multiples,
tsum_pos_pair_both_dvd_half_eq_inv_mersenne_sq. *Compared, runs 35643815458, 35715779405.*
- Proposition 6.34: tsum_pos_coprime_inv_mersenne_eq_one. *Compared, run 35715779405.*
- Proposition 6.35: divisibility_mass, cylinderMass_eq_divisibility_mass_mul,
cylinder_mediant_split, cylinder_root_values, normalised_split_probabilities,
stopping_transition_probabilities_sum_one, stopping_probability_ge_third.
Compared, runs 35624228171, 35643815458.
- Proposition 6.36: cylinderMass_children_le, sternBrocotDepthMass_error,
tendsto_sternBrocotDepthMass. *Compared, run 35715779405.*
- Proposition 6.37: mobiusNumeratorPolynomial_coeff,
mobiusNumeratorPolynomial_coeff_pos. *Compared, run 35624228171.*
- Proposition 6.38: mobiusNumeratorPolynomial_eval_two, numerator_eval_two_divisors.
Compared, run 35624228171.
- Proposition 6.39: radical_decomposition. *Compared, run 35624228171.*
- Proposition 6.40: cyclotomic_dvd_mobiusNumeratorPolynomial_sub,
mobiusNumerator_mod_cyclotomicEval, jordanTotientTwo_eq_prod_primeFactors.
Compared, run 35624228171.
- Proposition 6.41: cyclotomic_dvd_primeJump_new_fibre. *Compared, run 35624228171.*
- Proposition 6.42: cyclotomic_dvd_primeJump_old_fibre. *Compared, run 35624228171.*
- Proposition 6.43: lambertValue_eq_divisor_sum_series, alpha_divisor_sum_eq_totient,
lambertValue_alpha_eq_totientSeries. *Compared, runs 35624228171, 35643815458.*
- Proposition 6.45: transportResidueKernel_eq_mobiusTermKernel,
mobiusTermKernel_affine_in_multiplier, joint35ConeWindow_eq,
totientTail_enclosure, mobiusTermKernel_moment_annihilation,
joint35_mobiusTermKernel_zero, joint35_coefficient_moments,
joint35_truncation_error, joint35_nonintegral_of_separated_window. *Compared, runs 35624228171, 35643815458, 35674034595.*
- Proposition 6.46: composite_dilation_divisor_count,
composite_dilation_defect_eq_zero_of_prime_support,
composite_dilation_divisor_count_prime_support,

composite_dilation_defect_univ_six_one,
 composite_dilation_defect_univ_six_one_card, lambert_support_series,
 lambert_support_series_restricted, totient_convolution_weight_mul_zeta,
 sum_divisors_totient_ne_totient, totient_convolution_weight_prime,
 totient_convolution_weight_unbounded, totient_convolution_weight_not_periodic.

Compared, runs 35624228171, 35643815458.

Theorem 6.47: irrational_totient_series_iff_certificate_supply. *Compared, run* 35643815458.

Theorem 6.48: irrational_of_period_multiple_certificate_supply,
 period_multiple_certificate_at_one, rational_forces_period_multiple_integrality,
 period_multiple_certificate_supply_iff. *Compared, run* 35674034595.

Theorem 6.49: irrational_totient_series_iff_lcm_diagonal_certificate_supply.
Compared, run 35674034595.

Lemma 6.50: eq_prime_pow_of_not_dvd_periodLcm. *Compared, run* 35674034595.

Proposition 6.51: clean_lcm_ray_factorisation, unclean_lcm_ray_counterexample.
Compared, run 35674034595.

Theorem 6.52: lcm_grid_flatness, lcm_grid_fractional_parts,
 certificate_denominator_exclusion, lcm_grid_supply_iff. *Compared, run* 35674034595.

Theorem 6.53: lcm_grid_supply_iff, lcm_grid_multiplier_positive. *Compared, run* 35674034595.

Corollary 6.54: irrational_totient_series_iff_all_tail_diffs_nonintegral,
 periodLcm_diagonal_kill_iff_tail_diff_notMem_int,
 irrational_totient_series_of_lcm_diagonal_nonintegrality_supply,
 irrational_totient_series_of_lcm_cone_nonintegrality_supply. *Compared, runs* 35643815458, 35674034595.

Proposition 6.55: abs_tail_diff_scaled_sub_window_le, second_difference_error_bound,
 second_difference_certificate_sound, second_difference_cell_one_eight. *Compared, run* 35674034595.

Theorem 6.56: paperGridNumerator_eq, finite_grid_nonintegral_pair. *Compared, run* 35674034595.

Theorem 6.57: paperGridNumerator_eq, finite_grid_supply_irrational. *Compared, run* 35674034595.

Proposition 6.58: finite_carry_test_sound, finite_carry_candidate_count,
 finite_carry_true_orbit. *Compared, runs* 35643815458, 35674034595.

Theorem 6.59: irrational_of_first_harmonic_norm_gap,
 first_harmonic_re_bound_of_norm_bound,
 exists_certificate_of_first_harmonic_real_bound,
 exists_certificate_of_first_harmonic_norm_bound,
 nine_tenths_lt_cos_pi_div_eight. *Compared, runs* 35643815458, 35674034595.

Theorem 6.60: four_tail_combination_eq, four_tail_window_eq,
 windowDiscrepancy_diagonal_eq, four_tail_error_bound, totientTail_bounds,
 four_tail_criterion_sound, irrational_of_four_tail_supply,
 rational_forces_four_tail_diagonals_integral, four_tail_checked_instance.
Compared, run 35674034595.

Proposition 6.61: irrational_iff_diagonal_orbit_nonintegrality. *Compared, run* 35674034595.

Proposition 6.62: irrational_of_short_window_diagonal_supply,
 diagonal_certificate_unfolded, pointwise_completeness_supplies_some_depth.
Compared, run 35674034595.

- Proposition 6.63: `irrational_of_diagonal_orbit_separation_supply`, `oddGuarded_depth_eq_prescribed`, `abs_orbit_sub_rawApprox_lt`, `rawApprox_separation_of_orbit_separation`. *Compared*, run 35674034595.
- Proposition 6.64: `upper_endpoint_condition_iff`, `upper_endpoint_gap_nonintegral`, `integral_tail_forces_upper_endpoint_residue`, `irrational_of_upper_endpoint_gap_supply`. *Compared*, run 35674034595.
- Proposition 6.65: `te_chain_relations`. *Compared*, run 35674034595.
- Proposition 6.66: `endpoint_identity`, `integral_carry_strictly_between`, `endpoint_criterion_nonintegral`, `oddHalfCenteredLift_spec`, `centeredLift_range`. *Compared*, runs 35624228171, 35674034595.
- Proposition 6.67: `short_window_diagonal_through_six`, `short_window_diagonal_witnesses`. *Compared*, run 35674034595.
- Proposition 6.68: `extremal_order_curvature_pos`, `extremal_order_curvature_neg`, `extremal_order_curvature_ne_zero`. *Compared*, run 35643815458.
- Proposition 6.69: `directed_certificate_iff`, `directed_certificate_example`. *Compared*, run 35674034595.
- Proposition 6.70: `irrational_of_modFour_pulse_supply`, `rational_forces_pulse_class_integrality`. *Compared*, run 35674034595.
- Proposition 6.71: `irrational_totient_series_iff_certificate_supply`. *Compared*, run 35643815458.
- Proposition 6.72: `irrational_totient_series_iff_lcm_diagonal_certificate_supply`. *Compared*, run 35674034595.
- Proposition 6.73: `irrational_totient_series_of_lcm_cone_window_kill_supply`. *Compared*, run 35674034595.
- Proposition 6.74: `paperGridNumerator_eq`, `finite_grid_nonintegral_pair`, `finite_grid_supply_irrational`. *Compared*, run 35674034595.
- Proposition 6.75: `gapCertificate_window_1_240`, `irrational_totientSeries_of_fareyGapExclusionUnbounded`, `gapFareyBound_window_1_240`. *Compared*, run 35643815458.
- Proposition 6.76: `rank_floor_and_false_proposed_carryRank_ceiling`, `fiveQuarter_comparison_rational_with_carryRank_floor`, `no_generic_rationality_carryRank_ceiling`. *Compared*, runs 35624228171, 35674034595.
- Proposition 6.77: `finite_carry_test_sound`, `finite_carry_candidate_count`, `finite_carry_true_orbit`. *Compared*, runs 35643815458, 35674034595.
- Proposition 6.78: `lcmRayArithmeticLetter_eq_totient_difference`, `totient_mul_eq_totient_mul_gcd_div_totient_gcd`, `lcmRay_divisor_denominators_pos`, `lcmRay_divisor_product_formula`, `lcmRay_divisor_clean_formula`, `lcmRay_nondivisor_literal`, `lcmRay_divisor_gcd_example`. *Compared*, runs 35643815458, 35674034595.
- Proposition 6.79: `rough_integer_prime_count_and_totient_bound`. *Compared*, run 35643815458.
- Proposition 6.80: `shortWindow_totient_difference_pos`. *Compared*, run 35674034595.
- Proposition 6.81: `weighted_shortWindow_sum_eq_windowDiscrepancy`, `weighted_shortWindow_band_iff_certifiedKill`. *Compared*, run 35674034595.
- Proposition 6.82: `actualLcmTailOrbit_global_to_local`. *Compared*, run 35674034595.
- Proposition 6.83: `abs_actualLcmTailOrbit_sub_rawApprox_lt_explicit`, `tendsto_actualLcmRawErrorRadius_atTop_nhds_zero`. *Compared*, run 35674034595.
- Proposition 6.84: `actualLcm_tailDiff_shift_pos_paper`, `actualLcmTailOrbit_pos`. *Compared*, run 35674034595.

- Proposition 6.85: `actualLcm_integral_forces_topEdgeResidue_paper`. Compared, run 35674034595.
- Proposition 6.86: `penultimate_shortWindow_difference_eq_half`, `dyadicScale_unique_in_open_interval`. Compared, runs 35643815458, 35674034595.
- Proposition 6.87: `exists_certifiedKill_iff_twoBitResidueTest`, `dyadicMixedGuard_iff_twoBitBand`. Compared, runs 35643815458, 35674034595.
- Proposition 6.88: `two_mul_totient_dvd_totient_second_difference`, `fixedRank_cleanWindow_structure`. Compared, run 35674034595.
- Proposition 6.89: `carryShift_dvd_iff_tailDiff_integral`, `temperedCarry_eq_scaledTail_and_shift`. Compared, run 35624228171.
- Proposition 6.90: `rationality_forces_mod_period_and_unbounded_rank`. Compared, run 35624228171.
- Proposition 6.91: `exists_prime_totient_shift_four_mul_congr_two_mod_four`. Compared, run 35643815458.
- Proposition 6.92: `exists_prime_twoAdic_pulse_block`, `pulse_delta_of_divisor_data`, `exists_prime_integral_tailDiff_half_pulse`. Compared, run 35674034595.
- Proposition 6.93: `forwardMultiple_spec`, `forwardMultipleShift_least`, `forwardMultiple_enumeration`, `totientTail_eq_tsum_mobius_inversion`. Compared, run 35624228171.
- Proposition 6.94: `shortWindow_certificates_kill_omega_four_and_six`, `shortWindow_inequalities`, `actualLcmTailOrbit_eq_tail_difference`. Compared, runs 35643815458, 35674034595.
- Proposition 6.95: `certificate_logarithmic_depth`, `fixed_depth_bounds_indices`, `certifiedKill_depth_floor`. Compared, runs 35643815458, 35674034595.
- Proposition 6.96: `rational_tail_period_explicit_witnesses`, `totient_one_eq_one`, `eventual_tail_period_of_not_irrational`, `irrational_of_certificate_supply`, `tail_diff_notMem_int_of_irrational`, `exists_certifiedKill_iff_tail_diff_nonintegral`. Compared, runs 35643815458, 35674034595.
- Proposition 6.97: `lcm_grid_flatness`, `certificate_denominator_exclusion`, `lcm_grid_supply_iff`. Compared, run 35674034595.
- Proposition 6.98: `card_canonical_dyadic_index`, `linearIndependent_canonical_dyadic_family`, `canonical_family_basis_through_level`, `canonical_level_zero_channels`, `finrank_throughLevel_zero_eq_one`. Compared, run 35624228171.
- Proposition 6.99: `farey_window_1_240_exact_range`, `farey_window_1_240_range_magnitude`. Compared, run 35643815458.
- Proposition 6.100: `totient_series_ne_reduced_fraction_of_small_denominator`, `totient_series_ne_int_div_of_small_denominator`. Compared, run 35643815458.
- Proposition 6.101 (the Lean statement is stronger):
`irrational_of_den_mul_abs_sub_tendsto_zero`, `one_div_den_mul_den_le_abs_sub`,
`irrational_of_int_mul_near_int`, `irrational_of_pow_mul_near_int`,
`irrational_of_den_mul_error_tendsto_zero`, `irrational_of_dirichlet_gap`,
`one_div_den_mul_den_le_abs_diff`, `one_div_den_le_abs_int_combination`,
`irrational_of_basePower_dilation`, `basePower_dilation_not_universal`,
`irrational_xi`, `one_div_eight_le_dist_xi`. Compared, runs 35624228171, 35643815458.
- Proposition 6.103: `rational_gap_lower_bound`, `den_lower_bound_of_positive_error`,
`denominator_bound_tendsto_atTop_iff`,
`irrational_of_den_mul_error_product_tendsto_zero`,
`dyadic_prefix_denominator_bound_vacuous`, `dyadic_prefix_den_dvd`,

- `totientTail_le_add_two`, `dyadic_prefix_tail_le`. Compared, runs 35643815458, 35674034595.
- Proposition 6.104: `historical_table_size_and_initial_depths`, `historical_table_and_complete_band`. Compared, run 35674034595.
- Proposition 6.105: `card_visible_antidiagonal`, `totient_zero_eq_zero`, `visible_antidiagonal_one`, `positive_antidiagonal_one`, `tsum_pos_coprime_pairs_eq_series_sub_half`, `tsum_pos_coprime_pairs_product_form`. Compared, run 35643815458.
- Theorem (The values at exponents one and two): `mobiusMersenneTheta_one_and_two`. Compared, run 35643815458.
- Theorem (The first two summands give a positive Hankel gap): `twoAtom_hankel_gap`, `twoAtom_strict_logConcave`. Compared, run 35643815458.
- Theorem (Strict log-concavity for all integer $r \geq 1$): `theta_strict_logConcave`, `theta_hankel_det_neg`, `theta_hankel_two_neg`. Compared, run 35643815458.
- Theorem 6.125: `binaryCoeffSeries_rational_iff_exists_temperedBinaryOrbit`, `temperedBinaryOrbit_eq_scaledTail`. Compared, run 35643815458.
- Proposition (Uniqueness under the growth condition): `doubling_tempered_sequence_eq_zero`. Compared, run 35643815458.
- Proposition (An exact rational approximation formula): `tailDifference_sub_rationalApproximation`, `tailDifference_eq_coefficient_mul_series`, `totientSeries_pnat_form`, `totientSeries_eq_half_add_moebius_sq`. Compared, runs 35643815458, 35674034595.
- Proposition (A geometric tail bound): `abs_mobiusSquareTail_le_paper`, `abs_moebius_cast_le_one`, `mersenne_geometric_shift`, `tsum_quarter_geometric`. Compared, run 35643815458.
- Proposition (The sum over divisor indices): `sum_divisorIndices_mobius`, `sum_divisors_moebius_div_eq_totient_div`, `sum_divisorIndices_radical_form`, `squarefreeKernel_eq_prod_primeFactors`. Compared, runs 35624228171, 35643815458.
- Proposition (The divisor sum and its complement): `totientDifference_eq_divisorPart_add_complement`, `totient_eq_mobius_divisor_sum`, `divisorIndex_endpoint_behaviour`, `complementSummand_eq_phaseTerm`. Compared, run 35643815458.
- Theorem (Doubling the full totient difference): `totientDifference_doubling_seam`, `totient_two_mul_even`, `totient_two_mul_odd`. Compared, run 35643815458.
- Proposition (A doubling identity for two portions of the sum): `complementSummand_low_double_echo`. Compared, run 35643815458.
- Theorem (Positive coefficients of the numerator polynomial): `paperNumerator_eq_gcdWordForm`, `paperNumerator_coeff_pos`, `paperNumerator_coeff_eq_zero`, `paperNumerator_coeff_eq_gcd_divisor_sum`, `paperNumeratorPolynomial_eq`. Compared, run 35624228171.
- Proposition (The geometric remainder bound): `mersenneRemainder_identity_and_bound`, `one_sub_half_pow_ge`, `mersenneRemainderTail_le`, `tsum_eighth_pow_tail`, `four_thirds_tsum_eighth_pow_tail`. Compared, run 35624228171.
- Proposition (Squared distance from the phase one): `sum_sq_dist_from_phase_one`. Compared, run 35674034595.
- Lemma (Squared-distance bound for separated pairs): `card_mul_sq_le_pairwise_energy`. Compared, run 35643815458.
- Proposition (The finite divisor sum): `divisorChannels_sum_eq`, `residueOffset_of_dvd`, `residueKernel_increment_of_dvd`, `scaleExplicitShadow_eq_divisorChannels`. Compared, runs 35624228171, 35643815458.

Proposition (Separation larger than the error implies exclusion):

`tailDifference_not_integral_of_separation`, `projectedForeignDefect_paper`,
`foreignComplementBound_paper`. *Compared*, runs 35624228171, 35643815458.

Theorem (Coprime-pair counting and the totient): `card_coprime_antidiagonal`,
`boundary_pair_at_one`. *Compared*, run 35643815458.

Proposition (Two lattice sums): `tsum_totient_pow_shift`, `coprimeLattice_halfOpen_sum`,
`coprimeLattice_positive_sum`, `coprimeLattice_removed_pair`,
`coprimeLattice_gcd_layer_total`, `coprimeLattice_gcd_layer_total_eq_one_iff`.
Compared, run 35643815458.

Theorem (The classical coprime-pair Lambert identity): `coprimeLattice_lambert_identity`,
`coprimeLattice_lambert_rational`, `coprimeLattice_lambert_half_eq_one`,
`coprimeLattice_positive_sum`, `coprimeLattice_plain_half_eq_series_sub_half`,
`coprimeLattice_halfOpen_half_eq_series`. *Compared*, run 35643815458.

Theorem (Nondivisors in a short LCM window): `short_lcm_window_nondivisor`,
`clean_lcm_ray_factorisation`, `unclean_lcm_ray_counterexample`, `dvd_periodLcm`.
Compared, run 35674034595.

Theorem (Unbounded prime support in Mersenne factors):
`mersenneLayer_prime_divisor_order`, `mersenneLayer_unbounded_prime_support`.
Compared, run 35643815458.

Theorem (A prime satisfying the stated cyclotomic conditions):
`exists_clean_cyclotomic_anchor_paper`,
`cyclotomic_layer_prime_order_decomposition_paper`. *Compared*, run 35643815458.

Theorem (A sufficient order hypothesis for unbounded prime support):
`boundedDegreeOrderConsumer_unfolded`, `coprime_of_dvd_pow_sub_one`,
`dvd_pow_sub_one_iff_orderOf_dvd`, `boundedOrder_witness_iff_orderOf_le`,
`orderConsumer_index_lt_pow`, `orderConsumer_finite_prime_escape`,
`orderConsumer_unbounded_prime_divisors`,
`unbounded_prime_divisors_of_escape_of_nontrivial`,
`mersenneLayer_orderConsumer_instance`, `eventual_orderConsumer_conclusions`,
`binaryCyclotomicLayer_eventual_instance`, `binaryCyclotomic_allPrime_form_fails`.
Compared, run 35643815458.

Theorem (A uniform positive gap for rank-one quotients):
`rankOneSubrankQuotient_sub_theta_two_gt`, `mobiusMersenneTheta_ge_alpha`,
`mobiusMersenneTheta_lt_one`, `abs_mobiusMersenneTheta_sub_prefix_le`,
`mobiusMersenneTheta_two_eq_totient_offset`, `rankOne_denominator_pos`. *Compared*,
runs 35624228171, 35643815458, 35674034595.

Proposition (The bound is preserved by positive normalised averaging) (the Lean statement is stronger): `positive_direct_sum_sub_theta_two_gt`, `primitive_form_abs_gt`. *Compared*,
run 35643815458.

Theorem (Concatenation and specified period multiples):
`totientBlock_eq_paper_indexed_sum`, `totientBlock_concatenation`,
`totientBlock_doubling`, `cyclotomic_four_eval`,
`cyclotomic_four_two_pow_eq_cyclotomic_two`, `cyclotomic_three_two_pow`,
`orderOf_two_mod_seven`, `three_not_dvd_two_pow`,
`cyclotomic_three_eval_two_not_dvd_doubling_chain`. *Compared*, runs 35643815458,
35674034595.

Theorem 9.4: `windowFirstCos_unfolded`, `exists_certifiedKill_of_block_real_part_bound`.
Compared, run 35674034595.

Theorem 9.5: `exists_certifiedKill_of_subset_real_part_bound`,
`block_real_part_bound_of_subset_form`. *Compared*, run 35674034595.

- Theorem 9.6:** `blockNormCondition_unfolded`, `real_part_bound_of_norm_bound`, `exists_certifiedKill_of_block_norm_bound`, `irrational_of_blockNormCondition`. *Compared*, run 35674034595.
- Theorem (The proved implications):** `topEdgeResidueGap_unfolded`, `topEdgeResidueGap_forces_nonintegral`, `topEdgeResidueGap_orbit_nonintegral`, `irrational_of_topEdgeResidueGapSupply`. *Compared*, run 35674034595.
- Proposition 9.14:** `paperTeChain_item_one_unfolded`, `teChain_item_two_unfolded`, `teChain_item_three_unfolded`, `teChain_item_four_unfolded`, `teChain_item_five_unfolded`, `paperTeChain_five_sufficient_for_irrationality`, `paperTeChain_first_four_imply_topEdgeSupply`, `paperTeChain_fifth_gives_nonintegrality`, `paperTeChain_relations`, `topEdgeResidueGap_or_of_paperAdjacentSuffixMidband`, `powerTwoActualLcmTopEdgeResidueGapSupply_of_paperAdjacentSuffixMidband`, `irrational_of_paperAdjacentSuffixMidbandSupply`, `paperAdjacentSuffixMidbandSupply_of_oddGuardTopEdgeHalfWordBand`, `paperAdjacentSuffixMidbandSupply_of_flexibleActualTopEdgeMagnitude`, `paperAdjacentSuffixMidbandSupply_of_adjacentSuffixMidband`. *Compared*, run 35674034595.
- Theorem (The proved implication):** `terminalDominance_orbit_nonintegral`, `irrational_of_terminalDominanceSupply`. *Compared*, run 35674034595.
- Proposition (A sufficient extension):** `irrational_of_lower_escape_supply`, `corridor_escape_and_irrational_of_magnitude`, `teChain_item_four_unfolded`. *Compared*, run 35674034595.
- Proposition (A sufficient extension):** `irrational_of_logarithmicDepth_diagonal_supply`, `irrational_of_restrictedDepth_diagonal_supply`. *Compared*, run 35674034595.
- Proposition (Simultaneous certificates with unrestricted depth):** `exists_growingShift_simultaneous_certificate_iff_irrational`, `exists_simultaneous_depth_succ_of_irrational`, `exists_simultaneous_depth_of_irrational`. *Compared*, run 35674034595.
- Proposition (A sufficient extension):** `irrational_of_unbounded_window_one_gapCheck`, `gapCheck_window_one_excludes`. *Compared*, run 35643815458.
- Proposition (A sufficient accumulated-residue condition):** `irrational_of_accumulated_halfModulus_supply`, `certifiedKill_of_halfModulus_residue`. *Compared*, run 35674034595.
- Theorem (Arbitrarily large prime-power LCM jumps):** `exists_periodLcm_strict_jump_ge_paper`, `periodLcm_strict_jump_at_prime_pred`, `periodLcm_strict_jump_at_powerTwo_pred`, `periodLcm_zero_and_one_eq_one`. *Compared*, run 35674034595.
- Proposition (A sufficient inequality):** `irrational_of_powerTwo_postJump_slack_supply`, `canonicalAdjacentSuffixCentralSlack_paper_formula`. *Compared*, run 35674034595.
- Theorem (One explicit witness):** `periodLcm_four_eq_twelve`, `primeJump_witness_twelve_five_values`, `primeJumpTailCommutator_twelve_five_notMem_int`, `irrational_of_primeJumpSharp_supply`. *Compared*, run 35674034595.
- Proposition (The additional approximation hypothesis):** `irrational_totientSeries_of_rational_separation`, `den_mul_abs_sub_ge_one_div_den`. *Compared*, run 35643815458.
- Proposition (An additional hypothesis for a totient-specific rank argument) (the Lean statement is stronger):** `rationalControl_periodic_with_unbounded_carry_rank`, `rationality_gives_mod_period_and_unbounded_rank`. *Compared*, run 35624228171.

Proposition (A rational series preserving totient parity and the stated separation properties):

`parityCoboundaryWeight_le_six`, `parityCoboundaryWeight_le_self`,
`parityCoboundaryWeight_mod_two_eq_totient`,
`exists_later_arbitrarily_many_separated_parityCoboundaryWeight_carry_pairs`,
`tsum_parityCoboundaryWeight_eq_three_halves`. *Compared*, run 35624228171.

Lemma (Complement divisibility after multiplication): `complementDenominator_dvd_scalar`.
Compared, run 35643815458.

Lemma (Nonvanishing from a unique largest denominator exponent):

`signed_dyadic_clearing`, `signed_dyadic_sum_ne_zero`, `scaled_dyadic_sum_odd`.
Compared, runs 35624228171, 35643815458.

Proposition (A nonzero minor survives inverse-phase column weights):

`inversePhaseGauge_locks_row_and_preserves_minor`. *Compared*, run 35624228171.

Proposition (Tail integrality on an LCM grid): `lcm_grid_flatness`. *Compared*, run 35674034595.

Proposition (A finite-grid certificate gives a nonintegral pair): `paperGridNumerator_eq`,
`finite_grid_nonintegral_pair`. *Compared*, run 35674034595.

Theorem (Exact dyadic rank and infinite-dimensionality):

`canonicalTotientKernelFamily_entries`,
`canonicalTotientKernelFamily_independent_card_and_span`. *Compared*, run 35624228171.

Lemma 12.2 (the Lean statement is stronger): `orbit_tail_recurrence`, `orbit_tail_diff_eq`,
`orbit_tail_diff_sub_scaled_is_int`, `orbit_tail_diff_firstChar_eq`,
`orbit_tail_diff_fract_eq_doubling_orbit`, `doublingMap_iterate_apply`. *Compared*,
runs 35643815458, 35674034595.

Proposition 12.3: `irrational_totientSeries_of_block_cosine_gap`. *Compared*, run 35643815458.

Corollary 12.4: `irrational_totientSeries_of_quarterFarPhase_proportion`,
`quarterFarFromInt_iff_binaryDigitAt_change`,
`irrational_totientSeries_of_digitChange_count`,
`irrational_totientSeries_of_quarterFarPhase_count`,
`quarterFarFromInt_iff_floor_bounds`, `cos_nonpos_of_quarterFarFromInt`,
`tailOrbitFirstExp_re_eq`. *Compared*, run 35624228171.

Theorem 12.6 (the Lean statement is stronger): `lacCoef_bounds`,

`lacBeta_eq_factorial_series`, `irrational_lacBeta`, `lacunary_block_cos_gap`,
`lacunary_block_norm_fails`, `cos_pi_div_eight_gt`. *Compared*, run 35624228171.

Proposition 12.7 (the Lean statement is stronger): `prop_dickman`. *Compared*, run 35840809568.

Proposition 12.10: `certifiedKill_of_fullDepth_phase_separation`,
`bracket_of_two_sided_separation`. *Compared*, runs 35643815458, 35674034595.

Verification scope. Kernel checking validates a formal proof of its stated proposition relative to the declared axioms. It does not check that every prose summary has the same hypotheses or verify the literature comparisons. In particular, a checked implication with an unproved hypothesis is not a proof that Problem 249 is solved.

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References

- [1] J.-P. Allouche and J. Shallit, *The ring of k -regular sequences*, Theoret. Comput. Sci. **98** (1992), no. 2, 163–197, doi:[10.1016/0304-3975\(92\)90001-V](https://doi.org/10.1016/0304-3975(92)90001-V). Definitions and early statements are cited with the author preprint’s numbering.
- [2] M. Coons, *(Non)Automaticity of number theoretic functions*, J. Théor. Nombres Bordeaux **22** (2010), no. 2, 339–352, doi:[10.5802/jtnb.718](https://doi.org/10.5802/jtnb.718). Theorem 3.2, p. 348.
- [3] M. Coons, *Regular sequences and the joint spectral radius*, Internat. J. Found. Comput. Sci. **28** (2017), no. 2, 135–140, doi:[10.1142/S0129054117500095](https://doi.org/10.1142/S0129054117500095); arXiv:[1511.07535v1](https://arxiv.org/abs/1511.07535). Theorem 1, Proposition 4 and Corollary 7 are cited with the arXiv v1 numbering.
- [4] G. Martin, *Simultaneous inequalities among values of the Euler phi-function*, arXiv:[math/0603053v1](https://arxiv.org/abs/math/0603053), 2006. Theorem 1, pp. 1–2.
- [5] E. Wong, [answer 1211557](https://mathoverflow.net/a/1211557) to *An infinite sum based on the mod-parity of Euler’s totient function*, Mathematics Stack Exchange, 29 March 2015, accessed 15 September 2026.
- [6] A. Eldar, comment and formula added to OEIS A256936 (revisions 28 and 31), 15 March 2026, <https://oeis.org/history?seq=A256936>, accessed 16 September 2026.
- [7] S. Fan, comment on Erdős Problem #249, 16 May 2026, 19:01, [Erdős Problems discussion thread](https://mathoverflow.net/questions/478441/erdos-problem-249), accessed 16 September 2026.
- [8] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L’Enseignement Mathématique **28**, 1980, p. 61.
- [9] P. Erdős, *On arithmetical properties of Lambert series*, J. Indian Math. Soc. (N.S.) **12** (1948), 63–66, https://users.renyi.hu/~p_erdos/1948-04.pdf.
- [10] K. Postelmans and W. Van Assche, *Irrationality of $\zeta_q(1)$ and $\zeta_q(2)$* , J. Number Theory **126** (2007), no. 1, 119–154, doi:[10.1016/j.jnt.2006.11.011](https://doi.org/10.1016/j.jnt.2006.11.011); arXiv:[math/0604312v1](https://arxiv.org/abs/math/0604312). Theorem 1.3 is cited with the arXiv v1 pagination, p. 3.
- [11] Yu. V. Nesterenko, *Modular functions and transcendence questions*, Sb. Math. **187** (1996), no. 9, 1319–1348, doi:[10.1070/SM1996v187n09ABEH000158](https://doi.org/10.1070/SM1996v187n09ABEH000158). Corollary 2, p. 1320.
- [12] W. Schramm, *The Fourier transform of functions of the greatest common divisor*, Integers **8** (2008), #A50, <https://math.colgate.edu/~integers/i50/i50.pdf>. Theorem and equation (2), p. 2.
- [13] L. Tóth, *A survey of gcd-sum functions*, J. Integer Seq. **13** (2010), Article 10.8.1, <https://cs.uwaterloo.ca/journals/JIS/VOL13/Toth/toth10.pdf>. §1, equations (1)–(2).
- [14] M. Merca and M. D. Schmidt, *Generating special arithmetic functions by Lambert series factorizations*, Contrib. Discrete Math. **14** (2019), no. 1, 31–45, doi:[10.55016/ojs/cdm.v14i1.62425](https://doi.org/10.55016/ojs/cdm.v14i1.62425); arXiv:[1706.00393v2](https://arxiv.org/abs/1706.00393). Equation (1) is cited with the arXiv v2 pagination, p. 2.
- [15] M. Merca, *The Lambert series factorization theorem*, Ramanujan J. **44** (2017), 417–435, doi:[10.1007/s11139-016-9856-3](https://doi.org/10.1007/s11139-016-9856-3). Theorem 1.2, p. 420.
- [16] V. Kovač and T. Tao, *On several irrationality problems for Ahmes series*, Acta Math. Hungar. **175** (2025), no. 2, 572–608, doi:[10.1007/s10474-025-01528-0](https://doi.org/10.1007/s10474-025-01528-0); arXiv:[2406.17593v4](https://arxiv.org/abs/2406.17593). Theorem 2.3 is cited with the arXiv v4 pagination, p. 5.
- [17] R. Balasubramanian, S. Giri and P. Srivastav, *On correlations of certain multiplicative functions*, J. Number Theory **174** (2017), 221–238, doi:[10.1016/j.jnt.2016.10.001](https://doi.org/10.1016/j.jnt.2016.10.001); arXiv:[1511.02221v3](https://arxiv.org/abs/1511.02221). Theorem 2.2 is cited with the arXiv v3 pagination, p. 2.

- [18] A. Granville, *Smooth numbers: computational number theory and beyond*, in *Algorithmic Number Theory*, MSRI Publ. **44** (2008), 267–323, <https://library.slmath.org/books/Book44/files/09andrew.pdf>. Equations (1.1) and (1.3), p. 268.
- [19] I. Schoenberg, *Über die asymptotische Verteilung reeller Zahlen mod 1*, Math. Z. **28** (1928), 171–199, doi:[10.1007/BF01181156](https://doi.org/10.1007/BF01181156). §17, p. 193, for the distribution of $\varphi(n)/n$.
- [20] I. J. Schoenberg, *On asymptotic distributions of arithmetical functions*, Trans. Amer. Math. Soc. **39** (1936), no. 2, 315–330. Theorem 1, pp. 318–319; §8, p. 323.
- [21] N. Edeko, H. Kreidler and R. Nagel, *A dynamical proof of the van der Corput inequality*, Dyn. Syst. **37** (2022), no. 4, 648–665, doi:[10.1080/14689367.2022.2100244](https://doi.org/10.1080/14689367.2022.2100244); arXiv:[2106.11835v3](https://arxiv.org/abs/2106.11835). Theorem 2.1 is cited with the arXiv v3 pagination, p. 4.
- [22] K. Matomäki and M. Radziwiłł, *Multiplicative functions in short intervals*, Ann. of Math. **183** (2016), no. 3, 1015–1056, doi:[10.4007/annals.2016.183.3.6](https://doi.org/10.4007/annals.2016.183.3.6); arXiv:[1501.04585v4](https://arxiv.org/abs/1501.04585). Theorem 1 is cited with the arXiv v4 pagination, pp. 1–2.
- [23] T. Tao, *The logarithmically averaged Chowla and Elliott conjectures for two-point correlations*, Forum Math. Pi **4** (2016), e8, doi:[10.1017/fmp.2016.6](https://doi.org/10.1017/fmp.2016.6); arXiv:[1509.05422v4](https://arxiv.org/abs/1509.05422). Theorems 1.2–1.3 are cited with the arXiv v4 pagination, pp. 2 and 5.
- [24] T. Tao and J. Teräväinen, *Odd order cases of the logarithmically averaged Chowla conjecture*, J. Théor. Nombres Bordeaux **30** (2018), no. 3, 997–1015, doi:[10.5802/jtnb.1062](https://doi.org/10.5802/jtnb.1062); arXiv:[1710.02112v1](https://arxiv.org/abs/1710.02112). Theorem 1.1 is cited with the arXiv v1 pagination, p. 2.
- [25] B. Adamczewski and Y. Bugeaud, *On the complexity of algebraic numbers I. Expansions in integer bases*, Ann. of Math. **165** (2007), no. 2, 547–565. Theorem 1, p. 549.
- [26] G. Everest, A. J. van der Poorten, Y. Puri and T. Ward, *Integer sequences and periodic points*, J. Integer Seq. **5** (2002), Article 02.2.3. Lemma 3.1 is cited with the preprint pagination, p. 5.
- [27] P. Erdős and I. S. Gál, *On the law of the iterated logarithm. I*, Proc. Kon. Ned. Akad. Wetensch. Ser. A **58** = Indag. Math. **17** (1955), 65–76, https://www.renyi.hu/~p_erdos/1955-06.pdf.
- [28] S. Yazdani, *Multiplicative functions and k -automatic sequences*, J. Théor. Nombres Bordeaux **13** (2001), no. 2, 651–658, https://www.numdam.org/item/JTNB_2001__13_2_651_0/. Theorem 2 and its proof, pp. 652–653; Corollary 4, p. 654.
- [29] J.-P. Allouche, J. Shallit and R. Yassawi, *How to prove that a sequence is not automatic*, Exposition. Math. **40** (2022), no. 1, 1–22, doi:[10.1016/j.exmath.2021.08.001](https://doi.org/10.1016/j.exmath.2021.08.001); arXiv:[2104.13072v1](https://arxiv.org/abs/2104.13072). Theorem and example numbering follows the arXiv v1 (2021).
- [30] J. Bell and D. Smertnig, *Mahler series with multiplicative coefficient sequences*, arXiv:[2603.23456v1](https://arxiv.org/abs/2603.23456), 24 March 2026, preprint. Theorem 1.3, p. 2; Lemma 3.3, p. 9.
- [31] J. P. Bell, N. Bruin and M. Coons, *Transcendence of generating functions whose coefficients are multiplicative*, Trans. Amer. Math. Soc. **364** (2012), no. 2, 933–959, doi:[10.1090/S0002-9947-2011-05479-6](https://doi.org/10.1090/S0002-9947-2011-05479-6); arXiv:[1003.2221v2](https://arxiv.org/abs/1003.2221). Theorems 1.5–1.6 are cited with the arXiv v2 pagination, p. 2.
- [32] B. Adamczewski, M. Drmota and C. Müllner, *(Logarithmic) densities for automatic sequences along primes and squares*, Trans. Amer. Math. Soc. **375** (2022), no. 1, 455–499, doi:[10.1090/tran/8476](https://doi.org/10.1090/tran/8476); arXiv:[2009.14773v2](https://arxiv.org/abs/2009.14773). Theorem 1.4 and Remark 1.5, p. 4, and the proof at the end of Section 7, p. 19, refer to the arXiv v2 (2021).
- [33] B. Adamczewski, J. Bell and D. Smertnig, *A height gap theorem for coefficients of Mahler functions*, J. Eur. Math. Soc. **25** (2023), no. 7, 2525–2571, doi:[10.4171/JEMS/1244](https://doi.org/10.4171/JEMS/1244). Theorem 1.2(b), p. 2528; logarithmic height in Section 1.1.1, p. 2527.